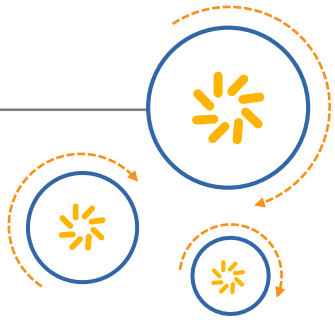
Qualcomm Technologies, Inc.



# QFE1100 xPT Software Design Guide

80-NG201-2 P

January 22, 2015

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# Revision history

| Revision | Date | Description |
|---|---|---|
| A | Jun 2013 | Initial release |
| B | Jul 2013 | Updated NV structure and added ESC Cal V4 algorithm |
| C | Jul 2013 | Updated Section 4.2.2.1 and Chapter 5 |
| D | Sep 2013 | Updated Sections 1.1, 2.1, 5.3.1, and 5.3.5, Figures 2-3, 4-12, 5-1, and 5-5, and Table C-1, and added Sections 3.2 and 4.3 |
| E | Sep 2013 | Updated Sections 5.3 and 5.5; added Section 2.6, Table 3-4, and Chapter 6 |
| F | Nov 2013 | Updated Sections 4.2, 5.2, and 5.4.1 and Figures 5-2, 5-5, 5-6, 5-15, and 5-20; added Section 4.2.5, Figures 5-8 and 5-13, and Table 4-2 |
| G | Nov 2013 | Updated Tables 1-1 and 4-2, Figures 3-1 and 3-2, Sections 2.3, 6.2.1, 6.2.5.1, and 7.2.4.1; added Sections 3.3, 3.5, 4.2.6, 4.2.7, 4.4, 5.3.1, 5.3.4, 5.4.4, 5.6, 6.2.4, 6.2.6.1, Table 3-10, Figures 3-8, 5-13 to 5-19, 6-15, and 6-16 |
| H | Dec 2013 | Updated Tables 3-2, 3-3, and 3-4 |
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| L | Feb 2014 | Updated Sections 4.1, 4.2.1.2, 4.4.2, 4.5, 5.1, 5.3.2.1, 5.3.2.2, 5.3.3, 6.3.1, 6.3.4, 6.3.6.1.1, and Table 4-2; added Sections 4.2.7, 4.6, 6.2, 6.2.22, and 6.2.3 |
| M | Jul 2014 | Updated Section 2.3, Figure 2-2, Table 3-1, Table 4-1, Section 4.1.2 to Section 4.1.4, Section 4.5; added references *Application Note: LTE B28 New Split Feature* (80-NM700-1), *QFE1101/QFE2101 APT Software Design Guide* (80-NG201-5), and *MPSS.BO.2.0 QFE Software Overview* (80-NN190-16); Section 3.2.4, 3.2.5; and Section 3.5.4 |
| N | Jul 2014 | Updated Sections 3.3.5 and 5.4.4; added Sections 3.2, 3.5, 3.8, 5.7.1, and Appendix E |
| P | Jan 2015 | Added Sections 3.5.13, 4.3, and 5.4.3; updated Section 4.5 and Table 3-1 |

**Note:** There is no Rev. I, O, Q, S, X, or Z per Mil. standards.

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# 1 Introduction

## 1.1 Purpose

This document describes the new Envelope Tracking (ET) design, which makes use of a QFE1100 to dynamically change the VCC to the Power Amplifier (PA) for reducing thermal effects and improving power consumption. Contents include calibration procedures, related NV items, and FTM APIs to enable this feature.

This document is intended for engineers integrating the ET RF feature with envelope tracking on MSM8974 and later chipsets.

## 1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `#include`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **`copy a:*.* b:`**.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

Shading indicates content that has been added or changed in this revision of the document.

## 1.3 Technical assistance

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If you do not have access to the CDMATech Support website, register for access or send email to support.cdmatech@qti.qualcomm.com.

# 2 System Design

## 2.1 System overview

Figure 2-1 is the overall system architecture.

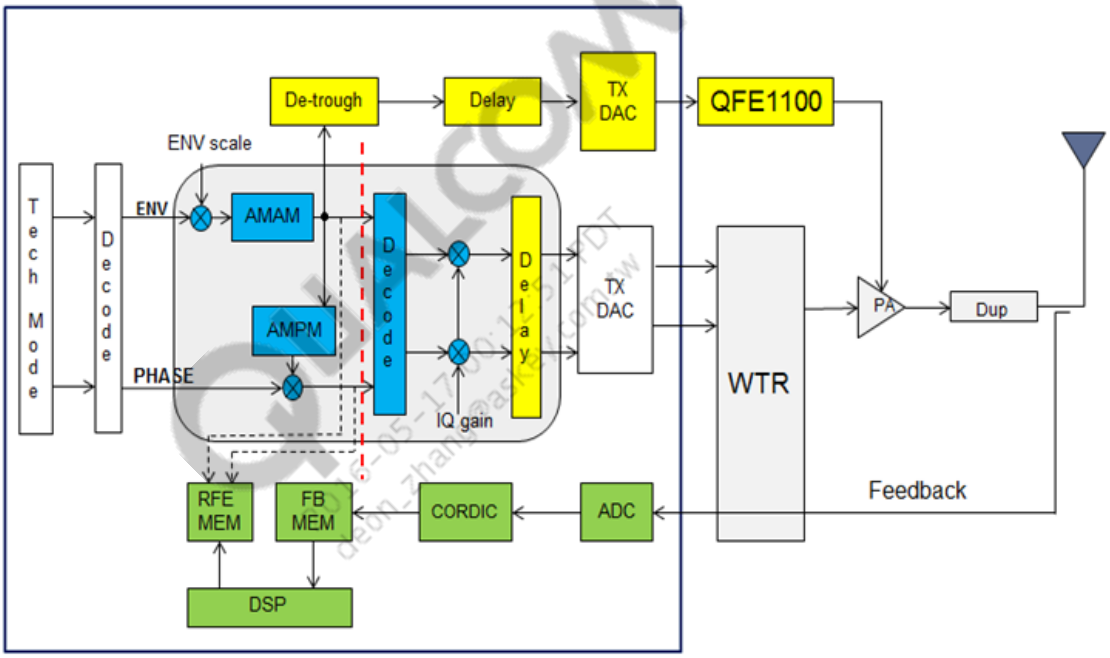



**Figure 2-1 System architecture**

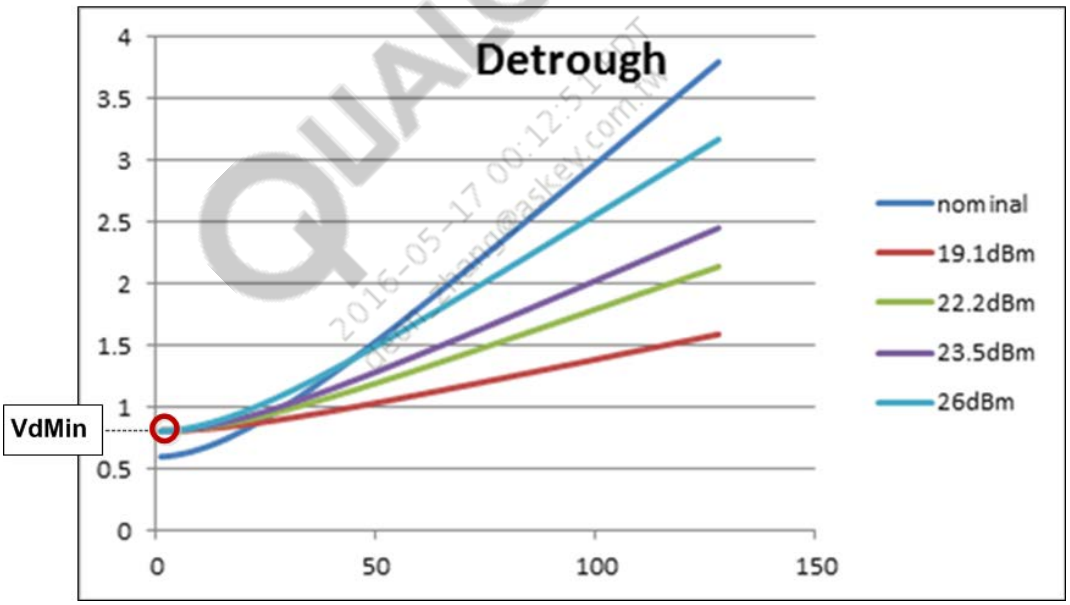
- ENV scale is used to adjust the envelope amplitude.
- IQ gain is used to adjust digital gain.
- Tx IQ capture is done after Digital Predistortion (DPD) (shown as a dotted red line in Figure 2-1).
- The output of AMAM is the input to the detrough block, AMPM, and decode.
- A detrough Lookup Table (LUT) is determined for each PA type/band through characterization, which is not phone dependent and which can be determined by the PA vendor alone.
- The output of detrough will go into a DAC, which goes to QFE1100 to generate PA bias for PA VCC input voltage.
- The feedback path is used for Rx IQ capture.
- Tx and Rx IQ captures are compared to construct the DPD table.

## 2.2 Simultaneous Tx and Rx IQ capture

- Tx capture
  - Envelope+phase after DPD (shown as a dotted red line in Figure 2-1)
- Tx capture+Rx capture to construct DPD
  - Load detrough
  - Program delay match (from characterization)
  - Simultaneously Tx and Rx capture
  - Construction DPD

## 2.3 Detrough table

An example of the detrough table is illustrated below. This is an LUT for envelope vs Vout. Based on the desired output power, Vmax is calculated from Magic Line (M-Line). A new detrough is scaled from the nominal detrough.




In the Nominal Detrough table, the maximum and minimum values are the QFE1100 maximum and minimum support voltages respectively. The VdMin value cannot be higher than 1.5 V.

## 2.4 xPT system flowchart per band

Customers need to fine-tune hardware matching before starting characterization. Hardware characterization includes three procedures: delay characterization (Section 5.3), RGI bias characterization (Section 5.2), and Pin Pout alignment characterization (Section 5.3.4). These procedures require the detrough table and ICQ value.

The detrough table, which is used only in the ET system, is from the PA vendor. The ICQ value is a DC current for PA operation to get best performance. This value is also from the PA vendor.

QTI provides tools to perform the three characterization procedures. The output of RGI bias characterization, delay characterization, and Pin Pout alignment characterization are used during calibration (see Chapter 4).

Figure 2-2 shows a system flowchart for xPT.

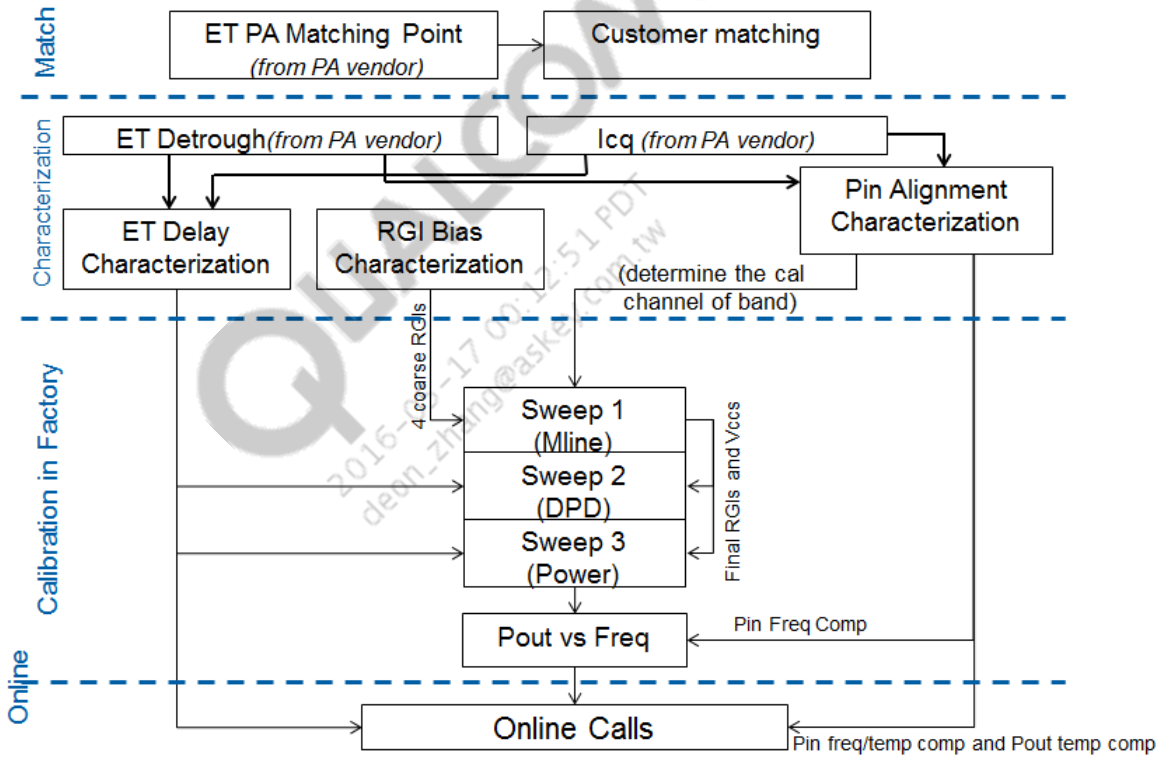



**Figure 2-2 xPT system flowchart**

# 2.5 QFE1100 ET/EPT

QFE1100 supports the following modes:

- Bypass mode – This fixes the VCC PA bias value.
- APT mode –The PA bias value can be adjusted based on Tx output power.
- Enhanced Power Tracking (EPT) mode – This is enhanced APT; a PA bias value that almost breaks ACLR is chosen.
- ET mode – Dynamic voltage by given a waveform can be provided.

Compare the power efficiency, as Figure 2-3 shows – ET mode→EPT mode→APT mode→ Bypass mode.

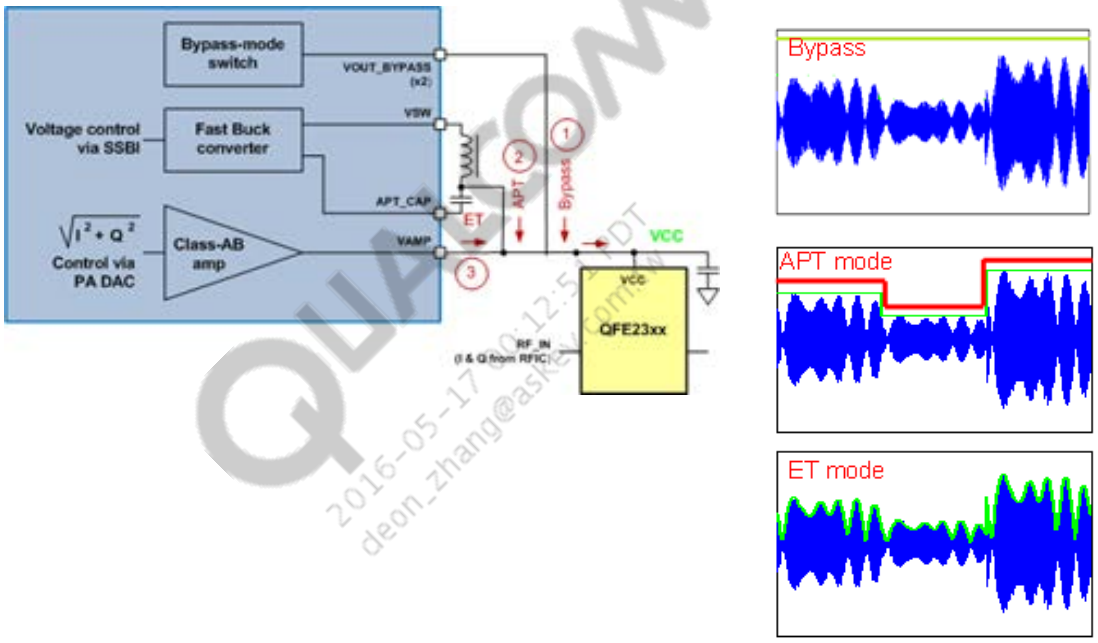



**Figure 2-3 QFE1100 operation mode**

A new MULTI_LIN_V3_NV, software, and calibration scheme is introduced to complete the design and implementation.

## 2.6 ET support status for SV and non-SV platforms

Table 2-1 lists an SV platform for 2 RF chips.

**Table 2-1 SV platform (2 RF chips)**

| | 1X | DO (SVDO) | DO (non-SVDO) | LTE |
|---|---|---|---|---|
| Operating chain | C2/C3 | C0/C1 | C2/C3 | C0/C1 |
| XPT support (standalone) | APT | ET | APT | ET |
| SVLTE mode | APT | — | — | Fall back to EPT |
| SVDO mode | APT | Fall back to EPT | — | — |

Table 2-2 lists non-SV platform for a single RF chip.

**Table 2-2 Non-SV platform (single RF chip)**

| | 1X | DO | LTE |
|---|---|---|---|
| Operating chain | C0/C1 | C0/C1 | C0/C1 |
| XPT support (standalone) | ET | ET | ET |

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# 3 Software Implementation

## 3.1 DPD configuration

There is one DPD configure NV item per band to store parameters for DPD software operation. All parameter changes should be discussed with QTI. Such changes must be characterized carefully in the QTI system design, and customers are not expected to make changes. Table 3-1 shows an example of how this NV appears.

**Table 3-1 Example for LTE DPD configuration NV item**

| Field offset | Name | Description |
|---|---|---|
| 0 | Xpt_config | ▪ 3 – User default<br>▪ 2 – ET<br>▪ 1 – EPT<br>▪ 0 – APT |
| 1 | Exp_scale | Reserved |
| 2 | Search_Center | Tx and Rx data are correlated in the window [searchCenter-searchWidth, searchCenter+searchWidth] |
| 3 | Search_Width | |
| 4 | Q_factor | Reserved |
| 5 | Lin_Tx_Mag | Reserved |
| 6 | Pm_order | Reserved |
| 7 | EPT_Env_scale_unity | EPT envelope scale – Provides freedom to adjust the tradeoff between power and ACLR/EVM; QTI has the recommended value, which customers cannot change |
| 8 | IQ_gain_Unity | IQ_gain |
| 9 | RGI_SEL_MAX_Power | Reserved |
| 10 | RGI_SEL_offset_low_0 | Reserved |
| 11 | RGI_SEL_offset_low_1 | Reserved |
| 12 | RGI_SEL_offset_low_2 | Reserved |
| 13 | REF_PIN_ET | Offset for ET Pin values |
| 14 | REF_PIN_EPT | Offset for EPT Pin values |
| 15 | RGI_SEL_offset_high_2 | Reserved |
| 16 | DPD_CFG_EXTRAP_BIN | DPD algorithm parameters for EPT and ET |
| 17 | DPD_CFG_FIRST _BIN | DPD algorithm parameters for EPT and ET |
| 18 | DPD_CFG_MIN_BIN_COUNT | DPD algorithm parameters for EPT and ET |
| 19 | DPD_CFG_MIN_FIRST_BIN_COUNT | DPD algorithm parameters for EPT and ET |
| 20 | DPD_CFG_MIN_LAST_BIN_COUNT | DPD algorithm parameters for EPT and ET |
| 21 | DPD_CFG_RX_COMPACT | FBRX |

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| Field offset | Name | Description |
|---|---|---|
| 22 | DPD_CFG_SPEC_INV | DPD algorithm parameters for EPT and ET |
| 23 | XPT_CFG_CMN_ENV_SCALE_ET_UNITY | ET envelope scale – Provides freedom to adjust the tradeoff between power and ACLR/EVM; QTI provides the recommended value, which cannot be changed |
| 24 | SELFTEST_CFG_SEARCH_CENTER | Correlation between Tx and Rx capture is needed in self-test EVM and VSWR processing |
| 25 | Reserved[8] | Reserved |
| 26 | Reserved[9] | Reserved |
| 27 | Reserved[10] | Reserved |
| 28 | Reserved[11] | Reserved |
| 29 | Reserved[12] | Reserved |
| 30 | Reserved[13] | Reserved |
| 31 | Reserved[14] | Reserved |

## 3.2 WMORE BB block clipper threshold for LTE

QFE33x5 with MSM8994/MDM9x35 enables the WMORE clipper in Signaling mode for LTE to help ACLR/RxBN/emission at the cost of EVM in only extreme cases. The WMORE threshold is applicable only to MPSS.BO (Bolt). See *MDM9x35+QFE Software Design Guide* (80-NK207-2). The block diagram of the WMORE clipper is shown in Figure 3-1.

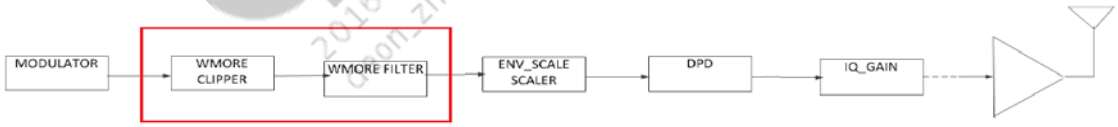



**Figure 3-1 Block diagram of WMORE clipper threshold**

The functionality of the WMORE clipper and filter is to clip the baseband signal so that the PAR is reduced before input to Env_Scale. The benefits of the WMORE clipper and filter are:

- Lowers the signal clipping threshold before Env_Scale without sacrificing out-of-band spectrum, thus ACLR/RxBN/emission can be improved with the same Env_Scale setting. However, the cost of the WMORE clipper is EVM degradation in extreme cases where TxAGC Env_scale back-off is small.
- Improves the average EVM by real-time clipping threshold adjustment in the Signaling mode for every individual band.

### 3.2.1 Clipper threshold

The WMORE clipping setting (available in MSM8974/MDM9x25 platforms) is hardcoded.

- CLIP_THRE_0 = 800
- CLIP_THRE_1 = 800
- BKOFTHRE_DYNAMIC_CLIP = 255

CLIP_THRE_0 is the clipper threshold for MPR0, CLIP_THRE_1 is the clipper threshold for MPR1 and beyond, and BKOFTHRE_DYNAMIC_CLIP is the env_scale back-off threshold (in 0.1 dB) for dynamic clipping threshold change.

The WMORE clipper is always enabled in FTM mode for characterization and calibration with a hardcoded value of 800.

To enable dynamic setting and customization for every individual band, new fields in EPT_DPD_NV are introduced to adjust the clipping threshold.

**Table 3-2 NV index in DPD_CONFIG_NV and default value**

| NV index | Field name | Sample value |
|---|---|---|
| 9 | CLIP_THRE_0 | 705 |
| 10 | CLIP_THRE_1 | 800 |
| 11 | BKOFTHRE_DYNAMIC_CLIP | 5 |

The value of CLIP_THRE_0/1 is the digital scale that describes how much clipping must be applied to the signal. The lower the threshold, the more clipping that must be applied in the baseband. Thus, the signal has smaller PAR but higher EVM. Since more clipping is introduced by the WMORE clipper with lower thresholds, less clipping is introduced with the same env_scale. Therefore, the ACLR margin can be improved. With enhanced WMORE clipping, we clip more for the 12RB signal to help improve the ACLR while the ACLR performance for a 50RB signal almost remains the same. The range of the clipping threshold can be from 650 to 1023, while 1023 indicates no clipping at all. It is recommended that the customer starts with a QTI-recommended setting and slightly modify the value for performance optimization.

## 3.2.2 Dynamic clipper threshold adjustment

The cost associated with WMORE clipping is the EVM degradation due to low PAR. To overcome this regression, the dynamic clipping threshold is introduced and is based on the amount of Env_Scale back-off. Because the Tx power in Signaling mode is based on Env_Scale back-off in ET mode, more headroom is left for Env_Scale to avoid clipping if the amount of Env_Scale back-off is large. If this occurs, the amount of clipping can be reduced in the WMORE clipper, which leads to a higher PAR for the output signal. To enable this dynamic adjustment, a threshold BKOFTHRE_DYNAMIC_CLIP (see Table 3-3) is introduced to allow less clipping if the Env_Scale back-off is beyond the threshold. As a result, the average EVM can be significantly improved by this dynamic threshold for clipping.


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Figure 3-2 illustrates the dynamic clipper threshold change.

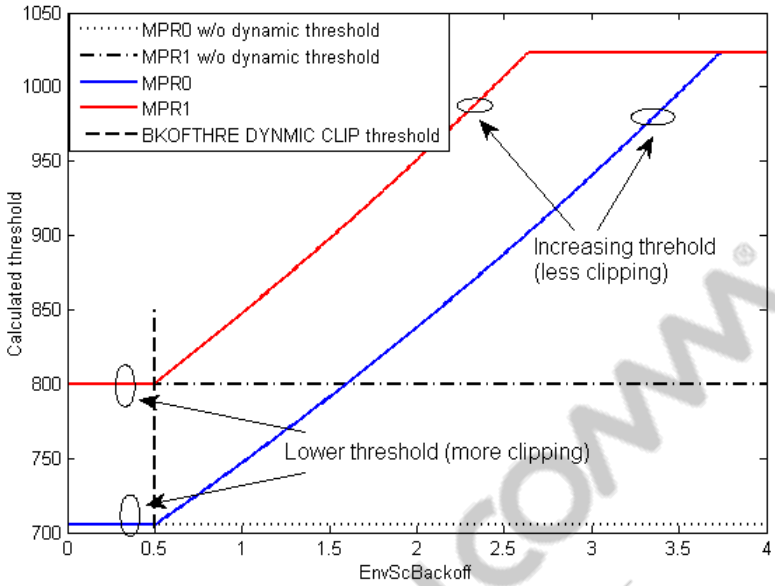



**Figure 3-2 Illustration of dynamic clipper threshold change**

If the Env_Scale back-off is small, the software retains a low clipper threshold so the output signal PAR is at a low level. When the Env_Scale back-off is large enough to pass the BKOFTHRE_DYNAMIC_CLIP threshold, the clipper threshold is increased, which reduces clipping to the signal. The clipper threshold increases to 1023 when there is no clipping. For the current ET design, Env_Scale back-off is used to cover the 2 dB to 3 dB power range. Therefore, as shown in Figure 3-2, the clipping threshold takes effect in most of the conditions (as the clipping threshold is less likely to increase to 1023 in most of the cases).

The dynamic clipping threshold can be modified by the customer. To disable the dynamic clipping, change BKOFTHRE_DYNAMIC_CLIP = 255. Thus, the software continues to use the clipping threshold in the NV.

### 3.2.3 Summary

- The suggested dynamic setting is already optimized by QTI. The customer can start using the suggested setting.
- The suggested setting enables the improvement of the ACLR, especially for 12RB signal.
- The WMORE clipping can be completed by changing the three elements in DPD_CONFIG NV to 0.
- If further modification is necessary, raise a case on the suggested setting.

# 3.3 Delay NV items

Three types of delay NV items are provided to control the delay time between modem-to-WTR path and modem-to-QFE1100 paths.

- Single delay
- Multiple delay
- Delay over frequency/temperature

For WCDMA and CDMA, customers can choose to use a single delay value or a multiple delay value. If both the multiple delay NV item and single delay NV item are enabled, the software applies the multiple delay values. The delay over frequency is used to compensate for the delay change over frequency based on the multiple delay value. Customers can decide to use it or not. When performing calibration, software reads delay value from the NV item.

## 3.3.1 Multiple delay NV items

The multiple delay NV items are:

- RFNV_LTE_<band>_TX_XPT_DELAY (Table 3-3)
- RFNV_WCDMA_<band>_TX_XPT_DELAY (Table 3-4)
- RFNV_CDMA_C2_<band>_TX_ET_DELAY_VS_FREQ (Table 3-5)

**Table 3-3 RFNV_LTE_<band>_TX_XPT_DELAY**

| Name | Description | Size |
|---|---|---|
| Num_active_entries | MSB bit indicates whether the UE has calibrated the delay value.<br>▪ 1 – Delay value is calibrated<br>▪ 0 – Delay value is from static NV<br>The other 7 bits represent the number of active entries, e.g., 144 means the UE has 16 active entries and the delay value is calibrated. | uint8 |
| Upper_Bound_channels | Array of 16 upper bounder channel | uint16 * 16 |
| Delay_LTEBW5Mhz | Array of 16 delay time | Int32 * 16 |
| Delay_LTEBW10Mhz | Array of 16 delay time for 10 MHz | Int32 * 16 |
| Delay_LTEBW20Mhz | Array of 16 delay time for 15 MHz and 20 MHz | Int32 * 16 |

**Table 3-4 RFNV_WCDMA_<band>_TX_XPT_DELAY**

| Name | Description | Size |
|---|---|---|
| Num_active_entries | MSB bit indicates whether the UE has calibrated the delay value.<br>▪ 1 – Delay value is calibrated<br>▪ 0 – Delay value is from static NV<br>The other 7 bits represent the number of active entries. | uint8 |
| Upper_Bound_channels | Array of 16 upper bounder channel | uint16 * 16 |
| Delay | Array of 16 delay time | Int32 * 16 |

**Table 3-5 RFNV_CDMA_C2_<band>_TX_ET_DELAY_VS_FREQ**

| Name | Description | Size |
|---|---|---|
| Num_active_entries | MSB bit indicates whether the UE has calibrated the delay value.<br>▪ 1 – Delay value is calibrated<br>▪ 0 – Delay value is from static NV<br>The other 7 bits represent the number of active entries. | uint8 |
| Upper_Bound_channels | Array of 16 upper bounder channel | uint32 * 16 |
| Delay | Array of 16 delay time | Int32 * 16 |

## 3.3.2 Single delay NV items

The single delay NV items are:

- RFNV_WCDMA_<band>_TX_ET_DELAY_I
- RFNV_CDMA_<Cx>_<band>_TX_ET_DELAY_I

The delay time is between the modem-to-WTR and modem-to-QFE1100 path. This NV item controls the delay time between the two paths in Online mode. The value must be the same as the value used in calibration and characterization. Data type is uint32.

## 3.3.3 Delay over frequency and temperature

The NV items for delay over frequency and temperature are:

- RFNV_LTE_Cx_<band>_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I
- RFNV_WCDMA_<band>_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I
- RFNV_CDMA_Cx_<band>_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I

The delay time is between the modem-to-WTR and modem-to-QFE1100 paths. This NV item controls the delay time between two paths in Online mode. The value must be the same as the value used in calibration and characterization. See Table 3-6.

**Table 3-6 Delay over frequency and temperature**

| Type | | Data | Size |
|---|---|---|---|
| Num_chans_Enable | | Number of enabled channels | uint8 |
| tx_chan[x] | | 16 channel list | unit16 |
| Temp[0] | Freq[0] | Offset value in minimum channels of interest | uint16 |
| | ... | ... | uint16 |
| | Freq[15] | Offset value in maximum channels of interest | uint16 |
| Temp[1] | Freq[0] | Offset value in minimum channels of interest | uint16 |
| | ... | ... | uint16 |
| | Freq[15] | Offset value in maximum channels of interest | uint16 |


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| Type | | Data | Size |
|---|---|---|---|
| Temp[2] | Freq[0] | Offset value in minimum channels of interest | uint16 |
| | ... | ... | uint16 |
| | Freq[15] | Offset value in maximum channels of interest | uint16 |
| ... | ... | ... | uint16 |
| Temp[7] | Freq[0] | Offset value in minimum channels of interest | uint16 |
| | ... | ... | uint16 |
| | Freq[15] | Offset value in maximum channels of interest | uint16 |

### 3.3.4 LTE split band

RFNV_LTE_B28_TX_XPT_DELAY_I contains the XPT delay values for all channels in LTE B28. Because there are overlapping channels with this method of band split, a new NV, RFNV_LTE_B28_B_C0_TX_XPT_DELAY_I, is introduced.

- RFNV_LTE_B28_TX_XPT_DELAY_I – This NV stores XPT delay char values for all possible channels through Dup A, irrespective of bandwidth.
- RFNV_LTE_B28_B_C0_TX_XPT_DELAY_I – This NV stores the XPT delay char values for all possible channels through Dup B, irrespective of bandwidth.

### 3.3.5 RFNV_LTE_<chain>_<band>_TX_XPT_BW_OFFSET_DELAY_I

A new delay, NV RFNV_LTE_<chain>_<band>_TX_XPT_BW_OFFSET_DELAY_I, contains the offset of the delay value with bandwidth other than 10 MHz. Note that 1.4 MHz and 3 MHz delays use the delay value of 5 MHz delay offset. Therefore, the 5 MHz delay offset value is duplicated three times in the NV. If this bandwidth offset NV does not exist in the phone, the software reads the absolute delay value for 5 MHz and 20 MHz in the TX_XPT_DELAY NV. The structure of the NV is summarized in Table 3-7. The NV is for Bolt only. See *MDM9x35+QFE Software Design Guide* (80-NK207-2) for more information.

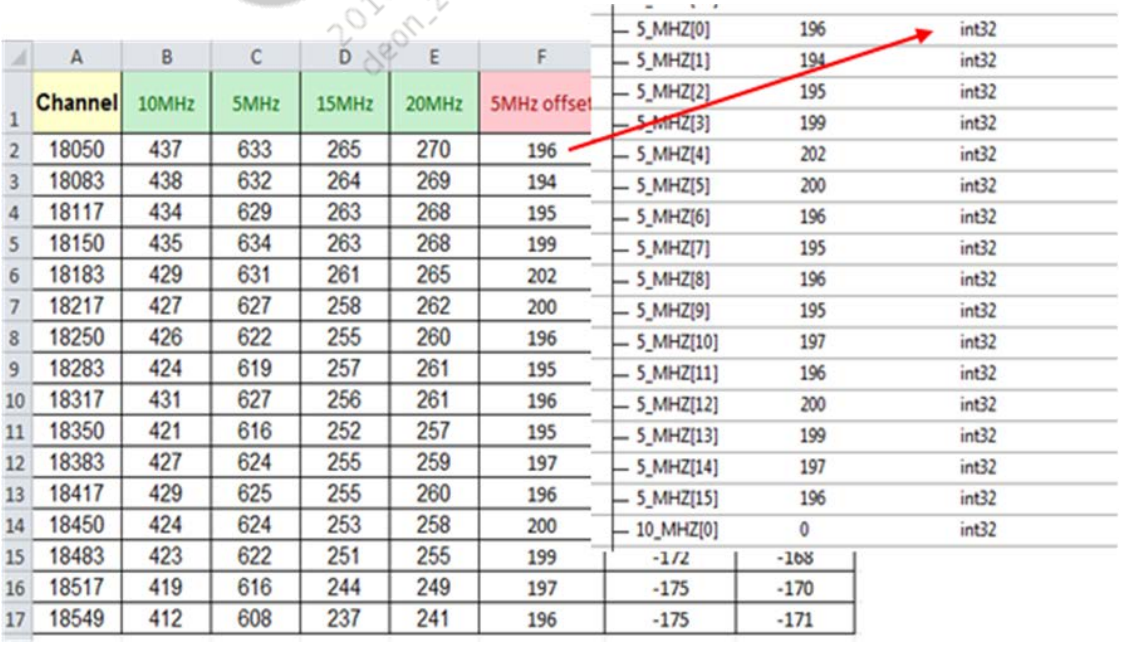
**Table 3-7 Structure of RFNV_LTE_<chain>_<band>_TX_XPT_BW_OFFSET_DELAY_I**

| Name | Description | Size |
|---|---|---|
| Header | Reserved | Int16*8 |
| 1.4 MHz delay offset | ET delay offset for 1.4 MHz in 0.1 ns with respect to 10 MHz | Int32*16 |
| 3 MHz delay offset | ET delay offset for 3 MHz in 0.1 ns with respect to 10 MHz | Int32*16 |
| 5 MHz delay offset | ET delay offset for 5 MHz in 0.1 ns with respect to 10 MHz | Int32*16 |
| 10 MHz delay offset | ET delay offset for 10 MHz in 0.1 ns with respect to 10 MHz | Int32*16 |
| 15 MHz delay offset | ET delay offset for 15 MHz in 0.1 ns with respect to 10 MHz | Int32*16 |
| 20 MHz delay offset | ET delay offset for 20 MHz in 0.1 ns with respect to 10 MHz | Int32*16 |
| 40 MHz delay offset | ET delay offset for 40 MHz in 0.1 ns with respect to 10 MHz | Int32*16 |

Currently, the delay offset NV needs to be filled by manual calculation. For example, a 5 MHz offset is calculated by 5 MHz delay minus 10 MHz delay. As shown in the illustration below, at channel 18050, the delay offset for 5 MHz is 633 to 437 = 196. For LTE 1.4 and LTE 3, the delay offset NV is the same as LTE 5. The value for LTE 5 can be directly copied to 1.4 MHz and 3 MHz bandwidth.

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Channel | 10MHz | 5MHz | 15MHz | 20MHz | 5MHz offset | 15MHz offset | 20MHz offset | |
| 2 | 18050 | 437 | 633 | 265 | 270 | 196 | -172 | -167 | |
| 3 | 18083 | 438 | 632 | 264 | 269 | 194 | -174 | -169 | |
| 4 | 18117 | 434 | 629 | 263 | 268 | 195 | -171 | -166 | |
| 5 | 18150 | 435 | 634 | 263 | 268 | 199 | -172 | -167 | |
| 6 | 18183 | 429 | 631 | 261 | 265 | 202 | -168 | -164 | |
| 7 | 18217 | 427 | 627 | 258 | 262 | 200 | -169 | -165 | |
| 8 | 18250 | 426 | 622 | 255 | 260 | 196 | -171 | -166 | |
| 9 | 18283 | 424 | 619 | 257 | 261 | 195 | -167 | -163 | |
| 10 | 18317 | 431 | 627 | 256 | 261 | 196 | -175 | -170 | |
| 11 | 18350 | 421 | 616 | 252 | 257 | 195 | -169 | -164 | |
| 12 | 18383 | 427 | 624 | 255 | 259 | 197 | -172 | -168 | |
| 13 | 18417 | 429 | 625 | 255 | 260 | 196 | -174 | -169 | |
| 14 | 18450 | 424 | 624 | 253 | 258 | 200 | -171 | -166 | |
| 15 | 18483 | 423 | 622 | 251 | 255 | 199 | -172 | -168 | |
| 16 | 18517 | 419 | 616 | 244 | 249 | 197 | -175 | -170 | |
| 17 | 18549 | 412 | 608 | 237 | 241 | 196 | -175 | -171 | |
| 18 | | | | | | | | | |
| 19 | | | | | | | | | |

After the offset calculation is done, write the offset value into the NV as shown.



| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | Channel | 10MHz | 5MHz | 15MHz | 20MHz | 5MHz offset |
| 2 | 18050 | 437 | 633 | 265 | 270 | 196 |
| 3 | 18083 | 438 | 632 | 264 | 269 | 194 |
| 4 | 18117 | 434 | 629 | 263 | 268 | 195 |
| 5 | 18150 | 435 | 634 | 263 | 268 | 199 |
| 6 | 18183 | 429 | 631 | 261 | 265 | 202 |
| 7 | 18217 | 427 | 627 | 258 | 262 | 200 |
| 8 | 18250 | 426 | 622 | 255 | 260 | 196 |
| 9 | 18283 | 424 | 619 | 257 | 261 | 195 |
| 10 | 18317 | 431 | 627 | 256 | 261 | 196 |
| 11 | 18350 | 421 | 616 | 252 | 257 | 195 |
| 12 | 18383 | 427 | 624 | 255 | 259 | 197 |
| 13 | 18417 | 429 | 625 | 255 | 260 | 196 |
| 14 | 18450 | 424 | 624 | 253 | 258 | 200 |
| 15 | 18483 | 423 | 622 | 251 | 255 | 199 |
| 16 | 18517 | 419 | 616 | 244 | 249 | 197 |
| 17 | 18549 | 412 | 608 | 237 | 241 | 196 |

| | | |
|---|---|---|
| 5_MHZ[0] | 196 | int32 |
| 5_MHZ[1] | 194 | int32 |
| 5_MHZ[2] | 195 | int32 |
| 5_MHZ[3] | 199 | int32 |
| 5_MHZ[4] | 202 | int32 |
| 5_MHZ[5] | 200 | int32 |
| 5_MHZ[6] | 196 | int32 |
| 5_MHZ[7] | 195 | int32 |
| 5_MHZ[8] | 196 | int32 |
| 5_MHZ[9] | 195 | int32 |
| 5_MHZ[10] | 197 | int32 |
| 5_MHZ[11] | 196 | int32 |
| 5_MHZ[12] | 200 | int32 |
| 5_MHZ[13] | 199 | int32 |
| 5_MHZ[14] | 197 | int32 |
| 5_MHZ[15] | 196 | int32 |
| 10_MHZ[0] | 0 | int32 |

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# 3.4 Pin/Pout NV items

These Pin/Pout NV items are similar to TX_LIN_VS_TEMP in legacy design. For ET systems, Cal V4 is a requirement that will store the calibration data to the MULTI_LIN_V3 NV item. This NV item is compatible with Pin/Pout NV in ET mode and the legacy TX_LIN_VS_TEMP NV in APT mode for temperature compensation. For legacy APT Cal (Cal V3), the calibration data is stored to the MULTI_LIN NV item. This NV item only works with the legacy TX_LIN_VS_TEMP NV item in APT mode. See Section 3.7.

## 3.4.1 <Tech>_<Band>_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I

This NV item is used for frequency and temperature compensation in PA power input for each PA state (R0, R1, R2, and R3). Table 3-8 lists the NV structure. Each PA state has its frequency offset value for each of eight temperature values, all at reference frequencies. The data is from Pin Pout alignment characterization.

**Table 3-8 <Tech>_<Band>_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I**

| Type | | | Data | Size |
|---|---|---|---|---|
| Enable | | | Reserve | uint8 |
| PA_state[x] | Temp[0] | Freq[0] | Offset value in maximum temperature of interest | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value in maximum temperature of interest | uint16 |
| | Temp[1] | Freq[0] | Offset value in maximum temperature of interest | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value at certain temperature | uint16 |
| | Temp[2] | Freq[0] | Offset value at certain temperature | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value at certain temperature | uint16 |
| | ... | ... | ... | uint16 |
| | Temp[7] | Freq[0] | Offset value in minimum temperature of interest | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value in minimum temperature of interest | uint16 |

### 3.4.2 <Tech>_<Band>_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I

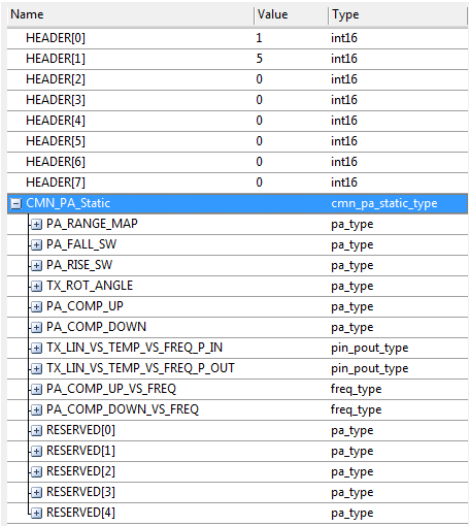
This NV item is used for frequency and temperature compensation in PA power input for each PA state (R0, R1, R2, and R3). Table 3-9 lists the NV structure. Each PA state has its frequency offset value for each of eight temperature values, all at reference frequencies. The data is from Pin Pout alignment characterization.

**Table 3-9 <Tech>_<Band>_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I**

| Type | | | Data | size |
|---|---|---|---|---|
| Enable | | | Reserve | uint8 |
| PA_state[x] | Temp[0] | Freq[0] | Offset value in maximum temperature of interest | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value in maximum temperature of interest | uint16 |
| | Temp[1] | Freq[0] | Offset value at certain temperature | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value at certain temperature | uint16 |
| | Temp[2] | Freq[0] | Offset value at certain temperature | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value at certain temperature | uint16 |
| | ... | ... | ... | uint16 |
| | Temp[7] | Freq[0] | Offset value in minimum temperature of interest | uint16 |
| | | ... | ... | uint16 |
| | | Freq[15] | Offset value in minimum temperature of interest | uint16 |

## 3.5 RFNV_<Tech>_<Chain>_<Band>_PA_STATIC_I

A new PA static NV item is introduced for CDMA, WCDMA, LTE, and TD-SCDMA technologies. The new static NV replaces numerous legacy PA-related NVs. The NV is for Bolt only.



| Name | Value | Type |
|---|---|---|
| HEADER[0] | 1 | int16 |
| HEADER[1] | 5 | int16 |
| HEADER[2] | 0 | int16 |
| HEADER[3] | 0 | int16 |
| HEADER[4] | 0 | int16 |
| HEADER[5] | 0 | int16 |
| HEADER[6] | 0 | int16 |
| HEADER[7] | 0 | int16 |
| CMN_PA_Static | | cmn_pa_static_type |
| PA_RANGE_MAP | | pa_type |
| PA_FALL_SW | | pa_type |
| PA_RISE_SW | | pa_type |
| TX_ROT_ANGLE | | pa_type |
| PA_COMP_UP | | pa_type |
| PA_COMP_DOWN | | pa_type |
| TX_LIN_VS_TEMP_VS_FREQ_P_IN | | pin_pout_type |
| TX_LIN_VS_TEMP_VS_FREQ_P_OUT | | pin_pout_type |
| PA_COMP_UP_VS_FREQ | | freq_type |
| PA_COMP_DOWN_VS_FREQ | | freq_type |
| RESERVED[0] | | pa_type |
| RESERVED[1] | | pa_type |
| RESERVED[2] | | pa_type |
| RESERVED[3] | | pa_type |
| RESERVED[4] | | pa_type |

## 3.5.1 Naming convention

The NV naming convention for the new PA static NV is RFNV_<Tech>_<Chain>_<Band>_PA_STATIC_I.

**Table 3-10 Sample NV names**

| Tech | NV |
|---|---|
| CDMA | RFNV_CDMA_C0_BC0_PA_STATIC_I |
| | RFNV_CDMA_C2_BC0_PA_STATIC_I |
| WCDMA | RFNV_WCDMA_C0_2100_PA_STATIC_I |
| LTE | RFNV_LTE_C0_B1_PA_STATIC_I |
| | RFNV_LTE_C1_B1_PA_STATIC_I |
| TD-SCDMA | RFNV_TDSCDMA_C0_B34_PA_STATIC_I |

## 3.5.2 NV structure

The first part of the NV structure is shared by all technologies (CDMA, WCDMA, LTE, and TD-SCDMA). Each column represents values for a particular PA state (0 to 7). The common part of the NV structure is the first 300 rows.

| | | | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Header | | 1 | 4 | | | | | | |
| 2 | PA_RANGE_MAP | | 3 | 2 | 2 | 0 | 0 | 0 | 0 | 0 |
| 3 | PA_FALL_SW | | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 4 | PA_RISE_SW | | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 5 | TX_ROT_ANGLE | | 00 | 01 | 10 | 11 | 0 | 0 | 0 | 0 |
| 6 | PA_COMP_UP | | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 7 | PA_COMP_DOWN | | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 8 | TX_LIN_VS_TEMP_VS_FREQ_P_IN | | | | | | 0 | 0 | 0 | 0 |
| | Temp[0] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | 8x16 | .... | | | | | 0 | 0 | 0 | 0 |
| | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| | Temp[1] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | .... | | | | | 0 | 0 | 0 | 0 |
| | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| | ... | ... | | | | | 0 | 0 | 0 | 0 |
| | Temp[7] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | .... | | | | | 0 | 0 | 0 | 0 |
| 135 | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| 136 | TX_LIN_VS_TEMP_VS_FREQ_P_OUT | | | | | | 0 | 0 | 0 | 0 |
| | Temp[0] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | 8x16 | .... | | | | | 0 | 0 | 0 | 0 |
| | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| | Temp[1] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | .... | | | | | 0 | 0 | 0 | 0 |
| | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| | ... | ... | | | | | 0 | 0 | 0 | 0 |
| | Temp[7] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | .... | | | | | 0 | 0 | 0 | 0 |
| 263 | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| 264 | PA_COMP_UP_VS_FREQ | | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| | 16 | Freq[0] | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| | | Freq[1] | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| | | ... | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 279 | | Freq[15] | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 280 | PA_COMP_DOWN_VS_FREQ | | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| | 16 | Freq[0] | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| | | Freq[1] | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| | | ... | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 295 | | Freq[15] | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 296 | Reserved | | | | | | | | | |
| 297 | 5 | | | | | | | | | |
| 298 | | | | | | | | | | |
| 299 | | | | | | | | | | |
| 300 | | | | | | | | | | |

The second part of the NV structure is CDMA-specific switchpoints. Rows 300 to 332 will only be present in CDMA NVs.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 301 | 1x_Access_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 302 | 1x_Access_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 303 | 1x_DATA_rate1_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 304 | 1x_DATA_rate1_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 305 | 1x_DATA_rate2_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 306 | 1x_DATA_rate2_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 307 | 1x_DATA_rate3_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 308 | 1x_DATA_rate3_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 309 | 1x_DATA_rate4_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 310 | 1x_DATA_rate4_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 311 | 1x_DATA_rate5_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 312 | 1x_DATA_rate5_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 313 | 1x_DATA_rate6_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 314 | 1x_DATA_rate6_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 315 | HDR_Access_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 316 | HDR_Access_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 317 | HDR_DOR0_RRI0_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 318 | HDR_DOR0_RRI0_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 319 | HDR_DOR0_RRI1_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 320 | HDR_DOR0_RRI1_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 321 | HDR_DOR0_RRI2_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 322 | HDR_DOR0_RRI2_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 323 | HDR_DOR0_RRI3_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 324 | HDR_DOR0_RRI3_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 325 | HDR_DOR0_RRI4_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 326 | HDR_DOR0_RRI4_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 327 | HDR_DOR0_RRI5_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 328 | HDR_DOR0_RRI5_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 329 | HDR_DORA_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 330 | HDR_DORA_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 331 | HDR_DORB_MC_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 332 | HDR_DORB_MC_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |

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## 3.5.3 NV size

The NV size is summarized in Table 3-11.

**Table 3-11 NV size for different technologies.**

| Technology | NV size |
|---|---|
| CDMA | 332 rows x 8 columns x 2 bytes (int16) = 5312 bytes |
| WCDMA | 300 rows x 8 columns x 2 bytes (int16) = 4800 bytes |
| LTE | 300 rows x 8 columns x 2 bytes (int16) = 4800 bytes |
| TD-SCDMA | 300 rows x 8 columns x 2 bytes (int16) = 4800 bytes |

### 3.5.4 Header information

The first row with eight elements provides the header information. The first element is NV version number and the second element denotes the number of active PA states (4 for legacy and 5 for SSKU ET, etc).

| Header | Details |
|---|---|
| 0 | Version No. |
| 1 | Active PA states |
| 2 | Reserved |
| 3 | Reserved |
| 4 | Reserved |
| 5 | Reserved |
| 6 | Reserved |
| 7 | Reserved |

### 3.5.5 PA range map

The second row denotes the PA_Range_Map values for the active PA states. The values denote physical PA range that is programmed in the device.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2 | PA_RANGE_MAP | 3 | 2 | 2 | 0 | 0 | 0 | 0 | 0 |

This part replaces legacy NV items, e.g.:

- NV_BCy_PA_R_MAP_I, RFNV_CDMA_C2_BCy_PA_R_MAP_I
- NV_WCDMA<band>_PA_RANGE_MAP_I
- NV_LTE_<band>_PA_RANGE_MAP_I
- RFNV_TDSCDMA_<band>_PA_RANGE_MAP_I

### 3.5.6 PA switchpoints and timer hysteresis values

The third and fourth rows denote the PA fall/rise switchpoints along with time hysteresis values (first element of the pa_fall_sw row).

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 3 | PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 4 | PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |

The switchpoint values are in dBm10 units for all technologies; e.g., a value of 135 denotes 13.5 dBm.

The timer hysteresis value is in µs. FED needs timer hysteresis in terms of the number of TxAGC counts to wait. The respective technology makes the conversion.Table 3-12 shows typical timer hysteresis values.

**Table 3-12 Timer hysteresis values**

| Technology | Timer hysteresis | FED units |
|---|---|---|
| CDMA | 2000 µs | TBD |
| WCDMA | 6670 µs | 6670 => 10 slots => 20 TxAGC counts |
| LTE | 2000 µs | 2000 => 2000/500 => 4 TxAGC counts |
| TD-SCDMA | TBD | TBD |

CDMA switchpoint NV items (row 301 to 332) have the same format as rise/fall switchpoints.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 301 | 1x_Access_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 302 | 1x_Access_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 303 | 1x_DATA_rate1_PA_FALL_SW | HY | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 304 | 1x_DATA_rate1_PA_RISE_SW | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |

This part replaces legacy NV items, e.g.:

- RFNV_CDMA_Cx_BCy_1X_RX_SWITCHPOINTS_I
- RFNV_CDMA_Cx_BCy_1X_TX_VOICE_SWITCHPOINTS_I
- RFNV_CDMA_Cx_BCy_1X_TX_DATA_SWITCHPOINTS_I
- NV_WCDMA<band>_R<x>_RISE_I
- NV_WCDMA<band>_R<x>_FALL_I
- NV_WCDMA<band>_TIMER_HYSTERISIS_I
- NV_LTE_<band>_TIMER_HYSTERISIS_I
- NV_LTE_<band>_PA_RISE_FALL_THRESHOLD_I
- RFNV_TDSCDMA_<band>_PA_RISE_FALL_I
- RFNV_TDSCDMA_<band>_TIMER_HYSTERISIS_I

## 3.5.7 Tx rotator angle values

The fifth row denotes TX_ROT_ANGLE NV values.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 5 | TX_ROT_ANGLE | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

This part replaces legacy NV items, e.g.:

- NV_WCDMA<band>_TX_ROT_ANGLE_PA_STATE_00_I
- NV_WCDMA<band>_TX_ROT_ANGLE_PA_STATE_01_I
- NV_WCDMA<band>_TX_ROT_ANGLE_PA_STATE_10_I
- NV_WCDMA<band>_TX_ROT_ANGLE_PA_STATE_11_I
- NV_LTE_<band>_TX_ROT_ANGLE_PA_STATE_I

- RFNV_TDSCDMA_<band>_TX_ROT_ANGLE_PA_STATE_I

## 3.5.8 PA compensate NVs for ILPC performance

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 6 | PA_COMP_UP | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
| 7 | PA_COMP_DOWN | | R1 | R2 | R3 | R4 | R5 | R6 | R7 |

This part replaces legacy NV items, e.g.:

- NV_WCDMA_<band>_PA_COMPENSATE_UP_I
- NV_WCDMA_<band>_PA_COMPENSATE_UP_R2_I
- NV_WCDMA_<band>_PA_COMPENSATE_UP_R3_I
- NV_WCDMA_<band>_PA_COMPENSATE_DOWN_I
- NV_WCDMA_<band>_PA_COMPENSATE_DOWN_R2_I
- NV_WCDMA_<band>_PA_COMPENSATE_DOWN_R3_I
- NV_LTE_<band>_PA_COMPENSATE_UP_DOWN_I
- RFNV_TDSCDMA_<band>_PA_COMPENSATE_UP_DOWN_I

## 3.5.9 Tx linearizer Pin/Pout values

Rows 8 through 135 contain Pin values per temperature, per frequency for 8 PA states. Similarly, rows 136 through 263 contain Pout values. The unused entries are filled with zeroes.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 8 | TX_LIN_VS_TEMP_VS_FREQ_P_IN | | | | | | 0 | 0 | 0 | 0 |
| | Temp[0] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | 8x16 .... | | | | | 0 | 0 | 0 | 0 |
| | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| | Temp[1] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | .... | | | | | 0 | 0 | 0 | 0 |
| | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| | ... | ... | | | | | 0 | 0 | 0 | 0 |
| | Temp[7] | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | .... | | | | | 0 | 0 | 0 | 0 |
| 135 | | Freq[15] | | | | | 0 | 0 | 0 | 0 |

This part replaces legacy NV items, e.g.:

- RFNV_CDMA_Cx_BCy_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I
- RFNV_CDMA_Cx_BCy_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I
- RFNV_WCDMA_<band>_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I
- RFNV_WCDMA_<band>_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I
- RFNV_LTE_<band>_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I
- RFNV_LTE_<band>_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I


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- RFNV_TDSCDMA_<band>_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I
- RFNV_TDSCDMA_<band>_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I

## 3.5.10 PA compensate vs frequency for ILPC performance

Rows 264 through 279 represent PA_compensate_up_vs_freq values and rows 280 through 295 represent PA_Compensate_down_vs_freq values.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 264 | **PA_COMP_UP_VS_FREQ** | | | | | | 0 | 0 | 0 | 0 |
| | **16** | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | Freq[1] | | | | | 0 | 0 | 0 | 0 |
| | | ... | | | | | 0 | 0 | 0 | 0 |
| 279 | | Freq[15] | | | | | 0 | 0 | 0 | 0 |
| 280 | **PA_COMP_DOWN_VS_FREQ** | | | | | | 0 | 0 | 0 | 0 |
| | **16** | Freq[0] | | | | | 0 | 0 | 0 | 0 |
| | | Freq[1] | | | | | 0 | 0 | 0 | 0 |
| | | ... | | | | | 0 | 0 | 0 | 0 |
| 295 | | Freq[15] | | | | | 0 | 0 | 0 | 0 |

This part replaces legacy NV items, e.g.:

- RFNV_WCDMA_<band>_PA_COMPENSATE_UP_R1_VS_FREQ_I
- RFNV_WCDMA_<band>_PA_COMPENSATE_UP_R2_VS_FREQ_I
- RFNV_WCDMA_<band>_PA_COMPENSATE_UP_R3_VS_FREQ_I
- RFNV_WCDMA_<band>_PA_COMPENSATE_DOWN_R1_VS_FREQ_I
- RFNV_WCDMA_<band>_PA_COMPENSATE_DOWN_R2_VS_FREQ_I
- RFNV_WCDMA_<band>_PA_COMPENSATE_DOWN_R3_VS_FREQ_I

## 3.5.11 Reserved rows

Rows 296 through 300 are reserved for some temporary expansion without a version increase.

## 3.5.12 CDMA-specific NVs

The second part of the NV structure is for CDMA-specific switchpoints, i.e., 1x_Access, 1x_DATA, HDR_Access, HDR_DOR0, HDR_DORA, HDR_DORB_MC switchpoints. Rows 300 through 332 are only available in CDMA NVs.Structure of RFNV_LTE

### 3.5.13 Compressed NV for PA static

NOTE: This section was added to this document revision.

The PA static NV is a fixed size and occupies approximately 4800 bytes per item. Due to limited EFS space, new RFNV_<tech>_<Band>_PA_STATIC_Z_I format NV items are introduced in Bolt. See *MDM9x35+QFE Software Design Guide* (80-NK207-2).

- The structure remains the same; only the items are compressed and written to EFS.
- Data content of the items remain the same before compression.
- The software uncompresses and reads the items.
- The rest of the functionality within the software is the same.
- Qualcomm Radio Control Toolkit (QRCT) must compress the item before writing to EFS.
- QRCT can read back the compressed NVs and display values like the uncompressed ones.
- The compression algorithm used is Zlib, which is the same algorithm used for other compressed items, such as MULTI_LIN_V3 NVs.

## 3.6 Tx MULTI_LIN_V3_NV items

This NV item enables the new multilinearizer ET feature of the RF driver. It allows independent xPT curves to be used for different regions of channels within a band.

In the QFE1100 ET/EPT design, there are a total of four gain states, and the first two highest gain states are for ET and EPT operation. Thus, NV items and calibration software contain certain changes, summarizing all elements, including those that are added. As with the MULTI_LIN_V2_NV item, there is one MULTI_LIN_V3_NV item per band. Table 3-13 shows the format definitions:

**Table 3-13 Data type of MULTI_LIN_V3_NV item**

| Block | Type | Size |
|---|---|---|
| 100 | TX_CAL_CHANS_U32_TYPE | uint16 |
| 101 | TX_FREQ_OFFSETS_TYPE | uint16 |
| 102 | TX_LIN_V3_TYPE | uint16 |
| 104 | TX_EPT_DPD_V2_TYPE | uint16 |
| 105 | TX_ET_DPD_TYPE | uint16 |
| 111 | LIN_TEMP_ADC_TYPE | uint16 |

### Table 3-14 MULTI_LIN_V3_NV structure

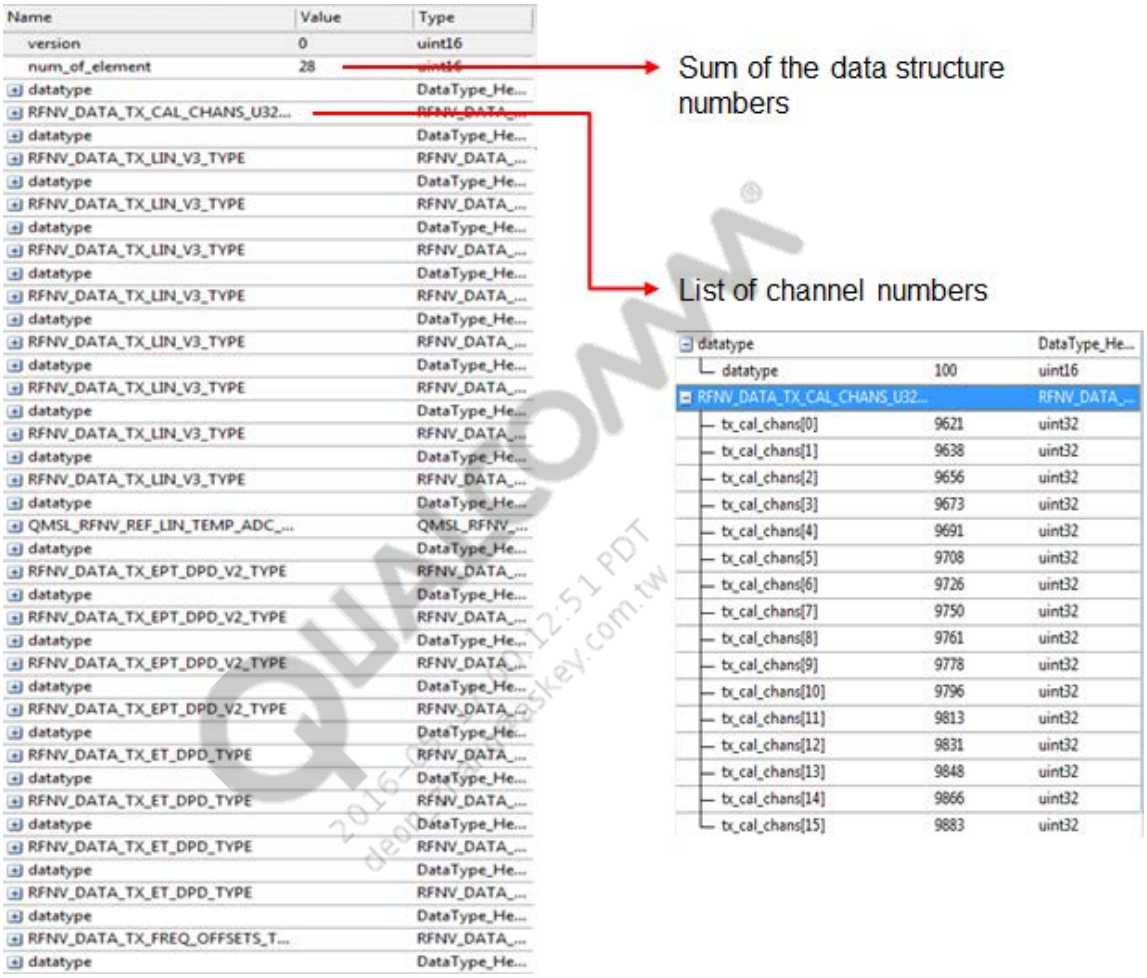
| Block | Type | | Data | Size |
|---|---|---|---|---|
| 100 | Tx cal channels | | | |
| | | | Array of 16, Tx calibration channel list | uint32 |
| 101 | Tx freq offset value | | | |
| | pa_state | | ▪ CDMA/WCDMA<br>▫ PA state 0,1 – APT<br>▫ PA state 2 – EPT<br>▫ PA state 3 – ET<br>▪ LTE<br>▫ PA state 0 – ET<br>▫ PA state 1 – EPT<br>▫ PA state 2,3 – APT | uint8 |
| | Fcomp_sweep_type | | ▪ 0 – xPT full bias<br>▪ 1 – xPT third sweep | uint8 |
| | Fcomp_idex | | Freq comp mapping table index<br>▪ 0 – APT<br>▪ 0 to 3 – EPT/ET | uint16 |
| | freq_offsets | | Array of 16 for 16 of Tx calibration channels; FreqComp element is a byte | 16 x uint8 |
| 102 | pa_state | | ▪ CDMA/WCDMA<br>▫ PA state 0,1 – APT<br>▫ PA state 2 – EPT<br>▫ PA state 3 – ET<br>▪ LTE<br>▫ PA state 0 – ET<br>▫ PA state 1 – EPT<br>▫ PA state 2,3 – APT | uint8 |
| | wave_type | | ▪ 0 – CDMA/WCDMA xPT<br>▪ 1 – LTE xPT<br>▪ 2 – MCDO (in ESC cal, the data in full bias (wave_type 3) is copied to the one in MCDO (wave_type 2); if MCDO is not used for reverse link, the data indicating as wave_type 2 is not used)<br>▪ 3 – Full bias | uint8 |
| | tx_chan | | Reference channel | uint32 |
| | upper_bound_tx_chan | | Upper bound channel | uint32 |
| | RGI | | Array of 64 RGI values | 64 x uint16 |
| | PWRs_db10 | | Array of 64 Tx power values | 64 x uint16 |
| | APTs | | Array of 64 APT values | 64 x uint32 |
| | Currents | | Array of 64; ICQ element is 2 bytes | 64 x uint16 |
| | IQ_Offset | | Array of 64; IQ offset element is 2 bytes | 64 x uint16 |
| | DPD_info | dpd_type_used | Use to recognize EPT DPD or ET DOD<br>▪ 0 – First sweep<br>▪ 2 – EPT third sweep<br>▪ 3 – ET third sweep | uint8 |

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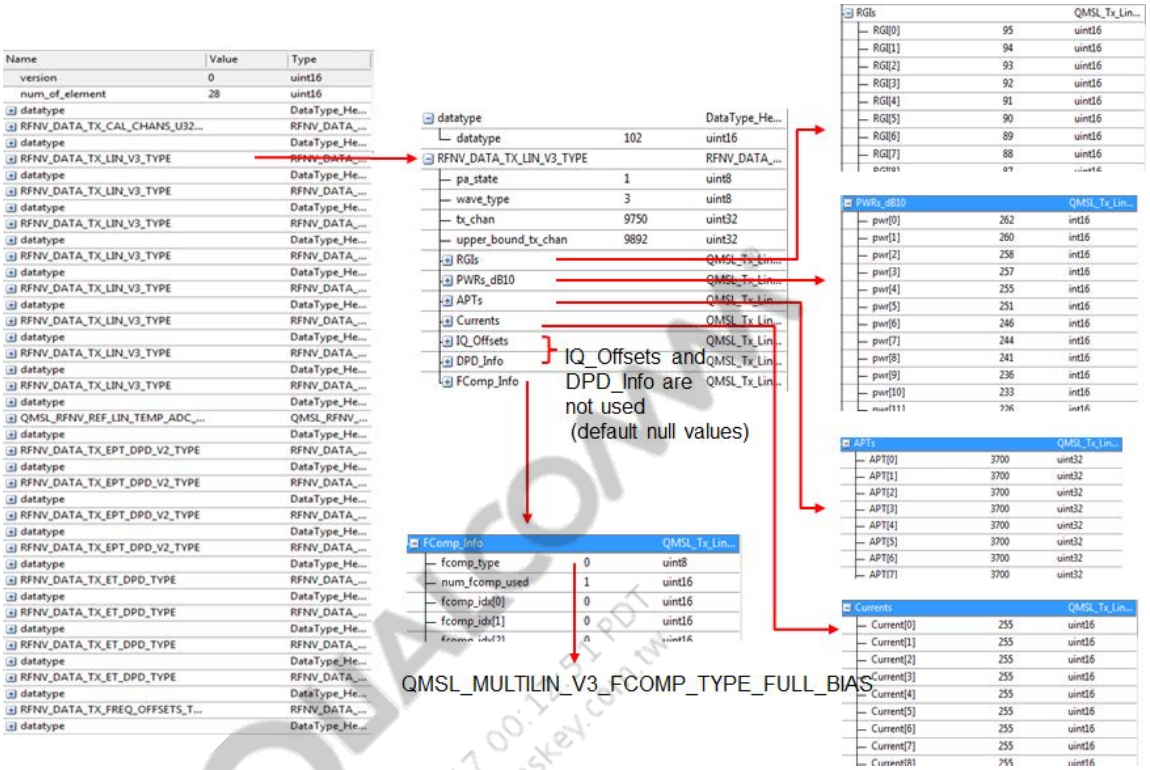
| Block | Type | | Data | Size |
|---|---|---|---|---|
| | | num_dpd_used | ▪ 0 – First sweep<br>▪ 4 – EPT/ET third sweep | uint16 |
| | | dpd_inx | ▪ Array of 64; DPD element is 2 bytes<br>▪ DPD index is maintained in linearizer and must match the indices in DPD data that follows; this is the mechanism to link the DPD data to a particular linearizer row | 64 x uint16 |
| | Fcomp_Info | Fcomp_type | Use to recognize sweep 1 or sweep 3<br>▪ 0 – xPT full bias<br>▪ 1 – xPT third sweep | uint8 |
| | | num_fcomp_used | ▪ 1 – xPT full bias<br>▪ 1 – APT<br>▪ 4 – EPT/ET | uint16 |
| | | Fcomp_idx | Fcomp index is maintained in linearizer and must match the indices in Freq offset table; this is the mechanism to link the freq offset data to a particular linearizer row. Unused elements need to be populated with maximum value of its type, e.g., uint16 type is 0xFFFF.<br>▪ Array of 64<br>▪ APT – Number of index depends on number of RGI in APT mode<br>▪ xPT full bias – Number of index depends on number of RGI in first sweep<br>▪ EPT – idx[0] to idx[7]<br>▪ ET – idx[0] to idx[3] | 64 x uint16 |
| 104 | TX_EPT_DPD_V2 | | | |
| | ept_dpd_index | | EPT DPD mapping table index: 0 to 3 | uint16 |
| | AMAM | | Array of 16 AMAM coefficient; AMAM element is 2 bytes | 16 x uint32 |
| | AMPM | | Array of 16 AMAP coefficient; AMPM element is 2 bytes | 16 x uint32 |
| 105 | TX_ET_DPD | | | |
| | et_dpd_index | | ET DPD mapping table index: 0 to 3 | uint16 |
| | AMAM | | Array of 16 AMAM coefficient; AMAM element is 2 bytes | 16 x uint32 |
| | AMPM | | Array of 16 AMAP coefficient; AMPM element is 2 bytes | 16 x uint32 |
| | EPDT | | Array of 16; EPDT element is 2 bytes | 16 x uint32 |
| 111 | temperature_Adc | | Therm ADC reading during calibration; the value can be converted to degrees based on the therm_temp and ENH_therm NV items | uint16 |

The following screenshots show how MULTI_LIN_V3 looks from the NV manager.

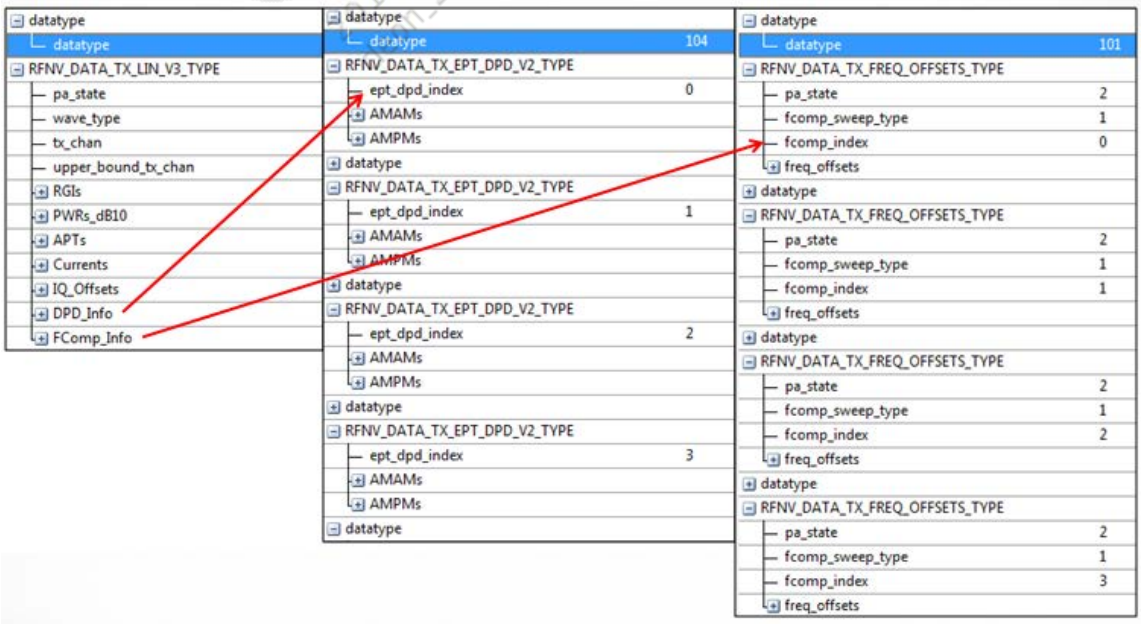
## MULTI_LIN_V3_NV structure

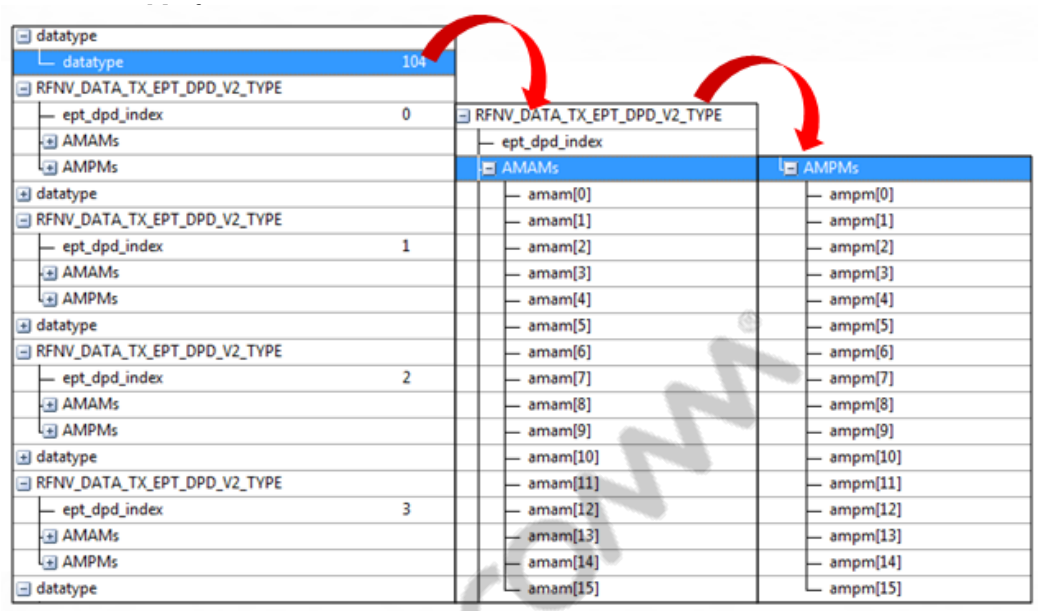


## MULTI_LIN_V3_NV Tx linearizer data

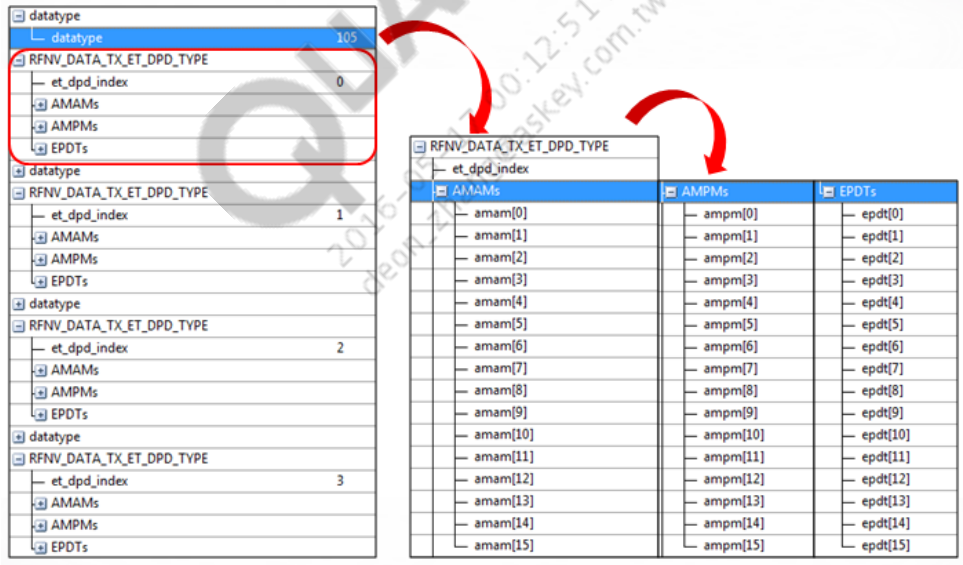


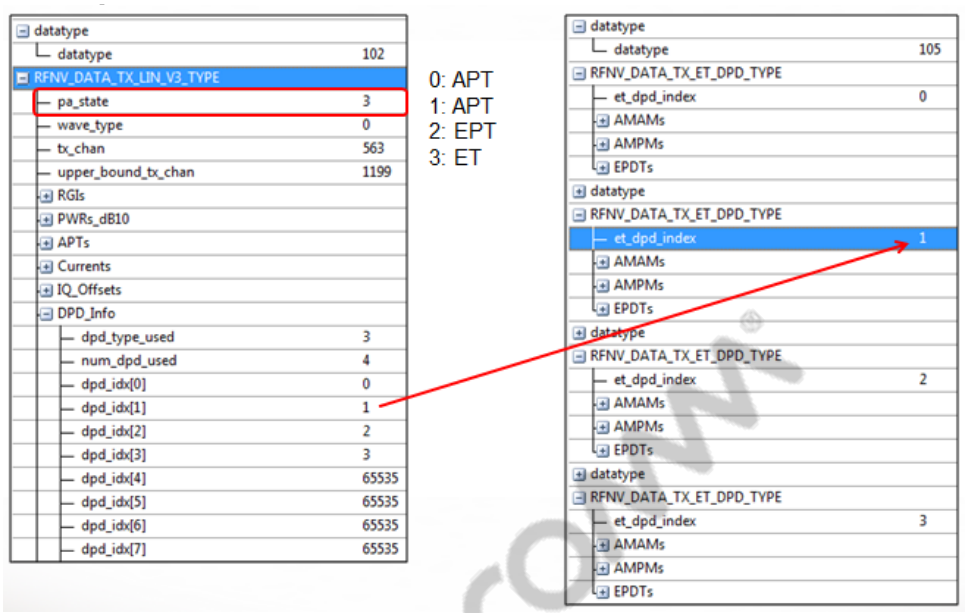

## MULTI_LIN_V3_NV – DPD index and Fcomp index mapping table



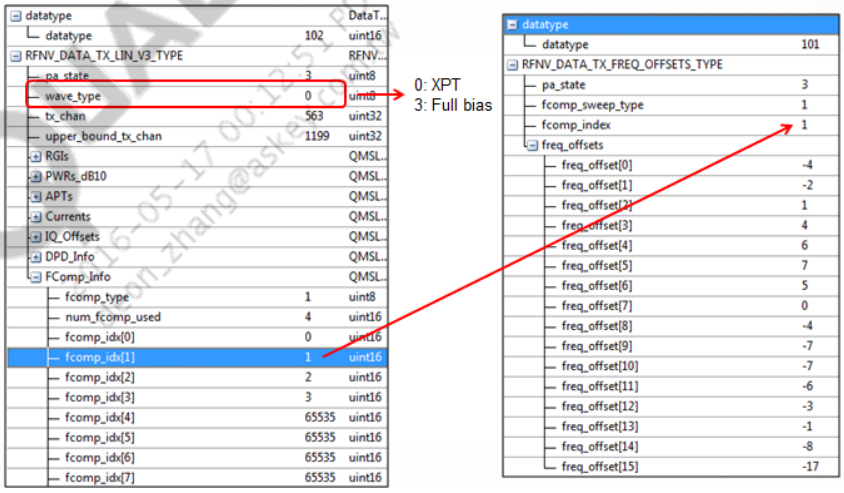



ET DPD table format






Fcomp index mapping



# 3.7 Online temperature and frequency compensation algorithms

This section describes the software implementation details for the xPT temperature and frequency compensation algorithms in accordance with the multilinearizer and temperature compensation NV items defined.

## 3.7.1 APT-only mode

In the APT-only mode of Tx functionality, the linearizer variation across temperatures was statically characterized to be a fixed offset at a given temperature. This was stored in legacy LIN_VS_TEMP/TX_PWR_TEMP_COMP NV items for WCDMA and CDMA. This fixed offset was used to compensate Pout (Tx power) in the linearizer column. The RGI/bias relationship was selectively controlled for hot/cold temperature compensation in order to optimally align the APT bias to the compensated output power value.

- The statically characterized fixed offset here was independent of frequency.
- This legacy temperature compensation algorithm is discarded for xPT operation. It is used by software for multi-lin V1 calibrated data.

## 3.7.2 xPT mode

xPT Tx operation, specifically in the DPD modes of operation (EPT and ET), has special requirements with regards to compensating for frequency and temperature variation. The legacy algorithms for compensating the calibrated linearizer across all PA states do not apply accurately to operation with DPD. RF software xPT temperature and frequency compensation algorithms are modified to provide both Pin and Pout linearizer compensation.

## 3.7.3 Temperature compensation algorithms

Table 3-15 lists the types of temperature compensation algorithms.

**Table 3-15 Temperature compensation algorithm summary**

| Multilinearizer version | xPT mode | Temperature compensation NV version | Compensation value compute | Linearizer compensation algorithm |
|---|---|---|---|---|
| V1 | APT only | Legacy | Legacy method | Pout compensation with APT bias align ALGO_MULTILIN_V1 |
| V3 | APT only | Legacy | Legacy method | Pin compensation only ALGO_MULTILIN_V3_LEGACY_NV |
| V3 | xPT | New Pin/Pout NV | New computation | Pin and Pout compensation algorithms ALGO_MULTILIN_V3 |


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Figure 3-3 shows the algorithm flow.

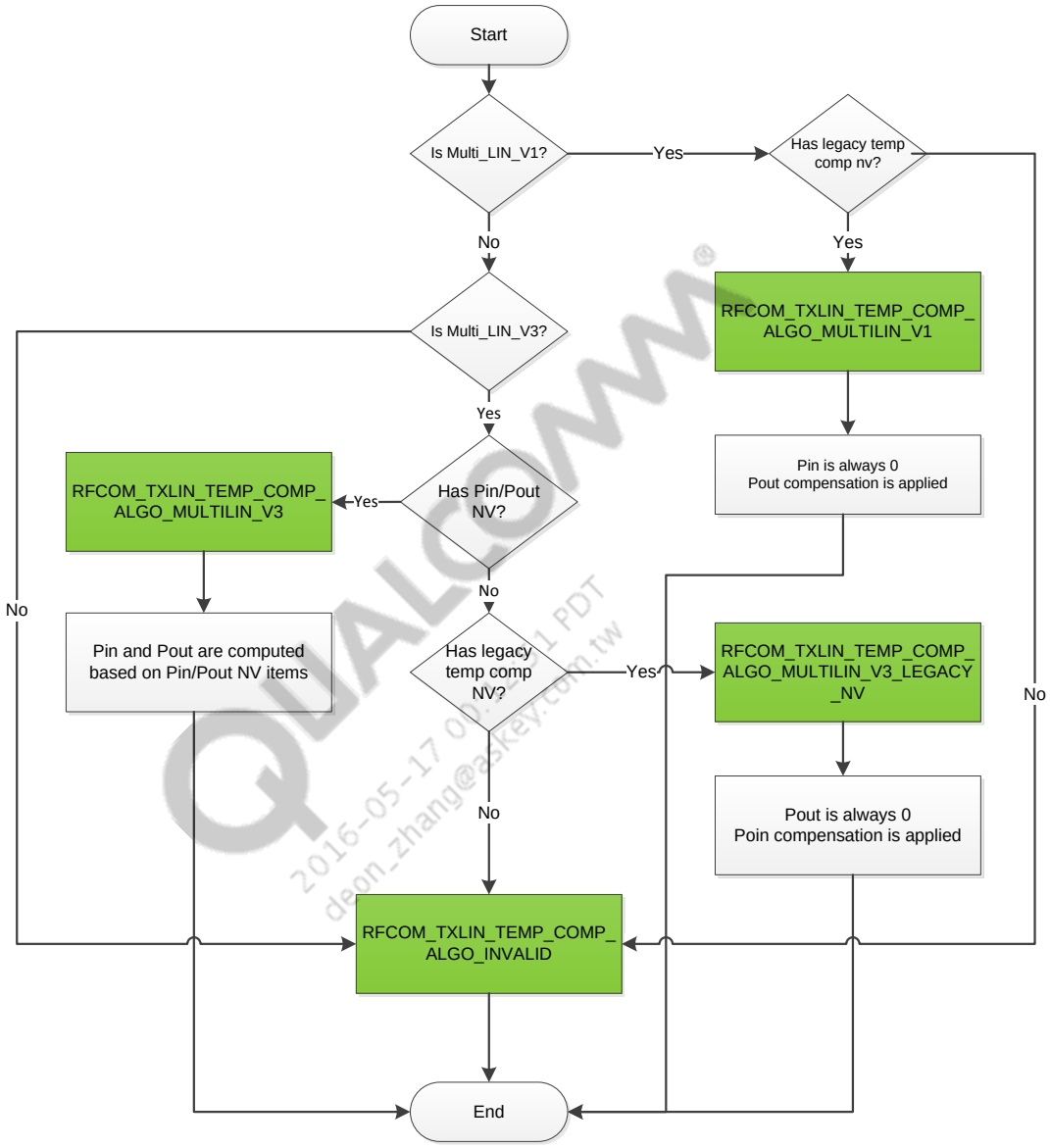



**Figure 3-3 Temperature compensation algorithm flow chart**

## 3.7.4 Pin/Pout freq/temp compensation in Online mode

This section provides details of online Pin/Pout freq/temp compensation.

### 3.7.4.1 Data involved

Table 3-16 lists data involved for online Pin/Pout freq/temp compensation.

**Table 3-16 Data**

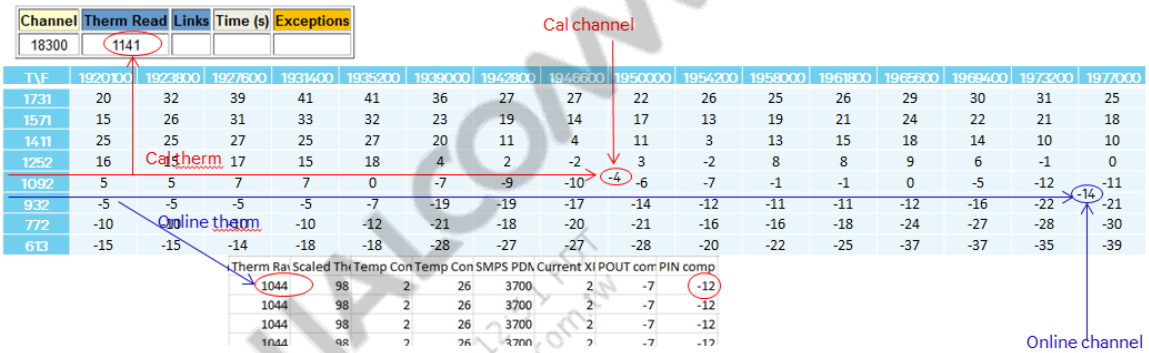
| Type | Data | Dimension | Note |
|---|---|---|---|
| Pin/Pout characterization | Pin | 8 temp * 16 freq<br>Per PA state/band/tech<br>(Section 3.4.1) | In 0.1 dB, positive number means Pin is higher and negative compensation is needed |
| | Pout | 8 temp * 16 freq<br>Per PA state/band/tech<br>(Section 3.4.2) | In 0.1 dB, positive number means Pout is higher and more TxAGC backoff is needed |
| Calibration | RGI delta sweep | Multi_lin_v3 waveform type3<br>Per band/tech<br>(Section 3.5) | In 0.1 dB – Obtained by low-IQ Gain sweep |
| | Linearizer data | ▪ Multi_lin_v3 waveform type0<br>▫ 4 per ET<br>▫ 4 per EPT<br>▫ Multiple per APT<br>▫ Per band/tech<br>(Section 3.5) | |
| | Cal thermal ADC | Multi_lin_v3 – Temperature_Adc<br>1 per band/tech<br>(Section 3.5) | Provide a reference thermal reading |
| | FreqComp | Multi_lin_v3 – Fcomp_type1<br>Per PA state/band/tech<br>(Section 3.5) | Obtained by FreqComp during calibration, and Pin is compensated |

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### 3.7.4.2 Pin for freq/temp comp

Linearly interpolate Pin value from Pin NVs according to the current frequency and temperature:

- Cal_Pin = Pin NV at the calibration channel and cal thermal ADC, e.g., calibration channel = 1950 MHz, Cal thermal ADC = 1141, Cal_Pin = -4
- Online_Pin = Pin NV at the online channel and online thermal ADC, e.g., online channel = 1975 MHz, online thermal ADC = 1044, Online_Pin = -14
- Pin which is shown in the QRCT log = Online_Pin – Cal_Pin = -10
- For Pin, WTR compensation for 50RB(-2) will also be applied to the final Pin value, which is -10-2 = -12




The main differences of Pout calculation between LTE and WCDMA/CDMA are:

- For WCDMA/CDMA, first interpolate to get the two adjacent temp comp arrays for the online channel and then interpolate between these two temp comp arrays to get the Online_Pout.
- There is no WTR compensation for WCDMA and CDMA.

### 3.7.4.3 Pin alignment to update RGI/IQ Gain in linearizers

Pin is used to update the RGI/IQ Gain used in linearizers.

- Data involved – Linearizer in Cal and RGI gain delta

**MEASUREMENTS: Tx XPT Swp3 - Linearizer**

| Channel | XPT Mode | PA State | TxAGC | Vcc | Exp Power | XPT Swp3 Power |
|---|---|---|---|---|---|---|
| 18300 | 2 | 0 | 84 | 2044 | 17 | 17.9 |
| 18300 | 2 | 0 | 85 | 2327 | 19 | 19.3 |
| 18300 | 2 | 0 | 87 | 2964 | 21 | 21.9 |
| 18300 | 2 | 0 | 89 | 3700 | 23.5 | 24 |

**SUREMENTS: Tx XPT Swp1 - RGI Delta Sweep**

| Channel | PA State | TxAGC | Vcc | Icq | XPT Swp1 RGI Delta S |
|---|---|---|---|---|---|
| 18300 | 1 | 77 | 3300 | 0 | 4.9 |
| 18300 | 1 | 78 | 3300 | 0 | 5.8 |
| 18300 | 1 | 79 | 3300 | 0 | 6.8 |
| 18300 | 1 | 80 | 3300 | 0 | 7.8 |
| 18300 | 1 | 81 | 3300 | 0 | 8.8 |
| 18300 | 1 | 82 | 3300 | 0 | 9.7 |
| 18300 | 1 | 83 | 3300 | 0 | 10.7 |
| 18300 | 1 | 84 | 3300 | 0 | 11.6 |
| 18300 | 1 | 85 | 3300 | 0 | 12.6 |
| 18300 | 1 | 86 | 3300 | 0 | 13.5 |
| 18300 | 1 | 87 | 3300 | 0 | 14.5 |
| 18300 | 1 | 88 | 3300 | 0 | 15.5 |
| 18300 | 1 | 89 | 3300 | 0 | 16.4 |
| 18300 | 1 | 90 | 3300 | 0 | 17.3 |
| 18300 | 1 | 91 | 3300 | 0 | 18.1 |
| 18300 | 1 | 92 | 3300 | 0 | 19.1 |
| 18300 | 1 | 93 | 3300 | 0 | 19.8 |
| 18300 | 1 | 94 | 3300 | 0 | 20.8 |

- Update linearizers based on Pin = -12
  - Pin = -12 means online should have 1.2 dB more Pin

| Channel | DPD # | Pout | RGI | IQ Gain | EnvScale | SMPS |
|---|---|---|---|---|---|---|
| 1975000 | 0 | 170 | 84 | 420 | 920 | 2044 |
| 1975000 | 1 | 190 | 85 | 420 | 920 | 2327 |
| 1975000 | 2 | 210 | 87 | 420 | 920 | 2964 |
| 1975000 | 3 | 235 | 89 | 420 | 920 | 3700 |

IQGain backoff 0.5 dB

Second

| Channel | DPD # | Pout | RGI | IQ Gain | EnvScale | SMPS |
|---|---|---|---|---|---|---|
| 1975000 | 0 | 170 | 86 | 401 | 920 | 2044 |
| 1975000 | 1 | 190 | 87 | 401 | 920 | 2327 |
| 1975000 | 2 | 210 | 89 | 401 | 920 | 2964 |
| 1975000 | 3 | 235 | 91 | 410 | 920 | 3700 |

**SUREMENTS: Tx XPT Swp1 - RGI Delta Sweep**

| Channel | PA State | TxAGC | Vcc | Icq | XPT Swp1 RGI Delta S |
|---|---|---|---|---|---|
| 18300 | 1 | 77 | 3300 | 0 | 4.9 |
| 18300 | 1 | 78 | 3300 | 0 | 5.8 |
| 18300 | 1 | 79 | 3300 | 0 | 6.8 |
| 18300 | 1 | 80 | 3300 | 0 | 7.8 |
| 18300 | 1 | 81 | 3300 | 0 | 8.8 |
| 18300 | 1 | 82 | 3300 | 0 | 9.7 |
| 18300 | 1 | 83 | 3300 | 0 | 10.7 |
| 18300 | 1 | 84 | 3300 | 0 | 11.6 |
| 18300 | 1 | 85 | 3300 | 0 | 12.6 |
| 18300 | 1 | 86 | 3300 | 0 | 13.5 |
| 18300 | 1 | 87 | 3300 | 0 | 14.5 |
| 18300 | 1 | 88 | 3300 | 0 | 15.5 |
| 18300 | 1 | 89 | 3300 | 0 | 16.4 |
| 18300 | 1 | 90 | 3300 | 0 | 17.3 |
| 18300 | 1 | 91 | 3300 | 0 | 18.1 |
| 18300 | 1 | 92 | 3300 | 0 | 19.1 |
| 18300 | 1 | 93 | 3300 | 0 | 19.8 |
| 18300 | 1 | 94 | 3300 | 0 | 20.8 |

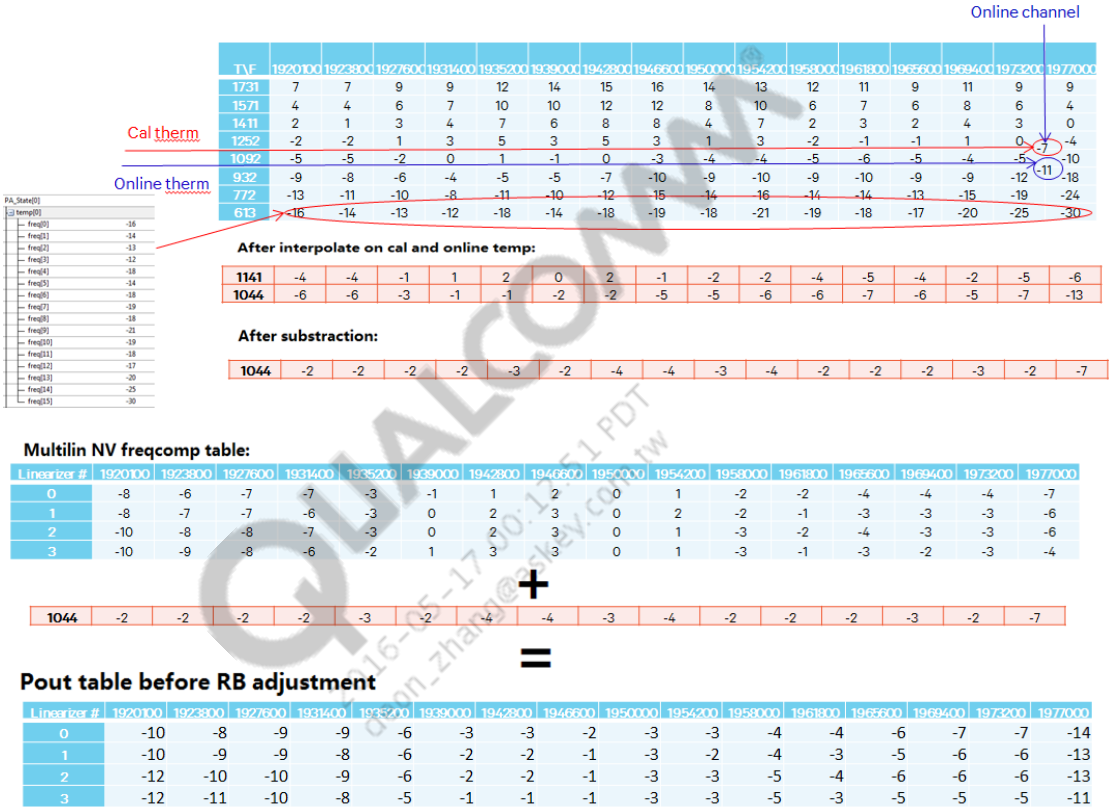
1.7 – 1.2 = 0.5dB

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### 3.7.4.4 Pout for freq/temp come

Pout NV is used for freq/temp comp.

- Pout_CalTherm = Pout NV at the online channel and cal thermal ADC, e.g., online channel = 1975 MHz, cal thermal ADC = 1141
- Pout_OnlineTherm = Pout NV at the online channel and online thermal ADC, e.g., online channel = 1975 MHz, online thermal ADC = 1044




| TVF | 1920100 | 1923800 | 1927600 | 1931400 | 1935200 | 1939000 | 1942800 | 1946600 | 1950000 | 1954200 | 1958000 | 1961800 | 1965600 | 1969400 | 1973200 | 1977000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1731 | 7 | 7 | 9 | 9 | 12 | 14 | 15 | 16 | 14 | 13 | 12 | 11 | 9 | 11 | 9 | 9 |
| 1571 | 4 | 4 | 6 | 7 | 10 | 10 | 12 | 12 | 8 | 10 | 6 | 7 | 6 | 8 | 6 | 4 |
| 1411 | 2 | 1 | 3 | 4 | 7 | 6 | 8 | 8 | 4 | 7 | 2 | 3 | 2 | 4 | 3 | 0 |
| 1252 | -2 | -2 | 1 | 3 | 5 | 3 | 5 | 3 | 1 | 3 | -2 | -1 | -1 | 1 | 0 | -4 |
| 1092 | -5 | -5 | -2 | 0 | 1 | -1 | 0 | -3 | -4 | -4 | -5 | -6 | -5 | -4 | -5 | -10 |
| 932 | -9 | -8 | -6 | -4 | -5 | -5 | -7 | -10 | -9 | -10 | -9 | -10 | -9 | -9 | -12 | -18 |
| 772 | -13 | -11 | -10 | -8 | -11 | -10 | -12 | -15 | -14 | -16 | -14 | -14 | -13 | -15 | -19 | -24 |
| 613 | -16 | -14 | -13 | -12 | -18 | -14 | -18 | -19 | -18 | -21 | -19 | -18 | -17 | -20 | -25 | -30 |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1141 | -4 | -4 | -1 | 1 | 2 | 0 | 2 | -1 | -2 | -2 | -4 | -5 | -4 | -2 | -5 | -6 |
| 1044 | -6 | -6 | -3 | -1 | -1 | -2 | -2 | -5 | -5 | -6 | -6 | -7 | -6 | -5 | -7 | -13 |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1044 | -2 | -2 | -2 | -2 | -3 | -2 | -4 | -4 | -3 | -4 | -2 | -2 | -2 | -3 | -2 | -7 |

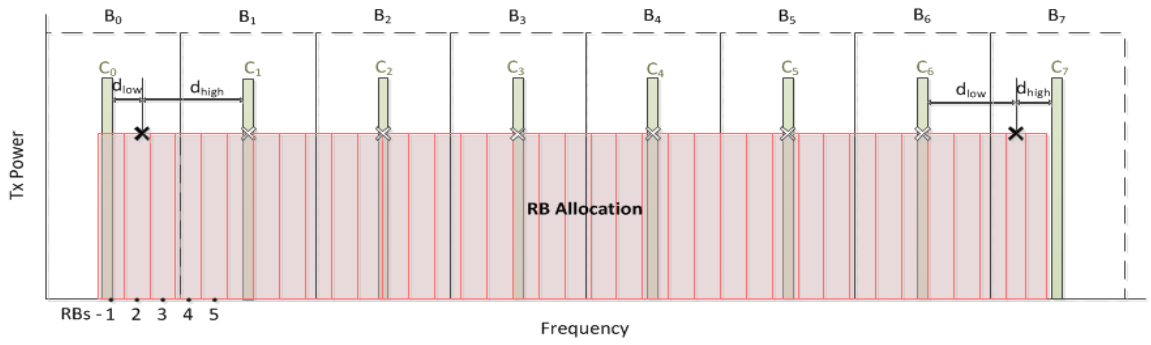
| Linearizer # | 1920100 | 1923800 | 1927600 | 1931400 | 1935200 | 1939000 | 1942800 | 1946600 | 1950000 | 1954200 | 1958000 | 1961800 | 1965600 | 1969400 | 1973200 | 1977000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | -8 | -6 | -7 | -7 | -3 | -1 | 1 | 2 | 0 | 1 | -2 | -2 | -4 | -4 | -4 | -7 |
| 1 | -8 | -7 | -7 | -6 | -3 | 0 | 2 | 3 | 0 | 2 | -2 | -1 | -3 | -3 | -3 | -6 |
| 2 | -10 | -8 | -8 | -7 | -3 | 0 | 2 | 3 | 0 | 1 | -3 | -2 | -4 | -3 | -3 | -6 |
| 3 | -10 | -9 | -8 | -6 | -2 | 1 | 3 | 3 | 0 | 1 | -3 | -1 | -3 | -2 | -3 | -4 |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1044 | -2 | -2 | -2 | -2 | -3 | -2 | -4 | -4 | -3 | -4 | -2 | -2 | -2 | -3 | -2 | -7 |

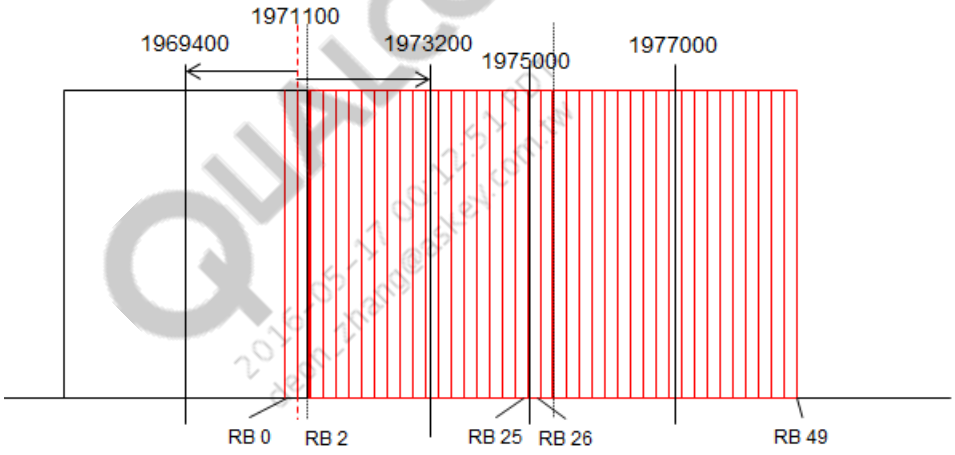
| Linearizer # | 1920100 | 1923800 | 1927600 | 1931400 | 1935200 | 1939000 | 1942800 | 1946600 | 1950000 | 1954200 | 1958000 | 1961800 | 1965600 | 1969400 | 1973200 | 1977000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | -10 | -8 | -9 | -9 | -6 | -3 | -3 | -2 | -3 | -3 | -4 | -4 | -6 | -7 | -7 | -14 |
| 1 | -10 | -9 | -9 | -8 | -6 | -2 | -2 | -1 | -3 | -2 | -4 | -3 | -5 | -6 | -6 | -13 |
| 2 | -12 | -10 | -10 | -9 | -6 | -2 | -2 | -1 | -3 | -3 | -5 | -4 | -6 | -6 | -6 | -13 |
| 3 | -12 | -11 | -10 | -8 | -5 | -1 | -1 | -1 | -3 | -3 | -5 | -3 | -5 | -5 | -5 | -11 |

The main differences of Pout calculation between LTE and WCDMA/CDMA are:

- For WCDMA/CDMA, we first interpolate to get the two adjacent temp comp arrays for the online channel and then do the interpolation between these two temp comp arrays to get the Online_Pout.
- There is no RB adjustment for WCDMA/CDMA.
- Adjust the Pout value based on the RB location and RB number (LTE only).



- Suppose there are 8 calibration channels, which means we can evenly divide the band to 8 bins.
- 1RB occupies 180 KHz and for LTE one channel occupies 100 KHz, thus, RB allocation might be across several bins.
- For those fully occupied bins (from C1 to C6), we just use the comp value of the calibration channel.
- For those partially occupied bins (C0, C7), we first determine the center channel of the partial part and perform the linear interpolation to get the proper comp value for the partial part.
- The overall comp value will be the weight-based average of all the allocated RBs, e.g.:
  - For one bin, it contains 41 channels, which totally occupies 41*100 KHz = 4.1 MHz.
  - 50 RB will occupy 50*180 KHz = 9 MHz bandwidth.
  - So, for channel 1975000 with start RB 0 and RB number 50, the RBs it occupies is shown here



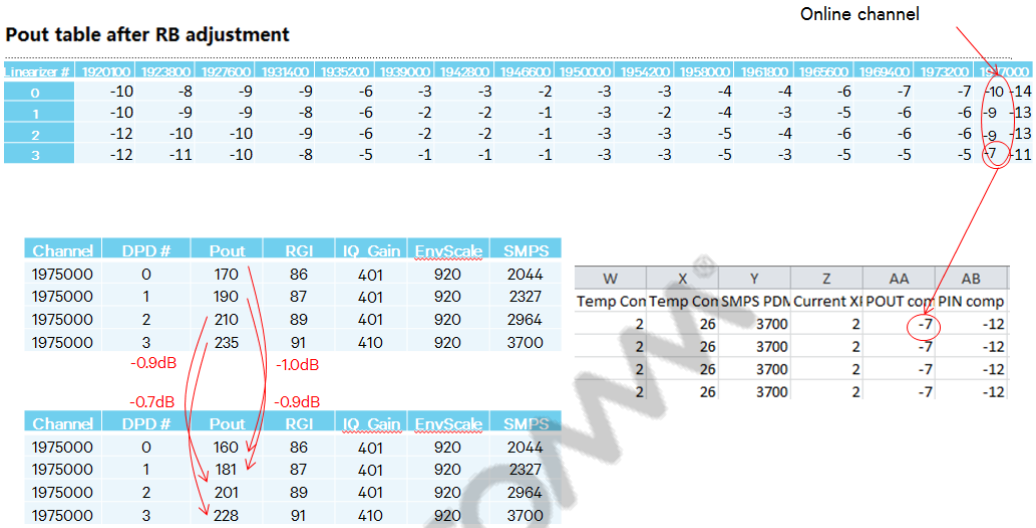

- Pick DPD index 0 as an example:
  - For RB 2 to RB 25, use offset value = -7, which is offset value at 1973200 in the Pout table before RB adjustment.
  - For RB 26 to RB 49, use offset value = -14, which is offset value at 1977000 in the Pout table before RB adjustment.
  - For RB 0 and 1, first find out the mid frequency of the partial part, which is 1971100.
  - Then do the linear interpolation to get the value for these two RBs; let the value be X.

    (-7 - (-7)) / (1973200 - 1969400) = (x - (-7)) / (1971100 - 1969400)

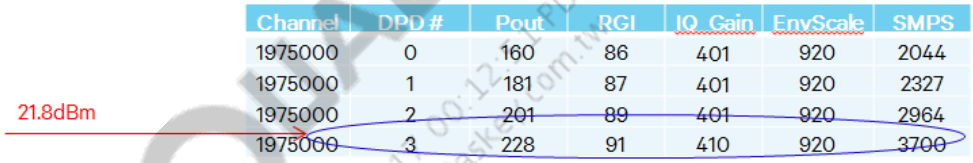
    To solve the equation: X = -7
  - The average for all the occupied RBs are as follows:

    (-7 (RB0) + -7 (RB1) + -7(RB2) +…+ -7(RB25) + -14(RB26) + -14(RB27) +…+ -14 (RB49)) / 50 = -10.22 = -10 (rounded off to integer value)

### 3.7.4.5 Pout compensation to update Pouts in linearizers




- For target Pout, pick up a linearizer and back off EnvScale, e.g., target Pout = 21.8 dBm




  - 19.9 < 21.8 < 22.8, so that Linearizer 3 is picked
  - EnvScale backoff 22.8 – 21.8 = 1 dB, so that EnvScale = 820
  - Final settings – RGI 91, IQ Gain 0.5 dB backoff, SMPS 3700, EnvScale 1 dB backoff, DPD #3

| Num RBs | Therm Ra | Temp Con | Temp Con | SMPS PDN | Current XI | POUT com | PIN comp | Selected F | ENV Scale | IQ Gain Ba | DPD Index | PA ICQ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 50 | 1044 | 2 | 26 | 3700 | 2 | -7 | -12 | 91 | 10 | 5 | 3 | 60 |
| 50 | 1044 | 2 | 26 | 3700 | 2 | -7 | -12 | 91 | 10 | 5 | 3 | 60 |
| 50 | 1044 | 2 | 26 | 3700 | 2 | -7 | -12 | 91 | 10 | 5 | 3 | 60 |
| 50 | 1044 | 2 | 26 | 3700 | 2 | -7 | -12 | 91 | 10 | 5 | 3 | 60 |

# 3.8 Non-0 dB Pin offset

Currently, the Pin applied for reference channel calibration temperature is always 0 dB. However, as we experience various PAs and hardware, some potential hardware issues overturn the decision to apply 0 dB for when DPD is constructed. Two instances of hardware issues with no easy hardware solution are:

- Thermal issue – Either the device heats up quickly, or the WTR thermal ADC reading cannot fully represent the true PA temperature during RFCal. It is observed that the best ACLR is achieved with a higher Pin than DPD construction settings. This is shown in either ACLR-based Pin char, or via manual FTM experiments.
- PA memory issue – The current LUT DPD solution cannot compensate PA memory effect that might drive Pin to non-0 dB for best ACLR.

Thus, adding the capability of applying non-0 dB Pin for reference channel calibration temperature to facilitate the deployment of the ET/EPT systems. The feature is only enabled on DimePM and Bolt modems. See *Presentation: QPA APT+DPD Software Overview* (80-NK201-5) for more information.

## 3.8.1 Procedure

### 3.8.1.1 Apply Pin offset value in DPD_CONFIG NV

To support the non-0 dB Pin, the bit fields 13 and 14 in the per-band per-tech static DPD_CONFIG NV for the offset of Pin values. In order to get higher RGI after applying Pin, a negative value must be used.

| | | |
|---|---|---|
| 13 | REF_PIN_ET | Offset for ET Pin values |
| 14 | REF_PIN_EPT | Offset for EPT Pin values |

### 3.8.1.2 Calibration changes

Calibration DLL version is required later than 17.75. Also, need to add the XPT_Swp3_Apply_Pin parameter in sweep3.

```
<Purpose_List>0,0,0</Purpose_List>
<Minimum_Delta_Power_dBm100>-100</Minimum_Delta_Power_dBm100>
<XPT_Swp3_Apply_Pin>TRUE</XPT_Swp3_Apply_Pin>
<XPT_Swp3_Channel_List>358</XPT_Swp3_Channel_List>
<EPT_Swp3_Meas_Pwr_List>18,20,22,23</EPT_Swp3_Meas_Pwr_List>
<EPT_Swp3_Calc_Pwr_List>17,19,21,22.5</EPT_Swp3_Calc_Pwr_List>
<XPT_Swp3_Deviation_From_Exp_Power_Min_dBm100>-500</XPT_Swp3_D
<XPT_Swp3_Deviation_From_Exp_Power_Max_dBm100>500</XPT_Swp3_De
<XPT_Swp3_Linearizer_Delta_Meas_Power_Min_dBm100>-50</XPT_Swp3_
<XPT_Swp3_Segment_Length>20000</XPT_Swp3_Segment_Length>
```

## 3.8.2 Calibration log

- The non-0 dB Pin offset does not impact M-Line result
- Swp3 power increases after applying the non-0 dB Pin offset

M-line Table:

| Channel | XPT Mode | TxAGC | Vcc (mV) | Pout (dBm) |
|---|---|---|---|---|
| 9400 | 1 | 74 | 937 | 6.17 |
| 9400 | 1 | 75 | 1063 | 8.78 |
| 9400 | 1 | 76 | 1152 | 10.25 |
| 9400 | 1 | 77 | 1369 | 13.03 |
| 9400 | 1 | 78 | 1518 | 14.53 |
| 9400 | 1 | 79 | 1703 | 16.09 |
| 9400 | 1 | 80 | 1910 | 17.56 |
| 9400 | 1 | 81 | 2118 | 18.83 |
| 9400 | 1 | 82 | 2432 | 20.44 |
| 9400 | 1 | 83 | 2760 | 21.85 |
| 9400 | 1 | 84 | 2991 | 22.73 |
| 9400 | 1 | 85 | 3242 | 23.59 |
| 9400 | 1 | 86 | 3474 | 24.31 |
| 9400 | 1 | 87 | 3809 | 25.26 |
| 9400 | 1 | 88 | 3988 | 25.73 |
| 9400 | 1 | 89 | 4176 | 26.2 |

0db pin in ref CH

-2db pin in ref CH

Swp3:

| Channel | XPT Mode | PA State | TxAGC | Vcc | Exp Power | XPT Swp3 Power |
|---|---|---|---|---|---|---|
| 9400 | 1 | 2 | 79 | 1703 | 15.5 | 16 |
| 9400 | 1 | 2 | 82 | 2432 | 19 | 20.4 |
| 9400 | 1 | 2 | 83 | 2760 | 20.5 | 21.8 |
| 9400 | 1 | 2 | 86 | 3474 | 23.7 | 24.2 |
| 9400 | 1 | 2 | 79 | 1703 | 15 | 15.9 |
| 9400 | 1 | 2 | 81 | 2118 | 18.5 | 18.8 |
| 9400 | 1 | 2 | 82 | 2432 | 20 | 20.4 |
| 9400 | 1 | 2 | 83 | 2760 | 21 | 21.8 |

| Channel | XPT Mode | PA State | TxAGC | Vcc | Exp Power | XPT Swp3 Power |
|---|---|---|---|---|---|---|
| 9400 | 1 | 2 | 79 | 1693 | 15.5 | 17.1 |
| 9400 | 1 | 2 | 82 | 2437 | 19 | 21.7 |
| 9400 | 1 | 2 | 83 | 2739 | 20.5 | 22.9 |
| 9400 | 1 | 2 | 86 | 3458 | 23.7 | 25.3 |
| 9400 | 1 | 2 | 79 | 1693 | 15 | 17 |
| 9400 | 1 | 2 | 81 | 2117 | 18.5 | 20 |
| 9400 | 1 | 2 | 82 | 2437 | 20 | 21.6 |
| 9400 | 1 | 2 | 83 | 2739 | 21 | 22.9 |

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# 4 xPT Calibration

This chapter describes the ESC V4 Cal, M-Line high-level algorithm, and xPT control API. ESC V4 uses the new opcode to send commands. M-Line calculation is done by firmware and software and it is not from callbox measurement. The xPT control API is used to debug the xPT system. Note the characterization needs that are required before calibration as indicated by green and orange in Figure 4-1.

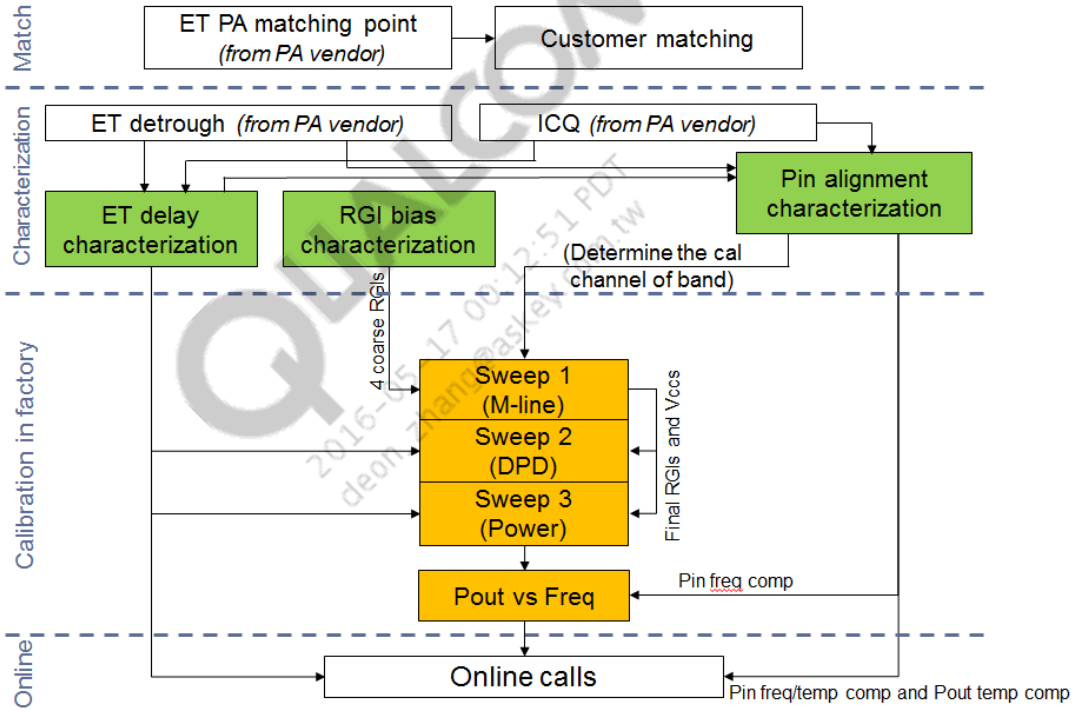



**Figure 4-1 xPT system flowchart**

# 4.1 ESC V4 calibration

Table 4-1 shows the FTM APIs to support ESC with new opcode calibration.

**Table 4-1 FTM APIs**

| ID | Command name | Description |
|---|---|---|
| 607 | QLIB_FTM_CFG_AND_EXEC_ESC_SWEEP | Configures and executes the calibration sweep with the passed-in data |
| 608 | QLIB_FTM_GET_ESC_SWEEP_RESULTS | Fetches the calibration results generated after the calibration sweep is complete |

Figure 4-2 shows the procedure for ESC V4 cal.

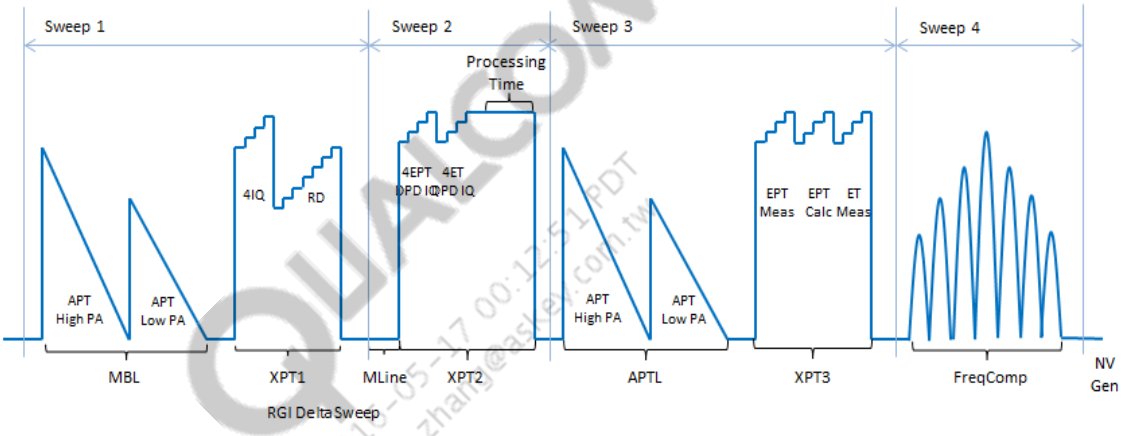
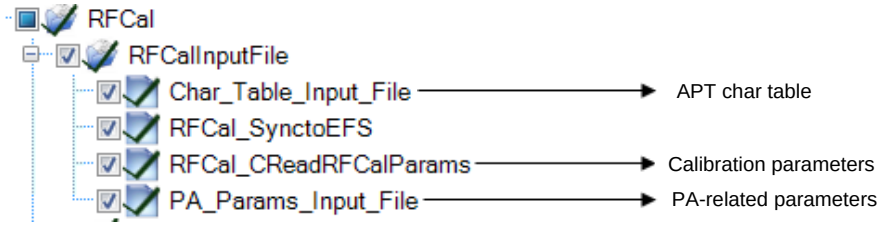



**Figure 4-2 ESC V4 Cal procedure**

## 4.1.1 ESC V4 cal input file parameters description

The ESC V4 Cal input file parameters described in Table 4-2 and Table 4-3.



#### 4.1.1.1 RF Cal parameters

Table 4-2 lists the input file parameters for ESC V4 cal.

**Table 4-2 ESC V4 Cal input file parameters**

| Sweeps | Description |
|---|---|
| ***Sweep 1*** | |
| Sweep_Type | Indicates which sweep to do; values are:<br>▪ 0 – MAX BIAS LINEARIZER<br>▪ 1 – APT LINEARIZER<br>▪ 2 – XPT SWP1<br>▪ 3 – MAX BIAS FREQCOMP<br>▪ 4 – APT FREQCOMP<br>▪ 5 – MULTI CHAN LINEARIZER<br>▪ 6 – XPT SWP2<br>▪ 7 – XPT SWP3<br>▪ 8 – XPT FREQCOMP<br>▪ 9 – XPT DELAY<br>▪ 10 – APT FREQCOMP STAGGER RXTX<br>▪ 11 – XPT FREQCOMP STAGGER RXTX<br>▪ 12 – XPT_FREQCOMP_SERIAL_TX( computed within the .dll for freq comp)<br>▪ 13 – XPT_FREQCOMP_SERIAL_RX (computed within the. dll for freq comp)<br>▪ 14 – INTRACA_FCOMP<br>▪ 15 – XPT_PIN_CAL |
| Presweep computation | Computations to be done before the sweep; values are:<br>▪ 0 – NO ACTION PRE COMPUTE TYPE<br>▪ 1 – RGI DELTAS PRE COMPUTE TYPE<br>▪ 2 – APT BIAS PRE COMPUTE TYPE<br>▪ 3 – FREQCOMP RGI PRE COMPUTE TYPE<br>▪ 4 – XPT SWP2 FROM SWP1 DATA COMPUTE TYPE<br>▪ 5 – LIM_FREQCOMP_RGI_PRE_COMPUTE_TYPE |
| Compute from sweep | Sweep from which the presweep compute should be performed; values are:<br>▪ 0 – Sweep 1<br>▪ 1 – Sweep 2<br>▪ 2 – Sweep 3<br>▪ 3 – Sweep 4<br>▪ 4 – Sweep 5 |
| Master device ID | List of all master device IDs |
| Slave device ID | List of all slave device IDs; enter -1 if no slave device is available for a master |
| NV device to chain map | One entry per device; format:<br>▪ Device1,Chain for Device1;<br>▪ Device2,Chain for Device2;<br>▪ .... |
| Tx device to calibrate | List the device ID for the Tx device to be calibrated |
| Rx device to calibrate | List the device ID for the Rx devices to be calibrated |

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| Sweeps | Description |
|---|---|
| Waveform type | ▪ 0 – Default waveform (CDMA or WCDMA)<br>▪ 1 – Continuous waveform<br>▪ 2 – Offset Continuous waveform<br>▪ 3 – LTE PUSCH waveform<br>▪ 4 – Enhanced UL waveform |
| Start Resource Block (RB) | Start RB for LTE waveform |
| Number of RBs | Number of RBs to transmit on the PUSCH channel |
| Tx BW | Tx bandwidth for LTE waveform; values are:<br>▪ 0 – BW_1P4MHz<br>▪ 1 – BW_3MHz<br>▪ 2 – BW_5MHz<br>▪ 3 – BW_10MHz<br>▪ 4 –BW_15MHz<br>▪ 5 –BW_20MHz |
| Rx BW | ▪ 0 – BW_1P4MHz<br>▪ 1 – BW_3MHz<br>▪ 2 – BW_5MHz<br>▪ 3 – BW_10MHz<br>▪ 4 –BW_15MHz<br>▪ 5 –BW_20MHz |
| EUL waveform type | EUL waveform type |
| Ref channel | Tx linearizer reference channel |
| Upper bound channel | Tx linearizer upper bound channel |
| PA state list | Tx linearizer PA state list (comma-separated)<br>**Note:** PA state is an index into the PA range map NV. PA range is the physical control word to the PA to set a state. PA state is always in the order of decreasing power for all technologies except LTE. 0 – Lo to 3 – High for all technologies except LTE. |
| RGI list | Tx linearizer RGI list; this parameter defines the Start Stop Step (SSS) range of values for the RGI list |
| Bias list | This parameter defines the bias list for each one of the RGI list values |
| PA quiescent current | Tx linearizer quiescent current list |
| IQ gain | Tx linearizer IQ gain list |
| Env scale | Tx linearizer envelope scale list |
| HDET avgs | Number of HDET averages |
| LPM HDET avgs | Number of LPM HDET averages |
| Segment length | Tx linearizer segment length in µs |
| Config segment length | Radio setup segment length in µs |
| Tuning segment length | Radio retune segment length in µs |
| Minimum start Pwr | Minimum start power that should be met for current linearizer |
| Maximum expected Pwr | Maximum power that should not be exceeded for current linearizer (is set as input level on the box) |
| Lowest end Pwr | Minimum end power that should be met for current linearizer |

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| Sweeps | Description |
|---|---|
| Purpose list | NV purpose list<br>▪ 0 – Voice linearizer<br>▪ 1 – Single carrier data linearizer<br>▪ 2 – Multicarrier data linearizer<br>▪ 3 – Maximum bias linearizer (first sweep data)<br>▪ 4 – Linearizer FreqComp<br>▪ 5 – Limiting FreqComp<br>▪ 6 – ILPC FreqComp<br>▪ 7 – Secondary FreqComp |
| XPT Swp1 channel list | Defines the reference channel |
| XPT Swp1 PA state | Defines the power amplifier state to be used for Sweep 1 |
| XPT Swp1 RGI list | Sweep 1 RGI list |
| XPT Swp1 bias list | Sweep 1 bias list |
| XPT Swp1 IQ gain | IQ gain – Use for all xPT sweeps |
| XPT Swp1 Env scale | Envelope scale<br>▪ This Env scale value is only used in xPT sweep 1 and xPT Sweep 2 DPD capture.<br>▪ The Env Scale value in NV is used for M-Line, xPT Swp3 and xPT Freq Comp.<br>▪ The NV has a separate Env Scale field EPT and ET, so based on which mode you are in, it will use the corresponding value. |
| XPT Swp1 Num IQ samples | Number of IQ samples |
| XPT Swp1 Cap segment length | Capture segment length in µs |
| XPT Swp1 RGI delta RGI List | Sweep 1 RGI delta list |
| XPT Swp1 RGI delta bias | Sweep 1 RGI delta bias |
| XPT_Swp1_Minimum_RGI_Delta_Power_dBm100 | Limit for the minimum RGI delta power for XPT Sweep 1; this parameter is scale by 1/100, e.g., to get a value of 0.25, the limit should be set to 25 (25/100). By default this parameter is set to zero. |
| XPT Swp1 RGI delta segment length | RGI Delta segment length in µs |
| Postsweep computation | ▪ 0 – NO_ACTION_POST_COMPUTE_TYPE<br>▪ 1 – XPT_SWP2_POST_COMPUTE_TYPE<br>▪ 2 – INT_CAL_OFFSET_COMPUTE_TYPE<br>▪ 3 – INT_CAL_POWER_COMPUTE_TYPE<br>▪ 4 – XPT_SWP2_SAVE_RGI_FOR_DELAY_SWP_TYPE<br>▪ 5 – XPT_SWP2_SAVE_BIAS_RGI_FOR_PIN_CAL_SWP_TYPE<br>▪ 6 – FBRX_CAL_FB_PATH_GAIN_COMPUTE_TYPE |
| Compute for sweep | Compute for which sweep |
| ***Sweep 2*** | |
| Sweep type | Sweep type |
| Presweep computation | Presweep computation |
| Compute from sweep | Computes from sweep |
| Master device ID | Master device ID |
| Slave device ID | Slave device ID |
| NV device to chain map | NV device to chain map |
| Tx device to calibrate | Tx device to calibrate |
| Rx device to calibrate | Rx device to calibrate |

| Sweeps | Description |
|---|---|
| Waveform type | Waveform type |
| Start RB | Start RB |
| Number of RBs | Number of RBs |
| Tx BW | Tx BW |
| Rx BW | Rx BW |
| EUL waveform type | EUL waveform type |
| XPT Swp2 channel list | Tx linearizer reference channel |
| XPT Swp2 EPT PA state | EPT Sweep 2 power amplifier state |
| XPT Swp2 ET PA state | ET Sweep 2 power amplifier state |
| EPT Swp2 Meas Pwr list | Expected measure power list values for EPT<br>**Note:** This list of values must be equal for all the sweeps. |
| EPT Swp2 Calc Pwr list | Interpolated power values for EPT<br>**Note:** This list of values must be equal for all the sweeps. |
| ET Swp2 Meas Pwr list | Expected measure power list values for ET<br>**Note:** This list of values must be equal for all the sweeps. |
| XPT Swp2 Num IQ samples | Number of IQ samples |
| XPT Swp2 Cap segment length | Capture segment length in µs |
| XPT Swp2 MLine bias option | ▪ 0 – Fail if M-Line calculated bias > VdMax<br>▪ 1 – Cap Bias to VdMax if M-Line calculated bias > VdMax<br>▪ 2 – Selects the closest M-Line to the target power except the highest target power, which will select the next higher M-Line |
| Postsweep computation | Postsweep computation |
| Compute for sweep | Compute for sweep |
| ***Sweep 3*** | |
| Sweep type | SweepType |
| Presweep computation | PreSweepComputation |
| Compute from sweep | ComputeFromSweep |
| Device selection→Master device ID | MasterDeviceID |
| Device selection→Slave device ID | SlaveDeviceID |
| Device selection→NV device to chain map | NVDeviceToChainMap |
| Device selection→Tx device to calibrate | TxDeviceToCalibrate |
| Device selection→Rx device to calibrate | RxDeviceToCalibrate |
| Waveform→Waveform type | WaveformType |
| Waveform→Start RB | WaveformStartRB |
| Waveform→Number of RBs | NumberOfRBs |
| Waveform→Tx BW | TxBW |
| Waveform→Rx BW | RxBW |
| Waveform→EUL waveform type | EULWaveformType |
| Ref channel | Tx linearizer reference channel |
| Upper bound channel | Tx linearizer upper bound channel |
| PA state list | PAStateList |

| Sweeps | Description |
|---|---|
| RGI list | RGIList |
| Bias list | BiasList |
| PA quiescent current | PAQuiescentCurrent |
| IQ gain | IQGain |
| Env scale | EnvScale |
| HDET avgs | HDETAvgs |
| LPM HDET avgs | LPMHDETAvgs |
| Segment length | SegmentLength |
| Config segment length | ConfigSegmentLength |
| Tuning segment length | TuningSegmentLength |
| Minimum start Pwr | Minimum start power for each PA State List mode |
| Maximum expected Pwr | Maximum expected power for each PA State List mode |
| Lowest end Pwr | The lowest expected power when the PA is operated in LPM |
| Purpose list | PurposeList |
| XPT Swp3 channel list | The channel on which to perform the linearization measurements; must be same with Ref_Channel |
| EPT Swp3 Meas Pwr list | Expected measure power list values for EPT<br>**Note:** This list of values must be equal for all the sweeps. |
| EPT Swp3 Calc Pwr list | Interpolated power values for EPT<br>**Note:** This list of values must be equal for all the sweeps. |
| ET Swp3 Meas Pwr list | Expected measure power list values for ET<br>**Note:** This list of values must be equal for all the sweeps. |
| XPT Swp3 deviation from Exp power Minimum dBm100 | Defines the minimum limit for the power value relative to the expected power; by default this parameter is equal to -300 |
| XPT Swp3 deviation from Exp power Maximum dBm100 | Defines the maximum limit for the power value relative to the expected power; by default this parameter is equal to 300 |
| XPT_Swp3_Linearizer_Delta_Meas_ Power_Min_ dBm100 | Limit for the minimum linearizer delta power for XPT Sweep 3; this parameter is scaled by 1/100, e.g., to get a value of 0.25, the limit should be set to 25 (25/100); by default this parameter is set to 0<br>**Note:** Negative deltas are allowed. |
| XPT Swp3 segment length | xPT sweep segment length in µs |
| Postsweep computation | PostSweepComputation |
| Compute for sweep | ComputeForSweep |
| XPT Swp3 deviation from Exp power Minimum dBm100 | Defines the minimum limit for the power value relative to the expected power; by default this parameter is equal to -300 |
| XPT Swp3 Deviation From Exp Power Maximum dBm100 | Defines the maximum limit for the power value relative to the expected power; by default this parameter is equal to 300 |
| XPT_Swp3_Linearizer_Delta_Meas_ Power_Min_dBm100 | Limit for the minimum linearizer delta power for XPT Sweep 3; this parameter is scaled by 1/100, e.g., to get a value of 0.25, the limit should be set to 25 (25/100); by default this parameter is set to 0<br>**Note:** Negative deltas are allowed. |
| XPT Swp3 segment length | xPT sweep segment length in µs |
| Postsweep computation | PostSweepComputation |
| Compute for sweep | ComputeForSweep |

| Sweeps | Description |
|---|---|
| ***Sweep 4*** | |
| Sweep type | SweepType |
| Pre-sweep computation | PreSweepComputation |
| Compute from sweep | ComputeFromSweep |
| Device selection→Master device ID | MasterDeviceID |
| Device selection→Slave device ID | SlaveDeviceID |
| Device selection→NV device to chain map | NVDeviceToChainMap |
| Device selection→Tx device to calibrate | TxDeviceToCalibrate |
| Device selection→Rx device to calibrate | RxDeviceToCalibrate |
| Waveform→Waveform type | WaveformType |
| Waveform→Start RB | WaveformStartRB |
| Waveform→Number of RBs | NumberOfRBs |
| Waveform→Tx BW | TxBW |
| Waveform→Rx BW | RxBW |
| Waveform→EUL waveform type | EULWaveformType |
| Meas Tx | Flag that enables Tx measurements |
| Tx channel list | List of channels for Freq Comp from source data |
| RGI threshold list | RGI thresholds for Lin Freq Comp<br>**Note:** There must be one value per element in PA_State_List. |
| Use RGI Pwr list | Specifies the method to determine the RGI/Bias used in Freq Comp; values are:<br>▪ 0 – Use FreqComp_Pwr_List/APT_FreqComp_Offset_List<br>▪ 1 – Use RGI_List/Bias_List |
| FreqComp Pwr list | Freq comp power list |
| APT FreqComp offset list | The APT_FreqComp_Offset_List is used to determine the bias value used in freq comp.<br>The target power for determining the bias is FreqComp_Pwr_List[i]+APT_FreqComp_Offset_List[i] so for the following:<br>▪ <FreqComp_Pwr_List> 16, 0.5 </FreqComp_Pwr_List><br>▪ <APT_FreqComp_Offset_List> 2, 2 </APT_FreqComp_Offset_List><br>The target power used to determine the RGI is 16 dBm.<br>The target power used to determine the bias is 16+2 dBm.<br>The reason for this is to have the ability to perform the freq comp at a higher bias to ensure that the PA is not being saturated during freq comp. |
| PA state list | PAStateList |
| RGI list | RGIList |
| Bias list | BiasList |
| PA quiescent current | PAQuiescentCurrent |
| IQ gain | IQGain |
| Env scale | EnvScale |

| Sweeps | Description |
|---|---|
| HDET Avgs | HDETAvgs |
| LPM HDET Avgs | LPMHDETAvgs |
| Minimum FreqComp HDET | Minimum value for the HDET register parameter in freq comp; default value is 0 |
| Maximum FreqComp HDET | Maximum value for the HDET register parameter in freq comp; default value is 4095 |
| Segment length | SegmentLength |
| Config segment length | ConfigSegmentLength |
| Tuning segment length | TuningSegmentLength |
| Maximum FreqComp Pwr | List that contains the maximum powers allows during the freq comp sweep |
| Min FreqComp Pwr | List that contains the minimum powers allows during the freq comp sweep |
| Purpose list | Controls the purpose of freq comp; values are:<br>▪ 0 – Trigger (No FreqComp)<br>▪ 4 – LinFreqComp<br>▪ 5 – LimFreqComp<br>▪ 6 – ILPC FreqComp<br>If the selected purpose is 0, the corresponding value in Use_RGI_Pwr_List should be 1.<br>**Note:** There must be one value per element in PA_State_List. |
| XPT Fcomp Ref channel list | Freq comp ref channel list |
| XPT Upper Bound channel list | Upper bound channel list for freq comp |
| EPT Fcomp Meas Pwr list | Expected measure power list values for EPT<br>**Note:** This list of values must be equal for all the sweeps. |
| EPT Fcomp Calc Pwr list | Interpolated power values for EPT<br>**Note:** This list of values must be equal for all the sweeps. |
| ET Fcomp Meas Pwr list | Expected measure power list values for ET<br>**Note:** This list of values must be equal for all the sweeps. |
| XPT Fcomp deviation from Exp power Min dBm100 | Defines the minimum limit for the power value relative to the expected power; by default this parameter is equal to -300 |
| XPT Fcomp deviation from Exp power Maximum dBm100 | Defines the maximum limit for the power value relative to the expected power; by default this parameter is equal to 300 |
| Meas Rx | Flag that enables Rx measurements |
| Rx channel list | Rx channel list |
| Rx actions list | Specifies the Rx action to take for each given Rx Level in DL_Level_List; should be only one value per Rx item to be actually calibrated; do not include 255 (the extra step between channels), it is automatically included; actions are:<br>▪ 0 – Calibrate LNA offset 0 (calibrate DVGA offset for CDMA/WCDMA)<br>▪ 1 – Calibrate LNA offset 1<br>▪ 2 – Calibrate LNA offset 2<br>▪ n – Calibrate LNA offset n |
| Meas Rx | Flag that enables Rx measurements |
| Rx channel list | Rx channel list |

| Sweeps | Description |
|---|---|
| Rx actions list | Specifies the Rx action to take for each given Rx Level in DL_Level_ List; should be only one value per Rx item to be actually calibrated; do not include 255 (the extra step between channels), it is automatically included; actions are:<br>▪ 0 – Calibrate LNA offset 0 (calibrate DVGA offset for CDMA/WCDMA)<br>▪ 1 – Calibrate LNA offset 1<br>▪ 2 – Calibrate LNA offset 2<br>▪ n – Calibrate LNA offset n |
| Rx PM list | Rx PA power mode |
| DL level list | List of output power sets in the call box |
| Rx C0 Maximum limits | Rx C0 maximum limits |
| Rx C0 Min limits | Rx C0 minimum limits |
| Rx C1 Maximum limits | Rx C1 maximum limits |
| Rx C1 Min limits | Rx C1 minimum limits |
| Rx C2 Maximum limits | Rx C2 maximum limits |
| Rx C2 Min limits | Rx C2 minimum limits |
| Rx C3 Maximum limits | Rx C3 maximum limits |
| Postsweep computation | PostSweepComputation |
| Compute for sweep | ComputeForSweep |
| ***Sweep 5*** | |
| Sweep type | Sweep type |
| Presweep computation | Presweep computation |
| Compute from sweep | Compute from sweep |
| Master device ID | Master device ID |
| Slave device ID | Slave device ID |
| NV device to chain map | NV device to chain map |
| Tx device to calibrate | Tx device to calibrate |
| Rx device to calibrate | Rx device to calibrate |
| Waveform type | Waveform type |
| Start RB | Start RB |
| Number of RBs | Number of RBs |
| Tx BW | Tx BW |
| Rx BW | Rx BW |
| EUL waveform type | EUL waveform type |
| XPT_Delay_Swp_Channel_List | Channel list for delay sweep |
| XPT_Delay_Swp_Upper_Bound_ Channel_List | Upper bound channel list for delay sweep |
| XPT_Delay_Swp_Num_RB_List | Start RB for LTE waveform |
| XPT_Delay_Swp_Start_RB_List | Number of RBs to transmit on the PUSCH channel |
| XPT_Delay_Swp_BW_List | BW for LTE waveform; 3 to 10 MHz |
| XPT_Delay_Swp_5MHz_Offset_List | Delay offset between 5 MHz and 10 MHz |
| XPT_Delay_Swp_10MHz_Offset_List | Delay offset between 10 MHz and 10 MHz |
| XPT_Delay_Swp_20MHz_Offset_List | Delay offset between 20 MHz and 10 MHz |

| Sweeps | Description |
|---|---|
| XPT_Delay_Swp_PA_State | PA state for delay sweep |
| XPT_Delay_Swp_RGI_Fudge_From_Mline | RGI fudge added to the last RGI from XPT linearizer |
| XPT_Delay_Swp_Bias | PA bias for delay sweep |
| XPT_Delay_Swp_IQ_Gain | IQ gain for delay sweep |
| XPT_Delay_Swp_Env_Scale | Envelope scale for delay sweep |
| XPT_Delay_Swp_PA_Quiescent_Current | PA current for delay sweep |
| XPT_Delay_Swp_Num_IQ_Samples | Number of IQ samples |
| XPT_Delay_Swp_Cap_Segment_Length | Capture segment length for delay sweep in µs |
| Segment_Length | Segment_Length |
| Config_Segment_Length | Config_Segment_Length |
| Tuning_Segment_Length | Tuning_Segment_Length |
| Post_Sweep_Computation | Post_Sweep_Computation |
| Compute_For_Sweep | Compute for sweep |

#### 4.1.1.2 Disabling or enabling debug messages

To disable or enable debug messages:

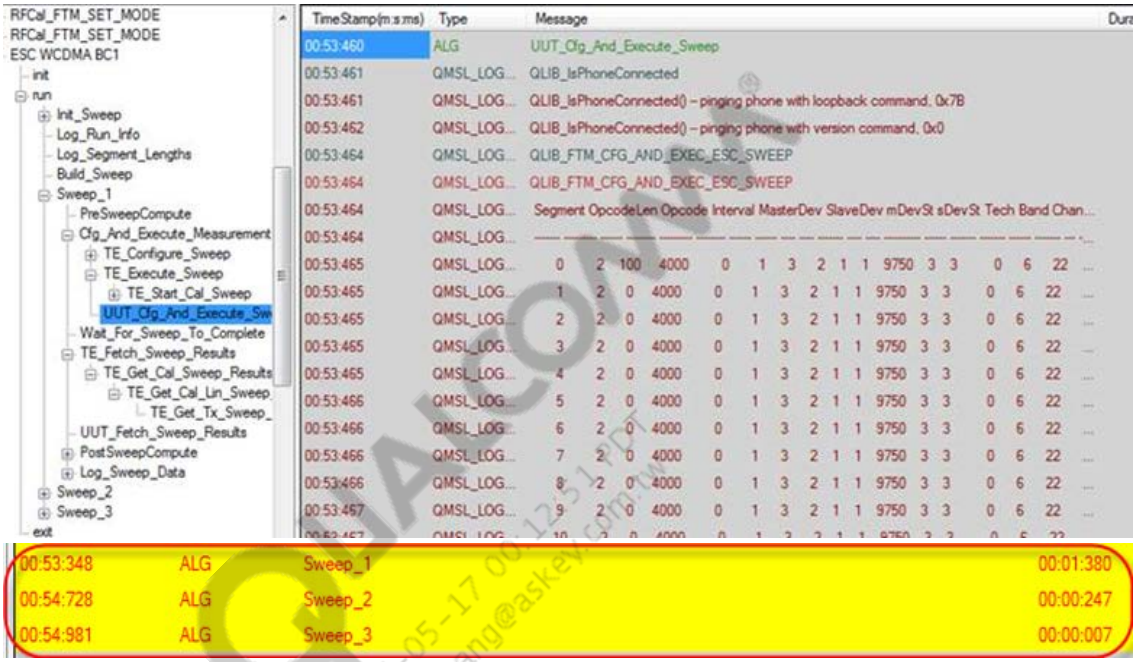
1. In the cal input file:
   - Disable – <En_Debug_Mode>0</En_Debug_Mode>
   - Enable – <En_Debug_Mode>1</En_Debug_Mode>
   - Increase Segment_Length for each XPT sweep
   - Increase <Segment_Length>4000,4000,4000,4000</Segment_Length> for all sweep

   For example:

```
<Node_Name name="LTE_B1_Dev0_0">
<RF_Mode>34</RF_Mode>
<En_Debug_Mode>1</En_Debug_Mode>
<NV_Generation_Params>
<NV_Mode>34</NV_Mode>
```

2. Multilevel TPL can be enabled by adding the following item in stationconfig.xml.

```
- <ItemGroup id="BoolItems">
    <ConfigItem value="1" name="EnableProfiling"/>
    <ConfigItem value="0" name="useLTEPersonalityForCal"/>
```




### 4.1.1.3 PA parameters input file

Table 4-3 lists PA parameters input file for ESC V4 PA.

**Table 4-3 ESC V4 PA_PARAMS_FILE**

| Parameters | Description |
|---|---|
| rf_mode | RF mode index, which uses different digital numbers to represent different bands, e.g.:<br>▪ WCDMA B1 rf_mode – 9<br>▪ LTE B1 rf_mode – 34 |
| IcqEt_table IcqEt; | Default Icq setting in xPT char such as RGI bias char, delay char, firmware delay char and Pin alignment char, etc. |
| IcqEpt_table IcqEpt; | Obsolete |
| IcqRgiDeltaSwp_table IcqRgiDeltaSwp | Icq setting used in ET calibration for both EPT/ET segment |
| MLineMaxPaSupplyEt_table MLineMaxPaSupplyEt | Defines maximal QPOET output voltage during ET calibration |
| MLineMaxPaSupplyEpt_table MLineMaxPaSupplyEpt | Obsolete; defines maximal QPOET output voltage during EPT calibration |

| Parameters | Description |
|---|---|
| RgiBiasCharCompPt_table RgiBiasCharCompPt | Defines RGI bias char compression point threshold; needs to be higher than the calibration compression point to provide some margin, e.g., if we a consider 3 dB compression point in ET calibration then we can use 5 dB as the threshold in RGI bias char |
| AclrDelayCharCompPoint_table AclrDelayCharCompPoint | Defines the compression point threshold in the ACLR delay char |
| AclrDelayCharVmax_table AclrDelayCharVmax | Defines the maximum PA bias setting in ACLR delay char |
| AclrDelayCharIqGain_table AclrDelayCharIqGain | Defines the optimal IQ gain settings in the ACLR delay char |
| StaticPinPoutCharRgiDeltaVmax_table StaticPinPoutCharRgiDeltaVmax | Defines the PA bias settings of the RGI delta sweep in the PinPout char |
| StaticPinPoutCharVmax_table StaticPinPoutCharVmax | Defines the PA bias settings for other sweeps except the RGI delta sweep in the PinPout char |
| FirmwareDelayIqGain_table FirmwareDelayIqGain | Defines the optimal IQ gain settings in the firmware delay char |
| FirmwareDelayCharRgiDetPaBias_table FirmwareDelayCharRgiDetPaBias | PA bias value for RGI determined in firmware delay char |
| FirmwareDelayVmax_table FirmwareDelayVmax | Defines the maximum PA bias setting in firmware delay char |
| FirmwareDelayVmin_table FirmwareDelayVmin | Defines the minimum PA bias setting in firmware delay char |
| FirmwareDelayCharCompPoint_table FirmwareDelayCharCompPoint | Defines the compression point threshold in the firmware delay char |
| IqGainRgiDeltaSwp_table IqGainRgiDeltaSwp | Defines the IQ gain setting of the RGI delta sweep in ET calibration; usually, this needs to set a low IQ gain for the RGI delta sweep, e.g.,150 for LTE, 400 for WCDMA, etc. |
| CalCompPtEpt_table CalCompPtEpt | Defines the EPT calibration compression point |
| CalCompPtEt_table CalCompPtEt | Defines the ET calibration and PIN char compression point |
| CharCompPtThresEt_table CharCompPtThresEt | (Obsolete) Defines the compression point threshold during Pin alignment char |
| MLineFitType_table MLineFitType | Defines the fit type for M-Line curve fitting in ET calibration; 0 – Linear fitting |
| MLineRegCoef_table MLineRegCoef | Parameter used in the M-Line linear fitting function; the default setting is acceptable |
| EtVmin_table EtVmin | Defines the lowest allowable PA bias value in ET calibration; PA bias cannot be allowed to be too low |
| Detrough_table | This is provided by the PA vendor and defines the mapping between the signal envelope and PA bias |

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## 4.1.2 xPT Sweep 1 – Full bias linearizer+XPT Swp1

xPT Sweepp1 includes full bias sweep in APT mode and xPT Sweep in EPT mode. The Full bias sweep is like bypass mode cal to learn the relationship between RGI and power by using fix bias value. cxPT Sweep is a new sweep for the ET/EPT system. In xPT Sweep1, the data will be used for calculating the M-Line.

### 4.1.2.1 Swp1 cal sequence

- Full bias linearizer at APT PA states
- XPT Swp1
  - Alignment sweep (four IQ captures for M-Line calculation) at EPT PA state
    - Four pairs of RGI and bias are from RGI Bias Char
    - RGI is high enough to well saturate PA to cover the M-Line n-dB compression point
    - Load AMAP/AMPM table
    - Firmware calculates four AMAM curves with proper scaling on Vout (Rx RMS calculation)
    - IQ captures at four predetermined (VCC, RGI) combinations
  - The four RGI values are filled in <XPT_Swp1_RGI_list> in the ESC input param file. In mass production, RGI values can be increased by 1 to 3 in order to provide margin on different phones.

    Figure 4-3 shows an alignment sweep.

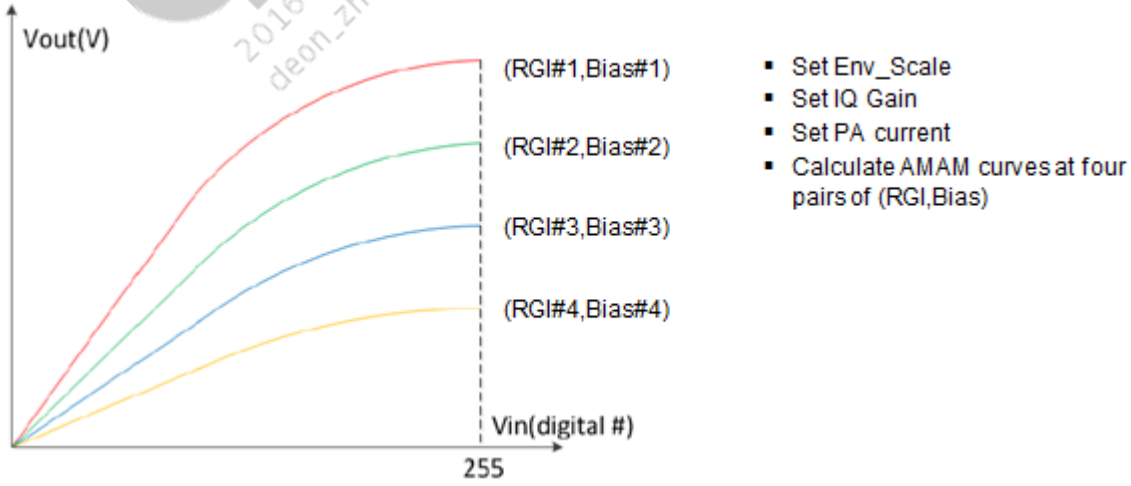



**Figure 4-3 Alignment sweep**

  - RGI delta sweep at APT PA state
    - Sweep RGI to learn power delta between the two RGI
    - PA operates in linear region
      - Use low IQ gain
    - Fix PA bias value

- Measure PA output power (dBm) at each RGI
  - RGI range in <XPT_Swp1_RGI_Delta_RGI_List> must be large enough to cover the four RGI values in <XPT_Swp1_RGI_list> and add a couple more RGI values

    Figure 4-4 shows ESC V4 calibration for Sweep 1.

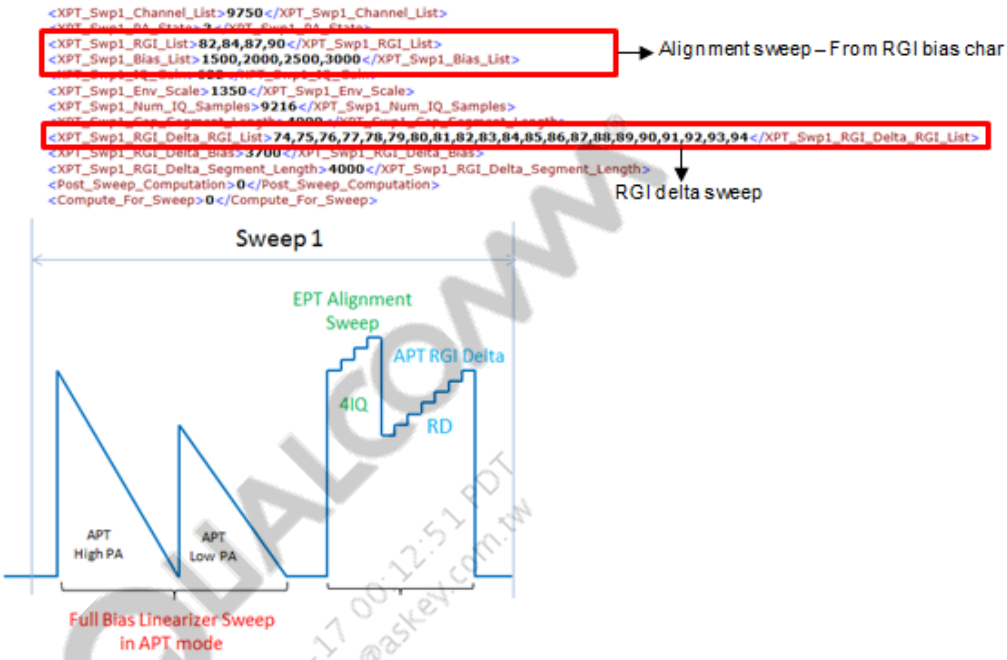



**Figure 4-4 ESC V4 Cal Sweep 1**

## 4.1.3 xPT Sweep 2

Digital Predistortion (DPD) will be done in Sweep2. To do IQ capture in four pairs of RGI and bias, firmware will pick up the RGI and bias from the M-Line based on target power. The M-Line table is calculated based on xPT Swp1 data (alignment sweep and RGI delta sweep).

### 4.1.3.1 Sweep2 cal sequence

- Principle of M-Line calculation for EPT and ET mode
  - ET – Calculate required RGI for desired output powers, scale nominal detrough
  - EPT – Calculate required (VCC, RGI) for desired output powers
  - Find n-dB compression points on each curve (from sweep1alignment sweep)
  - M-Line across m-points establish linear relationship among (Vin, Vout, VCC)
  - Software calculates m-point (Vin, Vout, VCC) and M-Line from properly scaled AMAM
  - Each RGI has its corresponding point on M-Line
- Cal sequence and parameter setting
  - <XPT_Swp2_MLine_Calc_Segment_Length> in the RF Cal input file is used to wait for finishing M-Line calculation in firmware

- Load Unity DPD (shown as a black line in the AMAM curve)
- Pick RGI and bias based on M-Line table
  - Pick the RGI and bias from M-Line table according to the power list in <ET_Swp2_Meas_Pwr_List>
- Measure ET and EPT power and do IQ capture
- EPT/ET DPD processing and contribute DPD table capture (shown as green, red, cyan, and magenta lines)

Figure 4-5 shows M-Line.

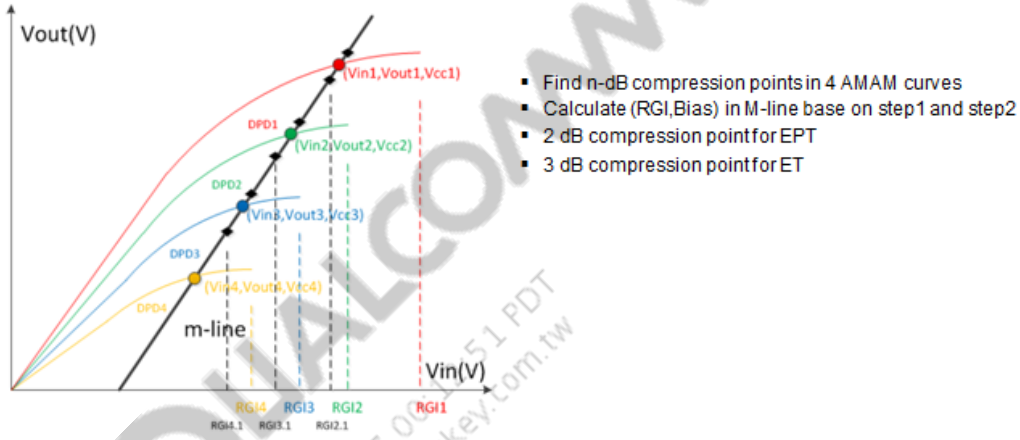
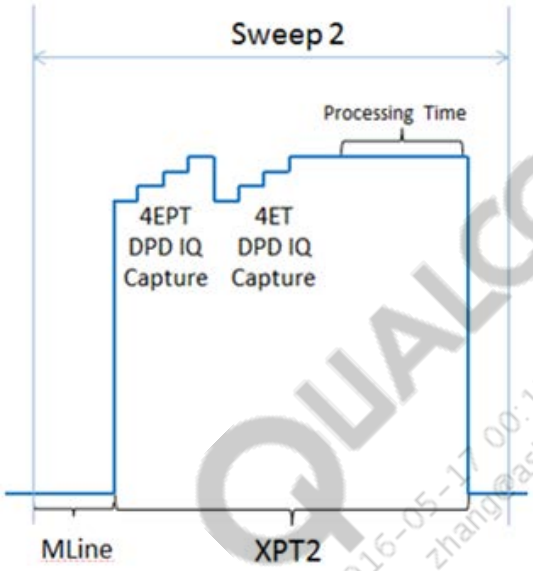



**Figure 4-5 M-Line**

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Figure 4-6 shows Esc V4 calibration for Sweep 2.

```
<XPT_Swp2_Channel_List>18900</XPT_Swp2_Channel_List>
<XPT_Swp2_EPT_PA_State>1</XPT_Swp2_EPT_PA_State>
<XPT_Swp2_ET_PA_State>0</XPT_Swp2_ET_PA_State>
<ET_Swp2_Meas_Pwr_List>18,20.5,23,25</ET_Swp2_Meas_Pwr_List>
<XPT_Swp2_Num_IQ_Samples>9216</XPT_Swp2_Num_IQ_Samples>
<XPT_Swp2_Cap_Segment_Length>1000</XPT_Swp2_Cap_Segment_Length>
<XPT_Swp2_MLine_Calc_Segment_Length>6000</XPT_Swp2_MLine_Calc_Segment_Length>
<XPT_Swp2_IQ_Processing_Segment_Length>24000</XPT_Swp2_IQ_Processing_Segment_Length>
<XPT_Swp2_MLine_Bias_Option>0</XPT_Swp2_MLine_Bias_Option>
<Post_Sweep_Computation>1</Post_Sweep_Computation>
<Compute_For_Sweep>2</Compute_For_Sweep>
```




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dawn_zhang@askey.com.tw

**Figure 4-6 ESC V4 Cal Sweep 2**

Figure 4-7 shows a Detrough table with Vdmin of 1.0 V.

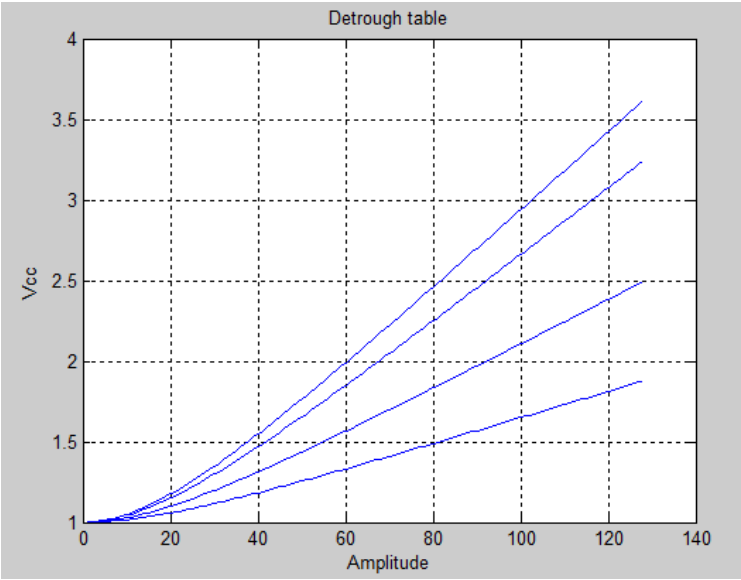



**Figure 4-7 Detrough table with Vdmin = 1.0 V**

## 4.1.4 xPT Sweep 3 – APT Sweep+XPT Swp3

Sweep3 includes the APT linearizer sweep and xPT Swp3. In APT linearizer sweep, the picked bias is associated with the APT char table and full bias sweep data. In xPT Swp3, the goal is to measure the Tx power after applying the DPDs, which are from Sweep2 for each target power.

### 4.1.4.1 Sweep3 cal sequence

- APT linearizer sweep
  - The RGI sweep range and step size must be the same as maximum bias linearizer at APT PA states (Sweep 1)
- XPT Swp3
  - Load EPT AMAM/AMPM and (RGI, VCC) from Sweep 2
  - Measure EPT linearizer power
  - Load ET AMAM/AMPM and (detrough, VCC, RGI) from Sweep 2
  - Measure ET linearizer power

Figure 4-8 shows ESC V4 calibration for Sweep 3.

```
<XPT_Swp3_Channel_List>18900</XPT_Swp3_Channel_List>
<ET_Swp3_Meas_Pwr_List>18,20.5,23,25</ET_Swp3_Meas_Pwr_List>
<XPT_Swp3_Segment_Length>1000</XPT_Swp3_Segment_Length>
<Post_Sweep_Computation>0</Post_Sweep_Computation>
<Compute_For_Sweep>0</Compute_For_Sweep>
```

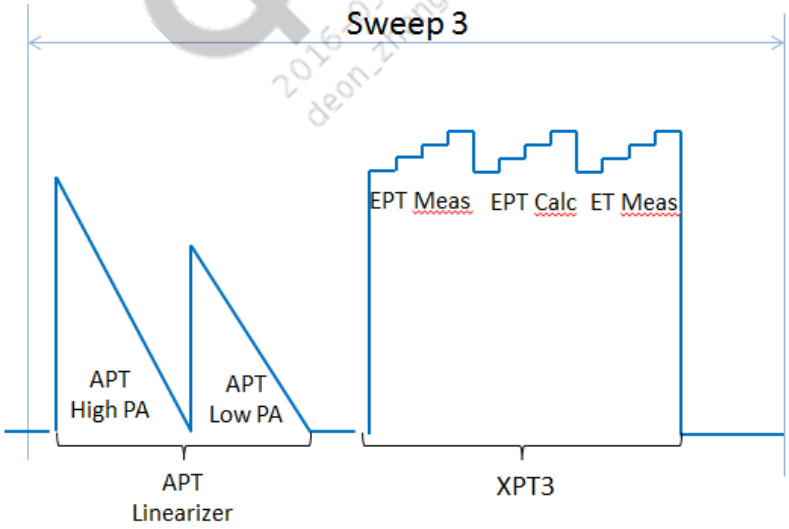



**Figure 4-8 ESC V4 Cal Sweep 3**

### 4.1.5 Freq comp sweep

Freq comp cal includes the APT freq comp and XPT freq comp. In XPT freq comp, the goal is the same as XPT Swp3, which is measure the power after applying DPD.

- APT freq comp
- Load EPT DPD tables (eight DPD tables from Sweep 2)
- Load pin offset data (pin offset data from Pin alignment characterization)
- Do EPT freq comp (with eight DPD tables)
- Load ET DPD tables (four DPD tables from Sweep 2)
- Load pin offset data (pin offset data from Pin alignment characterization)
- Do ET freq comp (with four DPD tables)

**NOTE:** The power list in Sweep 2/Sweep 3 and freq comp must be the same.

- Sweep 2 – <ET_Swp2_Meas_Pwr_List>
- Sweep 3 – <ET_Swp3_Meas_Pwr_List>
- Freq comp – <ET_Fcomp_Meas_Pwr_List>

**NOTE:** The channel list in freq comp and Pin alignment characterization must be the same. The channel list in TX_CAL_CHAN NV must be the same as the channel list in Pin alignment characterization.

Figure 4-9 shows ESC V4 calibration freq comp for Sweep 4.

```
<XPT_Fcomp_Ref_Channel_List>18900</XPT_Fcomp_Ref_Channel_List>
<XPT_Upper_Bound_Channel_List>19199</XPT_Upper_Bound_Channel_List>
<ET_Fcomp_Meas_Pwr_List>18,20.5,23,25</ET_Fcomp_Meas_Pwr_List>
```

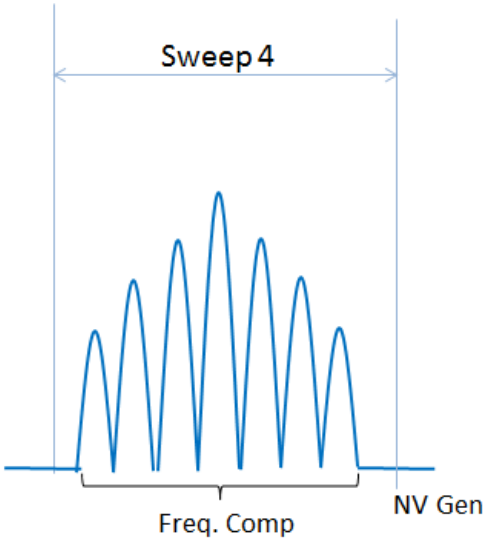



**Figure 4-9 ESC V4 Cal freq comp**

### 4.1.6 Delay calibration

- Delay calibration is not a requirement for each technology or band. It is optional.
- The customer can choose to enable/disable delay calibration as long as delay characterization data is sufficient for the customer to have enough performance margin.
- Delay calibration uses the same algorithm as firmware delay characterization; see Section 5.3.3. However, in delay calibration, RGI is from M-Line picked RGI+fudge factor. In delay characterization, RGI is from reaching the maximum target power or compression point.
- The RB settings listed in Waveform parameters are applied while doing delay IQ capture.
- The following XPT_Delay_Swp parameters are not used for delay calibration:
  - XPT_Delay_Swp_Num_RB_List
  - XPT_Delay_Swp_Start_RB_List
  - XPT_Delay_Swp_BW_List

Figure 4-10 shows a delay calibration.

```
<Sweep_Type>9</Sweep_Type>
<Pre_Sweep_Computation>0</Pre_Sweep_Computation>
<Compute_From_Sweep>0</Compute_From_Sweep>
<Device_Selection>
    <Master_Device_ID>0</Master_Device_ID>
    <Slave_Device_ID>1</Slave_Device_ID>
    <NV_Device_To_Chain_Map>0,0:1,1</NV_Device_To_Chain_Map>
    <Tx_Device_To_Calibrate>0</Tx_Device_To_Calibrate>
    <Rx_Device_To_Calibrate>0,1</Rx_Device_To_Calibrate>
</Device_Selection>
<Waveform>
    <Waveform_Type>3</Waveform_Type>
    <Start_RB>7</Start_RB>
    <Number_Of_RBs>36</Number_Of_RBs>
    <Tx_BW>3</Tx_BW>
    <Rx_BW>3</Rx_BW>
    <EUL_Waveform_Type>0</EUL_Waveform_Type>
</Waveform>
<XPT_Delay_Swp_Channel_List>18050,18083,18117,18150,18183,18217,18250,18283,18317,18350,18383,18417,18450
<XPT_Delay_Swp_Upper_Bound_Channel_List>18067,18100,18134,18167,18200,18234,18267,18300,18334,18367,18400
<XPT_Delay_Swp_Num_RB_List>36,36,36,36,36,36,36,36,36,36,36,36,36,36,36,36,36,36</XPT_Delay_Swp_Num_RB_List>
<XPT_Delay_Swp_Start_RB_List>7,7,7,7,7,7,7,7,7,7,7,7,7,7,7,7,7,7</XPT_Delay_Swp_Start_RB_List>
<XPT_Delay_Swp_BW_List>3,3,3,3,3,3,3,3,3,3,3,3,3,3,3,3,3,3</XPT_Delay_Swp_BW_List>
<XPT_Delay_Swp_5MHz_Offset_List>303,305,309,311,311,309,309,311,307,310,308,307,310,310,310,309</XPT_Dela
<XPT_Delay_Swp_10MHz_Offset_List>0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0</XPT_Delay_Swp_10MHz_Offset_List>
<XPT_Delay_Swp_20MHz_Offset_List>-605,-604,-600,-600,-599,-602,-602,-601,-603,-603,-603,-603,-602,-602,-6
<XPT_Delay_Swp_PA_State>0</XPT_Delay_Swp_PA_State>
<XPT_Delay_Swp_RGI_Fudge_From_MLine>0</XPT_Delay_Swp_RGI_Fudge_From_MLine>
<XPT_Delay_Swp_Bias>3800</XPT_Delay_Swp_Bias>
<XPT_Delay_Swp_IQ_Gain>200</XPT_Delay_Swp_IQ_Gain>
<XPT_Delay_Swp_Env_Scale>820</XPT_Delay_Swp_Env_Scale>
<XPT_Delay_Swp_PA_Quiescent_Current>156</XPT_Delay_Swp_PA_Quiescent_Current>
<XPT_Delay_Swp_Num_IQ_Samples>9216</XPT_Delay_Swp_Num_IQ_Samples>
<XPT_Delay_Swp_Cap_Segment_Length>2000</XPT_Delay_Swp_Cap_Segment_Length>
<Segment_Length>4000</Segment_Length>
<Config_Segment_Length>60000</Config_Segment_Length>
<Tuning_Segment_Length>18000</Tuning_Segment_Length>
<Post_Sweep_Computation>0</Post_Sweep_Computation>
<Compute_For_Sweep>0</Compute_For_Sweep>
```

**Figure 4-10 Delay calibration**

## 4.1.7 xPT calibration result

This section includes a calibration result and QXDM debug message. To enable the QXDM debug message, refer to add En_Debug_Mode in the Cal input file and enable the RF and FTM log in the QXDM F3 message.

In the Cal Input file, enable <En_Debug_Mode>1</En_Debug_Mode; for example:

```
<Node_Name name="LTE_B1_Dev0_0">
<RF_Mode>34</RF_Mode>
<En_Debug_Mode>1</En_Debug_Mode>
<NV_Generation_Params>
<NV_Mode>34</NV_Mode>
```

### 4.1.7.1 Sweep 1 – Alignment sweep

Figure 4-11 shows measurements for an alignment sweep.

<XPT_Swp1_Channel_List>**9750**</XPT_Swp1_Channel_List>
<XPT_Swp1_PA_State>**3**</XPT_Swp1_PA_State>
<XPT_Swp1_RGI_List>**81,82,84,86**</XPT_Swp1_RGI_List>
<XPT_Swp1_Bias_List>**1500,2000,2500,3000**</XPT_Swp1_Bias_List>
<XPT_Swp1_IQ_Gain>**622**</XPT_Swp1_IQ_Gain>
<XPT_Swp1_Env_Scale>**1350**</XPT_Swp1_Env_Scale>
<XPT_Swp1_Num_IQ_Samples>**9216**</XPT_Swp1_Num_IQ_Samples>
<XPT_Swp1_Cap_Segment_Length>**4000**</XPT_Swp1_Cap_Segment_Length>
<XPT_Swp1_RGI_Delta_RGI_List>**74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92**</XPT_Swp1_RGI_Delta_RGI_List>
<XPT_Swp1_RGI_Delta_Bias>**3700**</XPT_Swp1_RGI_Delta_Bias>
<XPT_Swp1_RGI_Delta_Segment_Length>**4000**</XPT_Swp1_RGI_Delta_Segment_Length>
<Post_Sweep_Computation>**0**</Post_Sweep_Computation>
<Compute_For_Sweep>**0**</Compute_For_Sweep>

**MEASUREMENTS: Tx XPT Swp1 - Alignment Sweep**

| Sweep Type | Channel | PA State | TxAGC | Vcc | cq | XPT Swp1 Power | Links | Time (s) | Exceptions |
|---|---|---|---|---|---|---|---|---|---|
| XPT_SWP1 | 9750 | 3 | 81 | 1500 | 56 | 19.4 | | | |
| XPT_SWP1 | 9750 | 3 | 82 | 2000 | 56 | 22.4 | | | |
| XPT_SWP1 | 9750 | 3 | 84 | 2500 | 56 | 24.9 | | | |
| XPT_SWP1 | 9750 | 3 | 86 | 3000 | 56 | 26.7 | | | |

**Figure 4-11 Alignment sweep table and measurements**


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Figure 4-12 shows Delta sweep table.

**MEASUREMENTS: Tx XPT Swp1 - RGI Delta Sweep**

| Sweep Type | Channel | PA State | TxAGC | Vcc | Icq | XPT Swp1 RGI Delta Swp Power |
|---|---|---|---|---|---|---|
| XPT_SWP1 | 9750 | 3 | 74 | 3700 | 255 | 9.4 |
| XPT_SWP1 | 9750 | 3 | 75 | 3700 | 255 | 10.8 |
| XPT_SWP1 | 9750 | 3 | 76 | 3700 | 255 | 11.6 |
| XPT_SWP1 | 9750 | 3 | 77 | 3700 | 255 | 12.6 |
| XPT_SWP1 | 9750 | 3 | 78 | 3700 | 255 | 13.4 |
| XPT_SWP1 | 9750 | 3 | 79 | 3700 | 255 | 14.7 |
| XPT_SWP1 | 9750 | 3 | 80 | 3700 | 255 | 15.8 |
| XPT_SWP1 | 9750 | 3 | 81 | 3700 | 255 | 17 |
| XPT_SWP1 | 9750 | 3 | 82 | 3700 | 255 | 18.4 |
| XPT_SWP1 | 9750 | 3 | 83 | 3700 | 255 | 19.2 |
| XPT_SWP1 | 9750 | 3 | 84 | 3700 | 255 | 20.5 |
| XPT_SWP1 | 9750 | 3 | 85 | 3700 | 255 | 21.5 |
| XPT_SWP1 | 9750 | 3 | 86 | 3700 | 255 | 21.8 |
| XPT_SWP1 | 9750 | 3 | 87 | 3700 | 255 | 22.5 |
| XPT_SWP1 | 9750 | 3 | 88 | 3700 | 255 | 23.2 |

**Figure 4-12 Delta sweep table**

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Figure 4-13 and Figure 4-14 show Sweep 1 log for QXDM Professional™ (QXDM Pro).

```
Num RGI Pwr Pairs = 19
=== RGI Pwr Table ===
RGI, Power = 74, 94
RGI, Power = 75, 108
RGI, Power = 76, 116
RGI, Power = 77, 126
RGI, Power = 78, 134
RGI, Power = 79, 147
RGI, Power = 80, 158
RGI, Power = 81, 170
RGI, Power = 82, 184
RGI, Power = 83, 192
RGI, Power = 84, 205
RGI, Power = 85, 215
RGI, Power = 86, 218
RGI, Power = 87, 225
RGI, Power = 88, 232
RGI, Power = 89, 235
RGI, Power = 90, 244
RGI, Power = 91, 255
RGI, Power = 92, 258
```

**Figure 4-13 QXDM Pro log – Sweep 1 (1 of 2)**

```
===== M Line Compute =====
Num XPT Cal Data = 4
Num RGI Delta Data = 19
===== M Line SP1 DATA =====
RGI   Vin   Pout  Vout  SMPS  Vcc  RMS(AMAM)
81    158   1940   208   1500   150   15667
82    185   2240   294   2000   200   15678
84    236   2490   393   2500   250   15597
86    275   2670   483   3000   300   15588
===== M Points =====
Index for ndB point : 205
M Point[0] = 12788, 25086, 15000
Index for ndB point : 210
M Point[1] = 15389, 35844, 20000
Index for ndB point : 194
M Point[2] = 18112, 46560, 25000
Index for ndB point : 188
M Point[3] = 20389, 56494, 30000
```

**Figure 4-14 QXDM Pro log – Sweep 1 (2 of 2)**


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### 4.1.7.2 Sweep 2 – IQ capture

Figure 4-15 and Figure 4-16 show log and measurements for IQ captures.

```
<XPT_Swp2_Channel_List>9750</XPT_Swp2_Channel_List>
<XPT_Swp2_EPT_PA_State>2</XPT_Swp2_EPT_PA_State>
<XPT_Swp2_ET_PA_State>3</XPT_Swp2_ET_PA_State>
<ET_Swp2_Meas_Pwr_List>19.5,21.5,23.5,25.5</ET_Swp2_Meas_Pwr_List>
<XPT_Swp2_Num_IQ_Samples>9216</XPT_Swp2_Num_IQ_Samples>
<XPT_Swp2_Cap_Segment_Length>4000</XPT_Swp2_Cap_Segment_Length>
<XPT_Swp2_MLine_Calc_Segment_Length>6000</XPT_Swp2_MLine_Calc_Segment_Length>
<XPT_Swp2_IQ_Processing_Segment_Length>30000</XPT_Swp2_IQ_Processing_Segment_Length>
<XPT_Swp2_MLine_Bias_Option>0</XPT_Swp2_MLine_Bias_Option>
<Post_Sweep_Computation>1,4</Post_Sweep_Computation>
<Compute_For_Sweep>2,4</Compute_For_Sweep>
```

**Figure 4-15 IQ capture**

```
MLine bias option is 0
ET MLine index beyond which MLine Vcc > VdMax(3800) is 11
====== Selection for Swp2 ET Meas Powers ======
PA   RGI   Vcc   Icq   IQ Gain  Env Scale  Delay  ReqPower(dB100) MLinePower(dB100)
3    80    1686  156   622      1350       0      1950            2018
3    81    2085  156   622      1350       0      2150            2237
3    82    2627  156   622      1350       0      2350            2467
3    83    2978  156   622      1350       0      2550            2589
```

**Figure 4-16 QXDM Pro log – IQ capture**

### 4.1.7.3 Sweep 3 – Linearizer

Figure 4-17 and Figure 4-18 show Linearizer measurements and table.

```
<XPT_Swp3_Channel_List>9750</XPT_Swp3_Channel_List>
<ET_Swp3_Meas_Pwr_List>19.5,21.5,23.5,25.5</ET_Swp3_Meas_Pwr_List>
<XPT_Swp3_Segment_Length>4000</XPT_Swp3_Segment_Length>
<Post_Sweep_Computation>0</Post_Sweep_Computation>
<Compute_For_Sweep>0</Compute_For_Sweep>
```

**Figure 4-17 Linearizer**

**MEASUREMENTS: Tx XPT Swp3 - Linearizer**

| Sweep Type | Channel | XPT Mode | PA State | TxAGC | Vcc | Exp Power | XPT Swp3 Power |
|---|---|---|---|---|---|---|---|
| XPT_SWP3 | 9750 | 2 | 3 | 80 | 1686 | 19.5 | 20.2 |
| XPT_SWP3 | 9750 | 2 | 3 | 81 | 2085 | 21.5 | 22.4 |
| XPT_SWP3 | 9750 | 2 | 3 | 82 | 2627 | 23.5 | 24.8 |
| XPT_SWP3 | 9750 | 2 | 3 | 83 | 2978 | 25.5 | 26 |

**Figure 4-18 Linearizer table**

#### 4.1.7.4 Sweep 4 – xPT freq comp

- Perform 16 channels xPT freq comp calibration.
- Apply DPD table corresponding to ET_Fcomp_Meas_Pwr_List for each channel:
  - There are four DPD tables in Sweep 2.
  - Each DPD table corresponds to Meas_Pwr.
  - The four Meas_Pwr which are listed in Meas_Pwr_List of Sweep 2, Sweep 3, and xPT freq comp are the same.
  - Apply four DPD tables for each table and get four freq comp values for each channel.

The following screenshots show DPD tables in 16 channels.

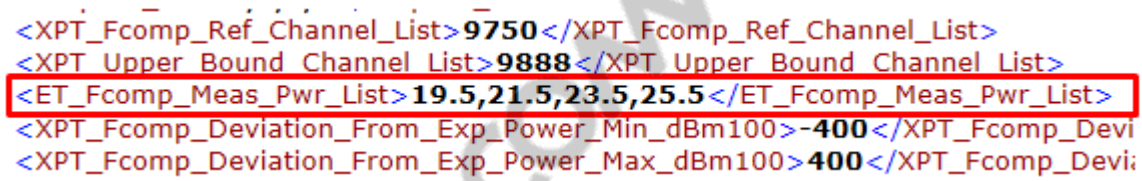



**Figure 4-19 xPT freq comp**

### Apply first DPD table in 16 channels

**MEASUREMENTS: Tx XPT - FreqComp**

| Sweep Type | Channel | Xpt Mode | Meas Exp Power | XPT FComp Power |
|---|---|---|---|---|
| XPT_FREQCOMP | 9621 | 2 | 1950 | 20.1 |
| XPT_FREQCOMP | 9638 | 2 | 1950 | 20.3 |
| XPT_FREQCOMP | 9656 | 2 | 1950 | 20.4 |
| XPT_FREQCOMP | 9673 | 2 | 1950 | 20.3 |
| XPT_FREQCOMP | 9691 | 2 | 1950 | 20.3 |
| XPT_FREQCOMP | 9708 | 2 | 1950 | 20.2 |
| XPT_FREQCOMP | 9726 | 2 | 1950 | 20.2 |
| XPT_FREQCOMP | 9750 | 2 | 1950 | 20.2 |
| XPT_FREQCOMP | 9761 | 2 | 1950 | 20.2 |

Apply second DPD table in 16 channels

| | | | | |
|---|---|---|---|---|
| XPT_FREQCOMP | 9621 | 2 | 2150 | 22.3 |
| XPT_FREQCOMP | 9638 | 2 | 2150 | 22.5 |
| XPT_FREQCOMP | 9656 | 2 | 2150 | 22.6 |
| XPT_FREQCOMP | 9673 | 2 | 2150 | 22.6 |
| XPT_FREQCOMP | 9691 | 2 | 2150 | 22.5 |
| XPT_FREQCOMP | 9708 | 2 | 2150 | 22.5 |
| XPT_FREQCOMP | 9726 | 2 | 2150 | 22.4 |
| XPT_FREQCOMP | 9750 | 2 | 2150 | 22.4 |
| XPT_FREQCOMP | 9761 | 2 | 2150 | 22.4 |

### Apply third DPD table in 16 channels

| | | | | |
|---|---|---|---|---|
| XPT_FREQCOMP | 9621 | 2 | 2350 | 24.6 |
| XPT_FREQCOMP | 9638 | 2 | 2350 | 24.8 |
| XPT_FREQCOMP | 9656 | 2 | 2350 | 24.9 |
| XPT_FREQCOMP | 9673 | 2 | 2350 | 24.9 |
| XPT_FREQCOMP | 9691 | 2 | 2350 | 24.8 |
| XPT_FREQCOMP | 9708 | 2 | 2350 | 24.8 |
| XPT_FREQCOMP | 9726 | 2 | 2350 | 24.8 |
| XPT_FREQCOMP | 9750 | 2 | 2350 | 24.7 |
| XPT_FREQCOMP | 9761 | 2 | 2350 | 24.7 |
| XPT_FREQCOMP | 9778 | 2 | 2350 | 24.7 |

Apply fourth DPD table in 16 channels

| | | | | |
|---|---|---|---|---|
| XPT_FREQCOMP | 9621 | 2 | 2550 | 25.8 |
| XPT_FREQCOMP | 9638 | 2 | 2550 | 26.1 |
| XPT_FREQCOMP | 9656 | 2 | 2550 | 26.1 |
| XPT_FREQCOMP | 9673 | 2 | 2550 | 26.1 |
| XPT_FREQCOMP | 9691 | 2 | 2550 | 26.1 |
| XPT_FREQCOMP | 9708 | 2 | 2550 | 26 |
| XPT_FREQCOMP | 9726 | 2 | 2550 | 26 |
| XPT_FREQCOMP | 9750 | 2 | 2550 | 26 |
| XPT_FREQCOMP | 9761 | 2 | 2550 | 26 |
| XPT_FREQCOMP | 9778 | 2 | 2550 | 25.9 |

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## 4.1.7.5 Enable M-Line debug information in the tool side

In order to get M-Line information from the tool side, <XPT_Swp2_Enable_Mline_Debug_Info> needs to be set to TRUE. The following shows how to enable M-Line debug.

```
<XPT_Swp2_Channel_List>9400</XPT_Swp2_Channel_List>
<XPT_Swp2_EPT_PA_State>2</XPT_Swp2_EPT_PA_State>
<XPT_Swp2_ET_PA_State>3</XPT_Swp2_ET_PA_State>
<ET_Swp2_Meas_Pwr_List>19.5,21.5,23.5,25.5</ET_Swp2_Meas_Pwr_List>
<XPT_Swp2_Num_IQ_Samples>9216</XPT_Swp2_Num_IQ_Samples>
<XPT_Swp2_Cap_Segment_Length>4000</XPT_Swp2_Cap_Segment_Length>
<XPT_Swp2_MLine_Calc_Segment_Length>8000</XPT_Swp2_MLine_Calc_Segment_Length>
<XPT_Swp2_IQ_Processing_Segment_Length>30000</XPT_Swp2_IQ_Processing_Segment_Length>
<XPT_Swp2_MLine_Bias_Option>0</XPT_Swp2_MLine_Bias_Option>
<XPT_Swp2_Enable_Mline_Debug_Info>TRUE</XPT_Swp2_Enable_Mline_Debug_Info>
<Post_Sweep_Computation>1</Post_Sweep_Computation>
<Compute_For_Sweep>2</Compute_For_Sweep>
```

#### 4.1.7.5.1 M-Line information in the calibration report

Figure 4-20 and Figure 4-21 show M-Line table is calculated from the firmware side based on the Sweep1 data (alignment sweep and RGI delta sweep). In the Sweep2, the tool side sends target power to modem software. Modem software will pick the RGI and bias for certain target power and does IQ capture.

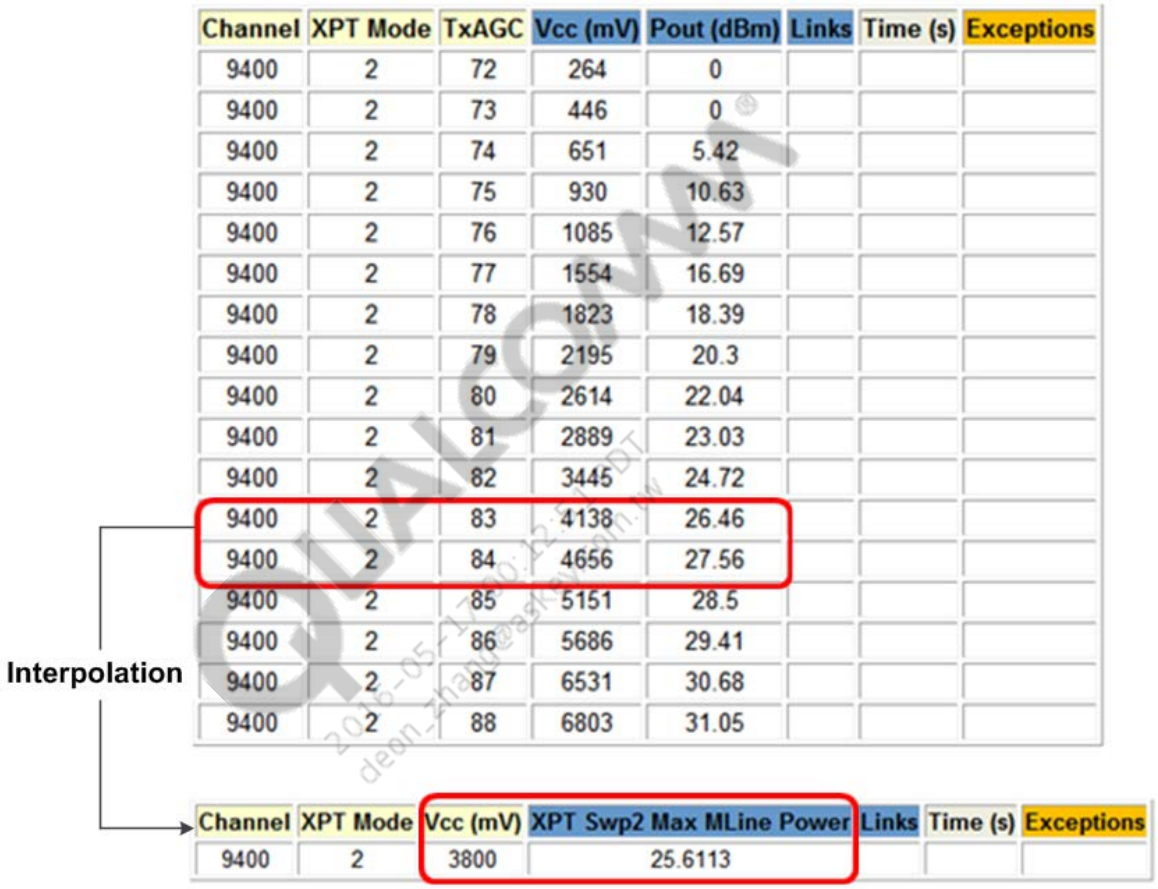



**Figure 4-20 Measurements – Tx XPT Swp2 and 3 – M-Line**

**NOTE:** The cal will fail if the target power is higher than the maximum M-Line power.

**MEASUREMENTS: Tx XPT Swp2 - ET M Line Selection Sweep Results**

| Channel | XPT Mode | Exp Power | Selection Valid | TxAGC | Vcc (mV) | Pout (dBm) | Links | Time (s) | Exceptions |
|---|---|---|---|---|---|---|---|---|---|
| 9400 | 2 | 19.5 | 1 | 79 | 2195 | 20.3 | | | |
| 9400 | 2 | 21.5 | 1 | 80 | 2614 | 22.04 | | | |
| 9400 | 2 | 23.5 | 1 | 82 | 3445 | 24.72 | | | |
| 9400 | 2 | 25.5 | 1 | 83 | 3300 | 26.46 | | | |

```
<XPT_Swp2_Channel_List>9400</XPT_Swp2_Channel_List>
<XPT_Swp2_EPT_PA_State>2</XPT_Swp2_EPT_PA_State>
<XPT_Swp2_ET_PA_State>3</XPT_Swp2_ET_PA_State>
<ET_Swp2_Meas_Pwr_List>19.5,21.5,23.5,25.5</ET_Swp2_Meas_Pwr_List>
<XPT_Swp2_Num_IQ_Samples>9216</XPT_Swp2_Num_IQ_Samples>
..... <XPT_Swp2_Cap_Segment_Length>4000</XPT_Swp2_Cap_Segment_Length>
```

**MEASUREMENTS: Tx XPT Swp2 - M Line RGI Compression**

| Channel | XPT Mode | TxAGC | Vcc (mV) | Compression (dB) | Compression Min | Links | Time (s) | Exceptions |
|---|---|---|---|---|---|---|---|---|
| 9400 | 2 | 81 | 1500 | 6.37518 | 2.5 | | | |
| 9400 | 2 | 83 | 2000 | 7.13095 | 2.5 | | | |
| 9400 | 2 | 84 | 2500 | 6.55804 | 2.5 | | | |
| 9400 | 2 | 85 | 3000 | 6.37518 | 2.5 | | | |

```
<XPT_Swp1_Channel_List>9400</XPT_Swp1_Channel_List>
<XPT_Swp1_PA_State>2</XPT_Swp1_PA_State>
<XPT_Swp1_RGI_List>81,83,84,85</XPT_Swp1_RGI_List>
<XPT_Swp1_Bias_List>1500,2000,2500,3000</XPT_Swp1_Bias_List>
<XPT_Swp1_IQ_Gain>622</XPT_Swp1_IQ_Gain>
<XPT_Swp1_Env_Scale>1350</XPT_Swp1_Env_Scale>
..... <XPT_Swp1_Num_IQ_Samples>9216</XPT_Swp1_Num_IQ_Samples>
```

**MEASUREMENTS: Tx XPT Swp2 - M Line Slope-Intercept**

| Channel | XPT Mode | a | b | d | e | Links | Time (s) | Exceptions |
|---|---|---|---|---|---|---|---|---|
| 9400 | 2 | 4.2939 | -2.5468 | 2.6034 | -1.2321 | | | |

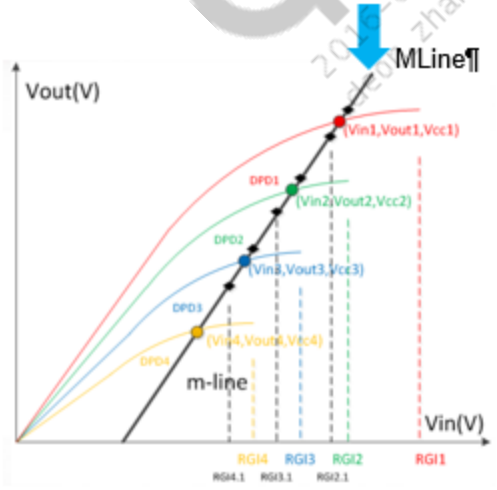



**Figure 4-21 Measurements – Tx XPT Swp2 – ETM line selection sweep and M-Line**

## 4.2 M-Line software design

Any signal peak on M-Line can share DPD and get best efficiency.

### 4.2.1 Parameters definition

- RGI – RGI value from RGI delta sweep
- Pout_RGI – Power measured for the RGI during the RGI delta sweep
- RGIVin(V) – Pout_RGI converted to voltage
- RGI_xpt – RGI value from alignment sweep
- BIAS_xpt – Bias value from alignment sweep
- VCC(V) – BIAS_xpt converted to voltage
- Pout – Power measured for the pair of (RGI_xpt, BIAS_xpt) during the alignment sweep
- Vout(V) – Pout converted to voltage
- Vout_n(V) – Vout(V) value in n-dB compression point
- (rgi, bias, pout) – RGI, bias, and power in M-Line

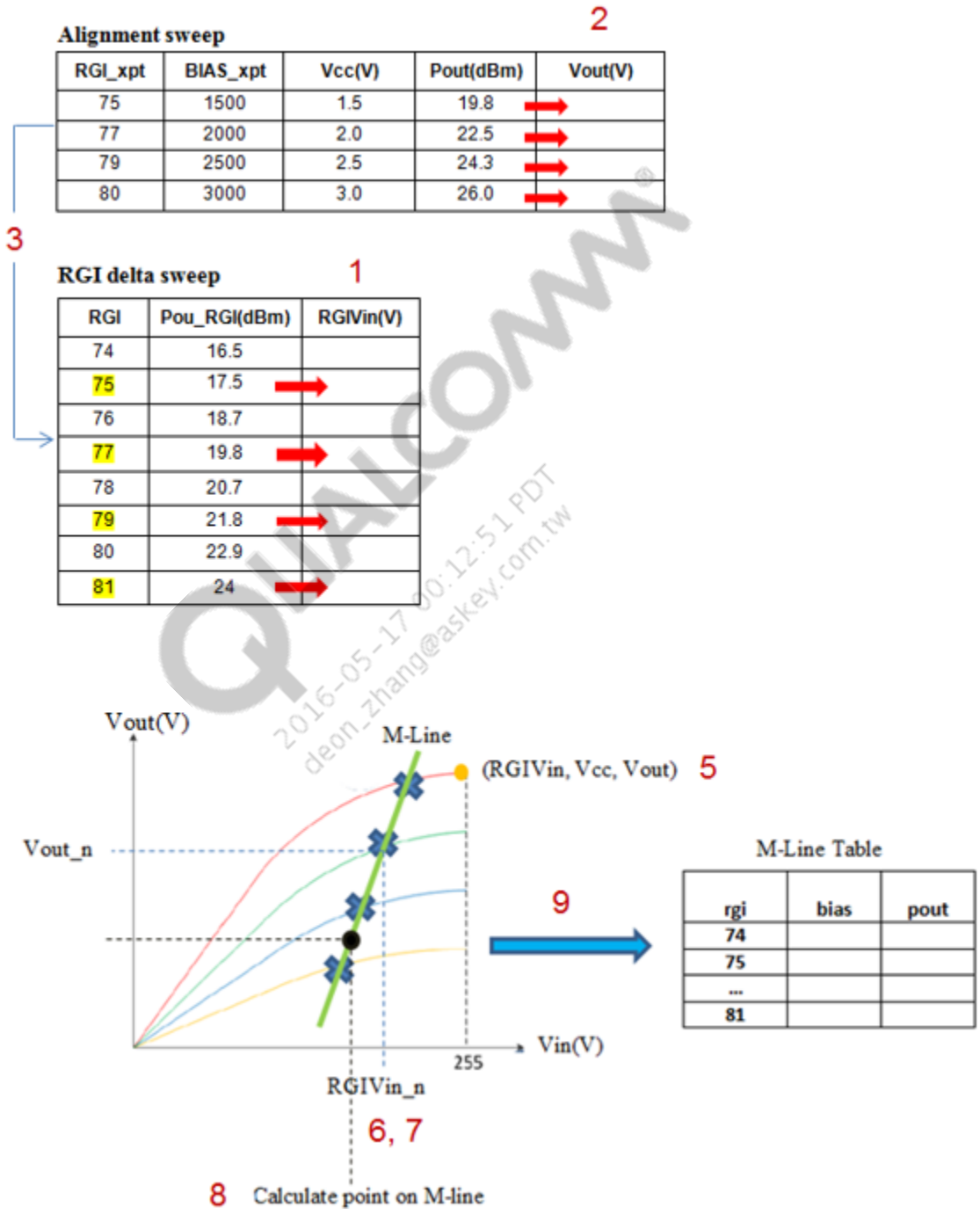
### 4.2.2 M-Line algorithm

1. Calculate RGIVin(V) based on Pout_RGI.
2. Calculate four points of the Vout(V) based on Pout.
3. For each RGI_xpt value, pick the RGIVin(V) from the RGI delta sweep and the Vout(V) from the alignment sweep. These four pairs of data form four points on an RGIVin vs Vout plane.
4. Scale RGIVin(V) and Vout(V) to 0~255. Get the relationship between (RGIVin, VCC, and Vout).
5. For each point, there is corresponding AMAM curve that goes through that point.
6. Find four points data of n-dB compression (RGIVin_n, VCC, and Vout_n) on the four curves. The M-Line goes through these four points.
7. Get the relationship between RGIVin_n(V), Vout_n(V), and VCC_n(V) for four data sets of n-dB compression point.
8. Calculate each point of (Vin, VCC, and Vout) in M-Line.
9. Convert (Vin, VCC, and Vout) to (rgi, bias, and Pout) for M-Line table.


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### 4.2.3 Example

The numbers in the illustration below correspond to steps provided in Section 4.2.2.




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# 4.3 Pin calibration

NOTE: This section was added to this document revision.

Pin variation over boards are larger, and Pin char data obtained from a couple boards may not be optimal in mass production. Use Pin calibration to align the Pin over freq for each board. The Pin calibration algorithm is similar to AMAM-based Pin char.

- Cal input file setting

  Add sweep type 15 for Pin calibration for WCDMA, CDMA, and LTE after the third sweep.

```
<Sweep_Params>
<Sweep_Type>15</Sweep_Type>
<Pre_Sweep_Computation>0</Pre_Sweep_Computation>
<Compute_From_Sweep>0</Compute_From_Sweep>
<Device_Selection>
<Waveform>
<Meas_Tx>1</Meas_Tx>
<XPT_PinCal_Swp_Channel_List>1315,1328,1341,1354,1367,1380,1393,1406,1412,1432,1445,1458,1471,1484,1497,1510</XPT_PinC
<XPT_PinCal_Swp_Ref_Channel>1412</XPT_PinCal_Swp_Ref_Channel>
<XPT_PinCal_Swp_PA_State_List>4</XPT_PinCal_Swp_PA_State_List>
<XPT_PinCal_Swp_XPT_Action_List>2</XPT_PinCal_Swp_XPT_Action_List>
<XPT_PinCal_Swp_IQ_Gain>622</XPT_PinCal_Swp_IQ_Gain>
<XPT_PinCal_Swp_Env_Scale>1350</XPT_PinCal_Swp_Env_Scale>
<XPT_PinCal_Swp_Num_IQ_Samples>9216</XPT_PinCal_Swp_Num_IQ_Samples>
<XPT_PinCal_Swp_Cap_Segment_Length>4000</XPT_PinCal_Swp_Cap_Segment_Length>
<XPT_PinCal_Swp_Proc_Segment_Length>1000</XPT_PinCal_Swp_Proc_Segment_Length>
<XPT_PinCal_Swp_Cap2Proc_Wait_Time>4000</XPT_PinCal_Swp_Cap2Proc_Wait_Time>
<XPT_PinCal_Swp_Target_CompressionDB10>30</XPT_PinCal_Swp_Target_CompressionDB10>
<Tuning_Segment_Length>10000</Tuning_Segment_Length>
<Post_Sweep_Computation>0</Post_Sweep_Computation>
<Compute_For_Sweep>0</Compute_For_Sweep>
</Sweep Params>
```

- Pin calibration is done after the third sweep and before the freq comp sweep.

```
Sweep 3
Logging Tx XPT Pin Cal Swp Measurements
Sweep Type: XPT_PIN_CAL
PA State    Channel PinBackOff_dB10 PinPeakComp_dB10
3           1315    11              65
3           1328    11              67
3           1341    7               71
3           1354    3               73
3           1367    -1              77
3           1380    -3              79
3           1393    -3              79
3           1406    -1              77
3           1412    0               75
3           1432    7               69
3           1445    12              65
3           1458    18              60
3           1471    21              58
3           1484    23              56
3           1497    21              58
3           1510    18              60
```

- Pin calibration XML log

**MEASUREMENTS: Tx XPT Pin Cal Swp**

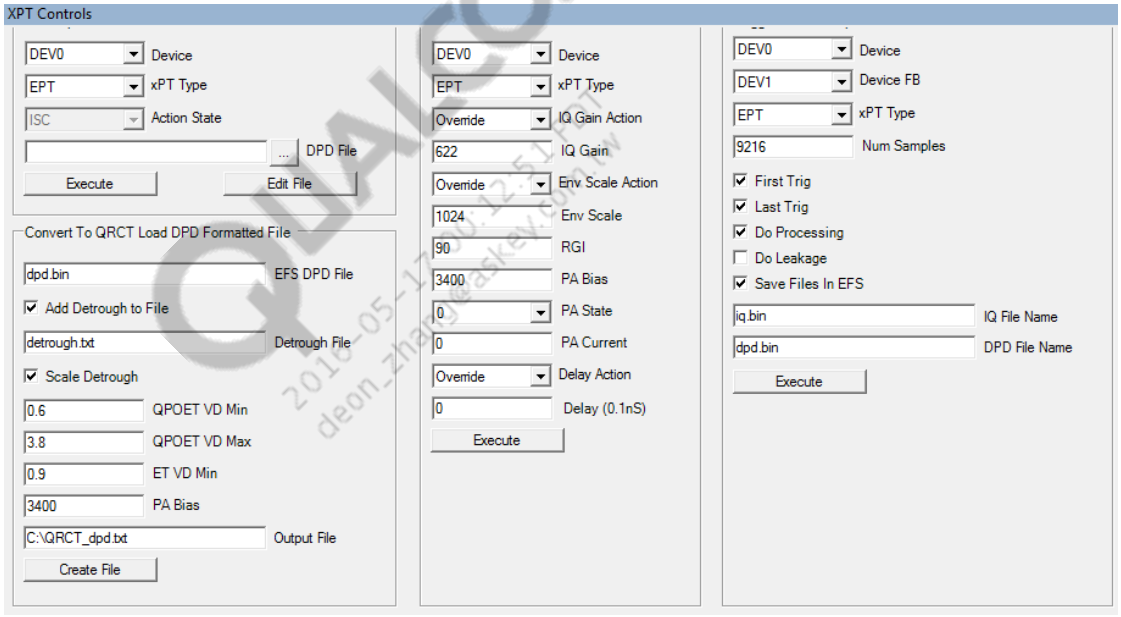
| PinCal PA State | PinCal Channel | PinCal RGI | PinCal Vcc | PinCal Icq | PinCal IQ Gain | PinCal Env Scale | PinCal Pin BackOff | PinCal Peak Compression | Pin |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 1315 | 78 | 2440 | 0 | 622 | 1350 | 11 | 65 | |
| 3 | 1328 | 78 | 2440 | 0 | 622 | 1350 | 11 | 67 | |
| 3 | 1341 | 78 | 2440 | 0 | 622 | 1350 | 7 | 71 | |
| 3 | 1354 | 78 | 2440 | 0 | 622 | 1350 | 3 | 73 | |
| 3 | 1367 | 78 | 2440 | 0 | 622 | 1350 | -1 | 77 | |
| 3 | 1380 | 78 | 2440 | 0 | 622 | 1350 | -3 | 79 | |
| 3 | 1393 | 78 | 2440 | 0 | 622 | 1350 | -3 | 79 | |
| 3 | 1406 | 78 | 2440 | 0 | 622 | 1350 | -1 | 77 | |
| 3 | 1412 | 78 | 2440 | 0 | 622 | 1350 | 0 | 75 | |
| 3 | 1432 | 78 | 2440 | 0 | 622 | 1350 | 7 | 69 | |
| 3 | 1445 | 78 | 2440 | 0 | 622 | 1350 | 12 | 65 | |
| 3 | 1458 | 78 | 2440 | 0 | 622 | 1350 | 18 | 60 | |
| 3 | 1471 | 78 | 2440 | 0 | 622 | 1350 | 21 | 58 | |
| 3 | 1484 | 78 | 2440 | 0 | 622 | 1350 | 23 | 56 | |
| 3 | 1497 | 78 | 2440 | 0 | 622 | 1350 | 21 | 58 | |
| 3 | 1510 | 78 | 2440 | 0 | 622 | 1350 | 18 | 60 | |

# 4.4 xPT control API

- QTI implemented a QRCT xPT control API to verify the calibration procedure. The location of this API is QRCT→FTM command→Common→XPT controls.
- The manual calibration test procedure is the same as ESC V4 calibration. See Section 4.1 and Section 4.2, respectively. The only difference is that the software goes through the FTM commands shown in Table 4-4, not through opcode.

**Table 4-4 xPT FTM command**

| Tool side | Function |
|---|---|
| QLIB_FTM_LOAD_DPD_TBLS | Loads AMAM/AMPM/Detrough table (for ET) |
| QLIB_FTM_XPT_TXAGC_OVERRIDE | Sets env_scale/IQ gain/RGI/PA gain/Current/mode |
| QLIB_FTM_TRIGGER_XPT_IQ_CAP | Triggers IQ capture |



### 4.4.1 Parameters description

Table 4-5 describes xPT control API parameters in detail.

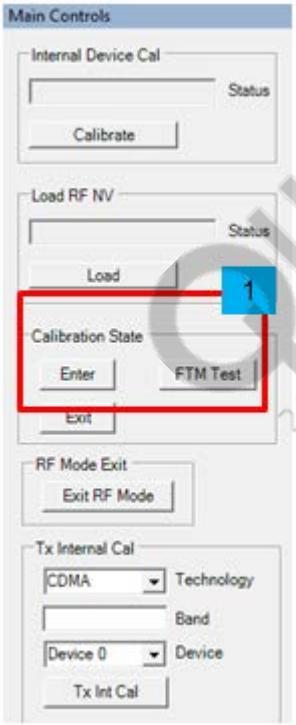
**Table 4-5 Parameters description**

| FTM command | Inputs | Description |
|---|---|---|
| Load DPD Table | Device | Dev0 to Dev4 |
| | XPT type | Specifies what type of DPD table is being loaded<br>▪ 0 – EPT<br>▪ 1 – ET |
| | Action state | Hardcoded – ISC (Init/Save/Commit) |
| | DPD file | DPD file format<br>▪ Num AMAM<br>▪ Num AMPM<br>▪ Num EPDT<br>▪ AMAM array<br>▪ AMPM array<br>▪ EPDT array |
| Tx AGC Override | Device | Dev0 to Dev4 |
| | XPT type | ▪ 0 – APT<br>▪ 1 – EPT<br>▪ 2 – ET |
| | IQ gain action | Override/skip/default |
| | IQ gain | IQ gain to override |
| | Envelope scale action | Override/skip/default |
| | Envelope scale | Envelope scale to override |
| | RGI | RGI to override |
| | PA Bias | PA bias to override |
| | PA state | PA state 0,1, 2, 3 to override |
| | Delay action | Override/skip/default |
| | Delay | Specifies the delay in ms between the Tx path and envelope path |
| Trigger xPT IQ Capture | Device | Dev0 to Dev4 |
| | Device FB | Dev0 to Dev4 (Dev 1 for non-SV, Dev3 for SV) |
| | XPT type | ▪ 0 – EPT<br>▪ 1 – ET |
| | Number of samples | Specifies the number of samples that must be captured for xPT IQ capture (9216) |
| | First trig | Default – TRUE |
| | Last trig | Default – TRUE |
| | Do processing | Hardcoded – TRUE |
| | Do leakage cal | Default – FALSE (Normal xPT IQ capture) |
| | Save file info to EFS | 1 – Save |
| | IQ filename | Iq.bin – Filename that is saved in UUT EFS) |
| | DPD filename | dpd.bin – Filename that is saved in UUT EFS |
| | EFS DPD File | dpd.bin from the EFS in the phone |

| FTM command | Inputs | Description |
|---|---|---|
| Convert to QRCT load DPD formatted file | Add detrough to file | For ET only |
| | Dethrough file | Specifies the detrough.txt location |
| | Scale detrough file | For ET only |
| | QPOET VD min | Minimum QPOET support voltage |
| | QPOET VD max | Maximal OPOET support voltage |
| | ET VD min | Minimum support voltage for PA |
| | PA bias | Need to use the same value as in Tx AGC Override |
| | Output file | C:\QRCT_dpd.txt |

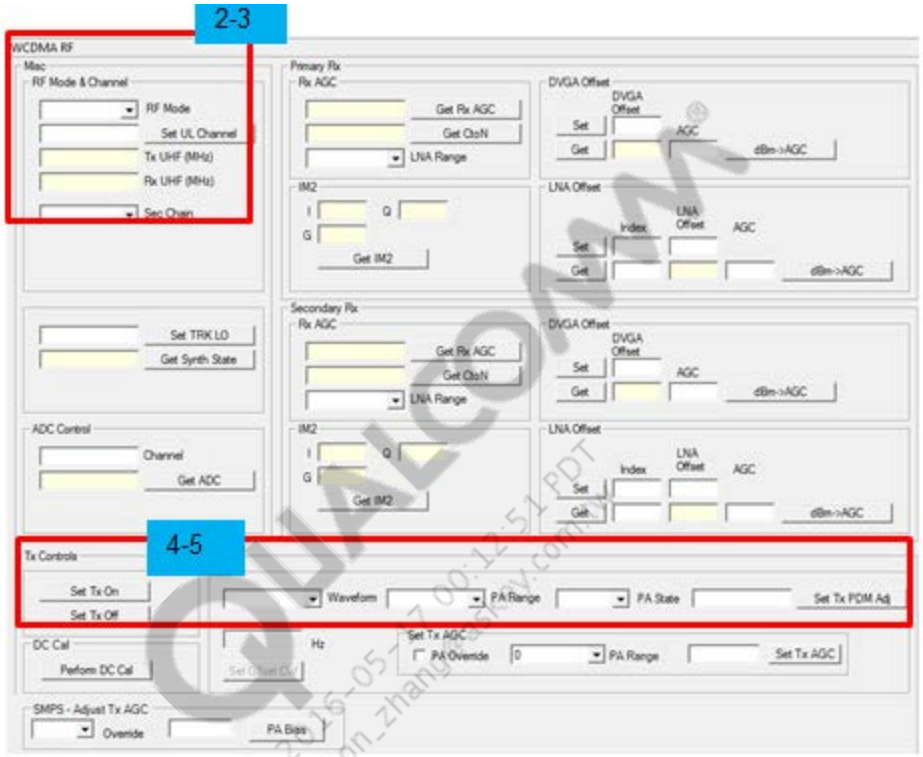
## 4.4.2 Manual test procedure

1. Set the FTM Calibration State.




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2. Set FTM RF Mode.
3. Set FTM Channel.
4. Set FTM Waveform.
5. Set Tx on.

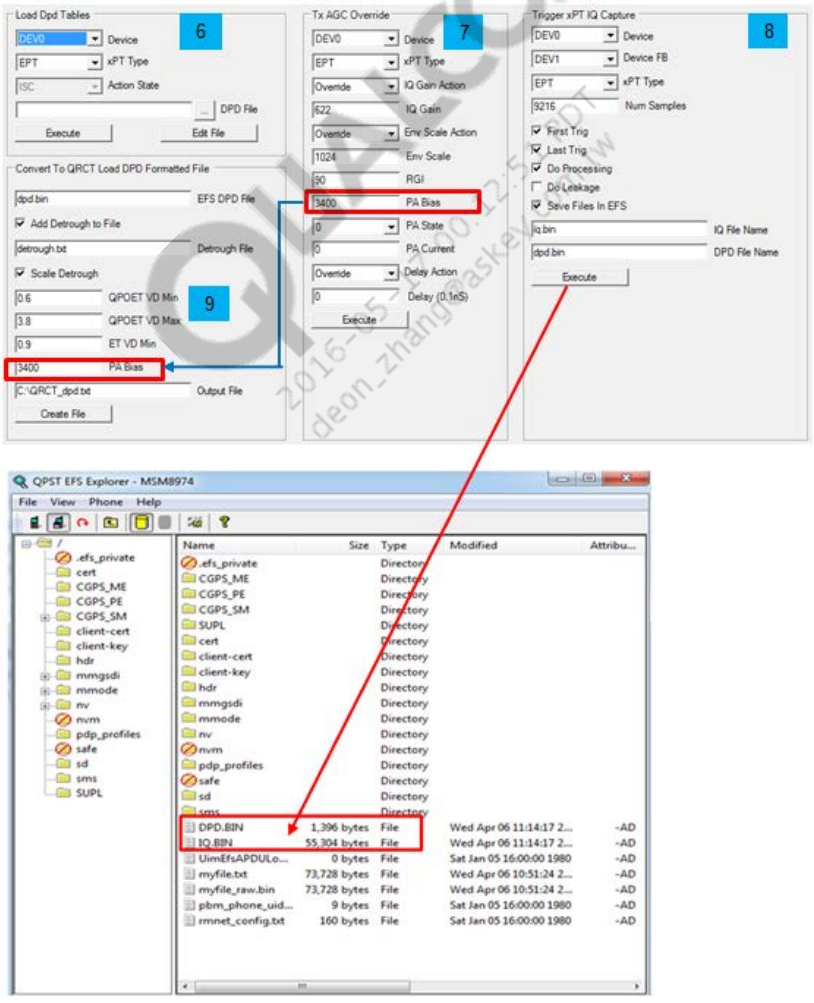


6. Load DPD – Unity DPD table (EPT/ET).

   Location of sample file – C:\Program Files (x86)\Qualcomm\QDART\Databases\QRCTSampleFiles

   - ET – sample_dpd_file_epdt.txt
   - EPT – sample_dpd_file.txt

7. Set required parameters for TxAGC Override.
8. Perform XPT IQ capture and provide the IQ and DPD filenames.
9. Save the DPD EFS file in the phone side and load it in the Convert to QRCT Load DPD Formatted File section.
   - The detrough table is located at C:\Program Files (x86)\Qualcomm\QDART\Databases\QRCTSampleFiles\SampleDetroughFile.txt.
   - Ensure that the detrough table is the same as the one in the Unity DPD table.




10. Load DPD from Step 9 (QRCT_DPD.txt).
11. TxAGC override occurs with the corresponding parameters.
12. In ET mode, you can get similar Tx power before/after reloading a new DPD file from Step 9.

## 4.4.3 Expand and convert to DPD file API

To debug Online mode performance issues, QTI created an API to help convert and expand 16 AMAM/AMPM data in MULTI_LIN_V3 NV to 64 AMAM/AMPM data. By using this API, customers can get exactly the same DPD table as Online mode used and compare Online mode and FTM mode performance.

### 4.4.3.1 Generate DPD file based on MULTI_LIN_V3

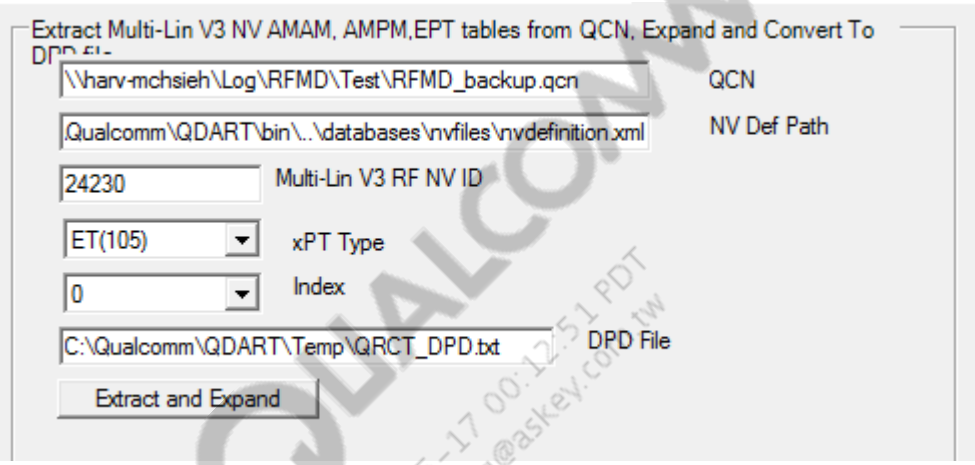
To create a DPD file based on the AMAM/AMPM data from MULTI_LIN_V3:

1. Provide a QCN which includes a MULTI_LIN_V3 NV for xPT.
2. Provide the NV definition path.
3. Provide the NV ID for the MULTI_LIN_V3 used in step 1.
4. Choose a DPD_index of MULTI_LIN_V3 to be generated to QRCT_DPD.txt.
5. Generate a QRCT_DPD.txt file.
   - In versions of QDART before 4.8.17, there are 64 data of AMAM and AMPM in this file, but no detrough table. Customers must follow Step 6 to add the detrough table in QRCT_DPD.txt.
   - In QDART 4.8.17 and later, the QRCT_DPD.txt includes AMAM/AMPM/Detrough. Customers can skip Step 6.
6. Add a scaled detrough table (corresponding to the DPD_index) to the QRCT_DPD.txt generated in Step 5.
   - DPD file format:
     - 64 AMAM array
     - 64 AMPM array
     - 128 detrough array

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**Table 4-6 Expand and convert to DPD file API**

| FTM command | Inputs | Description |
|---|---|---|
| Expand and Convert to DPD file | QCN | QCN with Multi-Lin V3 NV |
| | NV Def path | NV definition |
| | Multi-Lin V3 RF NV ID | Multi-Lin V3 RF NV ID |
| | DPD xPT Type | EPT or ET |
| | Index | DPD_index 0/1/2/3 from Multi-Lin V3 NV |
| | DPD file | Convert V3 NV to DPD file |




## 4.5 APT-only ESC V4 calibration

ESC V4 calibration can support APT and XPT calibration. Customers may follow the procedure below to modify the ESC V4 RF calibration input file for APT-only calibration.

- Calibrate the input file.
  - Change the sweep type number.
    - Sweep 1 – sweep_type 2→0
    - Sweep1 – Change the start RB to 22 and number RB to 6
    - Remove the entire sweep_type 6 and 9
    - Sweep 3 – sweep_type 7→1
    - Freq comp sweep – sweep_type 8→4
  - Delete XPT parameters.
  - Change <Compute_From_Sweep> in freq comp sweep from 2 to 1.
- Remove xPT-related NV items.
  - Delay NV items (see Section 3.2)
- Set the first element of the DPD NV item to APT-only mode (see Section 3.1).
- Change the PA rise/fall switchpoint for APT mode.

## 4.6 Full bias ESC V4 calibration

To modify the ET ESC V4 Cal input file to APT only, follow the procedure in Section 4.5. Based on the ESC V4 APT only cal input file, the following additional steps are used to enable full bias only calibration.

1. Remove sweep3 – <Sweep_Type>1</Sweep_Type>.
2. Change <Compute_From_Sweep> in freq comp sweep from 1 to 0.

**NOTE:** In MDM9x35 software, waveform type0 (voice sweep) and waveform type3 (full bias sweep) are required for updating Tx linearizer data in Online mode. Customers need to use APT with full bias cal for Bypass mode cal.

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# 5 ET Characterization

Each PA and frontend design will likely have a unique performance. To ensure that the ET system delivers specification-compliant performance, PA and RF frontends must be characterized. For proper performance, PA characterization should be performed in accordance with QTI suggestions. Note that ET has a different procedure from APT.

## 5.1 xPT characterization tool

MSM8974 with QFE1100 requires four characterization tools. The char input file is the same as Cal PA_PARAMS_FILE; see Table 4-3. Note that the delay characterization must be run before pin alignment characterization. For RFMD PA or QPA, QTI provides tips to run the char tool, see Section 6.3.4 .

- APT characterization
  - Perform in room temperature.
  - The highest power in the highest gain state needs to cover the maximum power.
  - The combination of ICQ and Bias in the lowest PA gain state needs to cover the min power.
- RGI bias characterization
  - Perform in room temperature.
  - The picked RGI and Bias need to cover n-dB compression point in M-Line.
- Delay characterization
  - Delay over-temperature is based on room temperature to calculate the delay offset value for other temperatures.
  - Perform delay char in room temperature in the beginning.
  - Do *not* close the char tree before finishing the temperature loop.
- Static pin alignment
  - Pin/Pout over-temperature is based on temperature to calculate the offset value.
  - Perform Pin char in room temperature in the beginning.
  - Do *not* close the char tree before finishing the temperature loop.
  - If the Pin char fails in a certain temperature, customers can run the pin char in that certain temperature again without closing the tree between the room temperature test and other temperature test but before the data process, since all of the offset values are based on room temperature data.

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Figure 5-1 is the ET char process flowchart.

- Green block – Run a Qualcomm Sequence Profiling Resource (QSPR) characterization tree
- Gray block – Data processing in characterization tree to generate *.xml file after collecting data in the room temperature (or full temperature)
- Blue block – Calculate average of data from generated *.xml from all phones
- Orange block – Modify the parameter setting in Cal input file

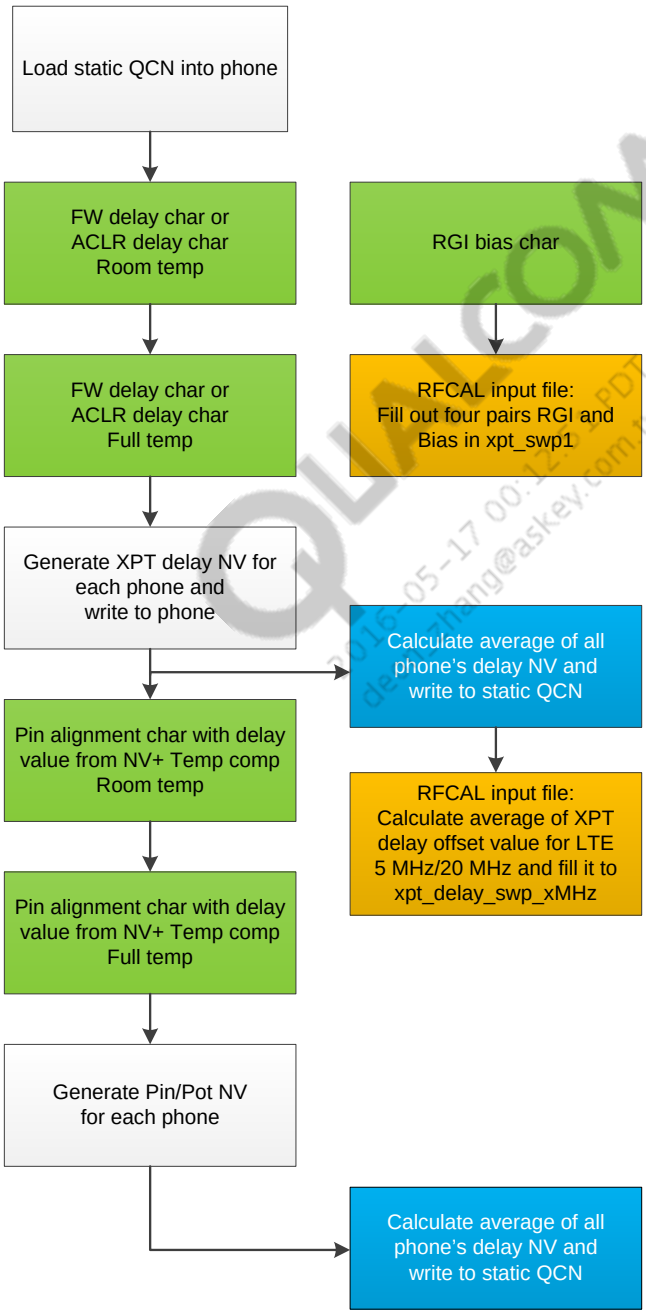
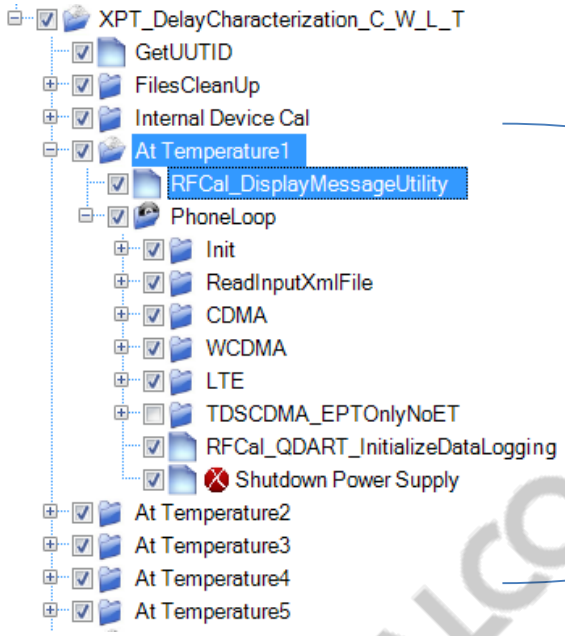
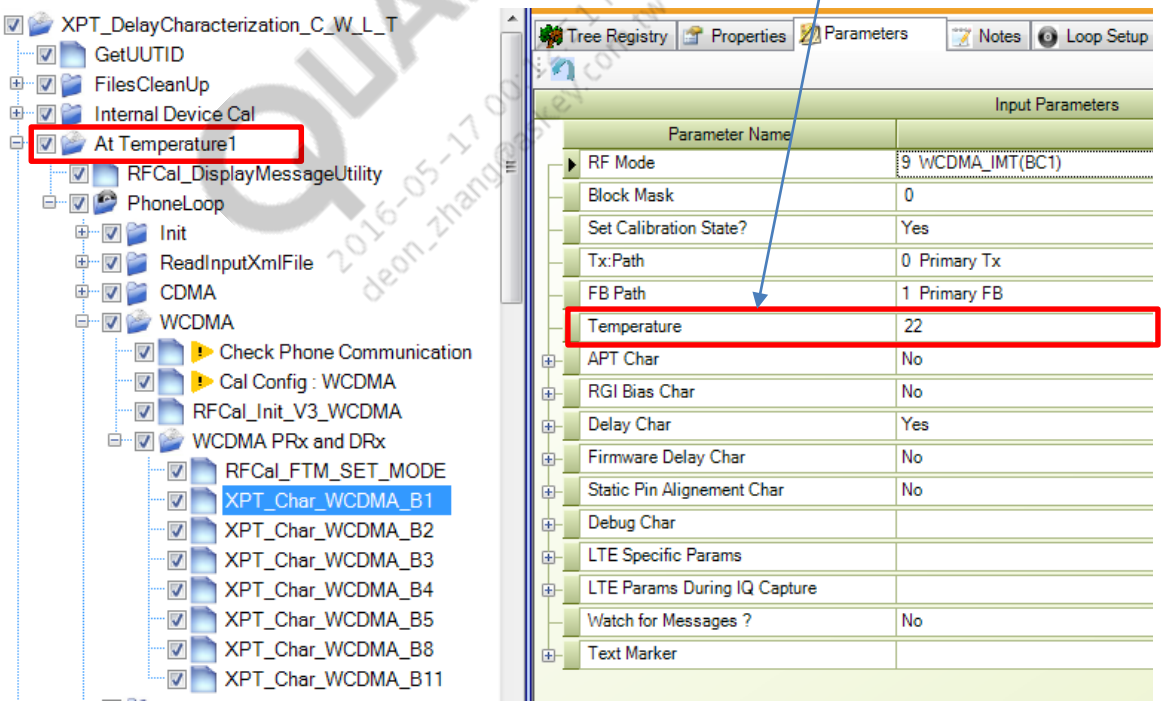



**Figure 5-1 ET char process flowchart**

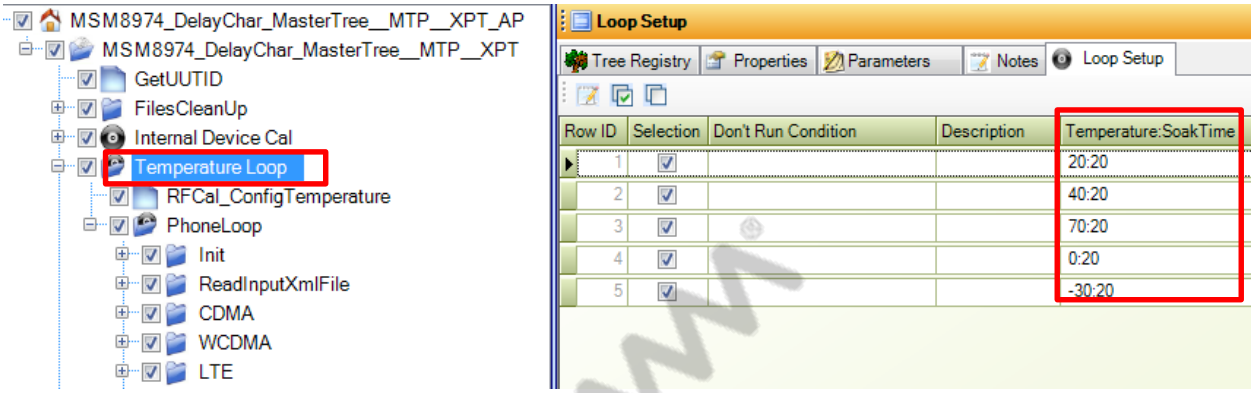
Two types of characterization trees are shown here; both of them support different temperatures.




Each temperature node has different temperature; the first node must be in room temperature.

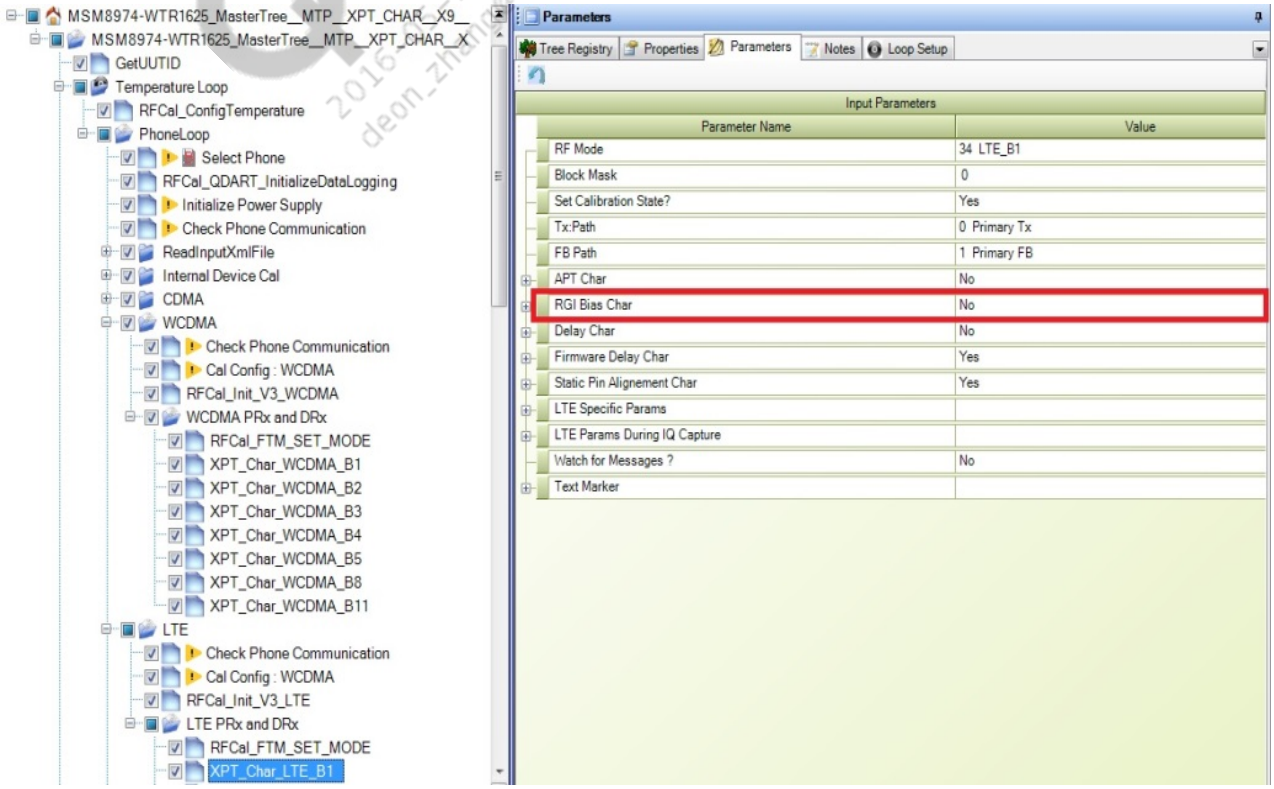


This shows Type2 how to control the temperature in temperature loop. The first temperature must be in room temperature.




## 5.2 xPT RGI bias characterization

QTI will supply the ET characterization tool as shown here to sweep RGI across various SMPS values and pick the four RGI bias pairs for ET/EPT calibration. The goal of RGI bias characterization is to determine PA bias and the corresponding RGI that meets the compression point threshold or ACLR threshold. This tool is similar to the APT characterization tool, but with a fixed PA current value, and it is used only in xPT mode. The output data of the RGI bias characterization is used in the first sweep of xPT calibration to compute the RGI and bias required for any desired output power.



## 5.2.1 RGI bias characterization parameters description

### 5.2.1.1 Common parameters

- RF mode – Select the desired band
- Set calibration state – Yes
- Tx path – 0 for primary path, 2 for SV path
- FB path – 1 for primary path, 3 for SV path

### 5.2.1.2 RGI bias characterization

RGI bias characterization parameters shown are as follows:

- List of channels – DLL supports multichannel RGI bias characterization. Enter comma-separated channels.
- List of RGI values and list of PA bias values – Values must be entered in (start, stop, step) manner and should be in *ascending order*.
- Expected power level – Set an expected power level on the call box to make the power measurements. Since there are no measurements for first bias and RGI combination, this parameter is explicitly listed in the tree.
- User margin – Since the RGI and bias are in ascending order, the power level will also be increasing from the current measurement to the next measurement. If the current power measurement is set as an expected level on the box, the call box might throw a level over failure for the next measurement. Therefore, a user margin is added to the current measurement and the resultant value is set as the expected level on the box for the next measurement.
- ACLR threshold, IQ gain, env scale gain values – These values will vary from one technology to the other.
- Optimize starting RGI – If enabled, start from last RGI value of the previous PA bias.

| Input Parameters | |
|---|---|
| Parameter Name | |
| RGI Bias Char | Yes |
| RGI Bias Char: List of Channels | 9400 |
| RGI Bias Char: PA State | 3 |
| RGI Bias Char: List of RGI Values | 60,96,1 |
| RGI Bias Char: List of PA Bias Values | 1500,3000,500 |
| RGI Bias Char: Expected Power Level For The First Bias and Rgi Pair | 25 |
| RGI Bias Char:User Margin | 5 |
| RGI Bias Char: ACLR Averages | 10 |
| RGI Bias Char: Threshold Type | 1 Compression |
| RGI Bias Char: Minimum Tx Power Needed for Compression Point Calculation | 10 |
| RGI Bias Char: ACLR Threshold | -32 |
| RGI Bias Char: IQ Gain | 622 |
| RGI Bias Char: Env Scale Gain | 1350 |
| RGI Bias Char: Optimize Starting RGI | Yes |
| RGI Bias Char: Save Output Files | No |

## 5.2.2 RGI bias characterization algorithm

Figure 5-2 shows an algorithm for RGI bias characterization and is as follows:

1. For the first bias, the RGI mentioned in the tree is used as a starting point.
2. From the next bias, the RGI that was selected for the previous bias is used as a starting point.
3. The RGI bias characterization result is logged as a failure if *none* of the measurements resulted in maxACLR greater than the ACLR threshold listed or if *none* of the compression is greater than the compression point threshold.
4. At the end of the RGI bias characterization, you will have the information about bias and RGI that meets the ACLR requirements or the compression point requirements.

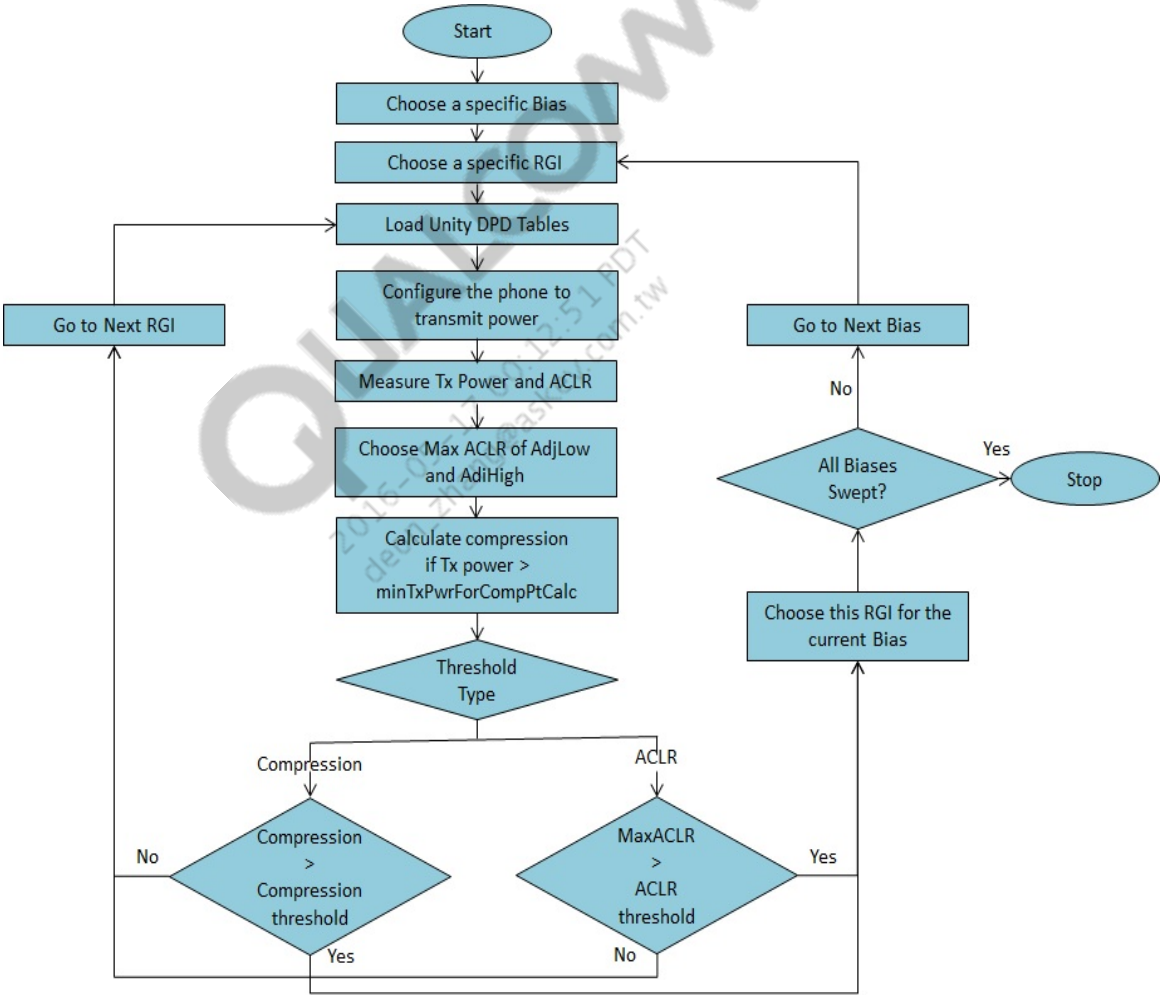



**Figure 5-2 RGI bias characterization algorithm**

# 5.3 Delay characterization

As shown in Figure 5-3, QFE1100 tracks the envelope of the waveform to be transmitted. It supplies the bias to the PA, based on the current amplitude of the envelope. The Tx path from the modem to the input of PA is different than the Tx path from the modem to QFE1100, which may result in inaccurate PA bias supply. If bias is not configured correctly, then the Tx power may not be accurate. Therefore, this delay value must be accurately characterized.

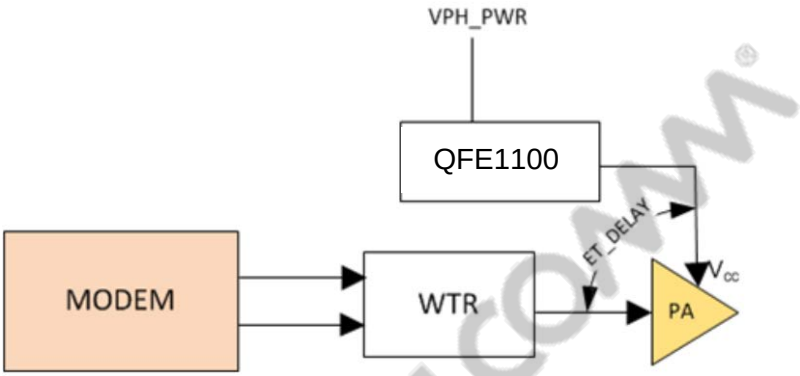



**Figure 5-3 Simple system architecture**

## 5.3.1 How to choose delay characterization tool

The two types of delay characterization tools to obtain the optimized delay value are firmware delay characterization and ACLR delay characterization. The delay value from firmware delay characterization is calculated from the firmware side, which is faster than ACLR delay characterization. However, ACLR delay characterization can show the relationship between delay value and ACLR performance. When the firmware delay provides the hardware delay, ACLR delay characterization selects a delay that provides symmetric ACLR. If PA ACLR performance is dominated by factors other than delay, the firmware delay will be different from the ACLR delay. Therefore, both tools are useful for getting optimized delay value as well as for debug purposes.

Two rules to choose the characterization tool are:

- For WCDMA and LTE
  - QTI recommends the firmware delay characterization tool since it is faster than the ACLR delay characterization tool
  - If there are factors other than hardware delay, the customer can use ACLR delay
- For CDMA
  - QTI does not support firmware delay in CDMA since CDMA is not sensitive to delay

## 5.3.2 ACLR delay characterization

### 5.3.2.1 Delay characterization parameters description

The following shows a description of delay characterization parameters and is listed as follows:

- List of RGI values – Values must be entered in (start, stop, step) manner and should be in *ascending order*. The delay char sweeps from the starting RGI and finds the RGI that meets the compression threshold or maximum allowed Tx power. It then uses this RGI value to find the corresponding delay time.
- Smooth ACLR curve – If enabled, ACLR is averaged by the adjacent eight values measured.
- Maximum Tx power allowed during RGI determination sweep – In the RGI determination sweep, if the power is larger than this value, the RGI is determined no matter whether the compression point criteria is met or not.
- List of channels and upper bound channels – Refer to the static QCN in each software release build.

The characterization input file is shown in Figure 5-8. Note that the setting of this input file depends on PA.

```
<AclrDelayCharCompPoint_table AclrDelayCharCompPoint="3"/>
<AclrDelayCharVmax_table AclrDelayCharVmax="3.4"/>
<AclrDelayCharIqGain_table AclrDelayCharIqGain="622"/>
```

| Input Parameters | |
|---|---|
| Parameter Name | Value |
| Delay Char | Yes |
| Delay Char:RGI Determination Sweep: EPT PA State | 3 |
| Delay Char:RGI Determination Sweep: List of RGI Values | 60,96,1 |
| Delay Char:RGI Determination Sweep: Minimum Tx Power Needed for Compression Point Ca | 10 |
| Delay Char:RGI Determination Sweep: Maximum Tx Power allowed during RGI Determination | 23 |
| Delay Char:RGI Determination Sweep: Expected Power Level For The First Bias and Rgi Pair | 20 |
| Delay Char: Delay Sweep: List of Channels | 9262,9281,9299,9318,9336,9354,9373,9391,9409,9427,9446,9464, |
| Delay Char: Delay Sweep: Upper Bound Channels | 9272,9290,9309,9327,9345,9364,9382,9400,9418,9437,9455,9473, |
| Delay Char: Delay Sweep: List of Delay Values(For all techs and for LTE 10MHz BW) | 100,500,10 |
| Delay Char: Delay Sweep: PA State | 3 |
| Delay Char: Delay Sweep: Expected Power Level For The First Bias and Rgi Pair | 25 |
| Delay Char: Delay Sweep: User Margin | 5 |
| Delay Char: Delay Sweep: ACLR Averages | 10 |
| Delay Char: Delay Sweep: Env Scale Gain | 1350 |
| Delay Char: Delay Sweep: Number of IQ Capture Samples | 9216 |
| Delay Char: Delay Sweep: Minimum Limit For Power Level (dBm) | 15 |
| Delay Char: Delay Sweep: Additional LTE BWs | |
| Delay Char: Delay Sweep: List of Delay Values for LTE 5MHz | 1500,1700,5 |
| Delay Char: Delay Sweep: List of Delay Values for LTE15 MHz | 500,700,5 |
| Delay Char: Delay Sweep: List of Delay Values for LTE20 MHz | 500,700,5 |
| Delay Char: Delay Sweep: Save IQ Capture Files | No |
| Delay Char: Delay Sweep: Smooth ACLR Curve | Yes |
| Delay Char: Delay Sweep: Delay Action Type | 5 Use Specified Value but add process variation |

### 5.3.2.2 Delay characterization algorithm

Figure 5-4 shows the algorithm for delay characterization and is listed as follows:

1. Select the RGI that meets the maximum target power or PA compression threshold. DLL 17.55 or later only use the PA compression point to pick RGI.
2. The delay value is selected from the parameters in the tree.
3. Since IQ capture is not currently supported for 20 MHz and 5 MHz bandwidths, the DPD data computed for 10 MHz bandwidth is saved for other bandwidths.
4. The delay time for different bandwidths might vary.

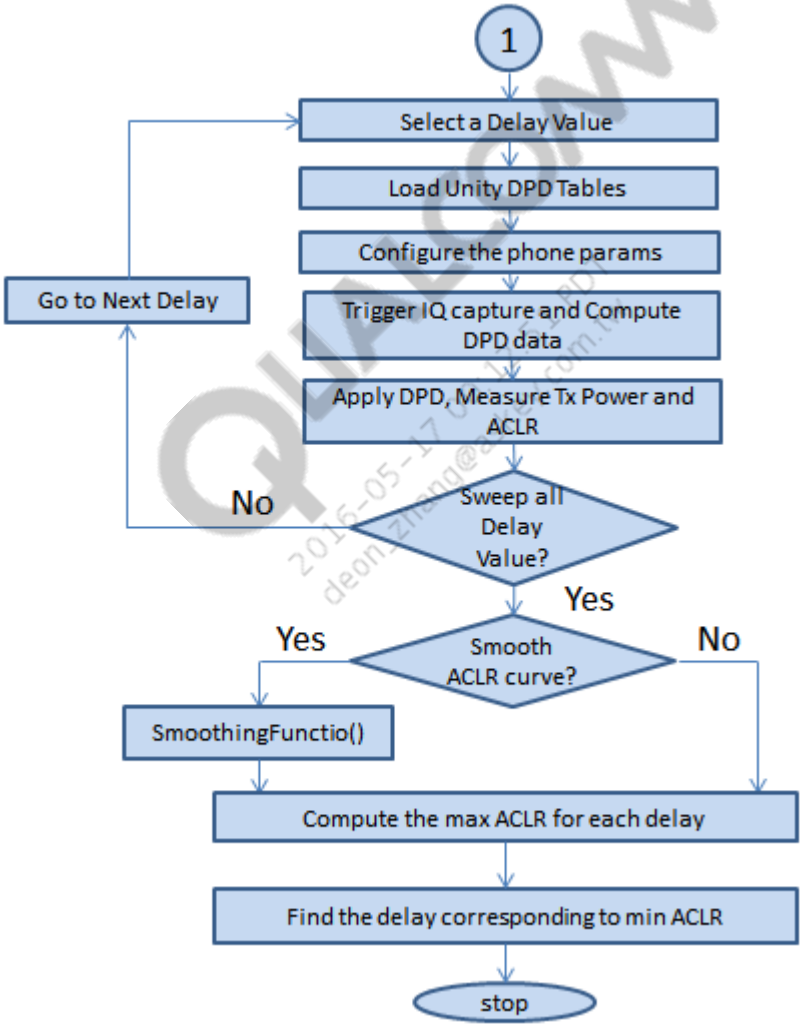



**Figure 5-4 Delay characterization algorithm**

As shown in Figure 5-5, in DLL17.59 and later disabling PA Bias fine tuning for time saving is supported. QPA and RFMA PA require enabling PA Bias fine tuning. For AVAGO PA and SKYWORKS PA, it is optional.

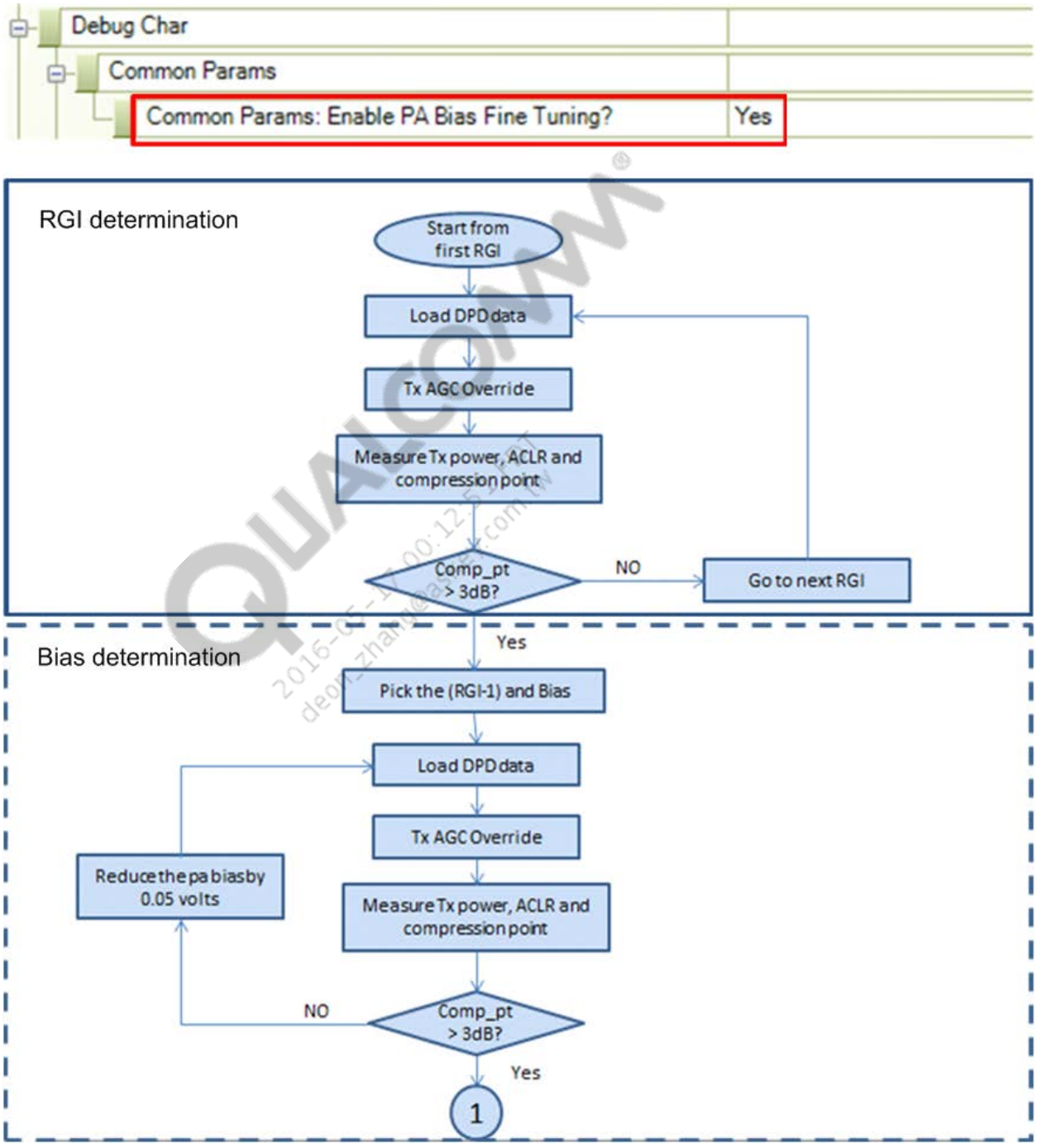



**Figure 5-5 ACLR delay char – RGI and PA Bias determination**

### 5.3.2.3 Delay char data analysis

As shown in Figure 5-6, ACLR over delay time has a V-shaped curve. The delay time is picked where the ACLR reaches the minimum value in V shape.

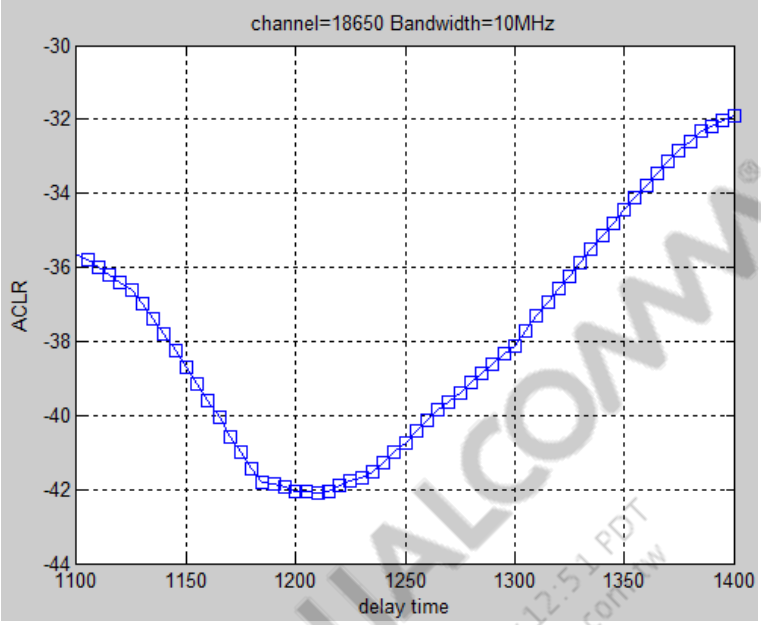



**Figure 5-6 ACLR over delay time**

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### 5.3.3 Firmware delay char

A list of channels and upper bound channels is shown here. Refer to the static QCN in each software release build.

| | |
|---|---|
| Firmware Delay Char | Yes |
| Firmware Delay Char:Common Params:Channel List | 18050,18083,18117,18150,18183,18217,18250, |
| Firmware Delay Char:Common Params:Upper Bound C | 18067,18100,18134,18167,18200,18234,18267, |
| Firmware Delay Char:Common Params:Env Scale Gain | 820 |
| Firmware Delay Char: Common Params: Save Output F | No |
| Firmware Delay Char:RGI Determination Sweep:EPT P | 0 |
| Firmware Delay Char:RGI Determination Sweep:List of | 60,96,1 |
| Firmware Delay Char:RGI Determination Sweep: Minim | 10 |
| Firmware Delay Char:RGI Determination Sweep: Maxim | 20 |
| Firmware Delay Char:RGI Determination Sweep: Expect | 20 |
| Firmware Delay Char:RGI Determination Sweep: User | 5 |
| Firmware Delay Char:Compute Delay Sweep:PA State | 0 |
| Firmware Delay Char:Compute Delay Sweep:Number of | 9216 |
| Firmware Delay Char:Compute Delay Sweep:Number of | 10 |
| Firmware Delay Char:Compute Delay Sweep:LTE BW L | 10,5,20 |
| Firmware Delay Char:Compute Delay Sweep:Num Pusc | 36,25,50 |
| Firmware Delay Char:Compute Delay Sweep:Start RB L | 7,0,25 |

As shown in here, common parameters are:

- Channel list
- IQ gain
- Env scale

| | |
|---|---|
| Firmware Delay Char:Common Params:Channel List | 18050,18083,18117,18150,18183,18217,18250,18283,18317,183 |
| Firmware Delay Char:Common Params:IQ Gain | 200 |
| Firmware Delay Char:Common Params:Env Scale Gain | 820 |
| Firmware Delay Char: Common Params: Save Output Files | No |

As shown in here, RGI determination sweep parameters are:

- EPT PA state – 1 for LTE
- RGI list – RGI sweep list
- Minimum Tx power needed for compression point calculation – Pmin compression point

| | |
|---|---|
| Firmware Delay Char RGI:Determination Sweep:EPT PA State | 1 |
| Firmware Delay Char RGI:Determination Sweep:List of RGI Values | 80,94,1 |
| Firmware Delay Char RGI:Determination Sweep: Minimum Tx Power N | 10 |

As shown in here, compute delay sweep as:

- PA state – ET PA state; 3 for WCDMA, 0 for LTE
- Number of IQ capture samples – 9216
- Number of delay capture – Total capture times to get average value
- LTE BW list – Supports 5 MHz, 10 MHz, and 20 MHz
- Number Pusch RBs list – Total of RB number
- Start RB list – Number of start RBs
- Firmware delay characterization is only supported in WCDMA and LTE because CDMA is not sensitive to delay

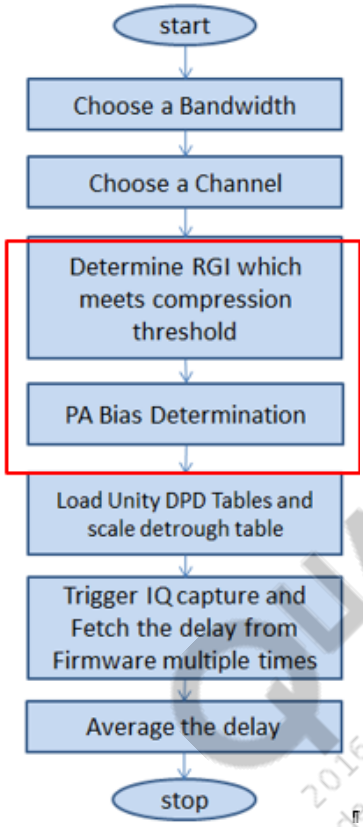
| | |
|---|---|
| Firmware Delay Char:Compute Delay Sweep:PA State | 0 |
| Firmware Delay Char:Compute Delay Sweep:Number of IQ Capture Sa | 9216 |
| Firmware Delay Char:Compute Delay Sweep:Number of Delay Capture | 10 |
| Firmware Delay Char:Compute Delay Sweep:LTE BW List | 10,5,20 |
| Firmware Delay Char:Compute Delay Sweep:Num Pusch RBs List | 36,25,50 |
| Firmware Delay Char:Compute Delay Sweep:Start RB List | 7,0,25 |

### 5.3.3.1 Firmware delay char flow

As shown in Figure 5-7, firmware delay char flow is:

1. RGI selection
   - Set PA state to EPT mode
   - IQ gain = 200; env_scale = 820; PA Bias = 3800
   - Select RGI such that PA compression point is 1 dB or Pout is higher than maximum power allowed during RGI determination sweep
   - Get RGI selection for all listed frequencies with the band
2. Sample capture and firmware delay processing
   - Sweep all supported LTE bandwidths
   - For each bandwidth, sample capture and firmware processing at multiple frequencies
   - Characterization input file is shown in; note that the setting of this file depends on PA

```
<FirmwareDelayIqGain_table FirmwareDelayIqGain="622"/>
<FirmwareDelayCharRgiDetPaBias_table FirmwareDelayCharRgiDetPaBias="3.3"/>
<FirmwareDelayVmax_table FirmwareDelayVmax="3.4"/>
<FirmwareDelayVmin_table FirmwareDelayVmin="0.6"/>
<FirmwareDelayCharCompPoint_table FirmwareDelayCharCompPoint="2"/>
```




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**Figure 5-7 Firmware delay char algorithm**

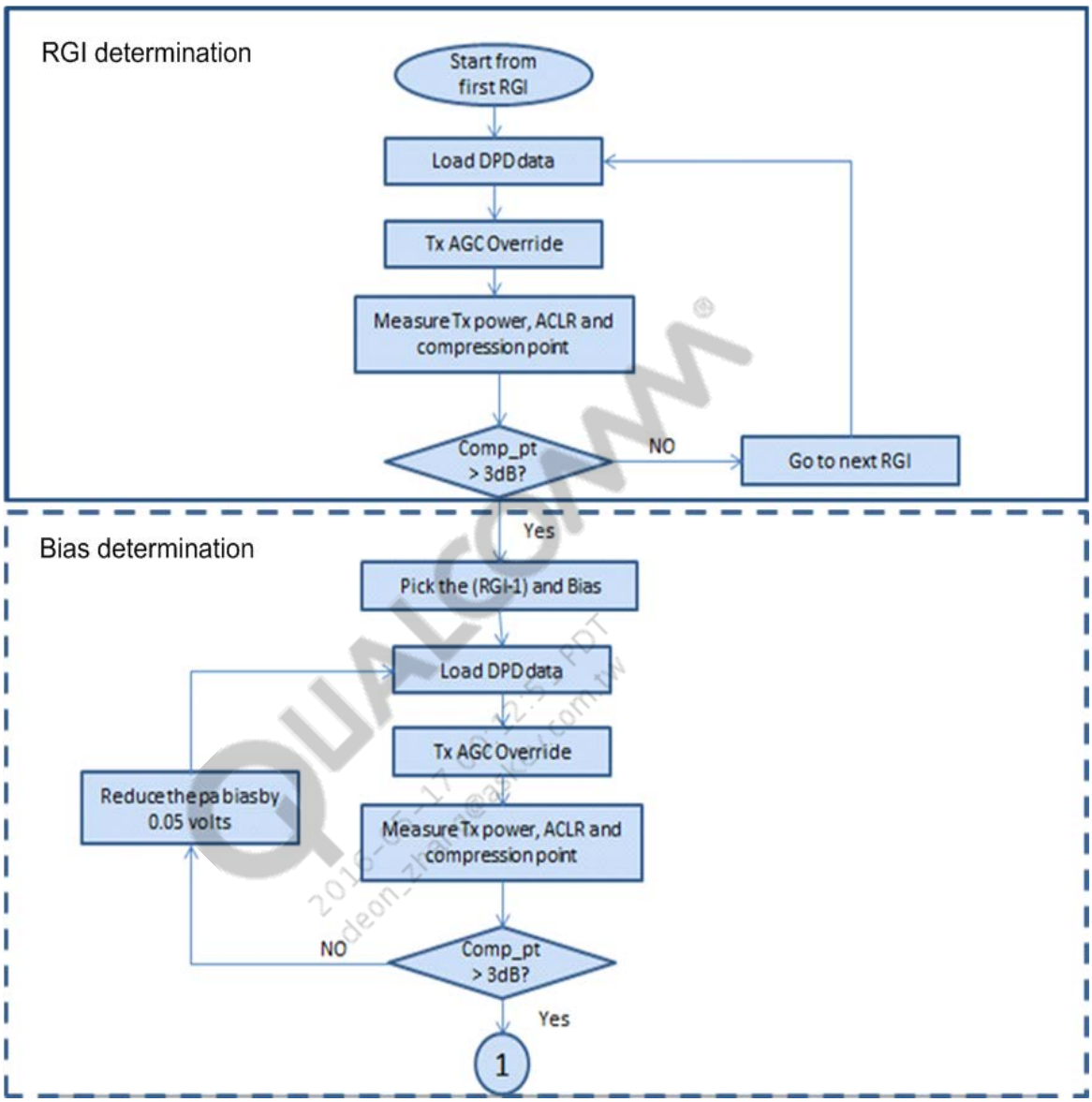
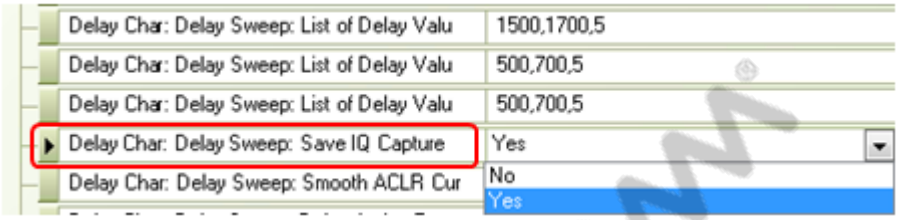



**Figure 5-8 Firmware delay char – RGI and PA bias determination**

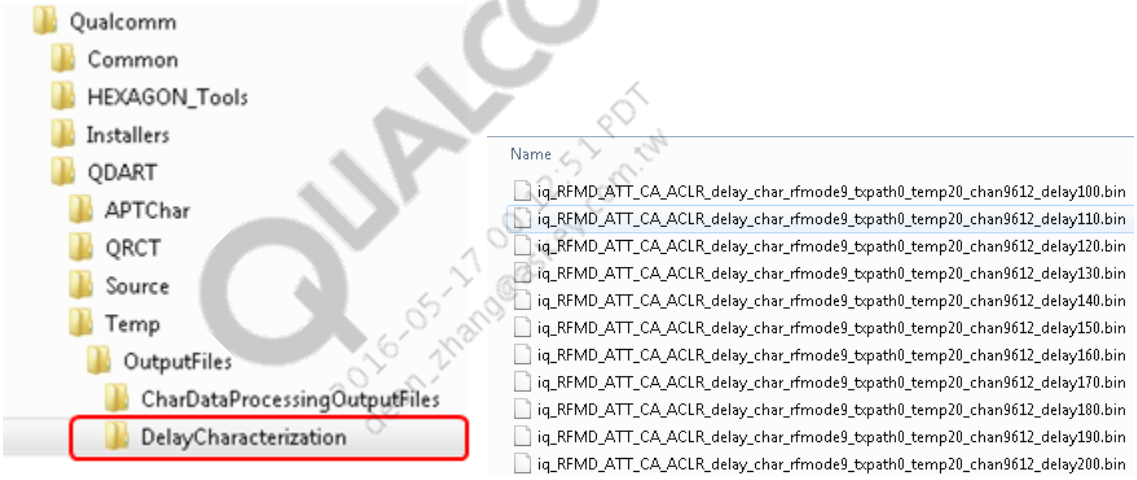
## 5.3.4 Output .xml file and IQ capture data

### 5.3.4.1 Output file for ACLR delay char

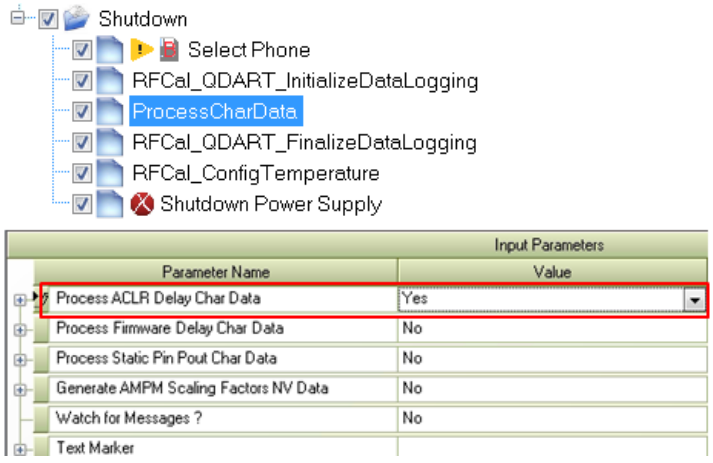
To save IQ capture data on ACLR delay characterization as shown, the Save IQ Capture flag in ACLR delay characterization must be enabled.



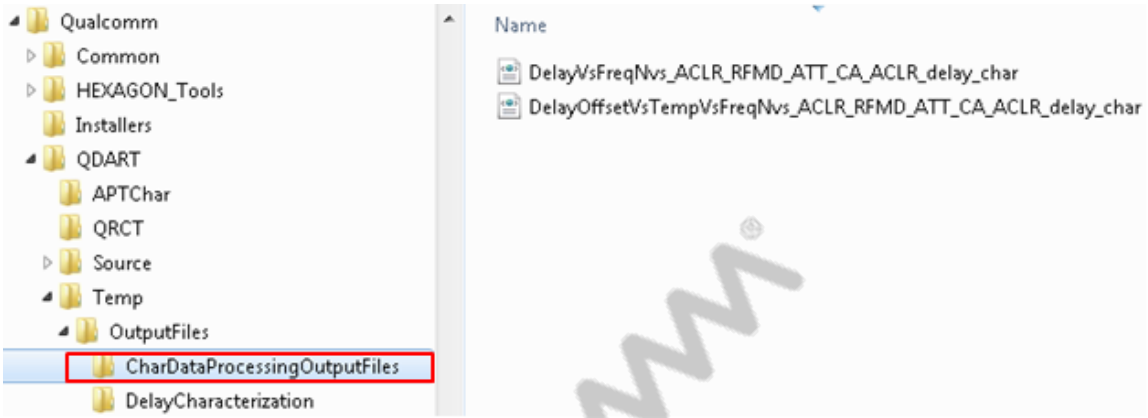

The output file is generated in C:\Qualcomm\QDART\Temp\OutputFiles\DelayCharacterization.




To generate the .xml file for the delay NV item, the Process ACLR Delay Char Data flag must be enabled.

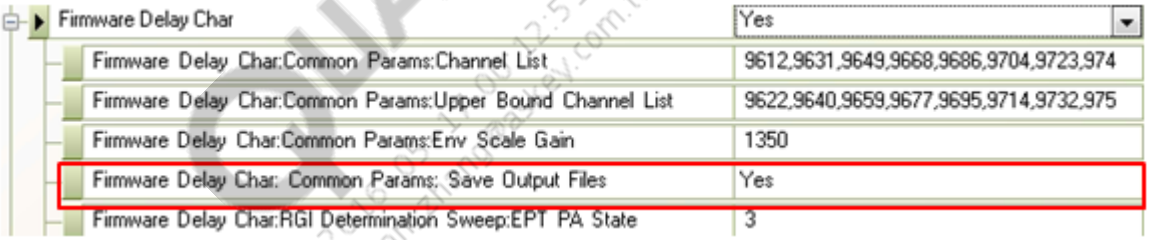


The output file is generated in C:\Qualcomm\QDART\Temp\OutputFiles\ CharDataProcessingOutputFiles.

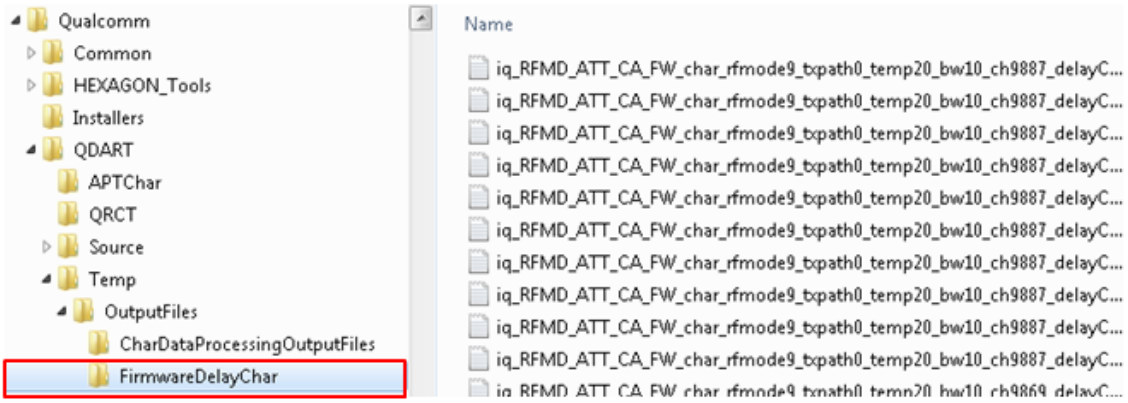



### 5.3.4.2 Output file for firmware delay characterization

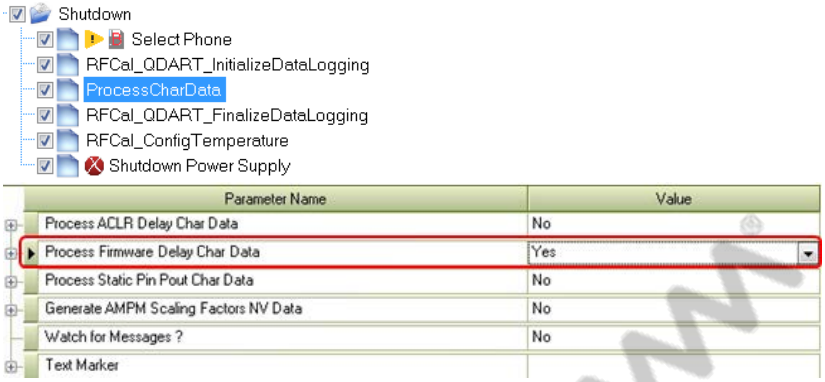
To save IQ capture data on firmware delay characterization, the Save IQ Capture flag in Firmware Delay Char must be enabled.



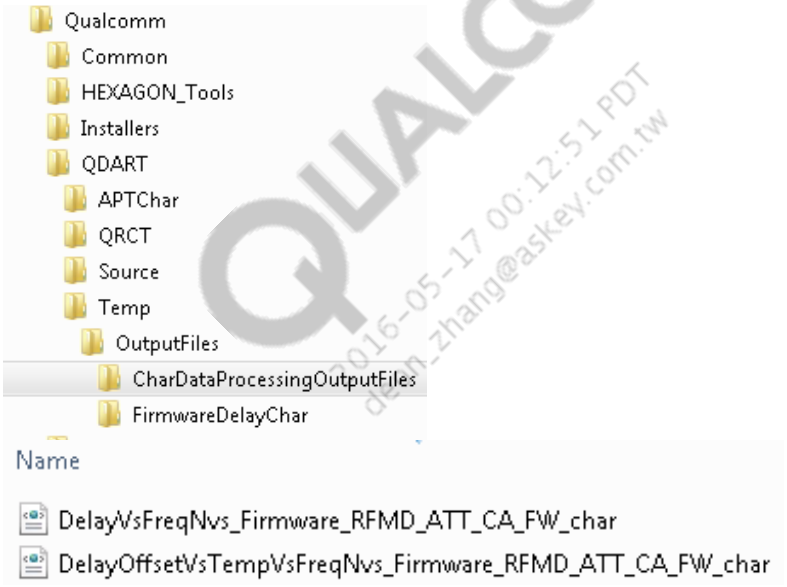

The output file is generated in C:\Qualcomm\QDART\Temp\OutputFiles\FirmwareDelayChar.



To generate the .xml file for the delay NV item, the Process ACLR Delay Char Data flag must be enabled.




The output file is generated in C:\Qualcomm\QDART\Temp\OutputFiles\ CharDataProcessingOutputFiles.




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## 5.4 Pin Pout characterization

DPD compensates for the PA's nonlinearity as shown in Figure 5-9. The predistorted signal is aligned with the PA's nonlinearity at calibration. The PA's nonlinearity changes due to duplexer changes across frequency and temperature. The PA's nonlinearity must be matched again. Fortunately, most PAs are scalable, so DPD does not need to be changed. Only adjust level of predistorted signal by IQ_gain and RGI.

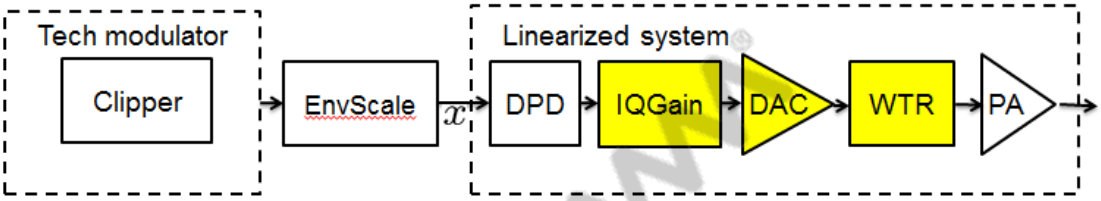



**Figure 5-9 Simple system architecture**

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## 5.4.1 Pin Pout characterization char parameters description

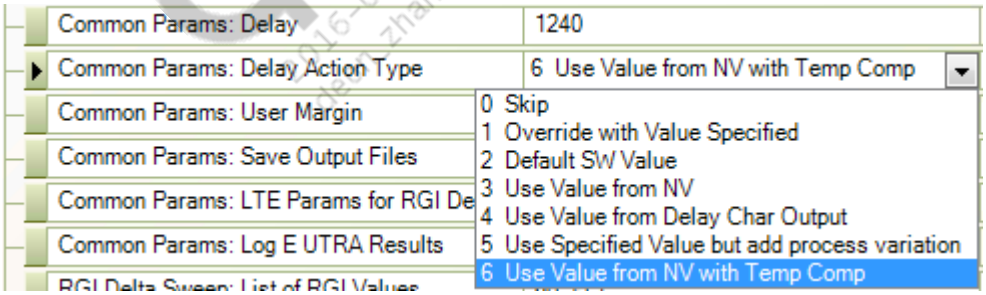
As shown here the char tool input file includes the detrough table, ICQ, compression point, and ET minimum voltage setting for each PA.

```
<RFCal_Data>
- <PA product_rev="0" product_id="1" manuf_id="263">
    <Description Revision="V1.0.26" ProductRev="0" ProductID="0x01" ManufID="0x107" name="ACPM_7600" />
  - <XptInput_V1 rf_mode="5" device="0">
      <IcqEt_table IcqEt="156" />
      <IcqEpt_table IcqEpt="36" />
      <IcqRgiDeltaSwp_table IcqRgiDeltaSwp="255" />
      <MLineMaxPaSupplyEt_table MLineMaxPaSupplyEt="3.8" />
      <MLineMaxPaSupplyEpt_table MLineMaxPaSupplyEpt="3.3" />
      <RgiBiasCharCompPt_table RgiBiasCharCompPt="5" />
      <AclrDelayCharCompPoint_table AclrDelayCharCompPoint="3" />
      <AclrDelayCharVmax_table AclrDelayCharVmax="3.4" />
      <AclrDelayCharIqGain_table AclrDelayCharIqGain="622" />
      <StaticPinPoutCharRgiDeltaVmax_table StaticPinPoutCharRgiDeltaVmax="3.3" />
      <StaticPinPoutCharVmax_table StaticPinPoutCharVmax="2.5" />
      <FirmwareDelayIqGain_table FirmwareDelayIqGain="622" />
      <FirmwareDelayCharRgiDetPaBias_table FirmwareDelayCharRgiDetPaBias="3.3" />
      <FirmwareDelayVmax_table FirmwareDelayVmax="3.4" />
      <FirmwareDelayVmin_table FirmwareDelayVmin="0.6" />
      <FirmwareDelayCharCompPoint_table FirmwareDelayCharCompPoint="2" />
      <IqGainRgiDeltaSwp_table IqGainRgiDeltaSwp="400" />
      <CalCompPtEpt_table CalCompPtEpt="2" />
      <CalCompPtEt_table CalCompPtEt="3" />
      <CharCompPtThresEt_table CharCompPtThresEt="2.5" />
      <MLineFitType_table MLineFitType="0" />
      <MLineRegCoef_table MLineRegCoef="0.5" />
      <EtVmin_table EtVmin="1" />
      <AptFirstSwpBiasHPMHigh_table AptFirstSwpBiasHPMHigh="3.7" />
      <AptFirstSwpIcqHPMHigh_table AptFirstSwpIcqHPMHigh="255" />
      <AptFirstSwpBiasHPMLow_table AptFirstSwpBiasHPMLow="3.7" />
      <AptFirstSwpIcqHPMLow_table AptFirstSwpIcqHPMLow="255" />
      <AptFirstSwpBiasLPM_table AptFirstSwpBiasLPM="3.7" />
      <AptFirstSwpIcqLPM_table AptFirstSwpIcqLPM="255" />

- <Detrough_table>
    <XptEpdtRec epdt="7" />
    <XptEpdtRec epdt="29" />
    <XptEpdtRec epdt="64" />
    <XptEpdtRec epdt="113" />
    <XptEpdtRec epdt="173" />
    <XptEpdtRec epdt="246" />
    <XptEpdtRec epdt="329" />
    <XptEpdtRec epdt="424" />
    <XptEpdtRec epdt="528" />
    <XptEpdtRec epdt="642" />
    <XptEpdtRec epdt="765" />
    <XptEpdtRec epdt="897" />
    <XptEpdtRec epdt="1037" />
    <XptEpdtRec epdt="1185" />
    <XptEpdtRec epdt="1341" />
    <XptEpdtRec epdt="1503" />
    <XptEpdtRec epdt="1673" />
    <XptEpdtRec epdt="1849" />
    <XptEpdtRec epdt="2031" />
    <XptEpdtRec epdt="2219" />
    <XptEpdtRec epdt="2413" />
    <XptEpdtRec epdt="2612" />
    <XptEpdtRec epdt="2816" />
    <XptEpdtRec epdt="3025" />
    <XptEpdtRec epdt="3238" />
    <XptEpdtRec epdt="3456" />
```

Common parameters are:

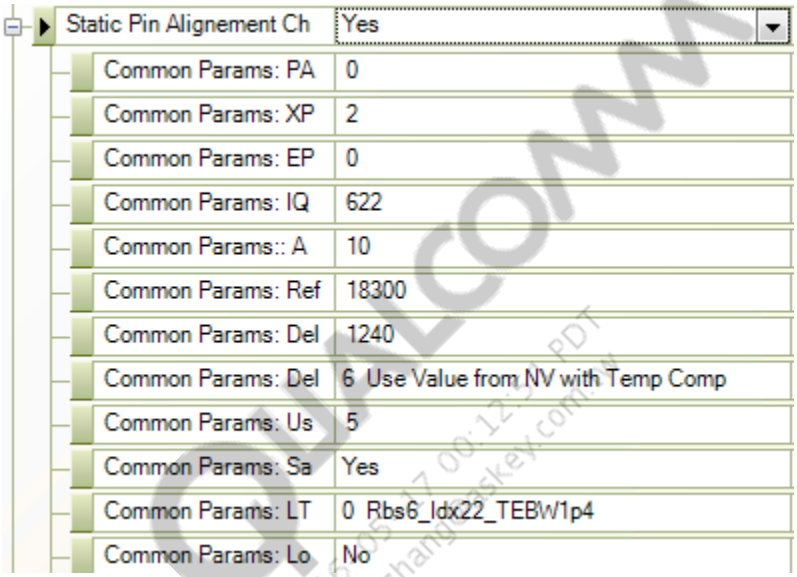
- PA state list
  - LTE – 0 for ET, 1 for EPT
  - WCDMA – 3 for ET, 2 for EPT
- XPT action list
  - 0 for APT, 1 for EPT, 2 for ET
- Delay action type
  - 0 – Skip
  - 1 – Override with the value specified in Common Params: Delay
  - 2 – Default software value
  - 3 – Use value from NV
  - 4 – Use the value from delay char output
    - Only takes the delay for the reference channel
    - If delay char does not include the reference channel, the char output delay value is 0
    - If delay char includes the reference channel, the delay equals the delay of the reference channel; the delay of other channels is not used
  - 5 – Use the specified value, but add process variation
  - 6 – Use the value from NV with temp comp





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- User margin
  - Similar as in RGI bias char
- Save output files
  - If yes, then the constructed DPD is saved at C:\Qualcomm\QDART\Temp
- Temperature
  - Is the default value in the log; this value is overridden if temperature chamber is set with test



| | |
|---|---|
| Static Pin Alignement Ch | Yes |
| Common Params: PA | 0 |
| Common Params: XP | 2 |
| Common Params: EP | 0 |
| Common Params: IQ | 622 |
| Common Params:: A | 10 |
| Common Params: Ref | 18300 |
| Common Params: Del | 1240 |
| Common Params: Del | 6 Use Value from NV with Temp Comp |
| Common Params: Us | 5 |
| Common Params: Sa | Yes |
| Common Params: LT | 0 Rbs6_Idx22_TEBW1p4 |
| Common Params: Lo | No |

RGI delta sweep and RGI determination sweep are shown here.

- List of RGI values – For RGI sweep
- Threshold type – Specifies the threshold type used to determine the RGI/bias pair
- Target ACLR – ACLR threshold to determine the RGI/bias pair
- Target compression – n-dB compression point to determine the RGI/bias pair

| | |
|---|---|
| RGI Delta Sweep: PA Bias | 3300 |
| RGI Delta Sweep: List of RGI Values | 60,98,1 |
| RGI Delta Sweep: Expected Power Level For Th | 25 |
| RGI Determination Sweep: List of RGI Values | 60,94,1 |
| RGI Determination Sweep: Env Scale Gain | 1350 |
| RGI Determination Sweep: Threshold Type | 1 Compression |
| RGI Determination Sweep: Minimum Tx Power N | 10 |
| RGI Determination Sweep: Target ACLR List | -36.0,-36.0 |
| RGI Determination Sweep: Expected Power Lev | 25 |
| IQ Capture and DPD Construction Sweep: Env S | 1350 |

Freq comp sweep and DPD parameters are shown here.

- Channel list – 16 calibration channels

**NOTE:** The channel list must be the same as freq comp calibration channel list. The channel list in the tree must use the same TX_CAL_CHAN NV.

- List of IQ gain values – Start from 622 to 322 with each Step size 5

| | |
|---|---|
| Freq Comp Sweep: Channel List | 9621,9638,9656,9673,9691,9708,9 |
| Freq Comp Sweep: List of IQ Gain Values | 622,322,-5 |
| Freq Comp Sweep: Env Scale Gain List | 1418,1350 |
| Freq Comp Sweep: Expected Power Level For T | 25 |

## 5.4.2 ACLR-based static Pin/Pout char tool procedure

Figure 5-10 and Figure 5-11 show an overall flowchart of static Pin alignment.

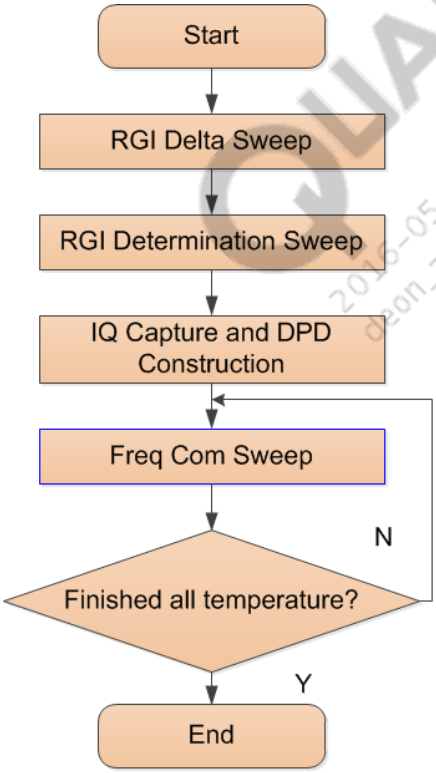



**Figure 5-10 Static Pin alignment flowchart**

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1. RGI delta sweep
   - Purpose – Measure gain deltas between RGIs
   - Procedure – Sweep RGI from low to high with fixed high PA bias, low IQ gain in APT mode at reference channel

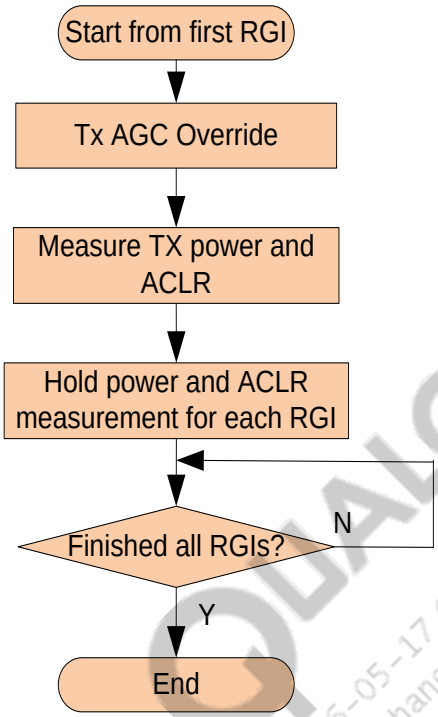



**Figure 5-11 RGI delta sweep**

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dec zhang@askey.com.tw

2. RGI determination sweep as shown in Figure 5-12 consists of:
   - Purpose – Find the RGI that gives practical peak-signal compression point for specified PA bias
   - Procedure
     - Sweep RGI from low to high
     - IQ capture samples
     - Calculate the compression point
     - Measure ACLR until the compression, ACLR hits the threshold

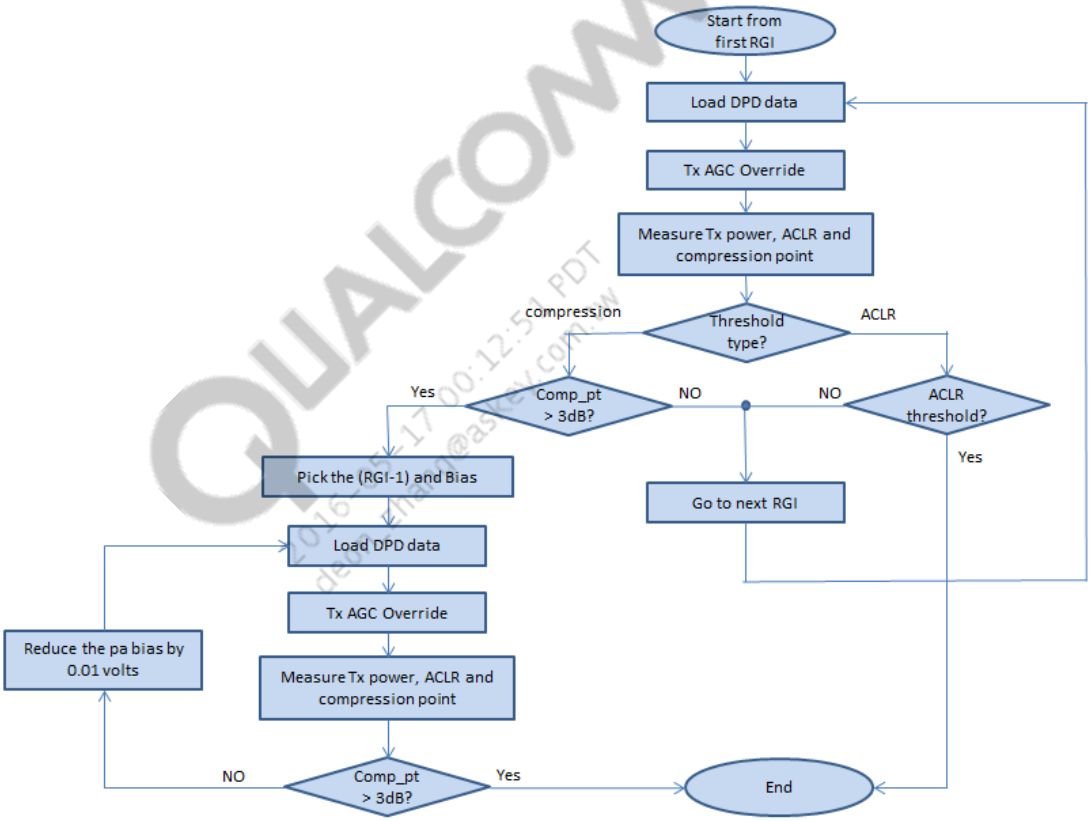



**Figure 5-12 RGI determination sweep**

3. IQ capture and DPD construction
   - Purpose – Calibrate the DPD at the determined RGI and PA bias from Step 2
   - Procedure
     - Do IQ capture at the determined RGI and PA bias from Step 2
     - Construct DPD table

4. Freq comp sweep as shown consists of:
   - Purpose – Measure ACLR and Pout across channels and Pin, and find the optimal Pin and Pout compensation
   - Procedure
     - Sweep 16 calibration channel
     - Apply the DPD, and the specified mode – EPT/ET, EnvScale, PA bias/detrough
     - Sweep Pin by – Use RGI = RGI_for_cal+Δ, and sweep IQGain in a range (622:-5:312)
     - Measure ACLR and Pout
     - Optimal Pin for a channel is found by the Pin that gives the best ACLR

```
Foreach PA state in PA state list
    do
    Get the room temperature DPD data for the current PA state
    Foreach channel in the channel list
        Foreach IQ gain in (622:-5:312)
            Load DPD data;
            TX AGC Override with (detRGI+fudge Factor);
            Measure Power and ACLR;
            Update IQ gain value that gives minimum ACLR for the current channel;
        End

        If the found optimal IQ gain is at the edge of IQ gain list
            If the found optimal IQ gain is at low edge
                Decrease the RGI used by 1;
            Else
                Increase the RGI used by 1;
            End
            Sweep the IQ gain list again with the new RGI;
        Else
            Calculate the pin value based on optimal IQ gain;
        End
    End
End
```

5. Repeat Step 4 – Pin sweep across all other seven temperatures
   - Purpose – Characterize the Pin/Pout across temperatures
   - Procedure
     - Repeat Step 4 for all temperatures
     - RGI delta, RGI determination, and DPD, which is given by room (reference) temperature only

## 5.4.3 AMAM-based static Pin/Pout char tool (fast Pin char)

NOTE: This section was added to this document revision.

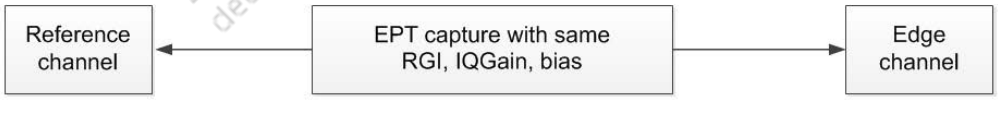
The AMAM-based Pin char is used to significantly shorten the char time for Pin char across frequency and to avoid sensitivity to ACLR measurement issues in Pin char. It is not to improve ACLR and relax requirements on component matching. Customers can compare the ACLR-based Pin char and AMAM-based Pin char only on good matching hardware. This feature is enabled only on DimePM and Bolt modems. For more information, see:

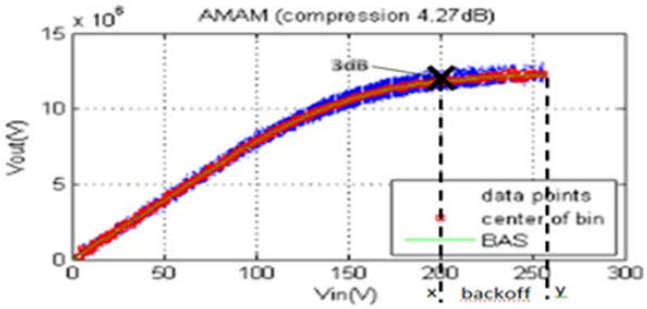
- *MDM9x35+QFE Software Design Guide* (80-NK207-2)
- *Presentation: MPSS.DPM.1.0 QFE APT+DPD Software Overview* (80-NK201-5)
- *Presentation: MPSS.DPM.2.0 QFE APT+DPD Software Overview* (80-NK202-5)

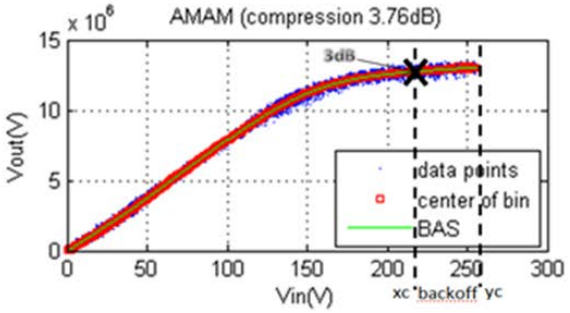
### 5.4.3.1 Principle and illustration

Pin values can be found based on the shifts of AMAM curves, which can be calculated based on sample captures, instead of ACLR.

- With the same RGI/IQGain/bias, EPT sample captures on reference channel and edge channel
  - Reference channel (left-hand side plot) gives peak compression 4.27 dB; 3 dB compression point has backoff db (x/y)
  - Edge channel (right-hand side plot) gives peak compression 3.76 dB; 3 dB compression point has backoff db (xc/yc)
- Edge channel Pin relative to reference channel is db (xc/yc) - db (x/y)



## Enable AMAM-based Pin char

| | |
|---|---|
| Static Pin Alignement Char | Yes |
| Common Params: PA State List | 2 |
| Common Params: XPT Action List | 1 |
| Common Params: EPT PA State for RGI Delta and RGI Determination S | 2 |
| Common Params: IQ Gain | 622 |
| Common Params:: ACLR Averages | 10 |
| Common Params: Reference Channel | 9750 |
| Common Params: Delay | 330 |
| Common Params: Delay Action Type | 6 Use Value from NV with Temp Comp |
| Common Params: User Margin | 5 |
| Common Params: Save Output Files | No |
| Common Params: LTE Params for RGI Determination and Freq Comp S | 0 Rbs6_Idx22_TEBW1p4 |
| Common Params: Log E UTRA Results | No |
| Common Params: Enable AMAM Based Pin Pout Char? | No |
| RGI Delta Sweep: List of RGI Values | No<br>Yes |
| RGI Delta Sweep: Expected Power Level For The First Bias and Rgi Co | 25 |
| RGI Determination Sweep: List of RGI Values | 60,94,1 |
| RGI Determination Sweep: Env Scale Gain | 1350 |
| RGI Determination Sweep: Threshold Type | 1 Compression |
| RGI Determination Sweep: Minimum Tx Power Needed for Compression | 10 |
| RGI Determination Sweep: Target ACLR List | -36.0 |

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Figure 5-13 shows an AMAM-based Pin char algorithm callflow.

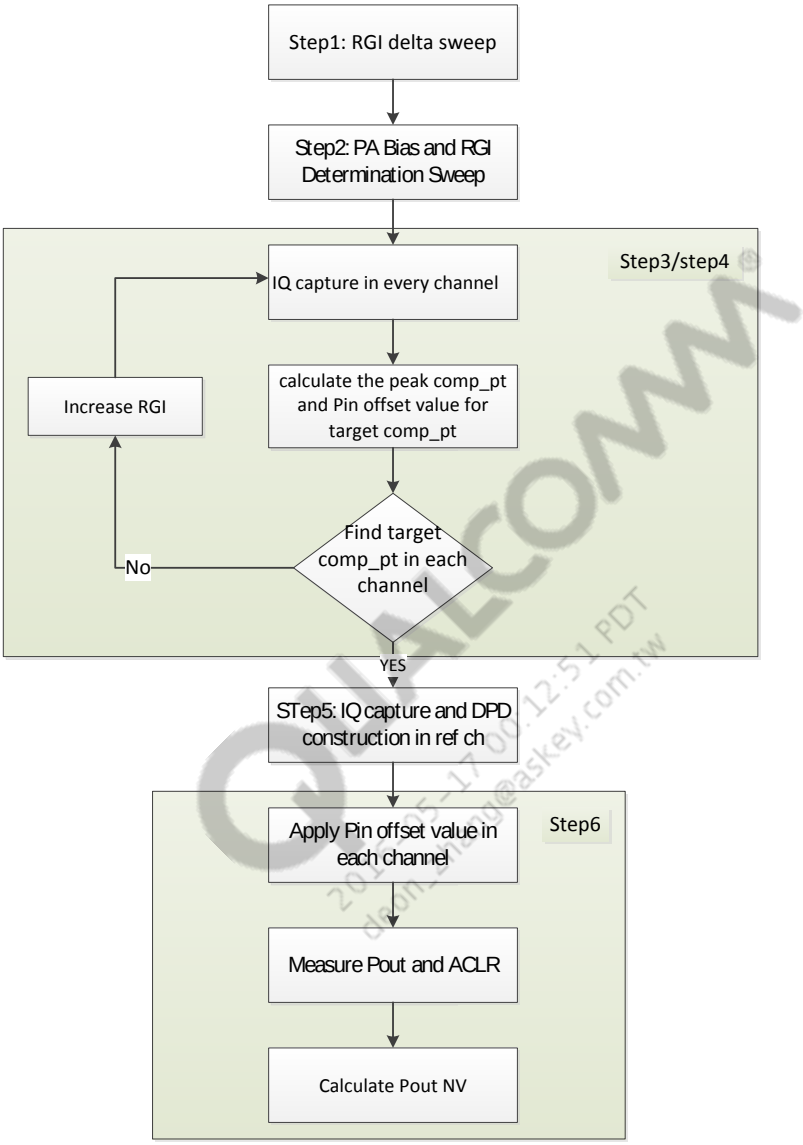



**Figure 5-13 AMAM-based Pin char algorithm callflow**

## 5.4.4 Pin alignment data analysis

Pin alignment and Pout each have a constant channel-wise shift due to duplexer breathing with temperature as shown in Figure 5-14 and Figure 5-15. After channel shifts, Pin alignment offsets and Pout stay constant across frequency, indicating that applying channel-independent Pin alignment and Pout offsets on top of static pin alignment and freq comp suffice.

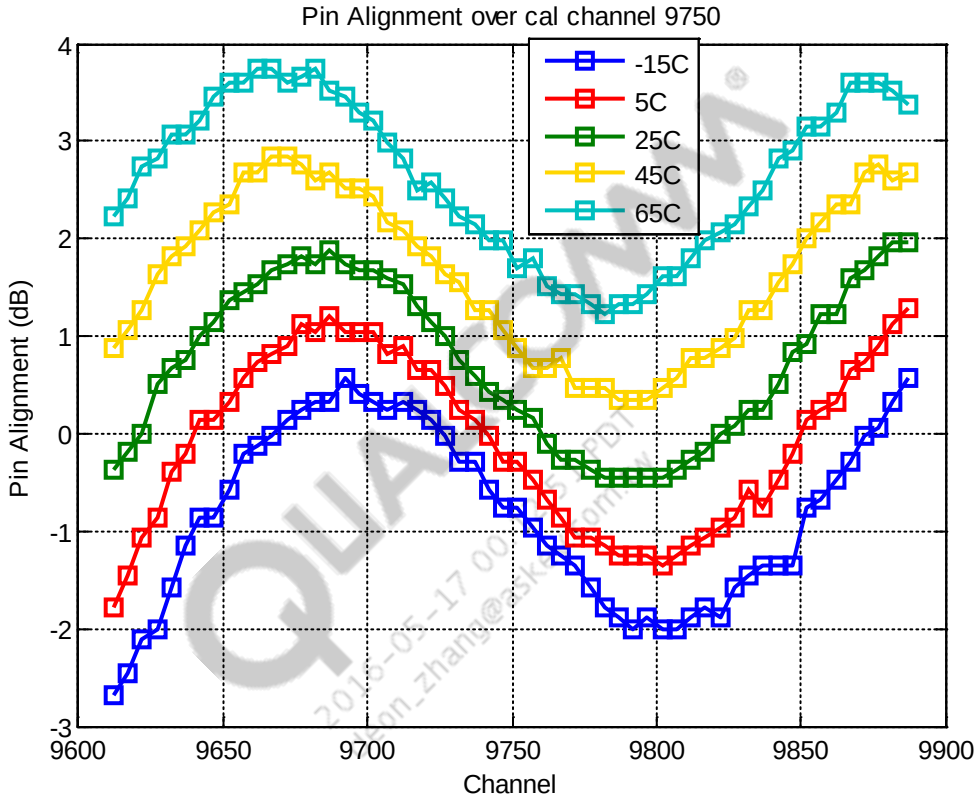



**Figure 5-14 Pin data over temperature and frequency**

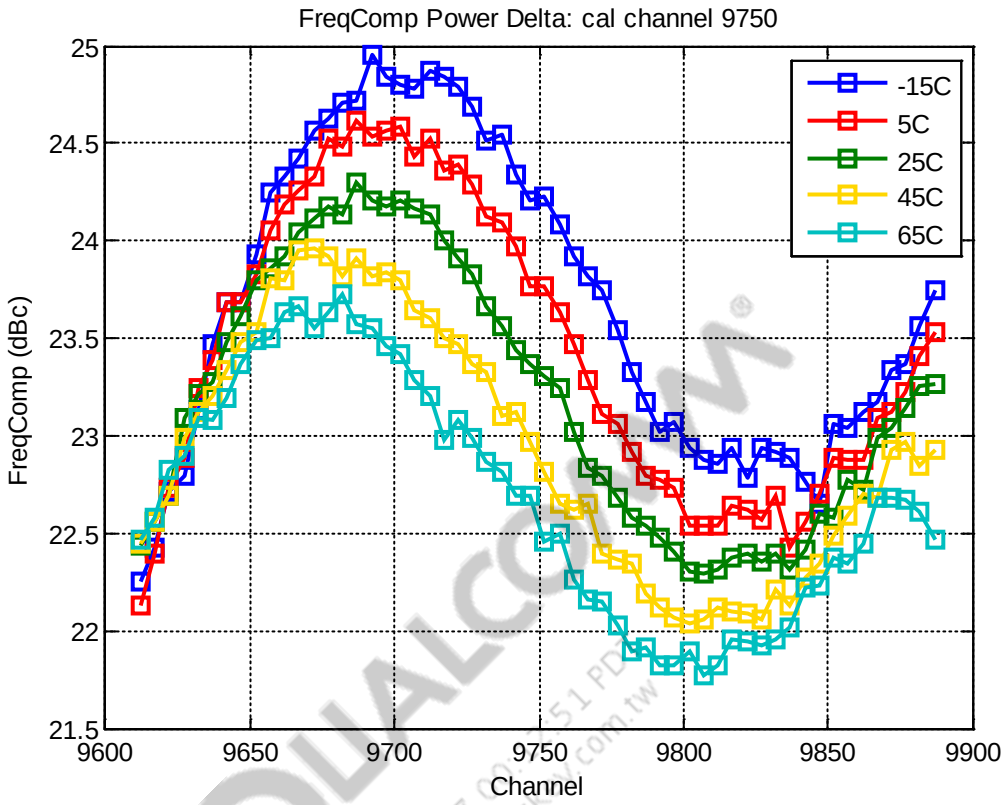


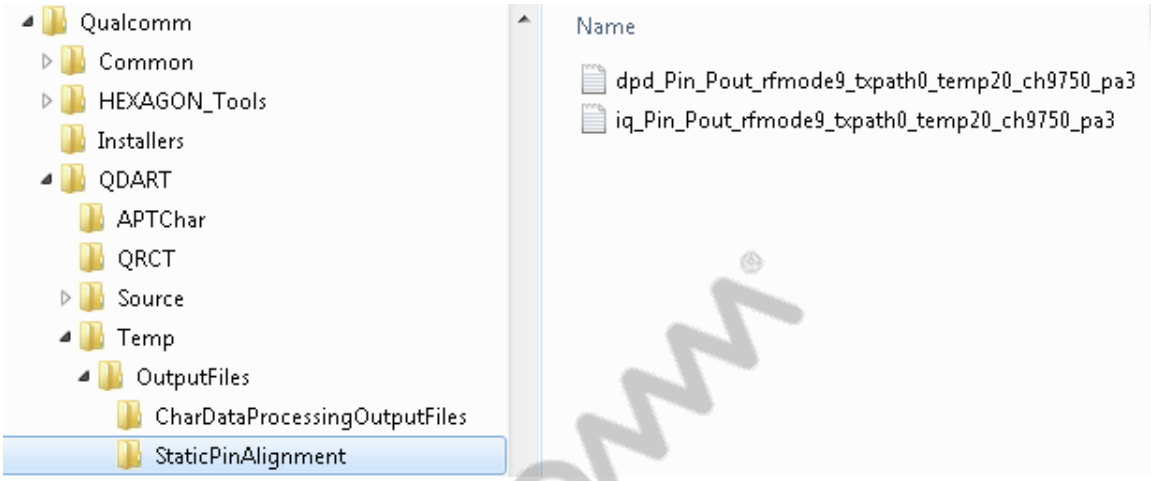

**Figure 5-15 Pout data over temperature**

## 5.4.5 Output .xml file and IQ capture data

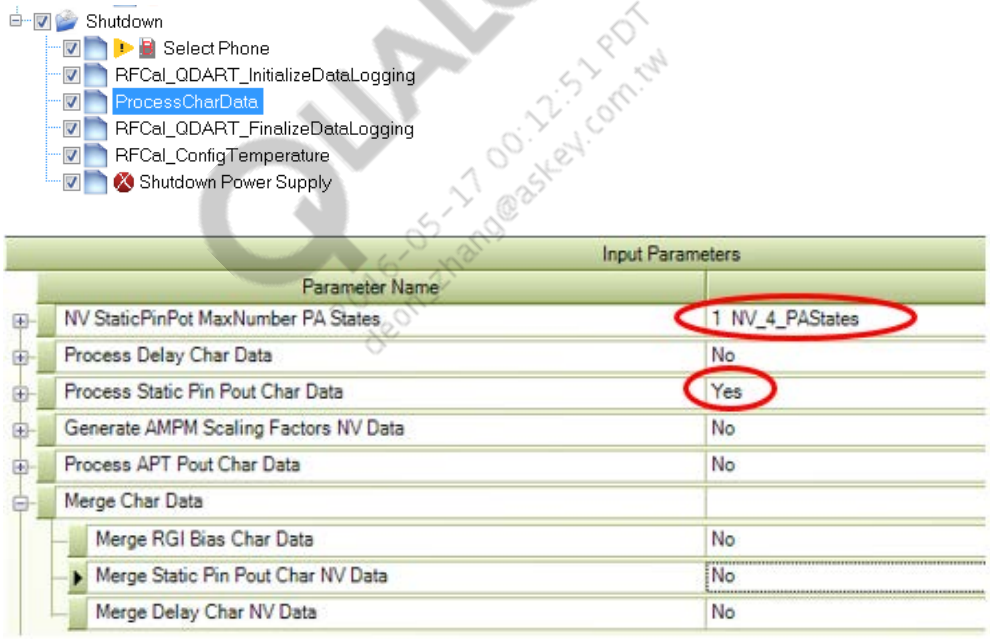
To save IQ capture data on Pin alignment characterization, the Save Output Files flag must be enabled.

| | |
|---|---|
| Static Pin Alignement Char | Yes |
| Common Params: PA State List | 3 |
| Common Params: XPT Action List | 2 |
| Common Params: EPT PA State for RGI Delta and RGI Determination | 3 |
| Common Params: IQ Gain | 622 |
| Common Params:: ACLR Averages | 10 |
| Common Params: Reference Channel | 9750 |
| Common Params: Delay | 330 |
| Common Params: Delay Action Type | 6 Use Value from NV with Temp Comp |
| Common Params: User Margin | 5 |
| Common Params: Save Output Files | Yes |

The output file is generated in C:\Qualcomm\QDART\Temp\OutputFiles\StaticPinAlignment.

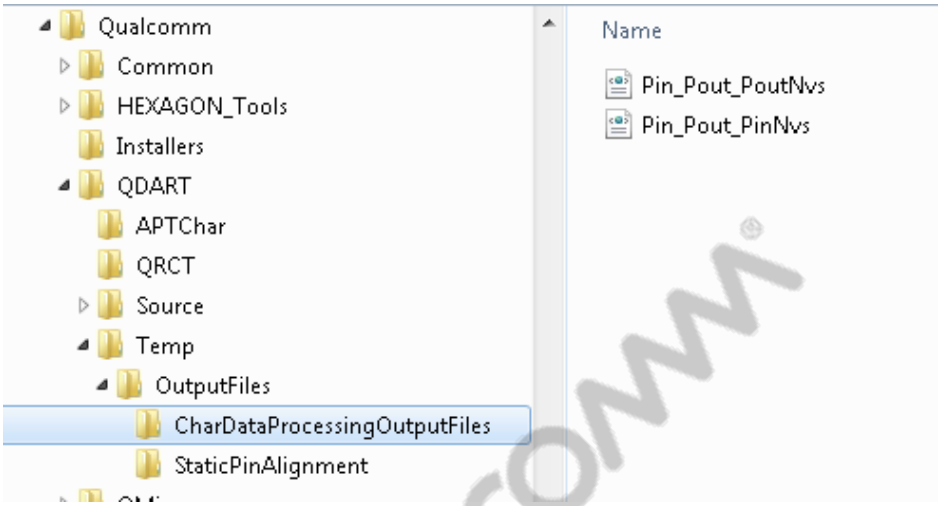



To generate the .xml file for the delay NV item, the Process Static Pin Pout Char Data flag must be enabled.




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The output file is generated in C:\Qualcomm\QDART\Temp\OutputFiles\ CharDataProcessingOutputFiles.




If the software uses new PA STATIC NV for 8 PA states, after the PinPout characterizations are done, execute the ProcessCharData to populate the NV. The illustration shows the parameters of ProcessCharData.

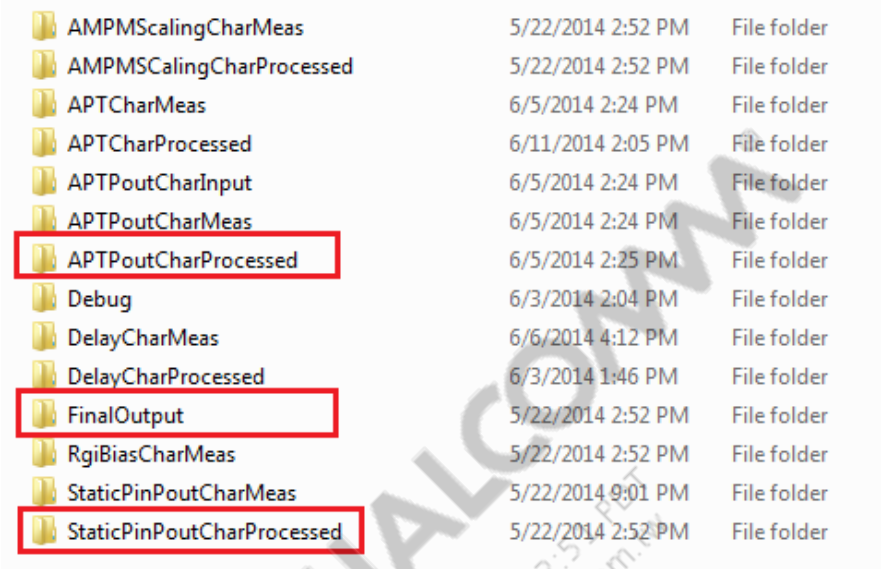
| Parameter Name | |
|---|---|
| NV StaticPinPot MaxNumber PA States | 2 NV_8_PAStates |
| Static NVs input folder. In case NV_8_PAStates | \\freeze\builds491\I |
| Process Delay Char Data | No |
| Process Static Pin Pout Char Data | Yes |
| Generate AMPM Scaling Factors NV Data | No |
| Process APT Pout Char Data | Yes |
| Merge Char Data | |
| Merge RGI Bias Char Data | No |
| Merge Static Pin Pout Char NV Data | Yes |
| Merge Delay Char NV Data | No |

1. Choose the NV_8_PAStates for the PA_STATIC NV with up to 8 PA states.
2. Specify the path that PA_STATIC NV is stored; the NV must be in XML format.
3. Enable Process PinPout char
4. Enable merge static Pinpout char NV data in merge char data

Generally, all the char results are stored in C:\Qualcomm\QDART\Temp\OutputFiles\XPTChar.

The NVs with APT Pout char and PinPout char are stored in APTPoutCharProcessed and StaticPinPoutCharProcessed, respectively.

The final merged PA_STATIC is in the FinalOutput folder. Download the PA_STATIC NV in the FinalOutput folder to the phone to finish the characterization.




## 5.5 IQ gain and Env_scale setting

Table 5-1 to Table 5-4 show settings for IQ gain and Env_scale.

**Table 5-1 IQ gain and Env_scale setting (1 of 4)**

| IQ gain | Calibration | | | | | |
|---|---|---|---|---|---|---|
| | Sweep 1 alignment | Sweep 1 RGI delta | Sweep 2 EPT | Sweep 2 ET | Sweep 3 EPT | Sweep 3 ET |
| CDMA | NV | Low IQ gain | 622 (NV) | 622 (NV) | 622 (NV) | 622 (NV) |
| WCDMA | NV | Low IQ gain | 622 (NV) | 622 (NV) | 622 (NV) | 622 (NV) |
| LTE | NV | Low IQ gain (150) | 420 (NV) | 420 (NV) | 420 (NV) | 420 (NV) |

**Table 5-2 IQ gain and Env_scale setting (2 of 4)**

| Env scale | Calibration | | | | | |
|---|---|---|---|---|---|---|
| | Sweep 1 alignment | Sweep 1 RGI delta | Sweep 2 EPT | Sweep 2 ET | Sweep 3 EPT | Sweep 3 ET |
| CDMA | 1024 | 1024 | 1024 | 1024 | 1126 (NV) | 1126 (NV) |
| WCDMA | 1350 | NA | 1350 | 1350 | 1418 (NV) | 1418 (NV) |
| LTE | 820 | NA | 820 | 820 | 920 (NV) | 920 (NV) |

**Table 5-3 IQ gain and Env_scale setting (3 of 4)**

| IQ gain | Characterization | | | | |
|---|---|---|---|---|---|
| | RGI/bias char | Pin char | Pin char IQ gain list | ACLR delay char | Firmware delay char |
| CDMA | 622 (NV) | 622 | 622,322,-5 | 622 | 622 |
| WCDMA | 622 (NV) | 622 | 622,322,-5 | 622 | 622 |
| LTE | 622 (NV) | 622 | 622,322,-5 | 420 | 250 (avago)<br>350 (QPA) |

**Table 5-4 IQ gain and Env_scale setting (4 of 4)**

| Env scale | Characterization | | | | |
|---|---|---|---|---|---|
| | RGI/bias char | Pin char steps 1 to 3 | Pin char Step 4 (sweep) | ACLR delay char | Firmware delay char |
| CDMA | 1024 | 1024 | NV | 1024 | 1024 |
| WCDMA | 1350 | 1350 | NV | 1350 | 1350 |
| LTE | 820 | 820 | 820 | 820 | 820 |

# 5.6 Number of RBs for IQ capture

Table 5-5 to Table 5-6 show number of RBs for IQ capture.

**Table 5-5 Number of RBs for IQ capture (1 of 2)**

| LTE | Calibration | | | | Delay cal |
|---|---|---|---|---|---|
| | Sweep 1 alignment | Sweep 2 ET | Sweep 3 ET | Sweep 4 Freq comp | |
| Waveform type | 3 (10 MHz) | 3 (10 MHz) | 3 (10 MHz) | 3 (10 MHz) | 3 (10 MHz) |
| Start RB | 19 | 19 | 22 | 22 | 7 |
| Num RB | 12 | 12 | 6 | 6 | 36 |

**Table 5-6 Number of RBs for IQ capture (2 of 2)**

| LTE | Characterization | | | | |
|---|---|---|---|---|---|
| | ACLR delay | Pin alignment | Firmware delay (10 MHz) | Firmware delay (5 MHz) | Firmware delay (20 MHz) |
| Start RB | 19 | 22 | 36 | 25 | 50 |
| Num RB | 12 | 6 | 7 | 0 | 25 |

# 5.7 APT Pout characterization

## 5.7.1 APT Pout Char tool

APT Pout characterization is only necessary for WCDMA while other technologies are optional. The purpose of APT Pout char is to characterize the Pout offset across temperature in APT mode. Since the Pout over temperature is characterized in EPT and ET mode, it is necessary to have the Pout over temperature in APT mode as well for passing ILPC at PA switchpoints.

The illustration shows the parameters for APT Pout characterization.

| | |
|---|---|
| APT Pout Char | No |
| APT Pout Char: Reference Channel | 9750 |
| APT Pout Char: List of Characterization Channels | 18001,18038,18076,18114,18152, |
| APT Pout Char: PA State List | 0,1,2 |
| APT Pout Char: Target Power List | 3,6,13 |
| APT Pout Char: Number of RGI/Bias Pairs | 3 |
| APT Pout Char: RGI/Bias Pairs Select Method | 0 low side |
| APT Pout Char: LTE Specific Params | |
| APT Pout Char: LTE Specific Params: Bandwi | 3 10MHz |
| APT Pout Char: LTE Specific Params: Num RB | 12 |
| APT Pout Char: LTE Specific Params: Start RB | 19 |

- PA state list – This is the list of PA states that perform APT Pout characterization.
- Target power list – For each PA state, this is the target power level to perform APT Pout characterization.
- Number of RGI/bias pairs – The number of power levels to perform APT Pout characterization. If more than one, RGI is picked descending from the target power RGI with 1 count per step.

The APT characterization and APT Pout characterization must be enabled simultaneously or the UUTID of APT Pout characterization is the same as the APT characterization table.

A sample log for APT Pout characterization is shown.

```
Tech = 10; RFMode = 34; Temp = 22; IsRoomTemp = 1; ThermADC = 1264; Channel = 18266; PA State = 1
RGI             ICQ             Bias        Pwr(dBm)
67              127             2490         4.5221
66              127             1990        2.66992
65              127             1750        1.17001
Tech = 10; RFMode = 34; Temp = 22; IsRoomTemp = 1; ThermADC = 1264; Channel = 18300; PA State = 1
RGI             ICQ             Bias        Pwr(dBm)
67              127             2490        4.88647
66              127             1990        3.05322
65              127             1750        1.54373
Tech = 10; RFMode = 34; Temp = 22; IsRoomTemp = 1; ThermADC = 1264; Channel = 18342; PA State = 1
RGI             ICQ             Bias        Pwr(dBm)
67              127             2490        5.16092
66              127             1990        3.32767
65              127             1750        1.83566
Tech = 10; RFMode = 34; Temp = 22; IsRoomTemp = 1; ThermADC = 1264; Channel = 18380; PA State = 1
RGI             ICQ             Bias        Pwr(dBm)
67              127             2490        5.13117
66              127             1990        3.29321
65              127             1750        1.81635
```

After APT Pout characterization is performed, the results fill into the Pout portion of the PA_STATIC NV, and it is performed after PinPout characterization is finished.


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## 5.7.2 Manual tune APT Pout data

In the legacy design, the LIN_VS_TEMP NV item contains eight adjustment values based on temperature. Each element corresponds to an adjustment value for a given temperature; each element in the NV_BCy_TX_LIN_x_I table is effectively adjusted by a constant amount, based on this table. In the MSM8974/MDM9x25 software design, the MULTI_LIN_V3 NV item is used to store the linearizer data and Pin/Pout are used to store the temp comp/freq comp offset value. Based on the new design with ESC V4 Cal, Pout NV in APT state is very important for power accuracy over temperature. The purpose of the APT Pout NV is the same as the legacy LIN_VS_TEMP NV item. Follow the procedure below to obtain the APT Pout data.

1. Clear all APT modes (PA states 0, 1, and 2) temperature compensation in the Tech_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I NV items. (Ensure that all temperature compensations are set to zero before running CalVer.)
2. Run RF Cal verification for the desired band.
   - Ensure that the correct temperature loop is set in the test tree.
3. Plot the Tx error for all temperatures.
4. Estimate the temp comp NVs for each PA state, i.e., 0, 1, and 2.
   - Eight groups bins (each group has 16 bins over freq) in each NV correspond to eight temperatures 85°C, 69°C, 53°C, 37°C, 20°C, 4°C, -13°C, and -30°C.
   - Set the NV bins to compensate the Tx error in the plot for each temperature. Note that 10 indicates -1 dB, 20 indicates -2 dB, -10 indicates 1 dB, etc.

Repeat Step 2 through Step 4 until the Tx error in the plot is less than 1 dB for all APT PA states.

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# 6 xPT Debug

Hardware matching is very important for the ET system. There are four characterization tools, which have been introduced in Chapter 5. The delay characterization tool and Pin alignment characterization tool can help users understand the duplexer and PA output matches.

## 6.1 Hardware validation

In general, delay characterization output power/ACLR measurement provides a clear picture of ET hardware performance limits. For a good qualified ET hardware, we shall see similar power/delay across multiple channels. Additionally, Pin alignment characterization with full RB provides a clear picture to predict the best ACLR that can be achieved in Online mode and tell the ET hardware performance limit.

Figure 6-1 is a flowchart of hardware validation.

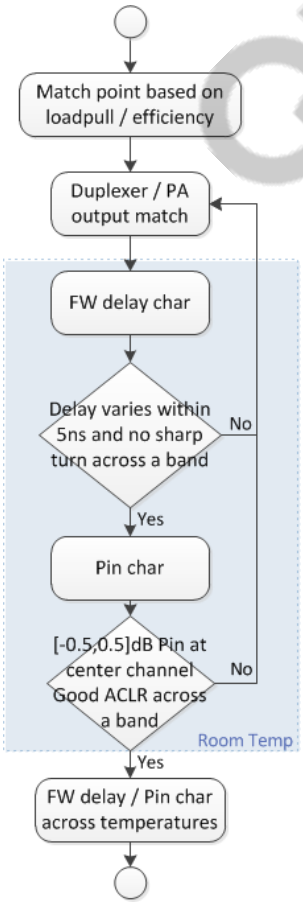


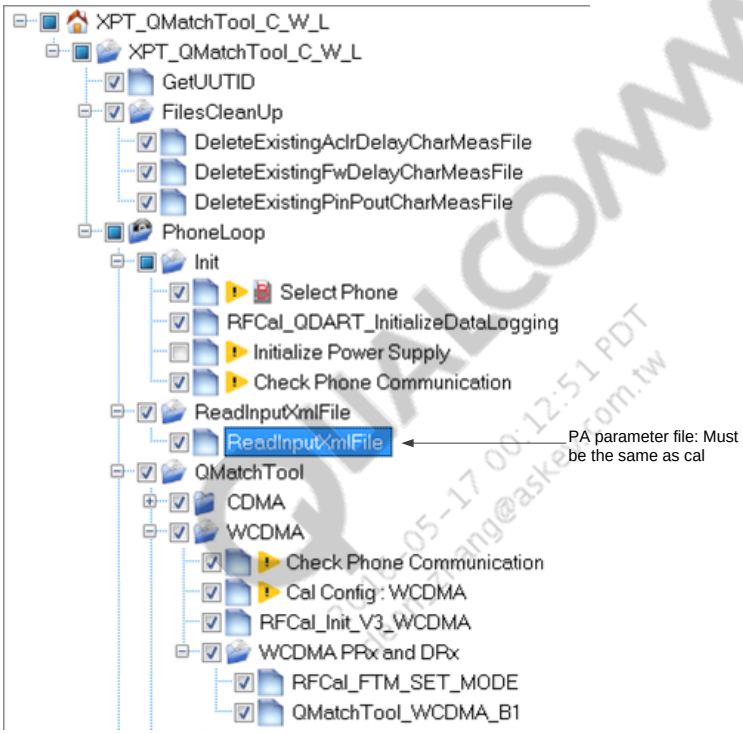

**Figure 6-1 Hardware validation flowchart**

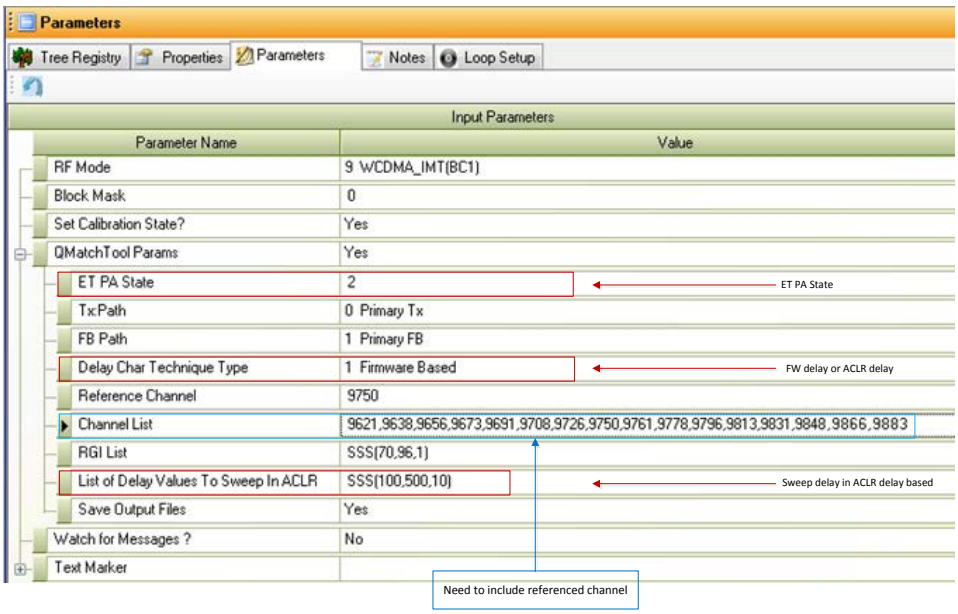
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2016-05-17 00:12:51 PDT
deon_zhang@askey.com.tw

# 6.2 QMatch tool

The QMatch tool design follows the hardware validation flowchart shown in Figure 6-1. The tool combines firmware/ACLR delay char and Pin char to facilitate hardware match process. Simplify the parameter interfaces as much as possible to ease the use. It only requires to be performed in room temperature since it is used to quickly check hardware validation.

In the QMatch tool, customers can choose firmware delay base or ACLR delay base char to get delay values across channels. Make sure the reference channel is part of the actual channel list. Customers can run the tool in high/mid/low channels for the first step of ET bringup.



### 6.2.1 Procedure

To perform matching procedures as outlined in Section 2.4:

1. Match point based on load pull and efficiency.
2. Perform duplexer and PA output matching.

To run delay characterization:

1. Perform ACLR delay characterization and firmware delay characterization on low/mid/high channels.
2. Compare firmware delay characterization results vs ACLR-based delay characterization results on one phone at room temperature. See Figure 6-2.

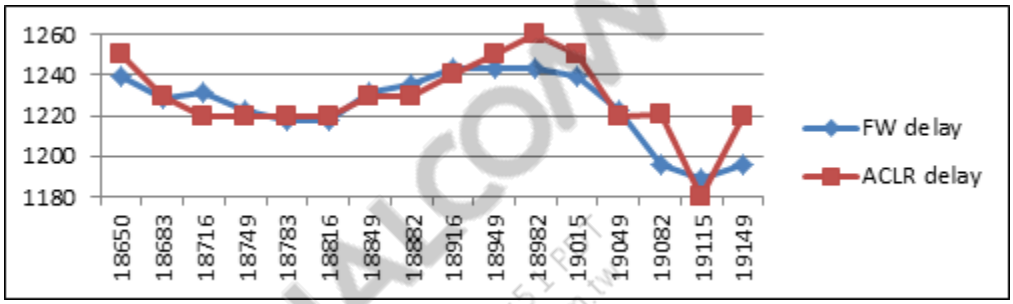



**Figure 6-2 Firmware delay vs ACLR-based delay – One phone, room temperature**

3. Usually firmware delay characterization and ACLR characterization can get similar delay values. If the delay values from two tools are different, the delay from ACLR delay characterization is not the real delay (which comes only from BB and QPOET registers setting).
4. Perform firmware delay characterization on multiple channels.

   If delay varies significantly across a band, this indicates the poorly matched hardware. See Figure 6-3.

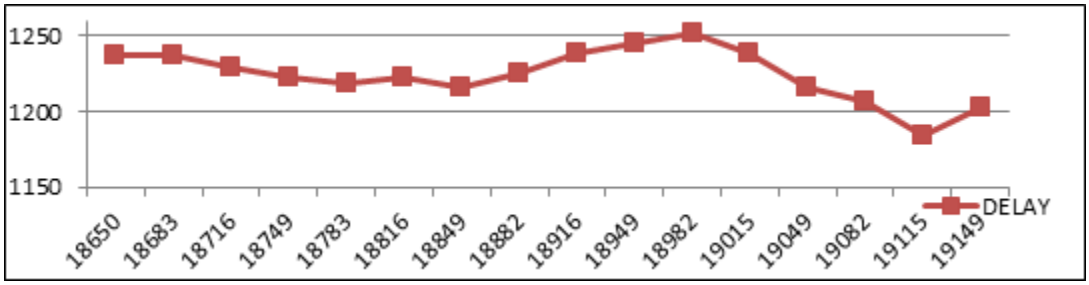



**Figure 6-3 Delay variation across a band**

To run Pin alignment characterization:

1. Perform Pin alignment characterization with full RB on 16 channels at room temperature.
2. Adjust bias value to get the power tested in Online mode.
3. Check ACLR on 16 channels and [-0.5, 0.5] dB Pin at the center channel. See Figure 6-4.

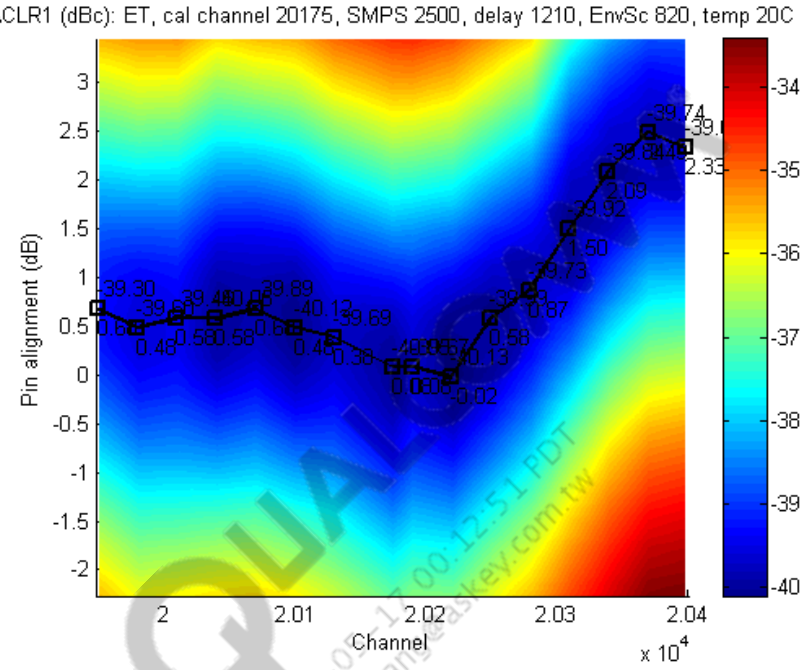



**Figure 6-4 Pin alignment characterization**

## 6.2.2 Performance check

To check performance:

1. Check all existing fail messages of the existing Pin/Delay char.
2. If the optimal pin at the reference channel is outside of [-0.5, 0.5] dB, the error is pin misalignment at the reference channel. After checking DPD healthiness, improve match starting from the reference channel.
3. If reference channel ACLR is worse than (LTE UTRA -40 dBc, WCDMA -40 dBc, cdma2000 -50 dBc), the error is ACLR at the reference channel is not leaving enough margin, improve match starting from the reference channel.
4. If some other channels have ACLR degradation more than 3dB compared with the reference channel, the error is ACLR across the band is not constantly good, improve match based on the ACLR results.

## 6.2.3 QMatch tool test results

QMatch tool test results are shown here.

### xPT firmware delay characterization and xPT static Pin alignment – RGI Delta sweep

**XPT Firmware Delay Characterization**

| RF Mode | Tx Path | PA State | PA Bias | RGI | PA Current | LTE Bandwidth | Channel | Temperature | Therm ADC | FW Based ET Path Delay | Links | Time (s) | Exceptions |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 34 | 0 | 0 | 3400 | 92 | 156 | 10 | 18299 | 22 | 1122 | 1186 | | | |
| 34 | 0 | 0 | 3400 | 92 | 156 | 10 | 18300 | 22 | 1122 | 1183 | | | |
| 34 | 0 | 0 | 3400 | 92 | 156 | 10 | 18301 | 22 | 1122 | 1183 | | | |

**XPT Static Pin Alignment: RGI Delta Sweep**

| RF Mode | Tx Path | Channel | PA State | PA Bias | RGI | IQ Gain | Env Scale | PA Current | Temperature | Therm ADC | TxPower_RgiDeltaSwp | AclrNeg1_RgiDeltaSwp | AclrPos1_RgiDeltaSwp | AclrNegEUTRA_RgiDeltaSwp | AclrPosEUTRA_RgiDeltaSwp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 34 | 0 | 18300 | 0 | 3300 | 70 | 150 | 0 | 255 | 22 | 1112 | -3.92 | -65.07 | -62.94 | -60.93 | -58.82 |
| 34 | 0 | 18300 | 0 | 3300 | 71 | 150 | 0 | 255 | 22 | 1112 | -3.12 | -65.73 | -62.87 | -61.33 | -58.96 |
| 34 | 0 | 18300 | 0 | 3300 | 72 | 150 | 0 | 255 | 22 | 1112 | -2.25 | -64.8 | -62.6 | -60.55 | -58.62 |
| 34 | 0 | 18300 | 0 | 3300 | 73 | 150 | 0 | 255 | 22 | 1112 | -1.09 | -64.24 | -62.47 | -60.2 | -58.54 |
| 34 | 0 | 18300 | 0 | 3300 | 74 | 150 | 0 | 255 | 22 | 1112 | 0.01 | -64.19 | -62.41 | -60.12 | -58.37 |
| 34 | 0 | 18300 | 0 | 3300 | 75 | 150 | 0 | 255 | 22 | 1112 | 0.97 | -64.36 | -62.5 | -60.21 | -58.51 |
| 34 | 0 | 18300 | 0 | 3300 | 76 | 150 | 0 | 255 | 22 | 1112 | 2 | -64.36 | -62.37 | -60.19 | -58.38 |
| 34 | 0 | 18300 | 0 | 3300 | 77 | 150 | 0 | 255 | 22 | 1112 | 2.92 | -64.27 | -62.24 | -60.05 | -58.29 |
| 34 | 0 | 18300 | 0 | 3300 | 78 | 150 | 0 | 255 | 22 | 1112 | 3.81 | -63.97 | -62.16 | -59.93 | -58.21 |

### xPT static Pin alignment – RGI determination sweep

**XPT Static Pin Alignment: RGI Determination Sweep**

| RF Mode | Tx Path | Channel | PA State | PA Bias | RGI | IQ Gain | Env Scale | PA Current | Temperature | Therm ADC | TxPower_RgiDetSwp | AclrNeg1_RgiDetSwp | AclrPos1_RgiDetSwp | AclrNegEUtra_RgiDetSwp | AclrPosEUtra_RgiDetSwp | Compression_RgiDetSwp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 34 | 0 | 18300 | 0 | 2500 | 70 | 622 | 820 | 156 | 22 | 1112 | 9.11 | -43.1 | -42.72 | -43.1 | -42.72 | -999 |
| 34 | 0 | 18300 | 0 | 2500 | 71 | 622 | 820 | 156 | 22 | 1112 | 10.13 | -42.68 | -42.1 | -42.68 | -42.1 | 0.0308559 |
| 34 | 0 | 18300 | 0 | 2500 | 72 | 622 | 820 | 156 | 22 | 1112 | 11.07 | -42.24 | -41.46 | -42.24 | -41.46 | -0.00997501 |
| 34 | 0 | 18300 | 0 | 2500 | 73 | 622 | 820 | 156 | 22 | 1112 | 12.25 | -41.83 | -40.82 | -41.83 | -40.82 | 0.0422884 |
| 34 | 0 | 18300 | 0 | 2500 | 74 | 622 | 820 | 156 | 22 | 1112 | 13.36 | -41.77 | -40.52 | -41.77 | -40.52 | -0.0101375 |
| 34 | 0 | 18300 | 0 | 2500 | 75 | 622 | 820 | 156 | 22 | 1112 | 14.42 | -41.06 | -39.55 | -41.06 | -39.55 | 0.0133925 |
| 34 | 0 | 18300 | 0 | 2500 | 76 | 622 | 820 | 156 | 22 | 1112 | 15.49 | -41.12 | -39.45 | -41.12 | -39.45 | 0.0146937 |

### XPT static Pin alignment – Freq comp sweep

**XPT Static Pin Alignment: Freq Comp Sweep**

| RF Mode | Tx Path | PA State | Channel | PA Bias | RGI | IQ Gain | Env Scale | PA Current | Temperature | Therm ADC | TxPower_FreqCompSwp | AclrNeg1_FreqCompSwp | AclrPos1_FreqCompSwp | AclrNegEUTRA_FreqCompSwp | AclrPosEUTRA_FreqCompSwp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 34 | 0 | 0 | 18299 | 2240 | 87 | 322 | 820 | 156 | 22 | 1112 | 19.2 | -37.9 | -34.97 | -37.9 | -34.97 |
| 34 | 0 | 0 | 18299 | 2240 | 87 | 332 | 820 | 156 | 22 | 1112 | 19.4 | -38.7 | -35.69 | -38.7 | -35.69 |
| 34 | 0 | 0 | 18299 | 2240 | 87 | 342 | 820 | 156 | 22 | 1112 | 19.57 | -39.55 | -36.43 | -39.55 | -36.43 |
| 34 | 0 | 0 | 18299 | 2240 | 87 | 352 | 820 | 156 | 22 | 1112 | 19.74 | -40.45 | -37.24 | -40.45 | -37.24 |
| 34 | 0 | 0 | 18299 | 2240 | 87 | 362 | 820 | 156 | 22 | 1112 | 19.89 | -41.34 | -38.11 | -41.34 | -38.11 |
| 34 | 0 | 0 | 18299 | 2240 | 87 | 372 | 820 | 156 | 22 | 1112 | 20.04 | -42.19 | -39.02 | -42.19 | -39.02 |
| 34 | 0 | 0 | 18299 | 2240 | 87 | 382 | 820 | 156 | 22 | 1112 | 20.18 | -42.9 | -39.99 | -42.9 | -39.99 |

### xPT static Pin Pout alignment – Computed Pin/Pout over freq

**XPT Static Pin/Pout Alignment: Computed Pin/Pout Over Freq**

| RF Mode | Tx Path | PA State | Channel | RGI | IQ Gain | Temperature | Therm ADC | Pin_SPinPoutAlign | Aclr_SPinPoutAlign | Pout_SPinPoutAlign | Links | Time (s) | Exceptions |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 34 | 0 | 0 | 18299 | 87 | 422 | 22 | 1112 | -0.409559 | -43.11 | 20.63 | | | |
| 34 | 0 | 0 | 18300 | 87 | 422 | 22 | 1112 | -0.409559 | -43.06 | 20.61 | | | |
| 34 | 0 | 0 | 18301 | 87 | 422 | 22 | 1112 | -0.409559 | -43.01 | 20.59 | | | |

# 6.3 Problem description and solution

## 6.3.1 ET delay characterization

Table 6-1 shows characterization for ET delay.

**Table 6-1 ET delay characterization**

| Problem | Possible cause | Action |
|---|---|---|
| Characterization abortion when loading DPD/change channel, etc. | General software/firmware stability of multiple tech/band/channel/ temperature when running the tool for a long time | File a case to problem area: RF→QFE_ET |
| Invalid delay value from ACLR delay characterization (or poor online performance when loaded in NV) | ▪ Improper PA state<br>▪ Improper compression point/power during mini-RGI bias sweep | 1. Set mini-RGI characterization to EPT PA state; set delay sweep to ET mode. Power must be 23 dBm or more during RGI sweep.<br>2. For some PAs, ACLR performance highly depends on compression point. Choose a characterization compression point close to what calibration chose. |
| Invalid delay value (extremely large, not repeatable) from firmware delay characterization | ▪ Improper PA state<br>▪ Check software/firmware version to see whether the build has required firmware/software | 1. Set mini-RGI characterization to EPT PA state; set delay sample capture to ET state<br>2. Firmware delay characterization supports WCDMA and LTE only. |
| Firmware delay is significantly different from ACLR delay (delta > 3 ns) | When the firmware delay provides the hardware delay, ACLR delay characterization selects a delay that provides symmetric ACLR. If PA ACLR performance is dominated by factors other than delay, the firmware delay will be different from the ACLR delay. This is generally true for QPA due to the PA memory effect and very asymmetrical left/right ACLR. | 1. Adjust the ACLR characterization comparison point, power to achieve the best achievable ACLR.<br>2. Check ACLR left and right to see whether they are symmetrical.<br>3. Firmware delay characterization usually matches ACLR characterization when ACLR left/right are symmetrical during delay sweep.<br>4. If firmware delay always differs from ACLR characterization by 3 ns or more, use ACLR delay characterization instead, since online performance is ACLR measurement. |

Figure 6-5 is an example of a significantly different ACLR delay vs a firmware delay.

- Left and right side ACLR are very asymmetrical.
- Best ACLR for either side is not achieved at the same delay, which is an indication of the memory effect.
- For this characterization result, ACLR characterization selected a delay of 1320 and firmware delay characterization calculated a delay of 1276.

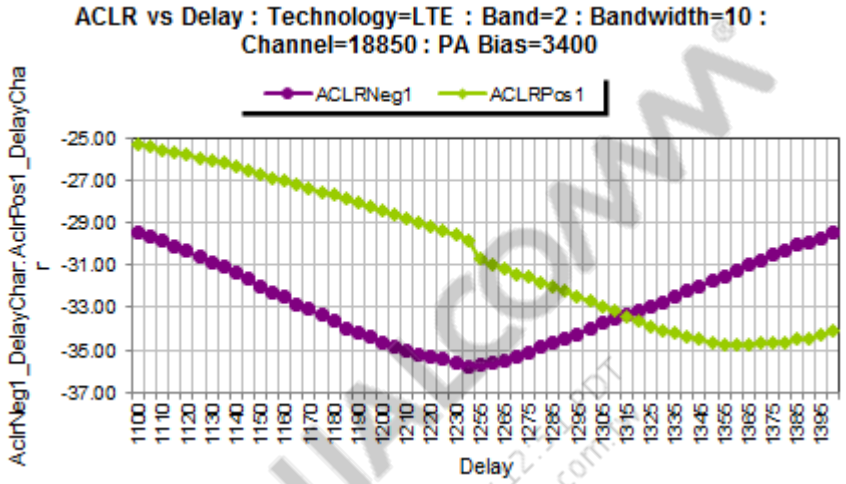



**Figure 6-5 ACLR delay vs firmware delay – Significant difference**

QUALCOMM
2016-05-17 20:12:
deon_zhang@asky.com

Table 6-2 and Figure 6-6 show a comparison for LTE delay shape.

**Table 6-2 LTE delay shape comparison**

| Problem | Possible cause | Action |
|---|---|---|
| LTE delay of different bandwidths does not hold same shape (see Figure 6-6) | The delay curve of LTE10/5/20 over multiple channels shall hold similar curvature, i.e., the offset between different LTE bandwidths shall be similar across channels. | 1. Check characterization tool settings (channel list, RB, etc.).<br>2. Check supported bandwidth(s) for the band.<br>3. The offset between LTE bandwidths shall hold similar values on multiple channels.<br>4. When matching is poor, delay varies greatly within the band; since each LTE bandwidth sees an average delay of different bandwidths (5/10/20MHz), delay offset at the same channel will vary slightly somehow.<br>5. Improve matching to maintain a relatively constant offset. |

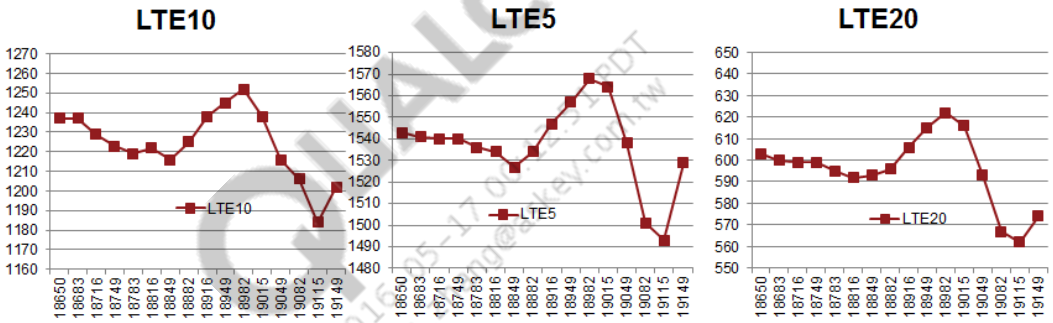



**Figure 6-6 LTE delay shape comparison**

Table 6-3 lists issues for delay varies significantly across a band.

**Table 6-3 Issue for Delay varies significantly across a band**

| Problem | Possible cause | Action |
|---|---|---|
| Delay varies significantly across a band | Hardware presents a difficult PA match for EPT/ET | Rematch hardware as shown in Figure 6-1.<br>Delay tolerance of each technology is roughly inversely proportional to signal bandwidth. Approximate tolerances in one channel of full RB are:<br>▪ WCDMA – ±6 ns<br>▪ LTE5 – ±5 ns<br>▪ LTE10 – ±3 ns<br>▪ LTE20 – ±1.5 ns<br>For the whole band – LTE10 delay tolerance is ±5 ns |

Figure 6-7 is an example of delay characterization output for well-matched hardware. Only the edge channel has noticeable delay variation due to duplexer edge response.

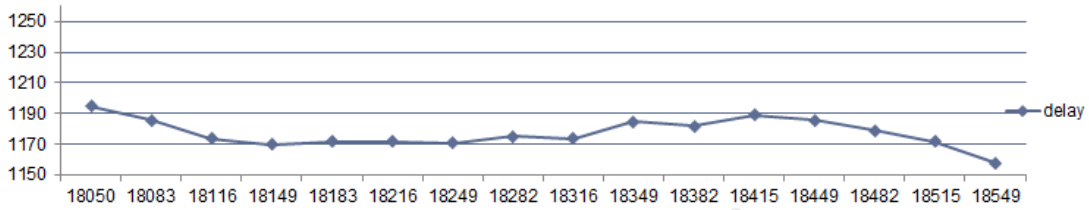



**Figure 6-7 Delay characterization output – Well-matched hardware**

Figure 6-8 is an example of delay characterization output for poorly matched hardware. Channel 19082 will have poor ACLR due to shape matching/delay change.

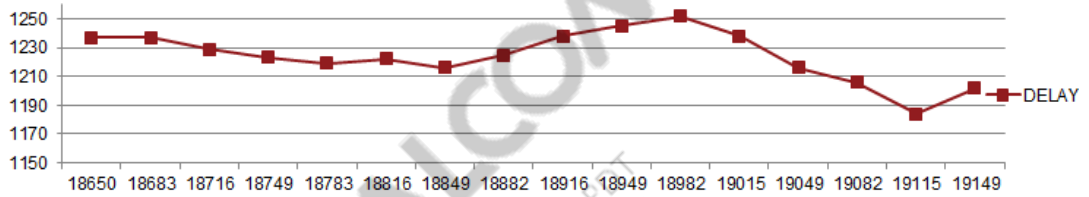



**Figure 6-8 Delay characterization output – Poorly matched hardware**

QUALCOMM®
2016-05-17 00:12:11 PDT
deon_zhang@askey.com.tw

Table 6-4 and Figure 6-9 show a delay variation across parts.

**Table 6-4 Delay variation across parts**

| Problem | Possible cause | Action |
|---|---|---|
| Delay varies significantly across parts (see Figure 6-9) | ▪ Hardware matching difference across parts<br>▪ Characterization tool software stability/ repeatability | 1. If one/a few parts stand out from the rest, check that hardware, and recharacterize the abnormal part(s) for repeatability or improve matching of the problematic part.<br>2. If most part delays vary significantly, then check matching of that hardware/band.<br>3. If all matching efforts still cannot get consistent delays, use delay calibration to take care of part-to-part variation. |

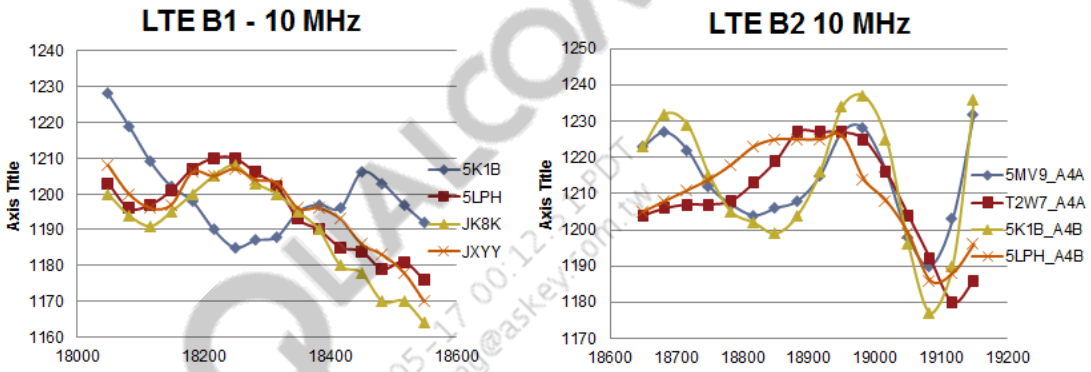



**Figure 6-9 Delay variation across parts**

Table 6-5 and Figure 6-10 show Sinusoid delay data across a band.

**Table 6-5 Sinusoid delay data across a band**

| Problem | Possible cause | Action |
|---|---|---|
| Delay data looks sinusoid across a band (pp > 5 ns; see Figure 6-10) | Matching between duplexer and antenna | Improve matching between duplexer and PA to reduce duplex reflection effect. |

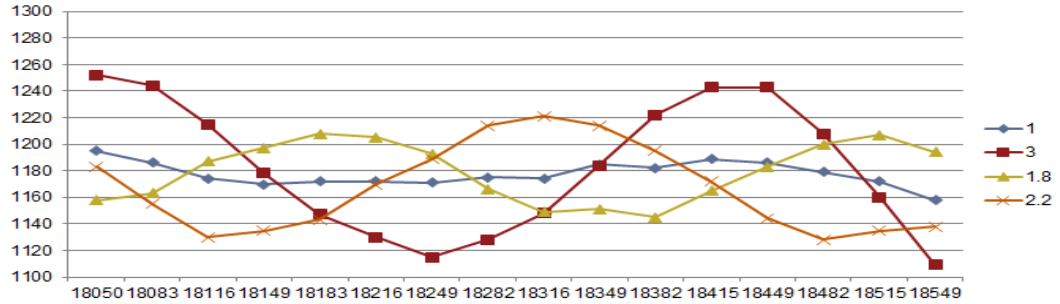



**Figure 6-10 Sinusoid delay data across a band**

## 6.3.2 ET delay characterization over temperature

- Delay change over temperature (see Figure 6-11)
  - Delay shifts mostly along frequency (left to right) when temperature changes due to duplexer response shift.
  - Delay shifts slightly up and down when temperature changes, due to WTR BB filter response.

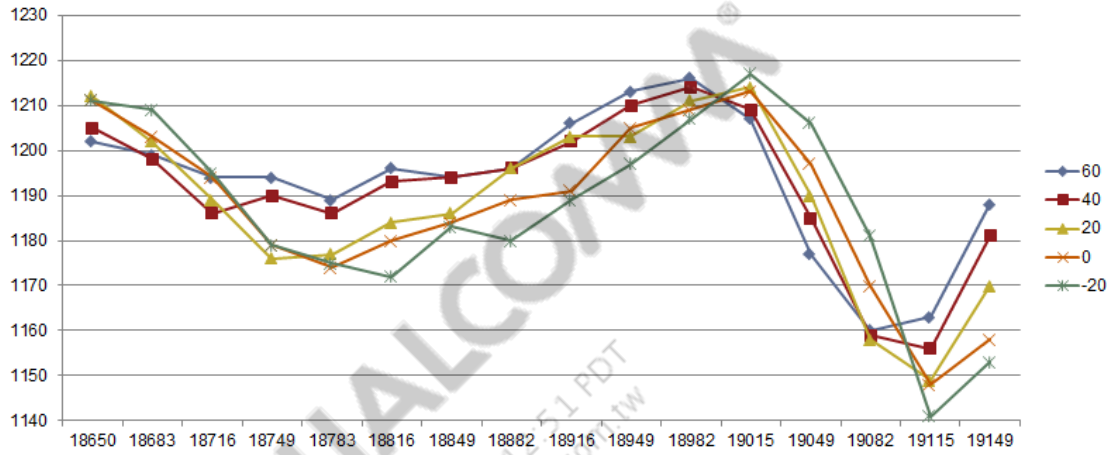



**Figure 6-11 Delay change over temperature**

- Delay offset NV
  - Due to the structure of the 2D delay offset NV, the NV holds only when multiple channel delay characterization data curvatures hold across multiple channels.
  - If multiple channel delay characterizations of multiple parts at room temperature have similar curves, delay offset NVs calculated from multiple parts are similar. In this case, delay offset NVs hold across parts, and delay offset NVs could help ET performance over temperature.
  - If multiple channel delay characterizations of multiple parts at room temperature have very different curves of dramatically different trends, delay offset NVs calculated from multiple parts could be very different. In this case, do not apply temperature delay offset NVs because applying the wrong offset does not benefit ET performance.

## 6.3.3 Pin alignment characterization

Table 6-6 and Figure 6-12 show unacceptable IQ gain.

**Table 6-6 Unacceptable IQ gain**

| Problem | Possible cause | Action |
|---|---|---|
| GENERAL_EXCEPTION – For RF mode 29 and PA state 3, RGI 85 is already tested and does not give IQ Gain within the acceptable range (see Figure 6-12) | Pin variation is greater than 6 dB across a band at the concerned temperature | ▪ Rematch hardware as shown in Figure 6-1.<br>▪ Workaround – Increase IQGain sweep range, e.g., from 622, 322, -5 to 622, 200, -5. |

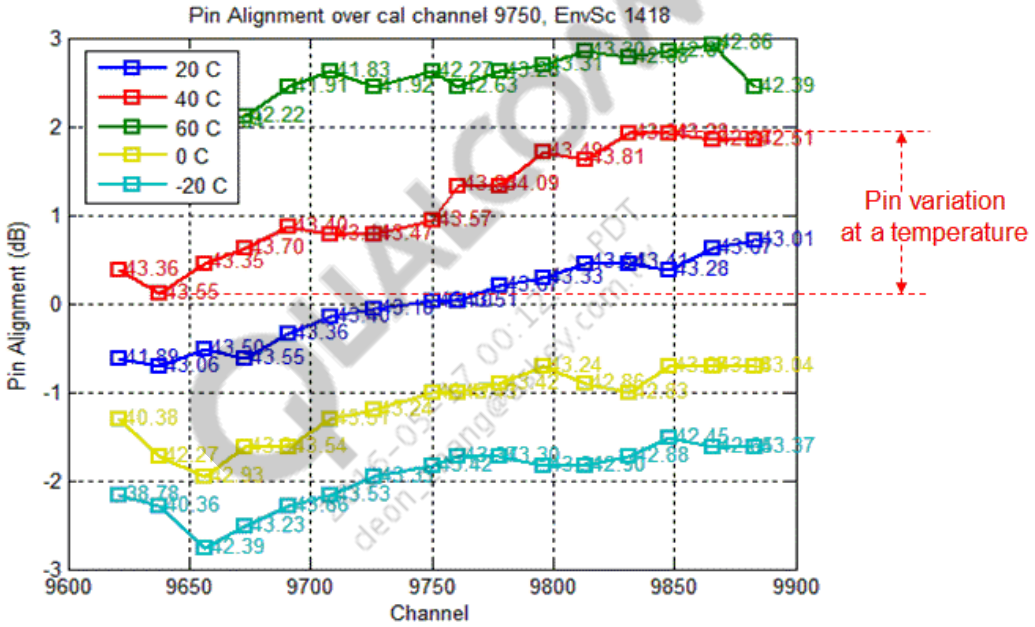



**Figure 6-12 Unacceptable IQ gain**

Table 6-7 and Figure 6-13 show bad ACLR.

**Table 6-7 Bad ACLR**

| Problem | Possible cause | Action |
|---|---|---|
| Bad ACLR | ▪ Hardware presents a difficult PA match for EPT/ET<br>▪ ET delay problem<br>▪ DPD problem | 1. For ET, ensure correct delay characterization.<br>2. Check IQ capture for DPD construction.<br>3. Rematch hardware as shown in Figure 6-1. |

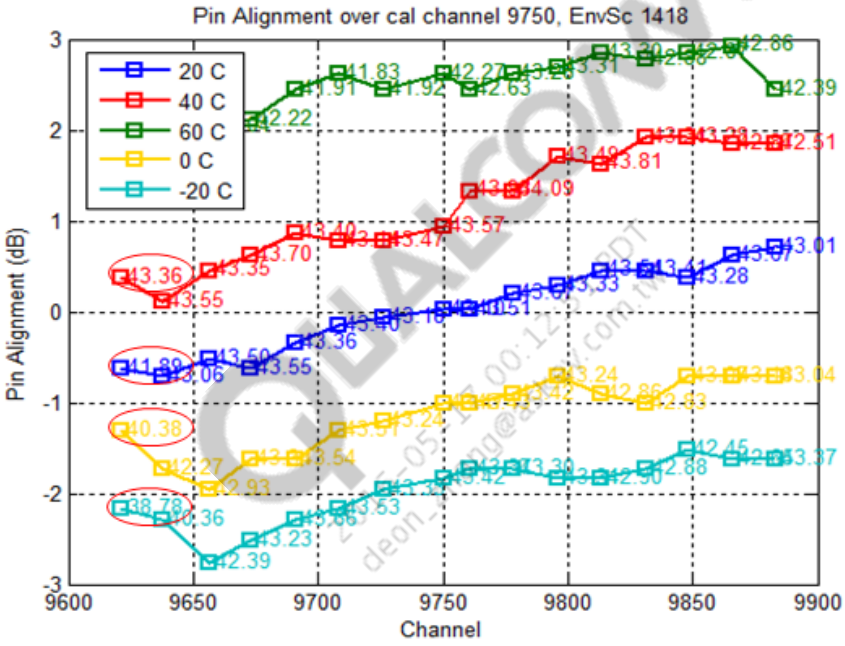



**Figure 6-13 Bad ACLR**

Table 6-8 and Figure 6-14 show optimal Pin outside [-0.5, 0.5] dB.

**Table 6-8 Optimal Pin outside [-0.5, 0.5] dB**

| Problem | Possible cause | Action |
|---|---|---|
| Optimal Pin for reference channel at room temperature is out of [-0.5,0.5] dB (see Figure 6-14) | ▪ Wrong characterization parameter(s)<br>▪ DPD problem<br>▪ ET delay problem<br>▪ Hardware presents difficulty even for reference channel | 1. Make sure IQGain parameter is correct.<br>2. For ET, ensure correct delay characterization.<br>3. Check IQ capture for DPD construction.<br>4. Re-match hardware as shown in Figure 6-1. |

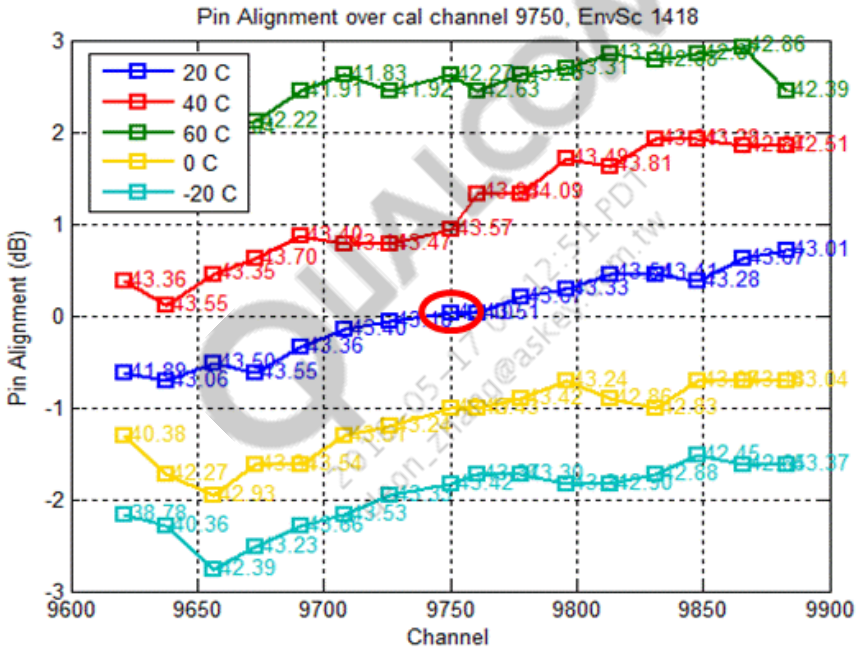



**Figure 6-14 Optimal Pin outside [-0.5, 0.5] dB**

Table 6-9 shows a common problem in Pin char.

**Table 6-9 Common problem in Pin char**

| Problem | Possible cause | Action |
|---|---|---|
| GENERAL_EXCEPTION – Could not find an optimal IQGain in 11 with RGI step size of 1 | ▪ Pin is > 10 dB or < -10 dB at certain channel or temperature<br>▪ DPD problem<br>▪ ET delay problem | 1. For ET, ensure correct delay characterization.<br>2. Check IQ capture for DPD construction.<br>3. May run out of RGI even for Online mode. Check WTR RGI spreadsheet. Consult modem/RF system. |
| Run out of RGI | Not enough RGI headroom for Pin alignment | 1. Ensure maximum available RGI is enlisted.<br>2. May run out of RGI even for Online mode. Check WTR RGI spreadsheet. Consult modem/RF system. |
| Fail at certain band/temp | | 1. There is no need to rerun all the tests.<br>2. Do not close and reopen QSPR.<br>3. If a 20°C (room temperature) run failed for a band, all temperatures for that band must be rerun.<br>4. If a different temperature failed for a band, only rerun the characterization for that particular temperature and band. NV generation will automatically pick up the data from successful runs. |

QUALCOMM® 2016-05-17 00:12:51 PDT deon_zhang@askey.com.tw

## 6.3.4 Run char tool for RFMD PA

In general, if the signal is strong and clean enough, the IQ capture data during the firmware delay characterization should lead to the correct delay value for Pin alignment characterization. However, there are some PAs whose delay values are very sensitive to signal level (Tx power). In this case, even a 1 dB delta in power will cause a different delay value in firmware delay characterization, which will lead to nonoptimal ACLR. Therefore, picking the correct RGI is important for ACLR performance.

The following approach will lead to a more accurate delay value:

1. Run Pin alignment characterization for each band. Note the bias/RGI value at the end of RGISweep.
2. Use the bias/RGI noted in Step 1 as the input to firmware delay characterization. Increase the maximum power limit for firmware delay characterization so that the 2 dB compression point is reached before the targeted power. Run firmware delay characterization.
3. Repeat Pin alignment characterization using the delay from Step 2.

**NOTE:** The DLL 17.55 or later can support fine tune bias value as Pin char RGI and Bias determination algorithm. Customers do not need to follow this procedure if DLL version is 17.55 or later.

## 6.3.5 Calibration

### 6.3.5.1 M-Line problem description

Table 6-10 lists the problem description for M-Line.

**Table 6-10 M-Line problem description**

| Problem | Possible cause | Action |
|---|---|---|
| Low Vdd-per-dBm in expected power (<1 V/6 dBm) | Wrong RGIs for alignment sweep | Check<br>1. RGI/bias characterization.<br>2. RGI/bias characterization sample captures. |
| Cannot find some target Pouts (cal fail) | ▪ RGI delta sweep list is not properly set<br>▪ Wrong RGIs for alignment sweep | Check<br>1. RGII delta sweep list.<br>2. RGI/bias characterization. |
| Cannot achieve maximum target Pout at Vd_max (cal fail) | ▪ Maximum target pout is set too high<br>▪ Hardware is not able to provide enough power | 1. Investigate maximum achievable power, considering PA specification, front-end loss, and temperature/frequency margin.<br>2. Consider maximum power vs ACLR tradeoff using EnvSc in NV. |
| Nonmonotonic M-Line predict power | Nonmonotonic RGI delta sweep | Future DLL changes will take care of this. |

Table 6-11 details a new opcode to provide M-Line error information.

**Table 6-11 CR568790 – xPT M-Line debug error code feature**

| Error code | Possible cause | Action |
|---|---|---|
| 1 | N dB compression point not found | The error means the 4 pars (RGI, bias) cannot cover ndB compression point. Rerun RGI bias characterization. |
| 2 | Cannot find the power in the M-Line table greater than requested power, i.e., expected power > maximum power in the M-Line table. (MEASURED) | The error means the highest power in the M-Line table cannot cover the measurement power. Try to increase RGI in RGI delta sweep if the PA is not saturated. |
| 3 | Cannot find a power in the M-Line table that is less than requested power, i.e., expected_power < min power in the M-Line table. (MEASURED) | The error means the lowest power in the M-Line table cannot cover the measurement power. Try to decrease RGI in RGI delta sweep. |
| 4 | M-Line selected bias > VdMax but interpolated bias < Vmax. (MEASURED) | M-Line selected bias is over VdMax but Interpolated bias is lower than Vmax. |
| 5 | M-Line selected bias > VdMax and interpolated bias > Vmax. (MEASURED) | M-Line selected bias is higher than VdMax and Interpolated bias is higher than Vmax. |

| Error code | Possible cause | Action |
|---|---|---|
| 6 | Cannot find the power in the M-Line table greater than requested power, i.e., expected power > maximum power in the M-Line table. (CALCULATED) | The error means the highest power in the M-Line table cannot cover the measurement power. Try to increase RGI in RGI delta sweep if the PA is not saturated. |
| 7 | Cannot find a power in the M-Line table that is less than requested power, i.e., expected_power < min power in the M-Line table. (CALCULATED) | The error means the lowest power in the M-Line table cannot cover the measurement power. Try to decrease RGI in RGI delta sweep. |
| 8 | M-Line selected bias > VdMax but interpolated bias < Vmax. (CALCULATED) | M-Line selected bias is over VdMax but interpolated bias is lower than Vmax. |
| 9 | M-Line selected bias > VdMax and interpolated bias > Vmax. (CALCULATED) | M-Line selected bias is higher than VdMax and Interpolated bias is higher than Vmax. |

#### 6.3.5.1.1 Example for troubleshooting

- Issue#1
  - Error message – GENERAL_EXCEPTION – DPD Valid FLag for Requested Power 1800 (lowest target power in sweep2) is FALSE. DPD Construction for segment number 132 failed. Check MLine results!
  - Solution
    - Check RGI delta sweep in QXDM F3log
    - Make sure the table has covered certain fail power
    - Adjust RGI in RGI delta sweep

- RGI sweep: check saturation power level

```
23:18:07.338  ftm_calibration_v3_data.c 00238 Num RGI Pwr Pairs = 23
23:18:07.338  ftm_calibration_v3_data.c 00260 === RGI Pwr Table ===
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 87, 108
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 88, 114
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 89, 118
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 90, 126
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 91, 138
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 92, 143
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 93, 146
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 94, 154
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 95, 161
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 96, 164
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 97, 170
23:18:07.338  ftm_calibration_v3_data.c 00274 RGI, Power = 98, 175
23:18:07.340  ftm_calibration_v3_data.c 00274 RGI, Power = 99, 178
```


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- Issue#2
  - Error Message – GENERAL_EXCEPTION – DPD Valid FLag for Requested Power 2500 (highest target power in sweep2) is FALSE. DPD Construction for segment number 132 failed. Check MLine results!
  - Solution
    - Check M-Line table in QXDM F3log
    - M-Line table need to contain a bias/RGI pair that meets the target output power but is < maximum bias
    - Reduce the maximum target output power if the maximum bias is exceeded to meet that power or increase the maximum RGI in the RGI delta list if not limited by bias
- Issue#3
  - Error Message – GENERAL_EXCEPTION – SWP2 fail or cannot find n-dB compression point
  - Solution
    - Check M-Line table in QXDM F3log
    - Check alignment sweep in first sweep
    - Make sure picked (RGI, bias) in sweep1 has covered n-dB compression point

Sweep Alignment: Check 4 pairs of (RGI, Bias). Need to cover n-dB compression points

```
14:15:54.678  ftm_common_xpt.c  00839  ===== M Points =====
14:15:54.678  ftm_common_xpt.c  00596  Index for ndB point not found. Set to Max
14:15:54.678  ftm_common_xpt.c  00857  M Point[0] = 26, 80, 15000
14:15:54.678  ftm_common_xpt.c  00596  Index for ndB point not found. Set to Max
14:15:54.678  ftm_common_xpt.c  00857  M Point[1] = 33, 110, 20000
14:15:54.678  ftm_common_xpt.c  00596  Index for ndB point not found. Set to Max
14:15:54.678  ftm_common_xpt.c  00857  M Point[2] = 41, 146, 25000
14:15:54.678  ftm_common_xpt.c  00857  M Point[3] = 13371, 39606, 30000
14:15:54.678  ftm_common_xpt.c  00870  RMS DBFS 1 [423953] = RMS DBFS 0 [66400] - 20log10(Env Scale [8200000]/622)
```

M-Line calculation data: make sure the lowest power and highest power cover sweep2 target power. There are two M-Line table. One is for EPT and the other is for ET.

```
14:15:54.678  ftm_calibration_v3_data.c  01223  ================= MLine EPT OUTPUTS =================
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 74, Bias = 2292, Power = 1009
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 75, Bias = 2332, Power = 1119
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 76, Bias = 2368, Power = 1209
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 77, Bias = 2409, Power = 1299
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 78, Bias = 2448, Power = 1379
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 79, Bias = 2503, Power = 1479
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 80, Bias = 2572, Power = 1589
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 81, Bias = 2635, Power = 1679
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 82, Bias = 2704, Power = 1769
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 83, Bias = 2791, Power = 1869
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 84, Bias = 2877, Power = 1959
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 85, Bias = 2996, Power = 2069
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 86, Bias = 3105, Power = 2159
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 87, Bias = 3255, Power = 2269
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 88, Bias = 3425, Power = 2379
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 89, Bias = 3599, Power = 2479
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 90, Bias = 3795, Power = 2579
14:15:54.678  ftm_calibration_v3_data.c  01229  MLine Outputs: RGI = 91, Bias = 3991, Power = 2669
```


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IQ capture in sweep2: Sometimes it failed in sweep2 because the max PA bias is limited.

```
14:15:54.681  ftm_calibration_v3_data.c 01353  PA  RGI  Vcc  Icq  IQ Gain  Env Scale  Delay  ReqPower(dB100) MLinePower(dB100)
14:15:54.681  ftm_calibration_v3_data.c 01454  1  83  2791  255  622  820  0  1800  1869
14:15:54.681  ftm_calibration_v3_data.c 01454  1  85  2996  255  622  820  0  2000  2069
14:15:54.681  ftm_calibration_v3_data.c 01454  1  87  3255  255  622  820  0  2200  2269
14:15:54.681  ftm_calibration_v3_data.c 01418  Interpolated Bias for Target power 2300 dbmx100 is 3302. Capping to 3300.
14:15:54.681  ftm_calibration_v3_data.c 01454  1  88  3300  255  622  820  0  2300  2379
14:15:54.681  ftm_calibration_v3_data.c 01464  ====== Selection for EPT Swp2 Calc Powers ======
14:15:54.681  ftm_calibration_v3_data.c 01466  PA  RGI  Vcc  Icq  IQ Gain  Env Scale  Delay  ReqPower(dB100) MLinePower(dB100)
14:15:54.681  ftm_calibration_v3_data.c 01542  1  82  2704  255  622  820  0  1700  1769
14:15:54.681  ftm_calibration_v3_data.c 01542  1  84  2877  255  622  820  0  1900  1959
14:15:54.681  ftm_calibration_v3_data.c 01542  1  86  3105  255  622  820  0  2100  2159
14:15:54.681  ftm_calibration_v3_data.c 01542  1  87  3255  255  622  820  0  2250  2269
14:15:54.681  ftm_calibration_v3_data.c 01566  ET MLine index beyond which MLine Vcc > VdMax(3800) is 255
14:15:54.681  ftm_calibration_v3_data.c 01570  ====== Selection for Swp2 ET Meas Powers ======
14:15:54.681  ftm_calibration_v3_data.c 01572  PA  RGI  Vcc  Icq  IQ Gain  Env Scale  Delay  ReqPower(dB100) MLinePower(dB100)
14:15:54.681  ftm_calibration_v3_data.c 01583  Cannot find ET power greater than Requested Meas Power of 1800. Max Power from MLine Outputs for VdMax(3800) is 0
14:15:54.681  ftm_calibration_v3_data.c 01583  Cannot find ET power greater than Requested Meas Power of 2000. Max Power from MLine Outputs for VdMax(3800) is 0
14:15:54.681  ftm_calibration_v3_data.c 01583  Cannot find ET power greater than Requested Meas Power of 2200. Max Power from MLine Outputs for VdMax(3800) is 0
14:15:54.681  ftm_calibration_v3_data.c 01583  Cannot find ET power greater than Requested Meas Power of 2300. Max Power from MLine Outputs for VdMax(3800) is 0
14:15:54.681  ftm_calibration_v3_data.c 01714  Cal v3 ftm_calibration_data_process_mline_calc_instr : Failed
```

- Issue#4
  - Based on the calibration algorithm, the RGI and bias chosen for Sweep 2 are from the M-Line table. However, if Sweep 2 IQ capture starts before M-Line calculation is complete, finding the target power for Sweep 2 may be difficult. Increasing XPT_Swp2_MLine_Calc_Segment_Length can solve this issue.

In the problem log shown, the red color means Sweep 2 is running while M-Line is still calculating.

```
00:01:29.203   ftm_common_xpt.c 00955  ===== M Line Result Table =====
00:01:29.203  ftm_calibration_v3_data.c 03748  QPOET Min Volt = 60, QPOET Max Volt = 380, ET Min Volt = 100
00:01:29.203   ftm_common_xpt.c 00956  RGI  Vin(V)  Vcc(V)  SMPS  Vout(V)  Pout(dBm)
00:01:29.203   ftm_common_xpt.c 00986  79  74  70  703  79  888
00:01:29.203  ftm_calibration_v3_data.c 03812  ====== LOAD CALC DPD ACTION LOAD UNITY ======
00:01:29.203   ftm_common_xpt.c 00986  80  83  86  860  111  1183
00:01:29.203   ftm_common_xpt.c 00986  81  86  92  928  125  1284
00:01:29.203   ftm_common_xpt.c 00986  82  98  113  1132  166  1533
00:01:29.203   ftm_common_xpt.c 00986  83  112  136  1364  214  1750
00:01:29.203   ftm_common_xpt.c 00986  84  121  152  1527  247  1875
00:01:29.203   ftm_common_xpt.c 00986  85  131  170  1705  283  1994
00:01:29.203  rfcommon_semaphore.c 00498  rfcommon_semaphore_pop_item: POP [9] Sem @ 0xba74804 {Items left 9}
00:01:29.203   ftm_common_xpt.c 00986  86  141  186  1868  316  2090
00:01:29.203   rfcommon_mdsp.c 02887  rfcommon_mdsp_xpt_txagc_override: sending chain 0
00:01:29.203   ftm_common_xpt.c 00986  87  156  213  2136  371  2228
00:01:29.203   rfcommon_mdsp.c 02992  rfcommon_mdsp_xpt_txagc_override: Delay value to FW in Q21: 0x7FFFFFFF
00:01:29.203   rfcommon_mdsp.c 03020  [CMN][RF->FW] TXAGC_ORIDE 0x2fd [ 0 | 765 ]
00:01:29.203   ftm_common_xpt.c 00986  88  163  226  2264  397  2287
00:01:29.203   ftm_common_xpt.c 00986  89  173  243  2433  432  2359
00:01:29.203   rfcommon_mdsp.c 03464  [CMN][FW->RF] TXAGC_ORIDE 0x2fd [ 0 | 765 ]
00:01:29.203   ftm_common_xpt.c 00986  90  185  264  2649  476  2444
00:01:29.203  rfcommon_semaphore.c 00598  rfcommon_semaphore_push_item: PUSH [9] Sem @ 0xba74804 {Items left 10}
00:01:29.203   rfcommon_mdsp.c 03478  rfcommon_mdsp_txagc_override_fwrsp_handler: CID.0x6800304 SNUM.0x2fd - Handled
00:01:29.203   ftm_common_xpt.c 00986  91  199  288  2880  523  2526
```

```
00:01:29.203   rfcommon_mdsp.c 03464 [CMN][FW->RF] TXAGC_ORIDE 0x2fd [ 0 | 765 ]
00:01:29.203  ftm_common_xpt.c 00986 90 185  264 2649 476  2444
00:01:29.203 rfcommon_semaphore.c 00598 rfcommon_semaphore_push_item: PUSH [9] Sem @ 0xba74804 {Items left 10}
00:01:29.203   rfcommon_mdsp.c 03478 rfcommon_mdsp_txagc_override_fwrsp_handler: CID.0x6800304 SNUM.0x2fd - Hand
00:01:29.203  ftm_common_xpt.c 00986 91 199  288 2880 523  2526
00:01:29.203   ftm_wcdma_ctl.c 06611 ftm_wcdma_xpt_override_txagc: XPT Mode 0, IQ Gain Action 0, IQ Gain -1, Env Scale Ac
00:01:29.203  ftm_common_xpt.c 00986 92 213  312 3128 573  2606
00:01:29.203   ftm_wcdma_ctl.c 06618 ftm_wcdma_xpt_override_txagc: RGI 65535, SMPS 2000, PA State 255, PA Curr 65535
00:01:29.203   ftm_wcdma_ctl.c 06622 set commom xpt txagc override, status = 1
00:01:29.203  ftm_common_xpt.c 00986 93 228  339 3394 627  2684
00:01:29.203 ftm_calibration_v3_data.c 03858 Return Value from Tx Override = 1
00:01:29.203  ftm_common_xpt.c 00986 94 248  372 3728 695  2773
00:01:29.203 ftm_calibration_v3_data.c 03865 Num Swp2 Meas Powers for Load Calc DPD = 4
00:01:29.203  ftm_common_xpt.c 00986 95 256  387 3879 726  2811
00:01:29.203 ftm_calibration_v3_data.c 03869 Update HW ID for Meas Power = 1900
00:01:29.203 ftm_calibration_data_module.c 01482 ftm_cal_data_set_hwid_by_power: Could not find power entry 1900
```

- Issue#5
  - IQ capture failure in sweep1 due to not enough IQ capture processing time, the RMS is zero even the TX output power is correct.
  - Solution – Increase XPT_swp1_cap_segment_Length

    The following shows an increase XPT_swp1_cap_segment_Length.

```
ftm_calibration_v3_data.c  01355  MLine Params: EPT Env Scale = 920
  ftm_common_xpt.c  00702  ===== M Line Compute =====
  ftm_common_xpt.c  00761  Num XPT Cal Data = 4
  ftm_common_xpt.c  00762  Num RGI Delta Data = 21
  ftm_common_xpt.c  00764  ===== M Line SP1 DATA =====
  ftm_common_xpt.c  00765  RGI   Vin   Pout  Vout  SMPS  Vcc  RMS(AMAM)
  ftm_common_xpt.c  00800  88   144   1910   201   1500   150   0
  ftm_common_xpt.c  00800  90   177   2240   294   2000   200   15453
  ftm_common_xpt.c  00800  91   197   2470   384   2500   250   15481
  ftm_common_xpt.c  00800  93   248   2650   472   3000   300   15439
  ftm_common_xpt.c  00868  ===== M Points =====
  ftm_common_xpt.c  00594  Index for max gain point not found. Using index zero
  ftm_common_xpt.c  00618  Index for ndB point not found. Setting it to Max = 255
  ftm_common_xpt.c  00892  M Point[0] = 14493, -1, 15000
  ftm_common_xpt.c  00613  Index for ndB point : 111
  ftm_common_xpt.c  00892  M Point[1] = 7801, 28503, 20000
  ftm_common_xpt.c  00613  Index for ndB point : 121
  ftm_common_xpt.c  00892  M Point[2] = 9425, 37934, 25000
  ftm_common_xpt.c  00613  Index for ndB point : 112
  ftm_common_xpt.c  00892  M Point[3] = 10990, 46217, 30000
  ftm_common_xpt.c  00925  RMS DBFS 1 [324005] = RMS DBFS 0 [66400] - 20log10(Env Scale
  ftm_common_xpt.c  00944  ===== M Line Curve Fitting =====
  ftm_common_xpt.c  00950  ===== Linear Fit =====
  ftm_common_xpt.c  00951  ===== Vout(Vin) =====
  ftm_common_xpt.c  00956  a = -1, b = -1
  ftm_common_xpt.c  00957  ===== Vcc(Vin) =====
  ftm_common_xpt.c  00962  d = 0, e = 32180
  ftm_common_xpt.c  00964  ===== M Line Result Table =====
  ftm_common_xpt.c  00965  RGI  Vin(V)  Vcc(V)  SMPS  Vout(V)  Pout(dBm)
  ftm_common_xpt.c  00998   78   50      275    2758   -1      -1
  ftm_common_xpt.c  00998   79   56      270    2708   -1      -1
  ftm_common_xpt.c  00998   80   63      264    2640   -1      -1
  ftm_common_xpt.c  00998   81   69      258    2584   -1      -1
  ftm_common_xpt.c  00998   82   77      251    2515   -1      -1
  ftm_common_xpt.c  00998   83   85      243    2438   -1      -1
  ftm_common_xpt.c  00998   84   94      236    2363   -1      -1
  ftm_common_xpt.c  00998   85   105      225    2258   -1      -1
  ftm_common_xpt.c  00998   86   116      216    2166   -1      -1
  ftm_common_xpt.c  00998   87   130      203    2038   -1      -1
```

- Issue#6
  - Measurement indicator error in CMW500 as shown – When we perform APT characterization on CMW500, sometimes there might be some measurement indicator errors

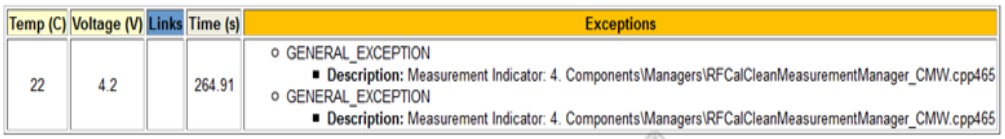



**Table 6-12 Measurement indicator error in CMW500**

| Measurement indicator value | Description | Action |
|---|---|---|
| 1 (Measurement Timeout) | Measurement results may be available, however, at least a part of the measurement provides only invalid results or has not completed the full statistic count. The measurement has been stopped after the configurable measurement timeout. | Check the RF connection |
| 3 (Overdriven) | Measurement results available, however, the accuracy of some results may be impaired because the input signal level was too high. | Check the RF connection<br>Check the output power level by QRCT or TPL log<br>Increase the User margin<br>Increase the Expected Power Level For The First Bias and Rgi Pair |
| 4 (Underdriven) | Measurement results available, however, the accuracy of some results may be impaired because the input signal level was too low. | Check the RF connection<br>Check the output power level by QRCT or TPL log<br>Decrease the User margin<br>Decrease the Expected Power Level For The First Bias and Rgi Pair |
| 6 (Trigger Timeout) | The measurement could not be started or continued because no trigger event was detected. | Check the RF connection |
| 7 (Acquisition Error) | The R&S CMW500 could not properly decode the RF input signal. | Check with latest DLL<br>If it is still not solved, file cases with detailed logs. |
| 8 (Sync Error) | The R&S CMW500 could not synchronize to the RF input signal. | Check with latest DLL<br>If it is still not solved, file cases with detailed logs. |
| 15 (Reference Frequency Error) | The instrument has been configured to use an external reference signal but the reference oscillator could not be phase locked to the external signal (e.g., signal level too low, frequency out of range or reference signal not available at all). | Check the RF connection<br>Check the output power level by QRCT or TPL log |

#### 6.3.5.2 Delay calibration

- Delay calibration development
  - Delay calibration sweep was executed as the last sweep of CAL V4, due to late development.
  - Delay characterization is still needed even if delay calibration is in place, since calibration Sweep 2 and Sweep 3 as well as Pin characterization need the delay value.
- Verification
  - Apply the same channel list and upper bound list as in delay characterization.
  - Compare delay calibration vs firmware delay characterization results. They should get very similar delays, because the same delay capture/processing is used in both delay calibration and firmware delay characterization.

### 6.3.6 Online problem analysis

Table 6-13 lists an analysis of an online problem.

**Table 6-13 Online problem analysis**

| Problem | Action |
|---|---|
| ACLR is bad across frequencies at room temperature | 1. Compare with Pin characterization ACLR results.<br>2. Make sure these results are close to 0 dB optimal Pin at reference channel and room temperature.<br>3. For ET, check delay.<br>4. Rematch hardware as shown in Figure 6-1. |
| ACLR bad across temperatures | 1. Possible wrong NV – Pin/Pout NV, missing characterization change list NV, etc.<br>2. Possible thermal ADC problem (provide QXDM Pro log with RF and FTM messaged for debugging).<br>3. Pin characterization on the faulty part to rule out part-to-part variation.<br>4. (Advanced) Run ProtoCal on the particular channel/temperature to obtain the best possible ACLR. |
| Maximum power fluctuation | 1. Calibration HDET issue; check calibration log HDET FreqComp.<br>2. Make sure calibration xPT FreqComp is normal; check calibration log xPT FreqComp. |
| Maximum power drops at edge channels and/or at hot temperatures | The maximum target Pout in calibration is not high enough. Set a higher maximum target Pout according to Pin characterization results (see Section 5.4.1) |
| Avago A4B-specific – APT calibration non_monotonic Pout at highest RGIs; desense and ACLR degradation, especially at edge channels and hot temperature | 1. Protection circuit, triggered especially with Vdd > 3.4 V, high power, and large VSWR.<br>2. Limit ET Vd maximum to 3.3 V, and lower maximum target Pout (and consequently achievable maximum power) accordingly. |

QUALCOMM® 2016-05-17 00:12:51 PDT deon_zhang@askey.com.tw

### 6.3.6.1 Performance debug procedure

General calibration and characterization problems have been covered in previous sections However, if customers have passed the calibration and characterization with good results but the Online mode performance is still not good, the next step is to check FTM mode performance. In this case, see Section 4.4.2 and follow the manual test procedure to debug performance issues.

#### 6.3.6.1.1 Checking procedure

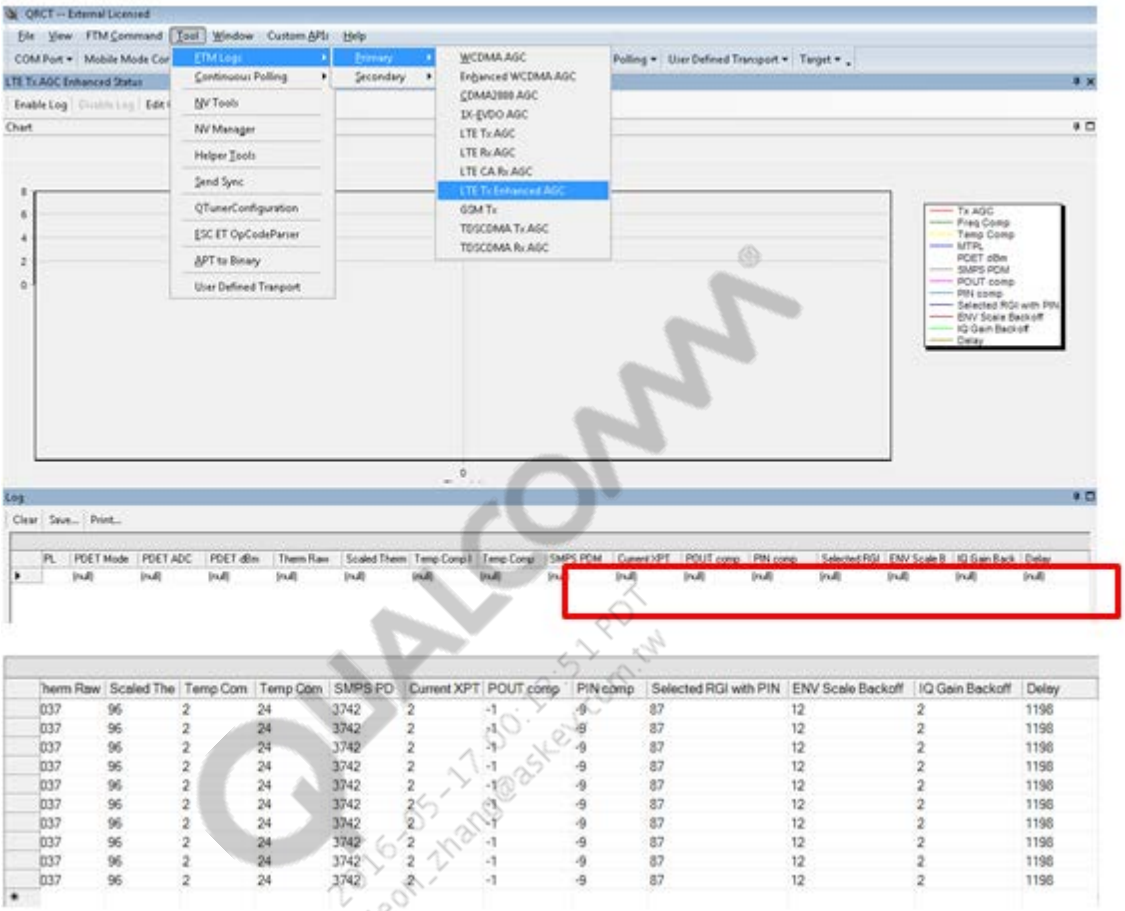
- Online mode performance issue
  - Check Pin, Pout, Delay, SMPS, IQ gain, EnvScale, and RGI by using QRCT FTM logs on the power which has ACLR performance issues.

**NOTE:** The RGI should use the selected RGI after applying Pin.

  - Check ACLR performance in FTM mode with the same setting by using the QRCT xPT control API; see Section 4.3.
  - If the ACLR result in FTM mode is the same as in Online mode:
    - Try to adjust each delay value, IQ gain, Env_scale, and RGI.
    - Adjust the delay value to get a V-shape to prove the delay value in Online mode is correct.
    - Adjust IQ gain to get an optimized value for ACLR. The IQ gain should be the same as the pin char result with the same test condition.
    - Adjust Env_scale if the power is too high with higher RGI.
  - Combining these parameters will get the optimized ACLR.
  - If the ACLR result in FTM mode is different from Online mode:
    - Use the method described in this section for FTM mode to check what parameter is off in Online mode.


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A checking of performance issues is shown here.




| Therm Raw | Scaled The | Temp Com | Temp Com | SMPS PD | Current XPT | POUT comp | PIN comp | Selected RGI with PIN | ENV Scale Backoff | IQ Gain Backoff | Delay |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |
| 037 | 96 | 2 | 24 | 3742 | 2 | -1 | -9 | 87 | 12 | 2 | 1198 |

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## 6.3.7 ET matching flow diagram summary

Figure 6-15 shows a flow diagram of ET matching.

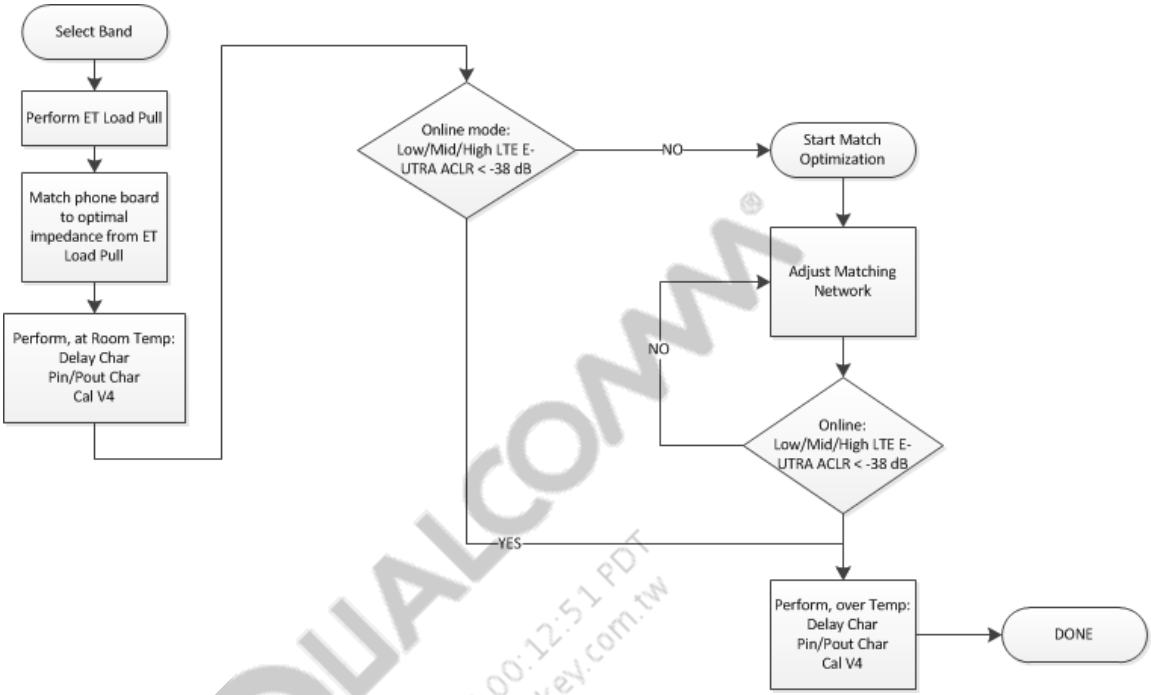



**Figure 6-15 ET matching flow diagram**

QUALCOMM®
2016-05-17 00:12:51 PDT
deon_zhang@skey.com.tw

# A MULTI_LIN_V3 NV Items

Table A-1 lists MULTI_LIN_V3 NV items generated after calibration.

**Table A-1 MULTI_LIN_V3 NV items**

| NV no. | Name |
|---|---|
| 24210 | RFNV_CDMA_C0_BC0_TX_MULTI_LIN_V3_DATA_I |
| 24211 | RFNV_CDMA_C0_BC1_TX_MULTI_LIN_V3_DATA_I |
| 24212 | RFNV_CDMA_C0_BC4_TX_MULTI_LIN_V3_DATA_I |
| 24213 | RFNV_CDMA_C0_BC5_TX_MULTI_LIN_V3_DATA_I |
| 24214 | RFNV_CDMA_C0_BC6_TX_MULTI_LIN_V3_DATA_I |
| 24215 | RFNV_CDMA_C0_BC10_TX_MULTI_LIN_V3_DATA_I |
| 24216 | RFNV_CDMA_C0_BC14_TX_MULTI_LIN_V3_DATA_I |
| 24217 | RFNV_CDMA_C0_BC15_TX_MULTI_LIN_V3_DATA_I |
| 24218 | RFNV_CDMA_C2_BC0_TX_MULTI_LIN_V3_DATA_I |
| 24219 | RFNV_CDMA_C2_BC1_TX_MULTI_LIN_V3_DATA_I |
| 24220 | RFNV_CDMA_C2_BC4_TX_MULTI_LIN_V3_DATA_I |
| 24221 | RFNV_CDMA_C2_BC5_TX_MULTI_LIN_V3_DATA_I |
| 24222 | RFNV_CDMA_C2_BC6_TX_MULTI_LIN_V3_DATA_I |
| 24223 | RFNV_CDMA_C2_BC10_TX_MULTI_LIN_V3_DATA_I |
| 24224 | RFNV_CDMA_C2_BC14_TX_MULTI_LIN_V3_DATA_I |
| 24225 | RFNV_CDMA_C2_BC15_TX_MULTI_LIN_V3_DATA_I |
| 24226 | RFNV_LTE_B1_TX_MULTI_LIN_V3_DATA_I |
| 24227 | RFNV_LTE_B2_TX_MULTI_LIN_V3_DATA_I |
| 24228 | RFNV_LTE_B3_TX_MULTI_LIN_V3_DATA_I |
| 24229 | RFNV_LTE_B4_TX_MULTI_LIN_V3_DATA_I |
| 24230 | RFNV_LTE_B5_TX_MULTI_LIN_V3_DATA_I |
| 24231 | RFNV_LTE_B6_TX_MULTI_LIN_V3_DATA_I |
| 24232 | RFNV_LTE_B7_TX_MULTI_LIN_V3_DATA_I |
| 24233 | RFNV_LTE_B8_TX_MULTI_LIN_V3_DATA_I |
| 24234 | RFNV_LTE_B9_TX_MULTI_LIN_V3_DATA_I |
| 24235 | RFNV_LTE_B10_TX_MULTI_LIN_V3_DATA_I |
| 24236 | RFNV_LTE_B11_TX_MULTI_LIN_V3_DATA_I |
| 24237 | RFNV_LTE_B12_TX_MULTI_LIN_V3_DATA_I |
| 24238 | RFNV_LTE_B13_TX_MULTI_LIN_V3_DATA_I |
| 24239 | RFNV_LTE_B14_TX_MULTI_LIN_V3_DATA_I |
| 24240 | RFNV_LTE_B17_TX_MULTI_LIN_V3_DATA_I |

| NV no. | Name |
|---|---|
| 24241 | RFNV_LTE_B18_TX_MULTI_LIN_V3_DATA_I |
| 24242 | RFNV_LTE_B19_TX_MULTI_LIN_V3_DATA_I |
| 24243 | RFNV_LTE_B20_TX_MULTI_LIN_V3_DATA_I |
| 24244 | RFNV_LTE_B21_TX_MULTI_LIN_V3_DATA_I |
| 24245 | RFNV_LTE_B23_TX_MULTI_LIN_V3_DATA_I |
| 24246 | RFNV_LTE_B25_TX_MULTI_LIN_V3_DATA_I |
| 24247 | RFNV_LTE_B34_TX_MULTI_LIN_V3_DATA_I |
| 24248 | RFNV_LTE_B38_TX_MULTI_LIN_V3_DATA_I |
| 24249 | RFNV_LTE_B39_TX_MULTI_LIN_V3_DATA_I |
| 24250 | RFNV_LTE_B40_TX_MULTI_LIN_V3_DATA_I |
| 24251 | RFNV_LTE_B41_TX_MULTI_LIN_V3_DATA_I |
| 24252 | RFNV_WCDMA_2100_TX_MULTI_LIN_V3_DATA_I |
| 24253 | RFNV_WCDMA_1900_TX_MULTI_LIN_V3_DATA_I |
| 24254 | RFNV_WCDMA_BC4_TX_MULTI_LIN_V3_DATA_I |
| 24255 | RFNV_WCDMA_800_TX_MULTI_LIN_V3_DATA_I |
| 24256 | RFNV_WCDMA_900_TX_MULTI_LIN_V3_DATA_I |
| 24257 | RFNV_WCDMA_1800_TX_MULTI_LIN_V3_DATA_I |
| 24258 | RFNV_WCDMA_1500_TX_MULTI_LIN_V3_DATA_I |
| 24545 | RFNV_LTE_B42_TX_MULTI_LIN_V3_DATA_I |
| 25481 | RFNV_LTE_B28_TX_MULTI_LIN_V3_DATA_I |
| 25721 | RFNV_LTE_B26_TX_MULTI_LIN_V3_DATA_I |
| 25722 | RFNV_TDSCDMA_B34_TX_MULTI_LIN_V3_DATA_I |
| 25723 | RFNV_TDSCDMA_B39_TX_MULTI_LIN_V3_DATA_I |
| 25724 | RFNV_TDSCDMA_B40_TX_MULTI_LIN_V3_DATA_I |

# B DPD Config NV Items

Table B-1 lists DPD config NV items.

**Table B-1 DPD config NV items**

| NV no. | Name |
|---|---|
| 24142 | RFNV_CDMA_C0_BC0_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24143 | RFNV_CDMA_C0_BC1_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24144 | RFNV_CDMA_C0_BC4_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24145 | RFNV_CDMA_C0_BC5_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24146 | RFNV_CDMA_C0_BC6_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24147 | RFNV_CDMA_C0_BC10_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24148 | RFNV_CDMA_C0_BC14_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24149 | RFNV_CDMA_C0_BC15_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24150 | RFNV_CDMA_C2_BC0_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24151 | RFNV_CDMA_C2_BC1_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24152 | RFNV_CDMA_C2_BC4_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24153 | RFNV_CDMA_C2_BC5_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24154 | RFNV_CDMA_C2_BC6_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24155 | RFNV_CDMA_C2_BC10_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24156 | RFNV_CDMA_C2_BC14_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24157 | RFNV_CDMA_C2_BC15_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24158 | RFNV_LTE_B1_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24159 | RFNV_LTE_B2_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24160 | RFNV_LTE_B3_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24161 | RFNV_LTE_B4_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24162 | RFNV_LTE_B5_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24163 | RFNV_LTE_B7_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24164 | RFNV_LTE_B8_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24165 | RFNV_LTE_B9_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24166 | RFNV_LTE_B11_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24167 | RFNV_LTE_B12_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24168 | RFNV_LTE_B13_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24169 | RFNV_LTE_B17_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24170 | RFNV_LTE_B18_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24171 | RFNV_LTE_B19_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24172 | RFNV_LTE_B20_TX_EPT_DPD_CONFIG_PARAMS_I |

| NV no. | Name |
|---|---|
| 24173 | RFNV_LTE_B21_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24174 | RFNV_LTE_B25_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24175 | RFNV_LTE_B34_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24176 | RFNV_LTE_B38_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24177 | RFNV_LTE_B39_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24178 | RFNV_LTE_B40_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24179 | RFNV_LTE_B41_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24180 | RFNV_WCDMA_2100_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24181 | RFNV_WCDMA_1900_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24182 | RFNV_WCDMA_BC4_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24183 | RFNV_WCDMA_800_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24184 | RFNV_WCDMA_900_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24185 | RFNV_WCDMA_1800_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24186 | RFNV_WCDMA_1500_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24187 | RFNV_TDSCDMA_B34_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24188 | RFNV_TDSCDMA_B39_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24189 | RFNV_TDSCDMA_B40_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24540 | RFNV_LTE_B6_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24541 | RFNV_LTE_B10_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24542 | RFNV_LTE_B14_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24543 | RFNV_LTE_B23_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24544 | RFNV_LTE_B42_TX_EPT_DPD_CONFIG_PARAMS_I |
| 24703 | RFNV_LTE_B26_TX_EPT_DPD_CONFIG_PARAMS_I |
| 25483 | RFNV_LTE_B28_TX_EPT_DPD_CONFIG_PARAMS_I |


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# C Delay Time Config NV Items

Table C-1 lists delay time config NV items.

**Table C-1 Delay time config NV items**

| NV no. | Name |
|---|---|
| 25532 | RFNV_LTE_B1_TX_XPT_DELAY_I |
| 25533 | RFNV_LTE_B2_TX_XPT_DELAY_I |
| 25534 | RFNV_LTE_B3_TX_XPT_DELAY_I |
| 25535 | RFNV_LTE_B4_TX_XPT_DELAY_I |
| 25536 | RFNV_LTE_B5_TX_XPT_DELAY_I |
| 25537 | RFNV_LTE_B6_TX_XPT_DELAY_I |
| 25538 | RFNV_LTE_B7_TX_XPT_DELAY_I |
| 25539 | RFNV_LTE_B8_TX_XPT_DELAY_I |
| 25540 | RFNV_LTE_B9_TX_XPT_DELAY_I |
| 25541 | RFNV_LTE_B10_TX_XPT_DELAY_I |
| 25542 | RFNV_LTE_B11_TX_XPT_DELAY_I |
| 25543 | RFNV_LTE_B12_TX_XPT_DELAY_I |
| 25544 | RFNV_LTE_B13_TX_XPT_DELAY_I |
| 25545 | RFNV_LTE_B14_TX_XPT_DELAY_I |
| 25546 | RFNV_LTE_B15_TX_XPT_DELAY_I |
| 25547 | RFNV_LTE_B16_TX_XPT_DELAY_I |
| 25548 | RFNV_LTE_B17_TX_XPT_DELAY_I |
| 25549 | RFNV_LTE_B18_TX_XPT_DELAY_I |
| 25550 | RFNV_LTE_B19_TX_XPT_DELAY_I |
| 25551 | RFNV_LTE_B20_TX_XPT_DELAY_I |
| 25552 | RFNV_LTE_B21_TX_XPT_DELAY_I |
| 25553 | RFNV_LTE_B22_TX_XPT_DELAY_I |
| 25554 | RFNV_LTE_B23_TX_XPT_DELAY_I |
| 25555 | RFNV_LTE_B24_TX_XPT_DELAY_I |
| 25556 | RFNV_LTE_B25_TX_XPT_DELAY_I |
| 25557 | RFNV_LTE_B26_TX_XPT_DELAY_I |
| 25558 | RFNV_LTE_B27_TX_XPT_DELAY_I |
| 25559 | RFNV_LTE_B28_TX_XPT_DELAY_I |
| 25560 | RFNV_LTE_B29_TX_XPT_DELAY_I |
| 25561 | RFNV_LTE_B30_TX_XPT_DELAY_I |
| 25562 | RFNV_LTE_B31_TX_XPT_DELAY_I |

| NV no. | Name |
|---|---|
| 25563 | RFNV_LTE_B32_TX_XPT_DELAY_I |
| 25564 | RFNV_LTE_B33_TX_XPT_DELAY_I |
| 25565 | RFNV_LTE_B34_TX_XPT_DELAY_I |
| 25566 | RFNV_LTE_B35_TX_XPT_DELAY_I |
| 25567 | RFNV_LTE_B36_TX_XPT_DELAY_I |
| 25568 | RFNV_LTE_B37_TX_XPT_DELAY_I |
| 25569 | RFNV_LTE_B38_TX_XPT_DELAY_I |
| 25570 | RFNV_LTE_B39_TX_XPT_DELAY_I |
| 25571 | RFNV_LTE_B40_TX_XPT_DELAY_I |
| 25572 | RFNV_LTE_B41_TX_XPT_DELAY_I |
| 25573 | RFNV_LTE_B42_TX_XPT_DELAY_I |
| 25574 | RFNV_LTE_B43_TX_XPT_DELAY_I |
| 25575 | RFNV_LTE_B44_TX_XPT_DELAY_I |
| 25507 | RFNV_CDMA_C0_BC0_TX_ET_DELAY_I |
| 25508 | RFNV_CDMA_C0_BC1_TX_ET_DELAY_I |
| 25509 | RFNV_CDMA_C0_BC4_TX_ET_DELAY_I |
| 25510 | RFNV_CDMA_C0_BC5_TX_ET_DELAY_I |
| 25511 | RFNV_CDMA_C0_BC6_TX_ET_DELAY_I |
| 25512 | RFNV_CDMA_C0_BC10_TX_ET_DELAY_I |
| 25513 | RFNV_CDMA_C0_BC14_TX_ET_DELAY_I |
| 25514 | RFNV_CDMA_C0_BC15_TX_ET_DELAY_I |
| 25515 | RFNV_CDMA_C2_BC0_TX_ET_DELAY_I |
| 25516 | RFNV_CDMA_C2_BC1_TX_ET_DELAY_I |
| 25517 | RFNV_CDMA_C2_BC4_TX_ET_DELAY_I |
| 25518 | RFNV_CDMA_C2_BC5_TX_ET_DELAY_I |
| 25519 | RFNV_CDMA_C2_BC6_TX_ET_DELAY_I |
| 25520 | RFNV_CDMA_C2_BC10_TX_ET_DELAY_I |
| 25521 | RFNV_CDMA_C2_BC14_TX_ET_DELAY_I |
| 25522 | RFNV_CDMA_C2_BC15_TX_ET_DELAY_I |
| 25523 | RFNV_WCDMA_2100_TX_ET_DELAY_I |
| 25524 | RFNV_WCDMA_1900_TX_ET_DELAY_I |
| 25525 | RFNV_WCDMA_BC4_TX_ET_DELAY_I |
| 25526 | RFNV_WCDMA_800_TX_ET_DELAY_I |
| 25527 | RFNV_WCDMA_900_TX_ET_DELAY_I |
| 25528 | RFNV_WCDMA_1800_TX_ET_DELAY_I |
| 25529 | RFNV_WCDMA_1500_TX_ET_DELAY_I |
| 26583 | RFNV_CDMA_C0_BC0_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26584 | RFNV_CDMA_C0_BC1_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26585 | RFNV_CDMA_C0_BC4_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26586 | RFNV_CDMA_C0_BC5_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26587 | RFNV_CDMA_C0_BC6_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26588 | RFNV_CDMA_C0_BC10_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |

| NV no. | Name |
|---|---|
| 26589 | RFNV_CDMA_C0_BC14_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26590 | RFNV_CDMA_C0_BC15_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26591 | RFNV_CDMA_C2_BC0_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26592 | RFNV_CDMA_C2_BC1_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26593 | RFNV_CDMA_C2_BC4_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26594 | RFNV_CDMA_C2_BC5_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26595 | RFNV_CDMA_C2_BC6_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26596 | RFNV_CDMA_C2_BC10_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26597 | RFNV_CDMA_C2_BC14_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26598 | RFNV_CDMA_C2_BC15_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26599 | RFNV_WCDMA_2100_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26600 | RFNV_WCDMA_1900_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26601 | RFNV_WCDMA_BC4_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26602 | RFNV_WCDMA_800_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26603 | RFNV_WCDMA_900_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26604 | RFNV_WCDMA_1800_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26605 | RFNV_WCDMA_1500_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26606 | RFNV_LTE_C0_B1_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26607 | RFNV_LTE_C0_B2_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26608 | RFNV_LTE_C0_B3_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26609 | RFNV_LTE_C0_B4_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26610 | RFNV_LTE_C0_B5_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26611 | RFNV_LTE_C0_B6_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26612 | RFNV_LTE_C0_B7_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26613 | RFNV_LTE_C0_B8_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26614 | RFNV_LTE_C0_B9_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26615 | RFNV_LTE_C0_B10_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26616 | RFNV_LTE_C0_B11_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26617 | RFNV_LTE_C0_B12_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26618 | RFNV_LTE_C0_B13_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26619 | RFNV_LTE_C0_B14_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26620 | RFNV_LTE_C0_B15_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26621 | RFNV_LTE_C0_B16_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26622 | RFNV_LTE_C0_B17_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26623 | RFNV_LTE_C0_B18_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26624 | RFNV_LTE_C0_B19_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26625 | RFNV_LTE_C0_B20_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26626 | RFNV_LTE_C0_B21_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26627 | RFNV_LTE_C0_B22_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26628 | RFNV_LTE_C0_B23_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26629 | RFNV_LTE_C0_B24_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26630 | RFNV_LTE_C0_B25_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |


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| NV no. | Name |
|---|---|
| 26621 | RFNV_LTE_C0_B16_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26622 | RFNV_LTE_C0_B17_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26633 | RFNV_LTE_C0_B28_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26634 | RFNV_LTE_C0_B29_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26635 | RFNV_LTE_C0_B30_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26636 | RFNV_LTE_C0_B31_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26637 | RFNV_LTE_C0_B32_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26638 | RFNV_LTE_C0_B33_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26639 | RFNV_LTE_C0_B34_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26640 | RFNV_LTE_C0_B35_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26641 | RFNV_LTE_C0_B36_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26642 | RFNV_LTE_C0_B37_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26643 | RFNV_LTE_C0_B38_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26644 | RFNV_LTE_C0_B39_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |
| 26645 | RFNV_LTE_C0_B40_TX_ET_DELAY_OFFSETS_VS_FREQ_VS_TEMP_I |

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# D Pin/Pout Table

Table D-1 lists Pin/Pout vs temp vs freq NV items.

**Table D-1 Pin/Pout vs temp vs freq NV items**

| NV no. | Name |
|---|---|
| 25585 | RFNV_CDMA_C0_BC0_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25586 | RFNV_CDMA_C0_BC1_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25587 | RFNV_CDMA_C0_BC4_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25588 | RFNV_CDMA_C0_BC5_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25589 | RFNV_CDMA_C0_BC6_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25590 | RFNV_CDMA_C0_BC10_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25591 | RFNV_CDMA_C0_BC14_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25592 | RFNV_CDMA_C0_BC15_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25593 | RFNV_CDMA_C0_BC0_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25594 | RFNV_CDMA_C0_BC1_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25595 | RFNV_CDMA_C0_BC4_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25596 | RFNV_CDMA_C0_BC5_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25597 | RFNV_CDMA_C0_BC6_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25598 | RFNV_CDMA_C0_BC10_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25599 | RFNV_CDMA_C0_BC14_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25600 | RFNV_CDMA_C0_BC15_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25601 | RFNV_CDMA_C2_BC0_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25602 | RFNV_CDMA_C2_BC1_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25603 | RFNV_CDMA_C2_BC4_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25604 | RFNV_CDMA_C2_BC5_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25605 | RFNV_CDMA_C2_BC6_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25606 | RFNV_CDMA_C2_BC10_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25607 | RFNV_CDMA_C2_BC14_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25608 | RFNV_CDMA_C2_BC15_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25609 | RFNV_CDMA_C2_BC0_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25610 | RFNV_CDMA_C2_BC1_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25611 | RFNV_CDMA_C2_BC4_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25612 | RFNV_CDMA_C2_BC5_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25613 | RFNV_CDMA_C2_BC6_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25614 | RFNV_CDMA_C2_BC10_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25615 | RFNV_CDMA_C2_BC14_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |

| NV no. | Name |
|---|---|
| 25616 | RFNV_CDMA_C2_BC15_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25617 | RFNV_WCDMA_2100_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25618 | RFNV_WCDMA_1900_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25619 | RFNV_WCDMA_BC4_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25620 | RFNV_WCDMA_800_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25621 | RFNV_WCDMA_900_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25622 | RFNV_WCDMA_1800_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25623 | RFNV_WCDMA_1500_TX_LIN_VS_TEMP_VS_FREQ_P_IN_I |
| 25624 | RFNV_WCDMA_2100_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25625 | RFNV_WCDMA_1900_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25626 | RFNV_WCDMA_BC4_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25627 | RFNV_WCDMA_800_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25628 | RFNV_WCDMA_900_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25629 | RFNV_WCDMA_1800_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |
| 25630 | RFNV_WCDMA_1500_TX_LIN_VS_TEMP_VS_FREQ_P_OUT_I |

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# E PA_STATIC NV Items

Table E-1 lists PA_STATIC NV items.

**Table E-1 PA_STATIC NV items**

| NV no. | Name |
|---|---|
| 27636 | RFNV_WCDMA_C0_2100_PA_STATIC_I |
| 27637 | RFNV_WCDMA_C0_1900_PA_STATIC_I |
| 27638 | RFNV_WCDMA_C0_BC4_PA_STATIC_I |
| 27639 | RFNV_WCDMA_C0_800_PA_STATIC_I |
| 27640 | RFNV_WCDMA_C0_900_PA_STATIC_I |
| 27641 | RFNV_WCDMA_C0_1800_PA_STATIC_I |
| 27642 | RFNV_WCDMA_C0_1500_PA_STATIC_I |
| 27649 | RFNV_CDMA_C0_BC0_PA_STATIC_I |
| 27650 | RFNV_CDMA_C0_BC1_PA_STATIC_I |
| 27651 | RFNV_CDMA_C0_BC4_PA_STATIC_I |
| 27652 | RFNV_CDMA_C0_BC5_PA_STATIC_I |
| 27653 | RFNV_CDMA_C0_BC6_PA_STATIC_I |
| 27654 | RFNV_CDMA_C0_BC10_PA_STATIC_I |
| 27655 | RFNV_CDMA_C0_BC14_PA_STATIC_I |
| 27656 | RFNV_CDMA_C0_BC15_PA_STATIC_I |
| 27657 | RFNV_CDMA_C2_BC0_PA_STATIC_I |
| 27658 | RFNV_CDMA_C2_BC1_PA_STATIC_I |
| 27659 | RFNV_CDMA_C2_BC4_PA_STATIC_I |
| 27660 | RFNV_CDMA_C2_BC5_PA_STATIC_I |
| 27661 | RFNV_CDMA_C2_BC6_PA_STATIC_I |
| 27662 | RFNV_CDMA_C2_BC10_PA_STATIC_I |
| 27663 | RFNV_CDMA_C2_BC14_PA_STATIC_I |
| 27664 | RFNV_CDMA_C2_BC15_PA_STATIC_I |
| 27753 | RFNV_TDSCDMA_C0_BC34_PA_STATIC_I |
| 27754 | RFNV_TDSCDMA_C0_BC39_PA_STATIC_I |
| 27755 | RFNV_TDSCDMA_C0_BC40_PA_STATIC_I |
| 27550 | RFNV_LTE_C0_B1_PA_STATIC_I |
| 27551 | RFNV_LTE_C0_B2_PA_STATIC_I |
| 27552 | RFNV_LTE_C0_B3_PA_STATIC_I |
| 27553 | RFNV_LTE_C0_B4_PA_STATIC_I |
| 27554 | RFNV_LTE_C0_B5_PA_STATIC_I |

| NV no. | Name |
|---|---|
| 27555 | RFNV_LTE_C0_B6_PA_STATIC_I |
| 27556 | RFNV_LTE_C0_B7_PA_STATIC_I |
| 27557 | RFNV_LTE_C0_B8_PA_STATIC_I |
| 27558 | RFNV_LTE_C0_B9_PA_STATIC_I |
| 27559 | RFNV_LTE_C0_B10_PA_STATIC_I |
| 27560 | RFNV_LTE_C0_B11_PA_STATIC_I |
| 27561 | RFNV_LTE_C0_B12_PA_STATIC_I |
| 27562 | RFNV_LTE_C0_B13_PA_STATIC_I |
| 27563 | RFNV_LTE_C0_B14_PA_STATIC_I |
| 27564 | RFNV_LTE_C0_B15_PA_STATIC_I |
| 27565 | RFNV_LTE_C0_B16_PA_STATIC_I |
| 27566 | RFNV_LTE_C0_B17_PA_STATIC_I |
| 27567 | RFNV_LTE_C0_B18_PA_STATIC_I |
| 27568 | RFNV_LTE_C0_B19_PA_STATIC_I |
| 27569 | RFNV_LTE_C0_B20_PA_STATIC_I |
| 27570 | RFNV_LTE_C0_B21_PA_STATIC_I |
| 27571 | RFNV_LTE_C0_B22_PA_STATIC_I |
| 27572 | RFNV_LTE_C0_B23_PA_STATIC_I |
| 27573 | RFNV_LTE_C0_B24_PA_STATIC_I |
| 27574 | RFNV_LTE_C0_B25_PA_STATIC_I |
| 27575 | RFNV_LTE_C0_B26_PA_STATIC_I |
| 27576 | RFNV_LTE_C0_B27_PA_STATIC_I |
| 27577 | RFNV_LTE_C0_B28_PA_STATIC_I |
| 27578 | RFNV_LTE_C0_B29_PA_STATIC_I |
| 27579 | RFNV_LTE_C0_B30_PA_STATIC_I |
| 27580 | RFNV_LTE_C0_B31_PA_STATIC_I |
| 27581 | RFNV_LTE_C0_B33_PA_STATIC_I |
| 27582 | RFNV_LTE_C0_B34_PA_STATIC_I |
| 27583 | RFNV_LTE_C0_B35_PA_STATIC_I |
| 27584 | RFNV_LTE_C0_B36_PA_STATIC_I |
| 27585 | RFNV_LTE_C0_B37_PA_STATIC_I |
| 27586 | RFNV_LTE_C0_B38_PA_STATIC_I |
| 27587 | RFNV_LTE_C0_B39_PA_STATIC_I |
| 27588 | RFNV_LTE_C0_B40_PA_STATIC_I |
| 27589 | RFNV_LTE_C0_B41_PA_STATIC_I |
| 27590 | RFNV_LTE_C0_B42_PA_STATIC_I |
| 27591 | RFNV_LTE_C0_B43_PA_STATIC_I |
| 27592 | RFNV_LTE_C0_B44_PA_STATIC_I |
| 27593 | RFNV_LTE_C1_B1_PA_STATIC_I |
| 27594 | RFNV_LTE_C1_B2_PA_STATIC_I |
| 27595 | RFNV_LTE_C1_B3_PA_STATIC_I |
| 27596 | RFNV_LTE_C1_B4_PA_STATIC_I |

| NV no. | Name |
|---|---|
| 27597 | RFNV_LTE_C1_B5_PA_STATIC_I |
| 27598 | RFNV_LTE_C1_B6_PA_STATIC_I |
| 27599 | RFNV_LTE_C1_B7_PA_STATIC_I |
| 27600 | RFNV_LTE_C1_B8_PA_STATIC_I |
| 27601 | RFNV_LTE_C1_B9_PA_STATIC_I |
| 27602 | RFNV_LTE_C1_B10_PA_STATIC_I |
| 27603 | RFNV_LTE_C1_B11_PA_STATIC_I |
| 27604 | RFNV_LTE_C1_B12_PA_STATIC_I |
| 27605 | RFNV_LTE_C1_B13_PA_STATIC_I |
| 27606 | RFNV_LTE_C1_B14_PA_STATIC_I |
| 27607 | RFNV_LTE_C1_B15_PA_STATIC_I |
| 27608 | RFNV_LTE_C1_B16_PA_STATIC_I |
| 27609 | RFNV_LTE_C1_B17_PA_STATIC_I |
| 27610 | RFNV_LTE_C1_B18_PA_STATIC_I |
| 27611 | RFNV_LTE_C1_B19_PA_STATIC_I |
| 27612 | RFNV_LTE_C1_B20_PA_STATIC_I |
| 27613 | RFNV_LTE_C1_B21_PA_STATIC_I |
| 27614 | RFNV_LTE_C1_B22_PA_STATIC_I |
| 27615 | RFNV_LTE_C1_B23_PA_STATIC_I |
| 27616 | RFNV_LTE_C1_B24_PA_STATIC_I |
| 27617 | RFNV_LTE_C1_B25_PA_STATIC_I |
| 27618 | RFNV_LTE_C1_B26_PA_STATIC_I |
| 27619 | RFNV_LTE_C1_B27_PA_STATIC_I |
| 27620 | RFNV_LTE_C1_B28_PA_STATIC_I |
| 27621 | RFNV_LTE_C1_B29_PA_STATIC_I |
| 27622 | RFNV_LTE_C1_B30_PA_STATIC_I |
| 27623 | RFNV_LTE_C1_B31_PA_STATIC_I |
| 27624 | RFNV_LTE_C1_B33_PA_STATIC_I |
| 27625 | RFNV_LTE_C1_B34_PA_STATIC_I |
| 27626 | RFNV_LTE_C1_B35_PA_STATIC_I |
| 27627 | RFNV_LTE_C1_B36_PA_STATIC_I |
| 27628 | RFNV_LTE_C1_B37_PA_STATIC_I |
| 27629 | RFNV_LTE_C1_B38_PA_STATIC_I |
| 27630 | RFNV_LTE_C1_B39_PA_STATIC_I |
| 27631 | RFNV_LTE_C1_B40_PA_STATIC_I |
| 27632 | RFNV_LTE_C1_B41_PA_STATIC_I |
| 27633 | RFNV_LTE_C1_B42_PA_STATIC_I |
| 27634 | RFNV_LTE_C1_B43_PA_STATIC_I |
| 27635 | RFNV_LTE_C1_B44_PA_STATIC_I |

# F References

## F.1 Related documents

| Documents | |
|---|---|
| ***Qualcomm Technologies, Inc.*** | |
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| *Presentation: MDM9x15/MSM8960 PA Power Management – Average Power Tracking (Software Topic)* | 80-N7908-1 |
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| *Presentation: MPSS.DPM.2.0 QFE APT+DPD Software Overview* | 80-NK202-5 |

## F.2 Acronyms and terms

| Term | Definition |
|---|---|
| ACLR | Adjacent Channel Leakage Power Ratio |
| ADC | Analog-To-Digital Converter |
| AGC | Automatic Gain Control, Analog Gain Controller |
| APT | Average Power Tracking |
| DAC | Digital-to-Analog Converter |
| DPD | Digital Predistortion |
| ET | Envelope Tracking |
| EPT | Enhanced Power Tracking |
| ESC | Ehanced Sweep Calibration |
| EUL | Enhanced Uplink |
| HDET | High-Power Detector |
| ILPC | Inner Loop Power Control |
| LPM | Low Power Mode |
| LUT | Lookup Table |
| MSB bit | Most Significant Bit (or Byte) |

| Term | Definition |
|---|---|
| M-Line | Magic Line |
| PA | Power Amplifier |
| PAR | Peak-to-Average Ratio |
| RFMD | RF Micro Devices |
| QDART | Qualcomm Development Acceleration Resource Toolkit |
| QFE | Qualcomm (Tunable) Front End |
| QPA | Qualcomm Power Amplifier |
| QRCT | Qualcomm Radio Control Toolkit |
| QSPR | Qualcomm Sequence Profiling Resource |
| RB | Resource Block |
| RMS | Root Mean Square |
| TPL | Tx Power Limit |
| UE | User Equipment |
| VCC | Voice Call Continuity |
| WTR | WLP Transmitter/Receiver |
| WTR BB | WLP Transmitter/Receiver Baseband |

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